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MPQ70165FS

MPSafe™ ASIL-B, 5.5V, PMIC 6 Buck Converters AEC-Q100 Qualified

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

DESCRIPTION

The MPQ70165FS is a Power Management IC designed to meet the power management requirements of processors used in various safety-relevant automotive systems.

Six integrated synchronous buck converters can be configured as 3 rails dual phase or 6 rails single phase converters. With dynamic voltage scaling, the output voltage of each converter can be changed during normal operation. The PMBus interface with PEC and integrated multi-page OTP memory (MOTP) allows for a high degree of configurability.

To prevent overshoot during start-up, soft-start functionality is featured on each converter. In addition, to allow proper discharge during shut-down, soft-stop functionality is featured on each converter. The slew rate for both functions is configurable through the digital registers.

A configurable windowed or Q&A watchdog and dedicated WDI pin, external voltage monitor, and sequence output pin are all featured.

2MHz fixed frequency peak current mode control is used to regulate output voltage, offers fast transient performance, and allows for a large reduction in external inductor and capacitor values. Frequency Spread Spectrum reduces EMI noises. A full protection suite is included and features under voltage lock-out (UVLO), under voltage protection (UVP), over voltage protection (OVP) and thermal shutdown.

Safety mechanisms such as built-in self-test (BIST) provide high diagnostic coverage, which help the system achieve its target ASIL rating. It is developed under MPS' advanced MPSafe™ Functional Safety Product Development process, which has been independently certified to meet ISO26262 guidelines.

The MPQ70165FS is available in 32-pin TQFN 5mmx5mm package.

FEATURES

- **Designed for Automotive Applications**
  - Input Voltage Range from 2.7V to 5.5V
  - Junction Temperature Rating of -40°C to +150°C
- **Reduces Board Size and BOM**
  - 2x 4A Bucks
  - 2x 3A Bucks
  - 2x 1A Bucks
  - Optional Multiphase Operation
  - Dynamic Voltage Scaling
- **Optimized for EMC/EMI**
  - 180° Phase Shift between Bucks 1 / 3 / 6 and Bucks 2 / 4 / 5
  - 2MHz Fixed Switching Frequency
  - Frequency Spread Spectrum
- **Additional Features**
  - Multi-Page OTP Memory
  - External Voltage Monitor
  - Windowed or Q&A Watchdog Timer
  - TQFN-32 (5mmx5mm) Package - Wettable Flank
  - PMBus Interface with PEC
  - AEC-Q100 Grade-1
- **Functional Safety**
  - ASIL-B Hardware Capable
  - Fail-Safe State Machine
  - Built-in Self-Test (BIST)
  - Clock Monitor
  - Voltage Reference Monitor
  - Protected Memory and Registers
  - Documents Available for ISO26262 System Design
  - Failure Modes Effects and Diagnostic Analysis (FMEDA)
  - Safety Manual

MPSafe Developed for Functional Safety Applications: ISO26262 Compliant

APPLICATIONS

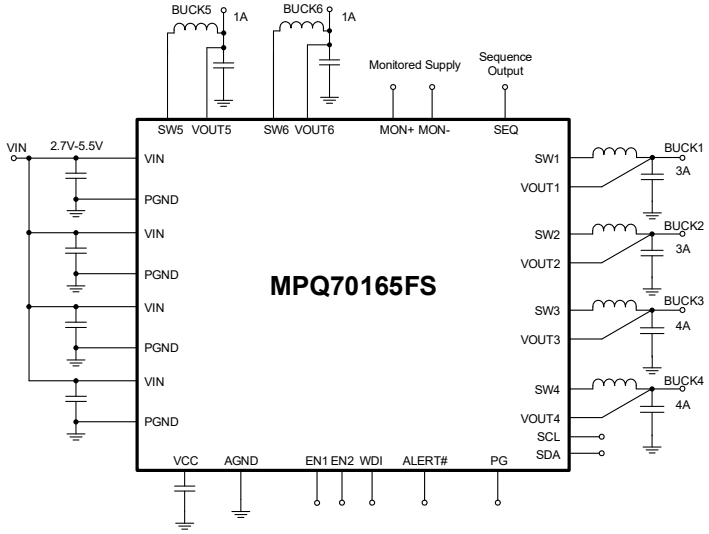
- Advanced Driver Assistance Systems
- ADAS Domain Controller
- Drive Assist ECU

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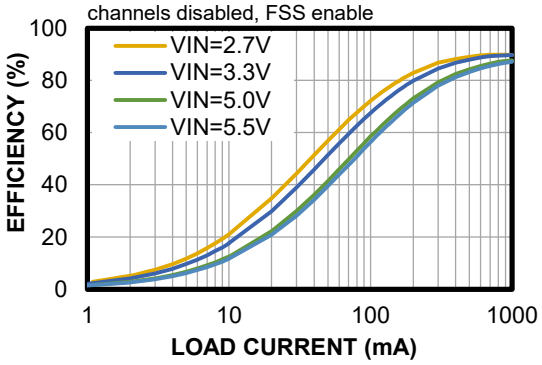
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TYPICAL APPLICATION



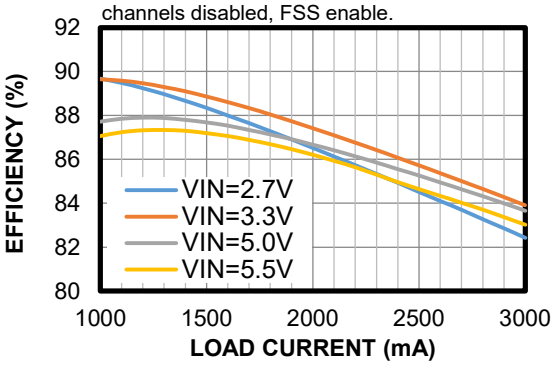
Efficiency vs. Load Current (Buck 1)

V_{OUT1} = 1.0V, f_{SW} = 2MHz, FCCM, L1 = 0.47µH, DCR = 6.3mΩ, VCHA042A-R47MS6, other channels disabled, FSS enable



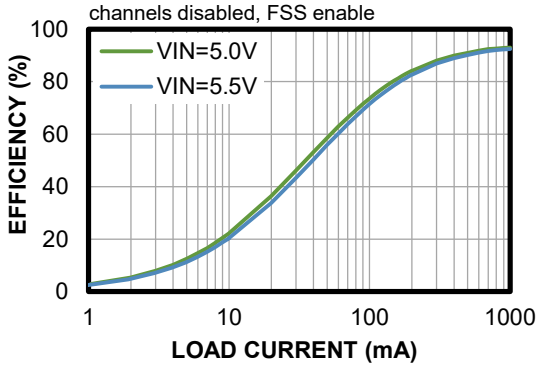
Efficiency vs. Load Current (Buck 1)

V_{OUT1} = 1.0V, f_{SW} = 2MHz, FCCM, L1 = 0.47µH, DCR = 6.3mΩ, VCHA042A-R47MS6, other channels disabled, FSS enable.



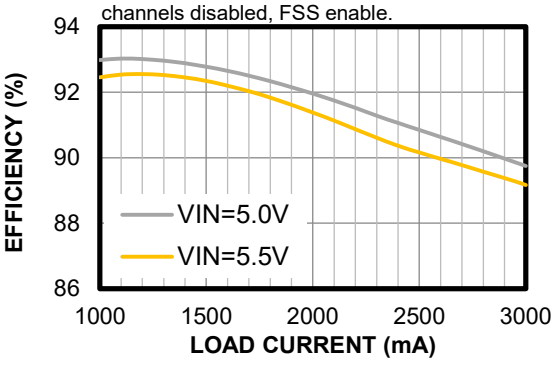
Efficiency vs. Load Current (Buck 2)

V_{OUT2} = 2.0V, f_{SW} =2MHz, FCCM, L2 = 0.68µH, DCR =8.6mΩ, VCHA042A-R68MS6, other channels disabled, FSS enable



Efficiency vs. Load Current (Buck 2)

V_{OUT2} = 2.0V, f_{SW} = 2MHz, FCCM, L2 = 0.68µH, DCR = 8.6mΩ, VCHA042A-R68MS6, other channels disabled, FSS enable.



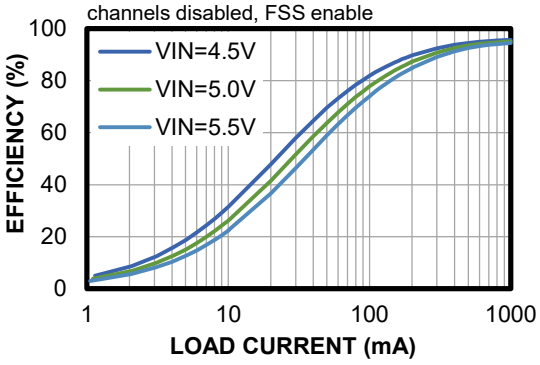
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TYPICAL APPLICATION *(continued)*

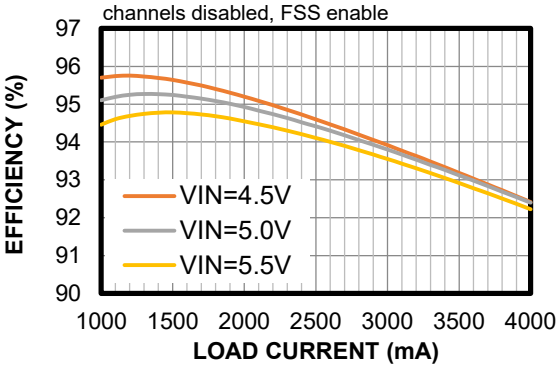
Efficiency vs. Load Current (Buck 3)

V_{OUT3v} = 3.3V, f_{SW} = 2MHz, FCCM, L3 = 0.47µH, DCR = 6.3mΩ, VCHA042A-R47MS6, other channels disabled, FSS enable



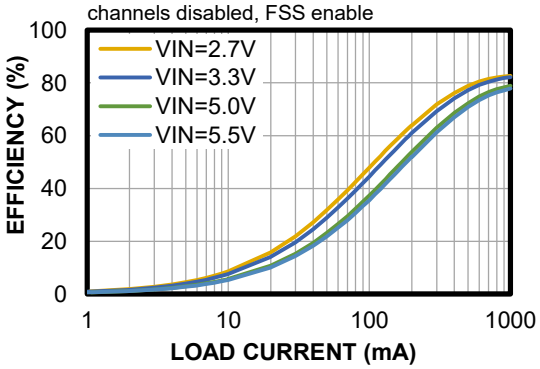
Efficiency vs. Load Current (Buck 3)

V_{OUT3} = 3.3V, f_{SW}=2MHz, FCCM, L3 = 0.47µH, DCR = 6.3mΩ, VCHA042A-R47MS6, other channels disabled, FSS enable



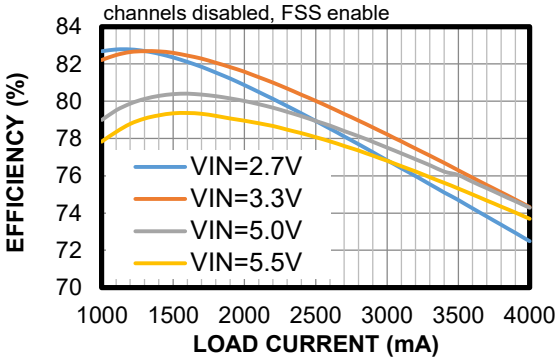
Efficiency vs. Load Current (Buck 4)

V_{OUT4} = 0.5V, f_{SW} = 2MHz, FCCM, L4 = 0.22µH, DCR = 3.4mΩ, XGL4015-221MEB, other channels disabled, FSS enable



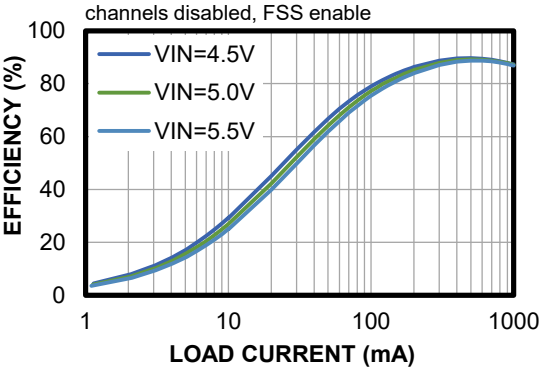
Efficiency vs. Load Current (Buck 4)

V_{OUT4} = 0.5V, f_{SW} = 2MHz, FCCM, L4 = 0.22µH, DCR = 3.4mΩ, XGL4015-221MEB, other channels disabled, FSS enable



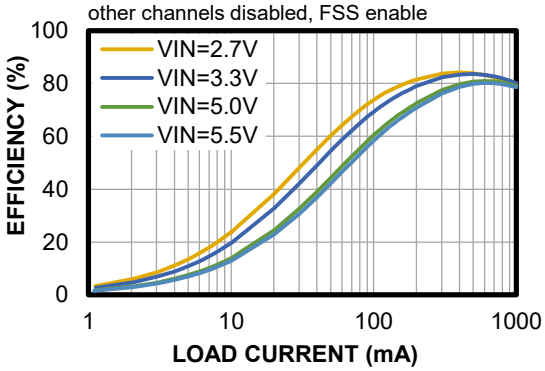
Efficiency vs. Load Current (Buck 5)

V_{OUT5} = 1.8V, f_{SW} = 2MHz, FCCM, L5 = 2.2µH, DCR = 70mΩ, VCTA25201B-2R2MS6, other channels disabled, FSS enable



Efficiency vs. Load Current (Buck 6)

V_{OUT6} = 0.8V, f_{SW} = 2MHz, FCCM, L6 = 1µH, DCR = 35mΩ, VCTA25201B-1R0MS6-99, other channels disabled, FSS enable



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ORDERING INFORMATION

Part Number*	Package	Top Marking	MSL Rating****
MPQ70165FSGUTE-xxxxAEC1**, ***	TQFN-32 (5mmx5mm)	*See Below*	1

* For Tape & Reel, add suffix -Z (e.g. MPQ70165FSGUTE-xxxxAEC1-Z)
** "xxxx" is the configuration code identifier for the register settings stored in the OTP register. Contact an MPS FAE to create this unique number.
*** Wettable Flank.
**** Moisture Sensitivity Level Rating

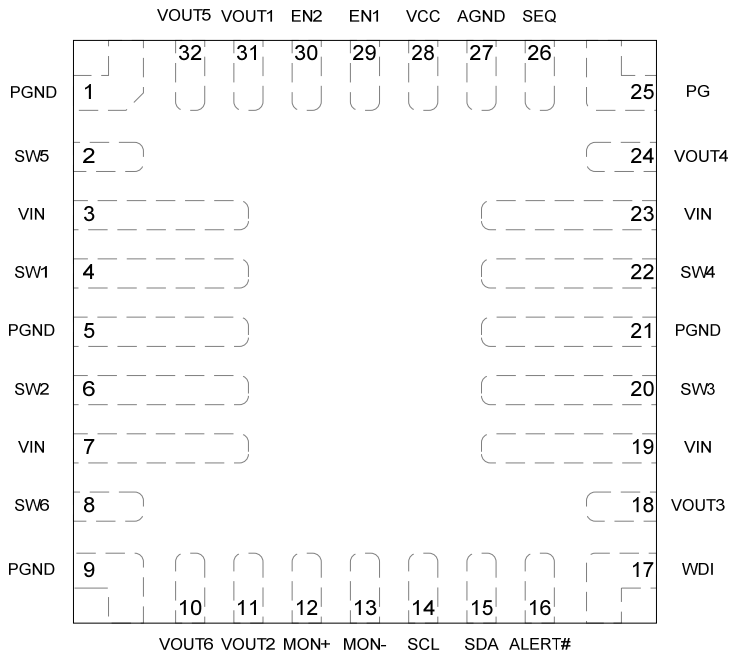
TOP MARKING

MPSYYWW
MP70165
LLLLLLL
E

MPS: MPS prefix
YY: Year code
WW: Week code
MP70165: Part number
LLLLLLL: Lot number
E: Wettable flank

PACKAGE REFERENCE

TOP VIEW



TQFN-32 (5mmx5mm)

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PIN FUNCTIONS

Pin #	Name	Description
1,5,9,21	PGND	**Power ground.** PGND pins are internally connected.
2	SW5	**Switch node 5.** SW5 is the switching node of the internal power switches for buck 5 converter. VIN pin 3 as input pin to buck 5. When buck 5 and buck 6 work in parallel, SW5 shall be connected to the buck 5 inductor. Float or connect a 100kΩ resistor between this pin and GND if not used.
3,7,19,23	VIN	**Input supply voltage.** VIN supplies power to some internal circuits and the power switches. A decoupling capacitor to ground must be placed close to each VIN to minimize switching spikes. VIN pins are not internally connected. It is not allowed to separate different VIN pins, connect VIN pins with copper pours and vias.
4	SW1	**Switch node 1.** SW1 is the switching node of the internal power switches for buck 1 converter. VIN pin 3 as input pin to buck 1. When buck 1 and buck 2 work in parallel, SW1 shall be connected to the buck 1 inductor. Float or connect a 100kΩ resistor between this pin and GND if not used.
6	SW2	**Switch node 2.** SW2 is the switching node of the internal power switches for buck 2 converter. VIN pin 7 as input pin to buck 2. When buck 1 and buck 2 work in parallel, SW2 shall be connected to the buck 2 inductor. Float or connect a 100kΩ resistor between this pin and GND if not used.
8	SW6	**Switch node 6.** SW6 is the switching node of the internal power switches for buck 6 converter. VIN pin 7 as input pin to buck 6. When buck 5 and buck 6 work in parallel, SW6 shall be connected to the buck 6 inductor. Float or connect a 100kΩ resistor between this pin and GND if not used.o
10	VOUT6	**Feedback input pin 6.** Feedback voltage input of buck 6 converter. Connect the VOUT6 pin directly to the buck 6 output. See the VOUT_COMMAND (21h) for V_{OUT6} setting. Float or connect a 100kΩ resistor between this pin and GND if not used.
11	VOUT2	**Feedback input pin 2.** Feedback voltage input of buck 2 converter. Connect the VOUT2 pin directly to the buck 2 output. See the VOUT_COMMAND (21h) for V_{OUT2} setting. Float or connect a 100kΩ resistor between this pin and GND if not used.
12	MON+	**Monitor positive differential input.** Monitor over-voltage and under-voltage for external voltages with programmable threshold. The MPQ70165FS receives the external voltage equal to VMON+ minus VMON-. There is an internal 2MΩ pull-down resistor. When VMON is working, an additional 1.19MΩ pull-down resistor inside. If not used, VMON_ENABLE (D4h bit2) shall be set to disable. In addition, connect this pin to GND if not used.
13	MON-	**Monitor negative differential input.** There is an internal 2MΩ pull-down resistor. It is recommended that connect this pin to GND during normal using. If not used, VMON_ENABLE (D4h bit2) shall be set to disable. In addition, please connect this pin to GND if not used.
14	SCL	**PMBus Serial Clock.** An external pull-up resistor (e.g.1.5kΩ) is needed to connect this pin to PMBus bus supply rail (e.g. V_{CC}). Connect this pin to GND if not used.
15	SDA	**PMBus Serial Data.** An external pull-up resistor (e.g.1.5kΩ) is needed to connect this pin to PMBus bus supply rail (e.g. V_{CC}). Connect this pin to GND if not used.
16	ALERT#	**Open drain fault output.** The output of ALERT# is an open drain of N-channel MOSFET. The default status is high. Connect a pulled-up resistor (e.g.100kΩ) to a stable pull-up supply (e.g. V_{CC}) if used. ALERT# will be pulled low if MPQ70165FS triggers fault. For detail fault, see the Table 1. Float or connect this pin to GND if not used.

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PIN FUNCTIONS *(continued)*

Pin #	Name	Description
17	WDI	**Window watchdog input.** Connect the watchdog trigger of micro controller to this pin in Window Watchdog mode. If select Question-Answer (QA) watchdog mode, the WDI pin will be reused to error input pin (ERRIN). When this pin is high level, MPQ70165FS will report ERRIN fault and shut down all channels. There is an internal 2MΩ pull-down resistor. If not use watchdog and ERRIN function, connect WDI pin to GND.
18	VOUT3	**Feedback input pin 3.** Feedback voltage input of buck 3 converter. Connect the VOUT3 pin directly to the buck 3 output. See the VOUT_COMMAND (21h) for V_{OUT3} setting. Float or connect a 100kΩ resistor between this pin and GND if not used.
20	SW3	**Switch node 3.** SW3 is the switching node of the internal power switches for buck 3 converter. VIN pin 19 as input pin to buck 3. When buck 3 and buck 4 work in parallel, SW3 shall be connected to the buck 3 inductor. Float or connect a 100kΩ resistor between this pin and GND if not used.
22	SW4	**Switch node 4.** SW4 is the switching node of the internal power switches for buck 4 converter. VIN pin 23 as input pin to buck 4. When buck 3 and buck 4 work in parallel, SW4 shall be connected to the buck 4 inductor. Float or connect a 100kΩ resistor between this pin and GND if not used.
24	VOUT4	**Feedback input pin 4.** Feedback voltage input of buck 4 converter. Connect the VOUT4 pin directly to the buck 4 output. See the VOUT_COMMAND (21h) V_{OUT4} setting. Float or connect a 100kΩ resistor between this pin and GND if not used.
25	PG	**Power good indicator.** The output of PG is an open drain of N-channel MOSFET. The default status is low. After all the output voltages are within +4.5%/-3.5% or +6.5%/-5.5% of the nominal voltages and no fault happens, PG status will be high. When MPQ70165FS triggers fault, PG will be asserted. For detail fault, please see the Table 1. Connect a pulled-up resistor (e.g. 100kΩ) to a stable pull-up supply (e.g. V_{CC}) if used. Float or connect this pin to GND if not used.
26	SEQ	**Sequencing digital output.** The output of SEQ is a push-pull structure. The SEQ will be pulled high to V_{CC} when SEQ is logical high. The SEQ will be pulled down to GND when SEQ is logical low. SEQ tracks EN group1 or EN group2 signal (configured via ENABLE_GROUP) with the adjustable delay time (configured via SEQ_DELAY). There is an internal 2MΩ pull-down resistor. If not used, the corresponding enable group (see ENABLE_GROUP) shall be pulled down. In addition, connect this pin to GND if not used.
27	AGND	**Analog ground.** AGND is the reference GND for the internal logic and signal circuit. AGND is internally connected to Power Ground. AGND should be connected to power ground in PCB.
28	VCC	**Internal Bias Supply.** V_{CC} is generated from VIN by the internal 3.35V LDO regulator. It supplies power to some internal circuit. Connect a 10uF capacitor from VCC to AGND with a trace as short as possible.
29	EN1	**Enable domain 1 input.** All bucks, VMON and SEQ can be assigned to enable group 1. Pull this pin below the specified threshold 1.3V shuts the enable group 1. Pull this pin up above the specified threshold 1.4V enables the enable group 1. There is an internal pull-down resistor depends on pin voltage. After V_{CC} is ready, the pull-down resistor is 1.95MΩ when pin voltage is higher than 1.4V and is 1.05MΩ when pin voltage is lower than 1.4V. Connect this pin to GND if not used.
30	EN2	**Enable domain 2 input.** All bucks, VMON and SEQ can be assigned to enable group 2. Pull this pin below the specified threshold 1.3V shuts the enable group 2. Pull this pin up above the specified threshold 1.4V enables the enable group 2. There is an internal pull-down resistor depends on pin voltage. After V_{CC} is ready, the pull-down resistor is 1.95MΩ when pin voltage is higher than 1.4V and is 1.05MΩ when pin voltage is lower than 1.4V. Connect this pin to GND if not used.

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PIN FUNCTIONS *(continued)*

Pin #	Name	Description
31	VOUT1	**Feedback input pin 1.** Feedback voltage input of buck 1 converter. Connect the VOUT1 pin directly to the buck1 output. See the VOUT_COMMAND (21h) for V_{OUT1} setting. Float or connect a 100kΩ resistor between this pin and GND if not used.
32	VOUT5	**Feedback input pin 5.** Feedback voltage input of buck 5 converter. Connect the VOUT5 pin directly to the buck 5 output. See the VOUT_COMMAND (21h) for V_{OUT5} setting. Float or connect a 100kΩ resistor between this pin and GND if not used.

ABSOLUTE MAXIMUM RATINGS (1)

VIN, SW_X.....–0.3V to +6.5V
EN_X, WDI .....–0.3V to +5.0V
All Other Pins .....–0.3V to +4.0V
Continuous Power Dissipation (T_A=+25°C)(2)(6)
TQFN-32(5mmx5mm)..... 5.78W
Operating Junction Temperature ..... +150°C
Lead Temperature ..... +260°C
Storage Temperature ..... -65°C to +150°C

Electrostatic Discharge (ESD) Rating

Human-body model (HBM) .....Class 2(3)
Charged-device model (CDM) .....Class C2b(4)

Recommended Operating Conditions

Supply Voltage V_{IN} .....2.7V to 5.5V
Buck1/2/5/6 Output Voltage..0.20625V to 0.4 x V_{IN}
Buck3/4 Output Voltage.....0.20625V to 3.6V
Operating Junction Temp. (T_J)..-40°C to +150°C

Thermal Resistance θ_{JA} θ_{JC}

TQFN-32 (5mmx5mm)		
JESD51-7	31.7	12 °C/W(5)
EVQ70165FS-UT-00A	21.6	0.88 °C/W(6)

Notes:

1) Exceeding these ratings may damage the device.
2) The maximum allowable power dissipation is a function of the maximum junction temperature T_J (MAX), the junction-to-ambient thermal resistance θ_{JA}, and the ambient temperature T_A. The maximum allowable continuous power dissipation at any ambient temperature is calculated by P_D (MAX) = (T_J (MAX)-T_A)/θ_{JA}. Exceeding the maximum allowable power dissipation will cause excessive die temperature, and the regulator will go into thermal shutdown. Internal thermal shutdown circuitry protects the device from permanent damage.
3) Per AEC-Q100-002.
4) Per AEC-Q100-011.
5) Measured on JESD51-7, 4-layer PCB. The values given in this table are only valid for comparison with other packages and cannot be used for design purposes. These values were calculated in accordance with JESD51-7, and simulated on a specified JEDEC board. They do not represent the performance obtained in an actual application, the value of θ_{JC} shows the thermal resistance from junction-to-case bottom.
6) Measured on MPS standard EVB of EVQ70165FS-UT-00A, 7.6cm*7.6cm, 2-Oz copper, 4-Layer PCB. The value of θ_{JC} shows the thermal resistance from junction-to-case top .

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ELECTRICAL CHARACTERISTICS

V_{IN} = 5V, V_{EN1} = 2V, V_{EN2} = 2V, T_J= -40°C to +150°C, unless otherwise noted. Typical values are at T_J= +25°C.

Parameter	Symbol	Condition	Min	Typ	Max	Units
Input Supply Voltage						
V_{IN} under-voltage lockout rising threshold	$V_{IN_UVLO_RISING}$		2.4	2.6	2.8	V
V_{IN} under-voltage lockout falling threshold	$V_{IN_UVLO_FALLING}$		2.25	2.45	2.65	V
V_{IN} quiescent current	I_{Q_ON}	All bucks on, no switching			20	mA
V_{IN} shutdown current	I_{Q_OFF}	All bucks off, all EN low			150	µA
VCC Regulator						
V_{CC} regulation voltage	V_{CC}	C = 2.2µF, I_{VCC} = 10mA	3.1	3.35	3.6	V
V_{CC} under-voltage lockout rising threshold	$V_{CC_UVLO_RISING}$			2.48		V
V_{CC} under-voltage lockout falling threshold	$V_{CC_UVLO_FALLING}$			2.35		V
Oscillator						
Switching frequency range(7)	f_{SW}		1.8	2	2.2	MHz
Minimum on time(7)	t_{ON_MIN}			60		ns
Minimum off time(7)	t_{OFF_MIN}			60		ns
Spread spectrum modulation frequency spread(7)	f_{SS_Spread}			15		%
Spread spectrum modulation frequency rate(7)	f_{SS_Rate}			9		kHz
Dynamic Output Voltage						
V_{OUT} output range(7)(8)	V_{OUT}	29h bit 0 = 1'b0 VOUT_SL = 1	0.20625		1.8	V
		29h bit 0 = 1'b1 VOUT_SL = 2	0.4125		3.6	V
V_{OUT} accuracy(9)	V_{OUT_ACC}	29h bit 0 = 1'b0 VOUT_SL = 1 V_{OUT} = 600mV, V_{CC} = 3.35V	-1.5		1.5	%
		29h bit 0 = 1'b1 VOUT_SL =1 V_{OUT} = 600mV, V_{CC} = 2.7V	-1.5		1.5	%
V_{OUT} startup slew rate range(7)(8)	$V_{OUTSLEW_ST}$	29h bit 0 = 1'b0 VOUT_SL = 1	1.25		10	mV/µs
		29h bit 0 = 1'b1 VOUT_SL = 2	2.5		20	
V_{OUT} startup slew rate accuracy(9)	$V_{OUTSLEW_ST_ACC}$	29h bit 0 = 1'b0 VOUT_SL = 1 A3h bit 1:0 = 2'b00 $V_{OUTSLEW_ST}$ = 1.25mV/µs	-15		15	%

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ELECTRICAL CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{EN1} = 2V, V_{EN2} = 2V, T_J= -40°C to +150°C, unless otherwise noted. Typical values are at T_J= +25°C.

Parameter	Symbol	Condition	Min	Typ	Max	Units
V_{OUT} shutdown slew rate range(7)(8)	$V_{OUTSLEW_SP}$	29h bit 0 = 1'b0, VOUT_SL = 1, soft stop enabled	1.25		10	mV/µs
		29h bit 0 = 1'b1, VOUT_SL = 2, soft stop enabled	2.5		20	
V_{OUT} shutdown slew rate accuracy(9)	$V_{OUTSLEW_SP_ACC}$	29h bit 0 = 1'b0 VOUT_SL = 1 A5h bit 1:0 = 2'b00 $V_{OUTSLEW_SP}$ = 1.25mV/µs	-15		15	%
V_{OUT} dynamic slew rate range(7)(8)	$V_{OUTSLEW}$	29h bit 0 = 1'b0, VOUT_SL = 1	2.5		20	mV/µs
		29h bit 0 = 1'b1, VOUT_SL = 2	5		40	
V_{OUT} dynamic slew rate accuracy(9)	$V_{OUTSLEW_ACC}$	29h bit 0 = 1'b0 VOUT_SL = 1 A7h bit 1:0 = 2'b00 $V_{OUTSLEW}$ = 2.5mV/µs	-15		15	%
Start up delay time range(7)(8)	t_{ON_DELAY}		0		7750	µs
Start up delay time accuracy(9)		60h bit 4:0 = 4'b0000 t_{ON_DELAY} = 0µs			275	µs
Shutdown delay time range(7)(8)	t_{OFF_DELAY}		0		7750	µs
Shutdown delay time accuracy(9)		61h bit 4:0 = 4'b0000 t_{OFF_DELAY} = 0µs			275	µs
Buck 1						
BUCK1 high side switch on resistance	$R_{DS_HS_BUCK1}$			40	90	mΩ
BUCK1 low side switch on resistance	$R_{DS_LS_BUCK1}$			20	45	mΩ
SW1 switch leakage current	I_{LEAK_BUCK1}				50	µA
SW1 peak current range(8)	$I_{LIM_HS_BUCK1}$		3.0		6.0	A
SW1 peak current accuracy(7)(9)		CEh = 0x00, current limit 100% scaling	4.0	6.2	9.5	A
SW1 valley current range(8)	$I_{LIM_LS_BUCK1}$		2.0		4.0	A
SW1 valley current accuracy(7)(9)		CEh = 0x00, current limit 100% scaling	3.0	3.8	5.1	A
SW1 ZCD current(7)	I_{ZCD_BUCK1}				300	mA
SW1 reverse current range(8)	I_{REV_BUCK1}		1.5		3.0	A
SW1 reverse current accuracy(7)(9)		CEh=0x00, current limit 100% scaling	2	3	4	A
V_{OUT1} impedance	R_{FB1_BUCK1}	29h bit 0 = 1'b0, VOUT_SL = 1	12			kΩ
	R_{FB2_BUCK1}	29h bit 0 = 1'b1, VOUT_SL = 2	12			kΩ
BUCK1 output discharge(9)	R_{DIS_BUCK1}	A9h bit0 = 1'b1, 100Ω output discharge enable		100		Ω

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ELECTRICAL CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{EN1} = 2V, V_{EN2} = 2V, T_J= -40°C to +150°C, unless otherwise noted. Typical values are at T_J= +25°C.

Parameter	Symbol	Condition	Min	Typ	Max	Units
Buck 2						
BUCK2 high side switch on resistance	$R_{DS_HS_BUCK2}$			40	90	mΩ
BUCK2 low side switch on resistance	$R_{DS_LS_BUCK2}$			20	45	mΩ
SW2 switch leakage current	I_{ILEAK_BUCK2}				50	μA
SW2 peak current range(8)	$I_{LIM_HS_BUCK2}$		3.0		6.0	A
SW2 peak current accuracy(7)(9)		CEh=0x00, current limit 100% scaling	4.0	6.2	9.5	A
SW2 valley current range(8)	$I_{LIM_LS_BUCK2}$		2.0		4.0	A
SW2 valley current accuracy(7)(9)		CEh=0x00, current limit 100% scaling	3.0	3.8	5.1	A
SW2 ZCD current(7)	I_{ZCD_BUCK2}				300	mA
SW2 reverse current range(8)	I_{REV_BUCK2}		1.5		3.0	A
SW2 reverse current accuracy(7)(9)		CEh=0x00, current limit 100% scaling	2	3	4	A
V_{OUT2} impedance	R_{FB1_BUCK2}	29h bit 0 = 1'b0, VOUT_SL = 1	12			kΩ
	R_{FB2_BUCK2}	29h bit 0 = 1'b1, VOUT_SL = 2	12			kΩ
BUCK2 output discharge(9)	R_{DIS_BUCK2}	A9h bit0 = 1'b1, 100Ω output discharge enable		100		Ω
Buck 3						
BUCK3 high side switch on resistance	$R_{DS_HS_BUCK3}$			30	55	mΩ
BUCK3 low side switch on resistance	$R_{DS_LS_BUCK3}$			16	32	mΩ
SW3 switch leakage current	I_{ILEAK_BUCK3}				60	μA
SW3 peak current range(8)	$I_{LIM_HS_BUCK3}$		4.5		9.0	A
SW3 peak current accuracy(7)(9)		CEh=0x00, current limit 100% scaling	5.3	8.5	12.7	A
SW3 valley current range(8)	$I_{LIM_LS_BUCK3}$		2.5		5.0	A
SW3 valley current accuracy(7)(9)		CEh=0x00, current limit 100% scaling	4.0	5.0	6.5	A
SW3 ZCD current(7)	I_{ZCD_BUCK3}				600	mA
SW3 reverse current range(8)	I_{REV_BUCK3}		2.0		4.0	A
SW3 reverse current accuracy(7)(9)	I_{REV_BUCK3}	CEh=0x00, current limit 100% scaling	2.5	3.8	5.5	A
V_{OUT3} impedance	R_{FB1_BUCK3}	29h bit 0 = 1'b0, VOUT_SL = 1	12			kΩ
	R_{FB2_BUCK3}	29h bit 0 = 1'b1, VOUT_SL = 2	12			kΩ
BUCK3 output discharge(9)	R_{DIS_BUCK3}	A9h bit0 = 1'b1, 100Ω output discharge enable		100		Ω

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ELECTRICAL CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{EN1} = 2V, V_{EN2} = 2V, T_J= -40°C to +150°C, unless otherwise noted. Typical values are at T_J= +25°C.

Parameter	Symbol	Condition	Min	Typ	Max	Units
Buck 4						
BUCK4 high side switch on resistance	$R_{DS_HS_BUCK4}$			30	55	mΩ
BUCK4 low side switch on resistance	$R_{DS_LS_BUCK4}$			16	32	mΩ
SW4 switch leakage current	I_{ILEAK_BUCK4}				60	µA
SW4 peak current range(8)	$I_{LIM_HS_BUCK4}$		4.5		9.0	A
SW4 peak current accuracy(7)(9)		CEh=0x00, current limit 100% scaling	5.3	8.5	12.7	A
SW4 valley current range(8)	$I_{LIM_LS_BUCK4}$		2.5		5.0	A
SW4 valley current accuracy(7)(9)		CEh=0x00, current limit 100% scaling	4.0	5.0	6.5	A
SW4 ZCD current(7)	I_{ZCD_BUCK4}				600	mA
SW4 reverse current range(8)	I_{REV_BUCK4}		2.0		4.0	A
SW4 reverse current accuracy(7)(9)		CEh=0x00, current limit 100% scaling	2.5	3.8	5.5	A
V_{OUT4} impedance	R_{FB1_BUCK4}	29h bit 0 = 1'b0, VOUT_SL = 1	12			kΩ
	R_{FB2_BUCK4}	29h bit 0 = 1'b1, VOUT_SL = 2	12			kΩ
BUCK4 output discharge(9)	R_{DIS_BUCK4}	A9h bit0 = 1'b1, 100Ω output discharge enable		100		Ω
Buck 5						
BUCK5 high side switch on resistance	$R_{DS_HS_BUCK5}$			130	250	mΩ
BUCK5 low side switch on resistance	$R_{DS_LS_BUCK5}$			75	140	mΩ
SW5 switch leakage current	I_{ILEAK_BUCK5}				20	µA
SW5 peak current range(8)	$I_{LIM_HS_BUCK5}$		0.9		1.8	A
SW5 peak current accuracy(7)(9)		CEh=0x00, current limit 100% scaling	1.2	2.0	2.7	A
SW5 valley current range(8)	$I_{LIM_LS_BUCK5}$		0.8		1.5	A
SW5 valley current accuracy(7)(9)		CEh=0x00, current limit 100% scaling	0.9	1.3	2.0	A
SW5 ZCD current(7)	I_{ZCD_BUCK5}				175	mA
SW5 reverse current range(8)	I_{REV_BUCK5}		0.7		1.1	A
SW5 reverse current accuracy(7)(9)		CEh=0x00, current limit 100% scaling	0.7	1.1	1.4	A
V_{OUT5} impedance	R_{FB1_BUCK5}	29h bit 0 = 1'b0, VOUT_SL = 1	12			kΩ
	$R_{FB2,BUCK5}$	29h bit 0 = 1'b1, VOUT_SL = 2	12			kΩ
BUCK5 output discharge(9)	R_{DIS_BUCK5}	A9h bit0 = 1'b1, 100Ω output discharge enable		100		Ω

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ELECTRICAL CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{EN1} = 2V, V_{EN2} = 2V, T_J= -40°C to +150°C, unless otherwise noted. Typical values are at T_J= +25°C.

Parameter	Symbol	Condition	Min	Typ	Max	Units
Buck 6						
BUCK6 high side switch on resistance	$R_{DS_HS_BUCK6}$			130	250	mΩ
BUCK6 low side switch on resistance	$R_{DS_LS_BUCK6}$			75	140	mΩ
SW6 switch leakage current	I_{ILEAK_BUCK6}				20	μA
SW6 peak current range(8)	$I_{LIM_HS_BUCK6}$		0.9		1.8	A
SW6 peak current accuracy(7)(9)		CEh=0x00, current limit 100% scaling	1.2	2.0	2.7	A
SW6 valley current range(8)	$I_{LIM_LS_BUCK6}$		0.8		1.5	A
SW6 valley current accuracy(7)(9)		CEh=0x00, current limit 100% scaling	0.9	1.3	2.0	A
SW6 ZCD current(7)	I_{ZCD_BUCK6}				175	mA
SW6 reverse current range(8)	I_{REV_BUCK6}		0.7		1.1	A
SW6 reverse current accuracy(7)(9)		CEh=0x00, current limit 100% scaling	0.7	1.1	1.4	A
V_{OUT6} impedance	R_{FB1_BUCK6}	29h bit 0 = 1'b0, VOUT_SL = 1	12			kΩ
	R_{FB2_BUCK6}	29h bit 0 = 1'b1, VOUT_SL = 2	12			kΩ
BUCK6 output discharge(9)	R_{DIS_BUCK6}	A9h bit0 = 1'b1, 100Ω output discharge enable		100		Ω
Thermal Protection						
Thermal warning(7)	T_{TW}		125	140	155	˚C
Thermal warning hysteresis(7)	T_{TW_HYS}			20		˚C
Thermal shutdown(7)	T_{TSD}		155	170	185	˚C
Thermal shutdown hysteresis(7)	T_{TSD_HYS}			20		˚C
Output Voltage Protection						
V_{OUT} over voltage rising threshold(9)	V_{OUT_OV}	29h bit 0 = 1'b0, VOUT_SL = 1, V_{OUT}=1.8V	112.5	115	117.5	%
V_{OUT} over voltage hysteresis(9)	$V_{OUT_OV_HYS}$	29h bit 0 = 1'b0, VOUT_SL = 1, V_{OUT}=1.8V		5.5		%
V_{OUT} under voltage falling threshold(9)	V_{OUT_UV}	29h bit 0 = 1'b0, VOUT_SL = 1, V_{OUT}=1.8V	72.5	75	77.5	%
V_{OUT} under voltage hysteresis(9)	$V_{OUT_UV_HYS}$	29h bit 0 = 1'b0, VOUT_SL = 1, V_{OUT}=1.8V		5		%
Hiccup time range(7)(8)	t_{HIC}		2		8	ms
Hiccup time accuracy(9)		A0h bit1:0 = 2'b00 t_{HIC} = 2ms	1.8	2.0	2.2	ms

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ELECTRICAL CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{EN1} = 2V, V_{EN2} = 2V, T_J= -40°C to +150°C, unless otherwise noted. Typical values are at T_J= +25°C.

Parameter	Symbol	Condition	Min	Typ	Max	Units
Enable						
ENx logic on threshold	V_{EN_LOG}		0.25	0.65	1	V
ENx rising threshold	V_{EN_RISING}		1.25	1.4	1.55	V
ENx falling threshold	$V_{EN_FALLING}$		1.15	1.3	1.45	V
ENx voltage hysteresis	V_{EN_HYS}			100		mV
ENx leakage	I_{EN}	V_{ENx} = 3.3V			12	µA
Built-in Self-Test (BIST)						
EN on to PRE-DIAG delay(7)	t_{EN_DELAY}	EN on detected to PRE-DIAG done			10	ms
EN on to 2nd RESTORE OTP delay(7)	t_{LOGIC_DELAY}	EN on detected to 2nd RESTORE OTP done			6	ms
PRE-DIAG time	$t_{PRE-DIAG}$	Include Clock check, ABIST, OTP CRC check			4	ms
ABIST time	t_{ABIST}				2.5	ms
Power Good						
PG over voltage threshold, rising(9)	$V_{OUT_PG_OV}$	29h bit 0 = 1'b0, VOUT_SL = 1, V_{OUT}=1.8V PG_THRESHOLD[0] = 1, percentage of actual V_{REF}	102.5	104.5	106.5	%
		29h bit 0 = 1'b0, VOUT_SL = 1, V_{OUT}=1.8V PG_THRESHOLD[0] = 0, percentage of actual V_{REF}	104.5	106.5	108.5	%
PG over voltage threshold, hysteresis(9)	$V_{OUT_PG_OV_HYS}$	29h bit 0 = 1'b0, VOUT_SL = 1, V_{OUT}=1.8V		0.5		%
PG under voltage threshold, falling(9)	$V_{OUT_PG_UV}$	29h bit 0 = 1'b0, VOUT_SL = 1, V_{OUT}=1.8V PG_THRESHOLD[0] = 1, percentage of actual V_{REF}	94.5	96.5	98.5	%
		29h bit 0 = 1'b0, VOUT_SL = 1, V_{OUT}=1.8V PG_THRESHOLD[0] = 0, percentage of actual V_{REF}	92.5	94.5	96.5	%
PG under voltage threshold, hysteresis(9)	$V_{OUT_PG_UV_HYS}$	29h bit 0 = 1'b0, VOUT_SL = 1, V_{OUT}=1.8V		0.5		%
PG sink capability	V_{PG}	I_{PG} = 4mA			300	mV
PG leakage	I_{PG}	V_{PG} = 3.3V			200	nA
PG delay range(7)(8)	t_{PG_DELAY}		0		10	ms
PG delay time accuracy(9)		C8h bit1:0 = 2'b01 t_{PG_DELAY} = 2ms	1.8	2.0	2.2	ms
ALERT#						
ALERT# sink capability	$V_{ALERT\#}$	I_{RESET} = 4mA			300	mV
ALERT# leakage	$I_{ALERT\#}$	$V_{ALERT\#}$ = 3.3V			200	nA
ALERT# delay range(7)(8)	$t_{ALERT\#_DELAY}$		0		10	ms

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ELECTRICAL CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{EN1} = 2V, V_{EN2} = 2V, T_J= -40°C to +150°C, unless otherwise noted. Typical values are at T_J= +25°C.

Parameter	Symbol	Condition	Min	Typ	Max	Units
ALERT# delay time accuracy(9)		C8h bit3:2 = 2'b10 $t_{ALERT\#_DELAY}$ = 5ms	4.5	5.0	5.5	ms
Reference Monitor						
Reference over voltage threshold(9)	V_{REF_OV}	VOUT_SL = 1, V_{OUT} = 1.8V	102	103	104	%
Reference under voltage threshold(9)	V_{REF_UV}	VOUT_SL = 1, V_{OUT} = 1.8V	96	97	98	%
System Clock(7)						
System clock fast monitor threshold	f_{CLK_FAST}		25	30	35	%
System clock slow monitor threshold	f_{CLK_SLOW}		25	30	35	%
Watch Dog						
Watch dog open window range(7)(8)	t_{WD_OPEN}		1		12	ms
Watch dog open window accuracy	$t_{WD_OPEN_ACC}$		-10		10	%
Watch dog closed window range(7)(8)	t_{WD_CLS}		1		12	ms
Watch dog closed window accuracy	$t_{WD_CLS_ACC}$		-10		10	%
WDI deglitch time(7)	t_{WD_DG}		20	30	40	µs
WDI high threshold	V_{WDIH}		2			V
WDI low threshold	V_{WDIL}				0.8	V
WDI leakage	I_{WDI}	V_{WDI} = 3.3V			10	µA
Auxiliary Monitor						
V_{MON} range(7)(8)	V_{MON}	D4h bit 0 = 1'b1	0.2		3.3875	V
		D4h bit 0 = 1'b0	0.4125		3.6	V
V_{MON} accuracy(9)	V_{MON_ACC}	D4h bit 0 = 1'b0, V_{MON} threshold ≥ 1.3V	-2.5		2.5	%
		D4h bit 0 = 1'b0, V_{MON} threshold < 1.3V	-32.5		32.5	mV
MON pin impedance	R_{MON}	Original buffer factor	0.4			MΩ
Sequencer Output						
Sequencer out high voltage	V_{SEQH}	I_{SEQ} = 1mA	V_{CC} - 200mV		V_{CC}	
Sequencer out low voltage	V_{SEQL}	I_{SEQ} = -1mA	0		150	mV
Sequencer delay time range(8)	t_{SEQ_DELAY}		0		7750	µs

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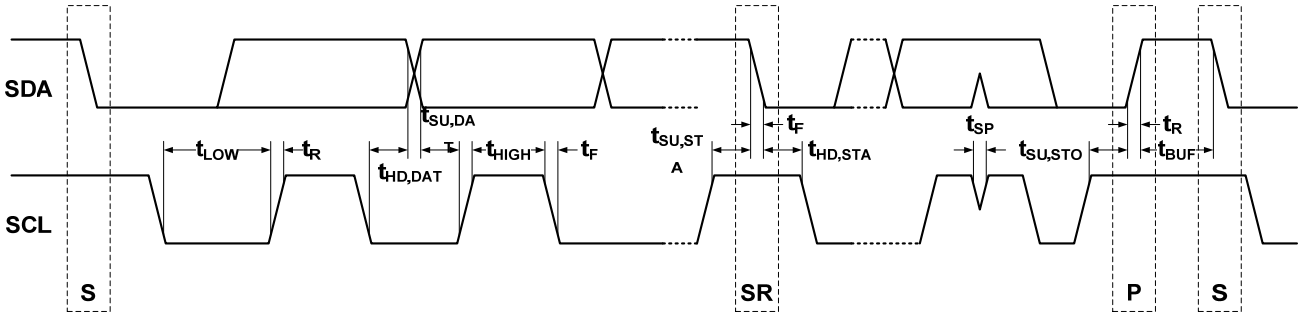
ELECTRICAL CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{EN1} = 2V, V_{EN2} = 2V, T_J= -40°C to +150°C, unless otherwise noted. Typical values are at T_J= +25°C.

Parameter	Symbol	Condition	Min	Typ	Max	Units
PMBus Logic Interface(7)						
SCL/SDA input logic low	V_{IL}		0		0.4	V
SCL/SDA input logic high	V_{IH}		1.2			V
SDA output logic low	V_{OL}	I_{LOAD} = 3mA			0.4	V
SCL clock frequency	f_{SCL}				1000	kHz
SCL high time	t_{HIGH}		0.26			µs
SCL low time	t_{LOW}		0.5			µs
Data setup time	t_{SU_DAT}		50			ns
Data hold time	t_{HD_DAT}		0			µs
Setup time for repeated start	t_{SU_STA}		0.26			µs
Hold time for start	t_{HD_STA}		0.26			µs
Bus free time between a start and a stop condition	t_{BUF}		0.5			µs
Setup time for stop condition	t_{SU_STO}		0.26			µs
Rise time of SCL/SDA	t_R				120	ns
Fall time of SCL/SDA	t_F	V_{DD} supplies the digital interface	20 x (V_{DD} / 5.5V)		120	ns
Pulse width of suppressed spike	t_{SP}		0		50	ns
Capacitance bus for each bus line	C_B				550	pF

Notes:

7) Derived from bench characterization, not tested in production.
8) Configurable via the PMBus interface.
9) Not tested in the entire option range. Only guaranteed for the specified option, which has been listed in column of condition.



S: Start Condition
SR: Repeated Start Condition
P: Stop Condition

Figure 1: PMBus Compatible Interface Timing Diagram

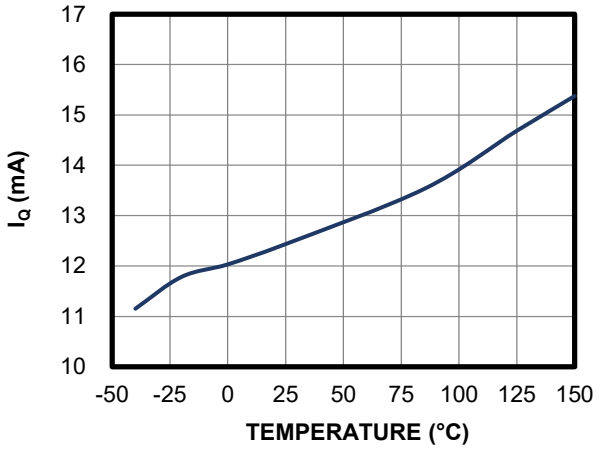
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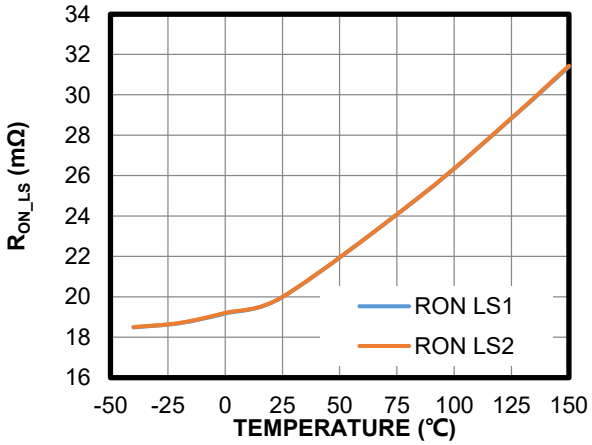
TYPICAL CHARACTERISTICS

V_{IN} = 5V, T_J = -40°C to +150°C, unless otherwise noted.

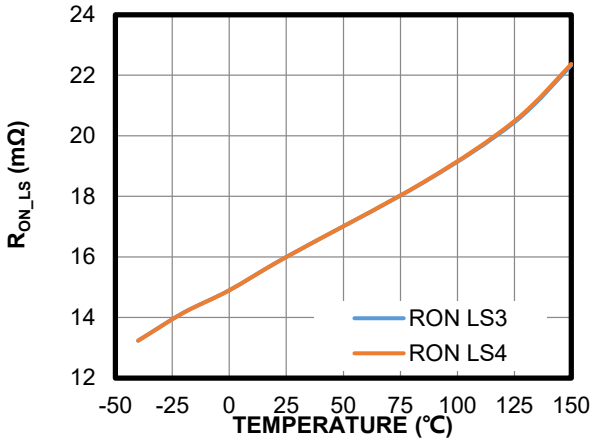
I_Q vs. Temperature



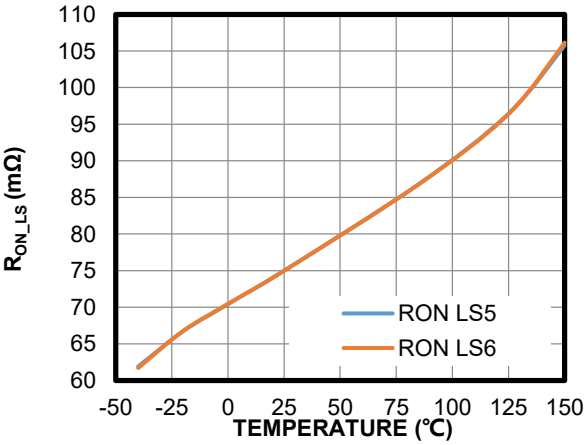
R_{LS_ON} vs. Temperature (Buck1/2)



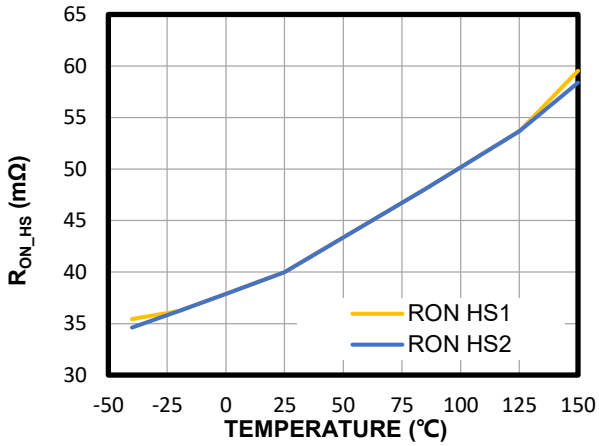
R_{LS_ON} vs. Temperature (Buck3/4)



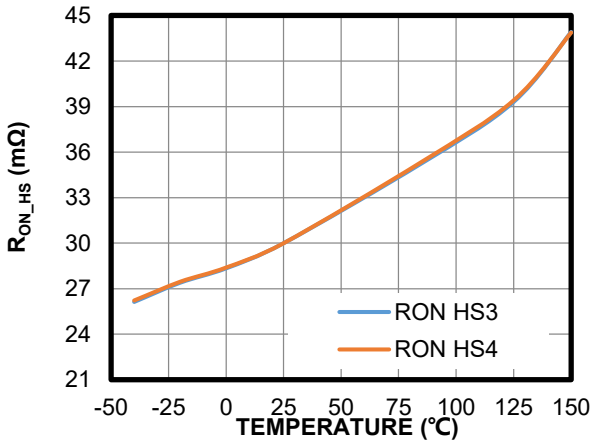
R_{LS_ON} vs. Temperature (Buck5/6)



R_{HS_ON} vs. Temperature (Buck1/2)



R_{HS_ON} vs. Temperature (Buck3/4)



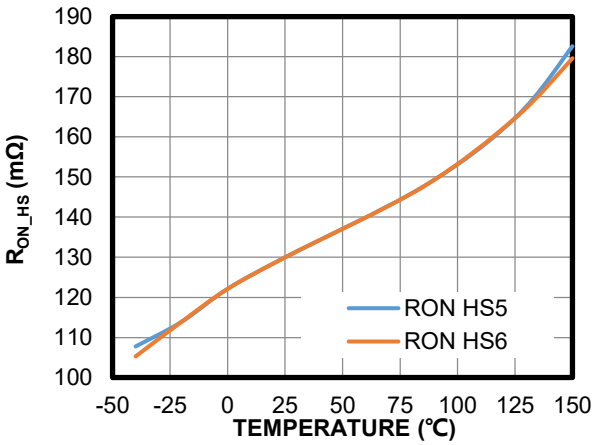
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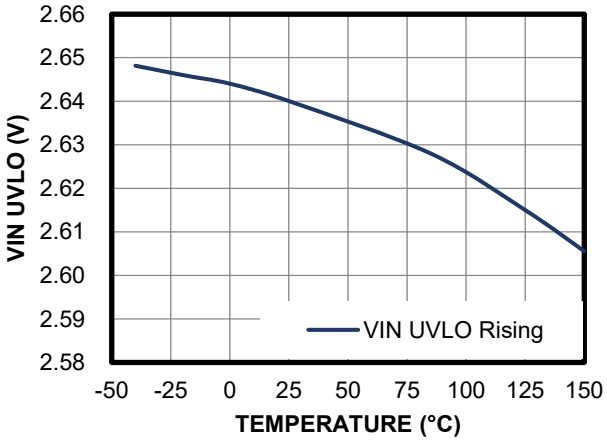
TYPICAL CHARACTERISTICS *(continued)*

V_{IN} = 5V, T_J = -40°C to +150°C, unless otherwise noted.

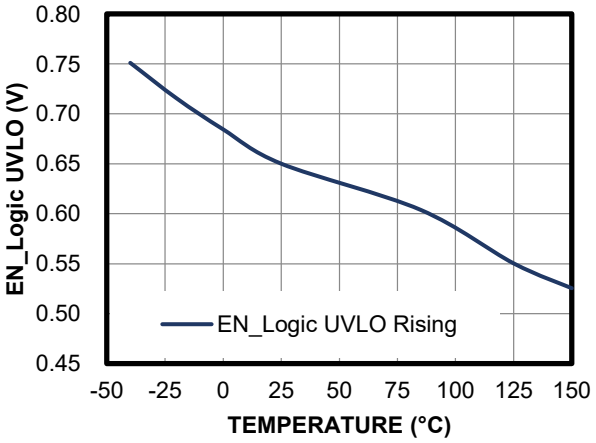
R_{HS_ON} vs. Temperature (Buck5/6)



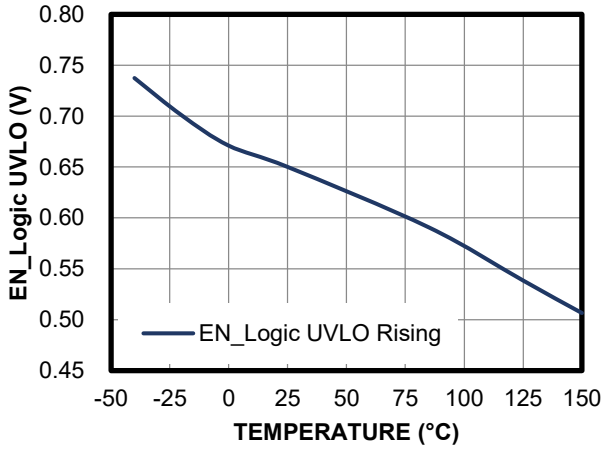
VIN UVLO Threshold vs. Temperature



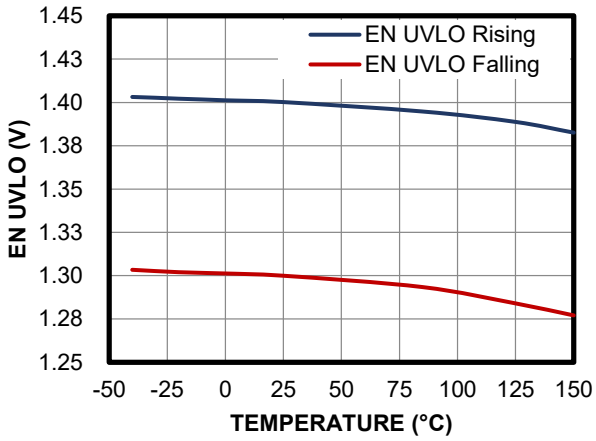
EN1_Logic UVLO Threshold vs. Temperature



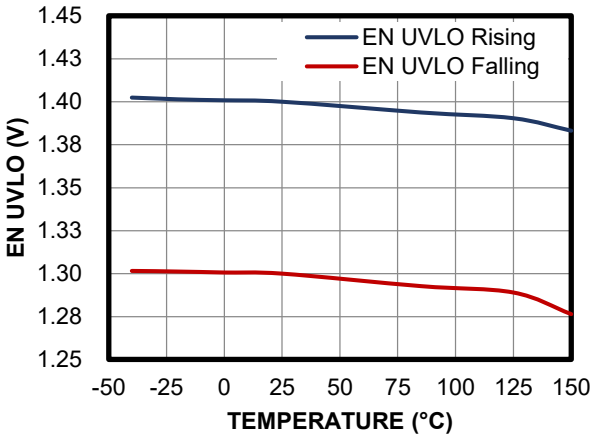
EN2_Logic UVLO Threshold vs. Temperature



EN1 UVLO Threshold vs. Temperature



EN2 UVLO Threshold vs. Temperature



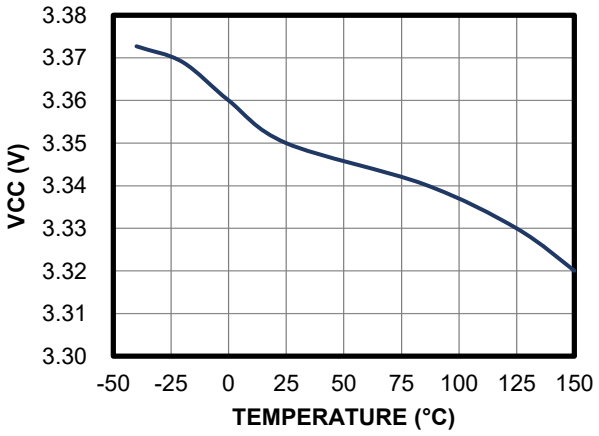
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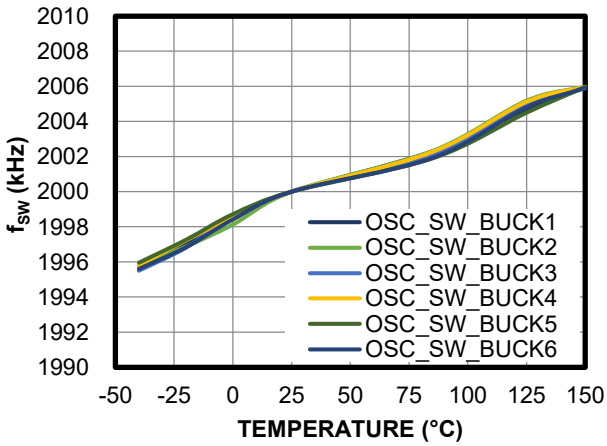
TYPICAL CHARACTERISTICS *(continued)*

V_{IN} = 5V, T_J = -40°C to +150°C, unless otherwise noted.

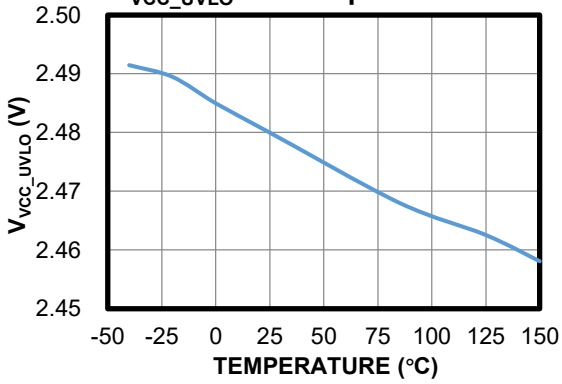
VCC vs. Temperature



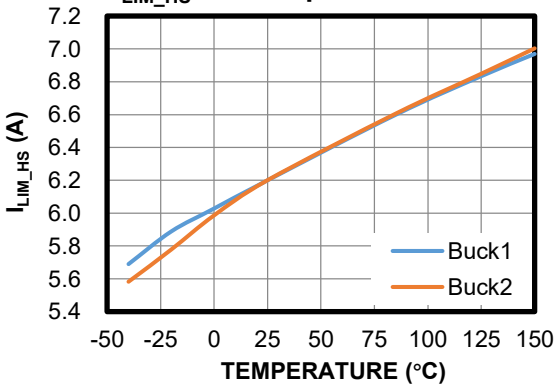
f_{SW} vs. Temperature



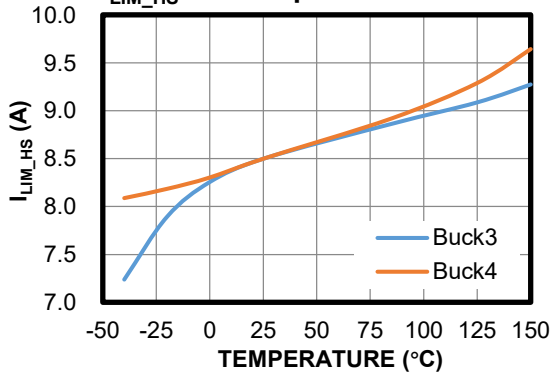
V_{VCC_UVLO} vs. Temperature



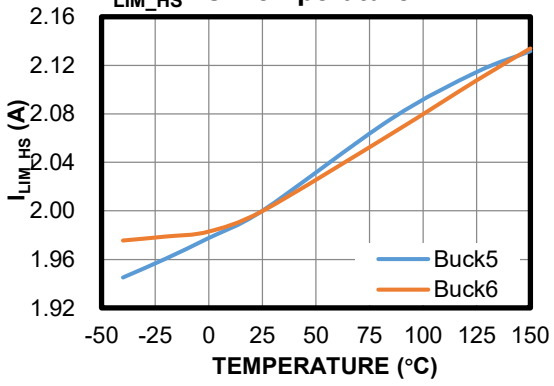
I_{LIM_HS} vs. Temperature



I_{LIM_HS} vs. Temperature



I_{LIM_HS} vs. Temperature

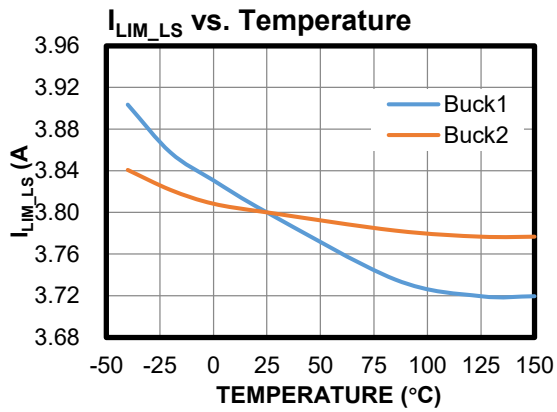


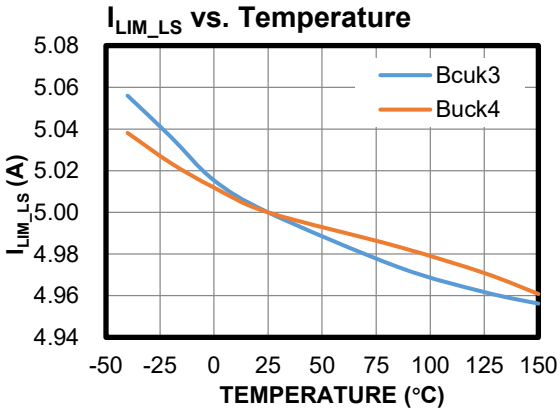
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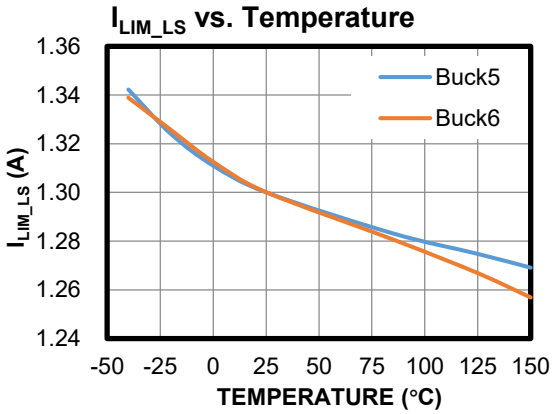
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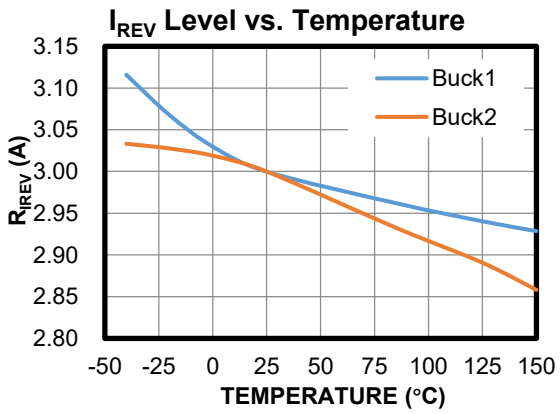
TYPICAL CHARACTERISTICS *(continued)*

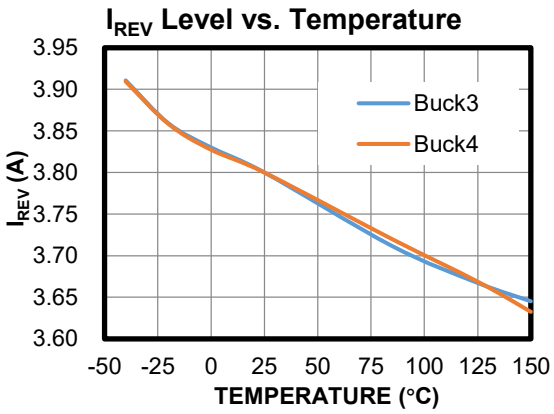
V_{IN} = 5V, T_J = -40°C to +150°C, unless otherwise noted.

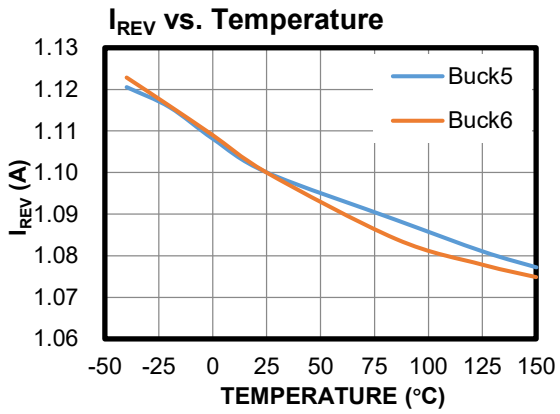












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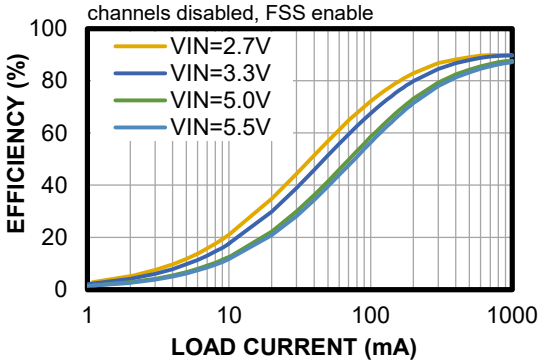
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TYPICAL PERFORMANCE CHARACTERISTICS

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

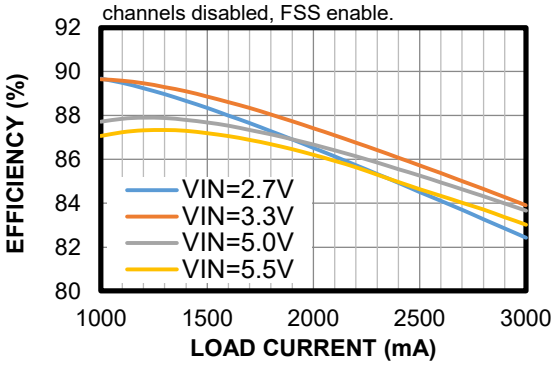
Efficiency vs. Load Current (Buck 1)

V_{OUT1} = 1.0V, f_{SW} = 2MHz, FCCM, L1 = 0.47µH, DCR = 6.3mΩ, VCHA042A-R47MS6, other channels disabled, FSS enable



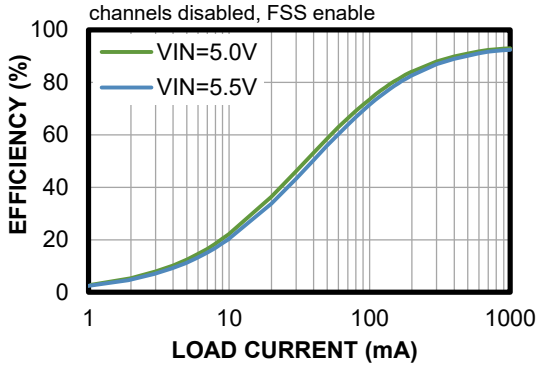
Efficiency vs. Load Current (Buck 1)

V_{OUT1} = 1.0V, f_{SW} = 2MHz, FCCM, L1 = 0.47µH, DCR = 6.3mΩ, VCHA042A-R47MS6, other channels disabled, FSS enable.



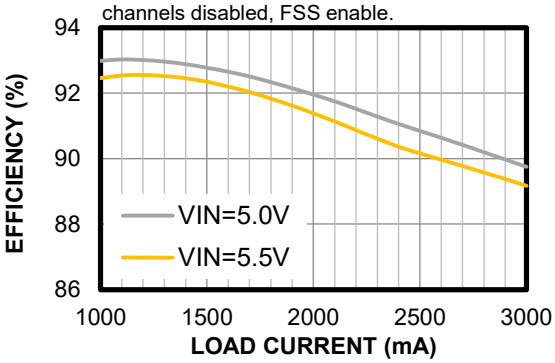
Efficiency vs. Load Current (Buck 2)

V_{OUT2} = 2.0V, f_{SW} =2MHz, FCCM, L2 = 0.68µH, DCR =8.6mΩ, VCHA042A-R68MS6, other channels disabled, FSS enable



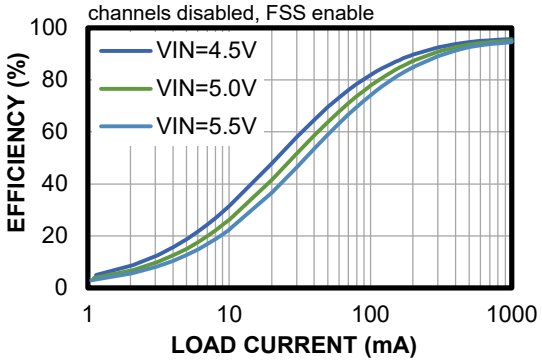
Efficiency vs. Load Current (Buck 2)

V_{OUT2} = 2.0V, f_{SW} = 2MHz, FCCM, L2 = 0.68µH, DCR = 8.6mΩ, VCHA042A-R68MS6, other channels disabled, FSS enable.



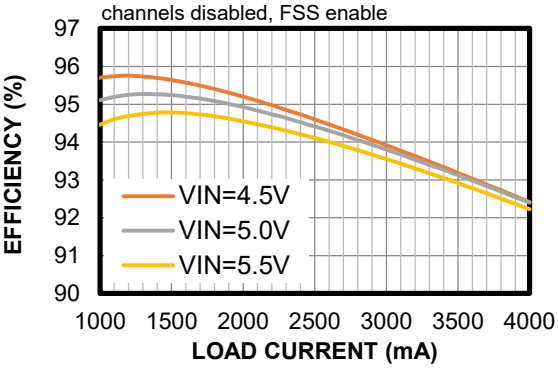
Efficiency vs. Load Current (Buck 3)

V_{OUT3v}= 3.3V, f_{SW} = 2MHz, FCCM, L3 = 0.47µH, DCR = 6.3mΩ, VCHA042A-R47MS6, other channels disabled, FSS enable



Efficiency vs. Load Current (Buck 3)

V_{OUT3} = 3.3V, f_{SW}=2MHz, FCCM, L3 = 0.47µH, DCR = 6.3mΩ, VCHA042A-R47MS6, other channels disabled, FSS enable



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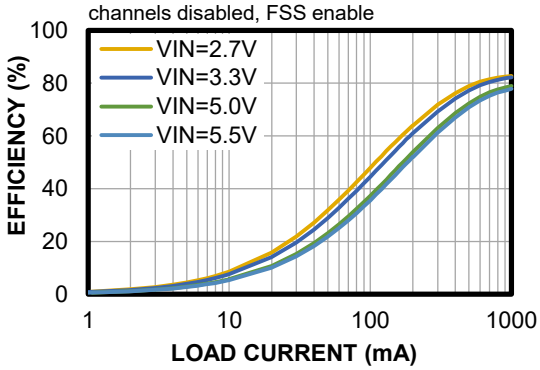
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

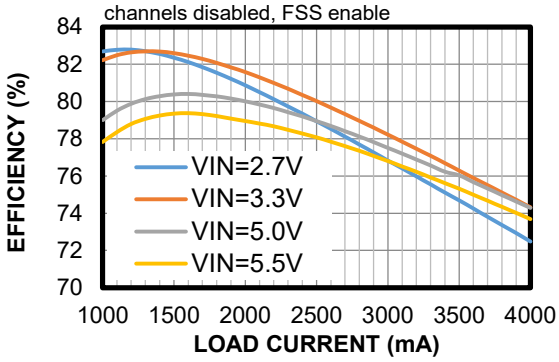
Efficiency vs. Load Current (Buck 4)

V_{OUT4} = 0.5V, f_{SW} = 2MHz, FCCM, L4 = 0.22µH, DCR = 3.4mΩ, XGL4015-221MEB, other channels disabled, FSS enable



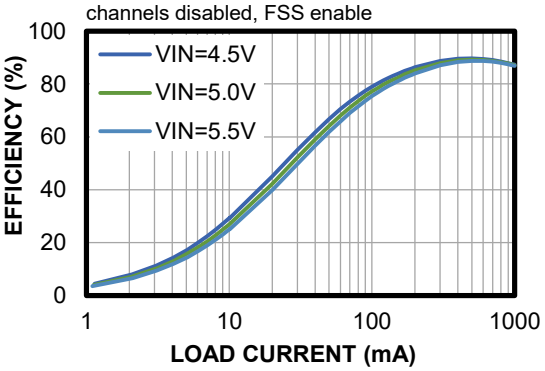
Efficiency vs. Load Current (Buck 4)

V_{OUT4} = 0.5V, f_{SW} = 2MHz, FCCM, L4 = 0.22µH, DCR = 3.4mΩ, XGL4015-221MEB, other channels disabled, FSS enable



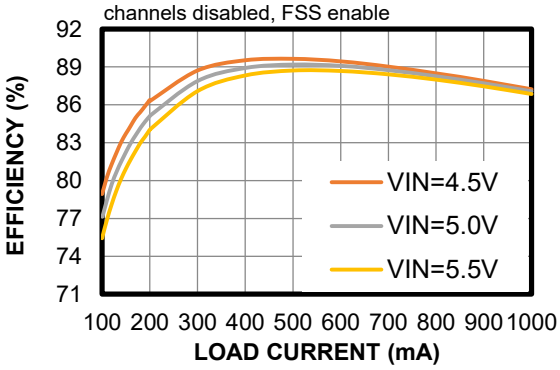
Efficiency vs. Load Current (Buck 5)

V_{OUT5} = 1.8V, f_{SW} = 2MHz, FCCM, L5 = 2.2µH, DCR = 70mΩ, VCTA25201B-2R2MS6, other channels disabled, FSS enable



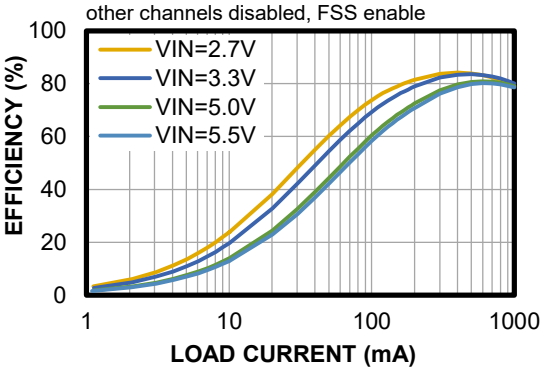
Efficiency vs. Load Current (Buck 5)

V_{OUT5} = 1.8V, f_{SW} = 2MHz, FCCM, L5 = 2.2µH, DCR = 70mΩ, VCTA25201B-2R2MS6, other channels disabled, FSS enable



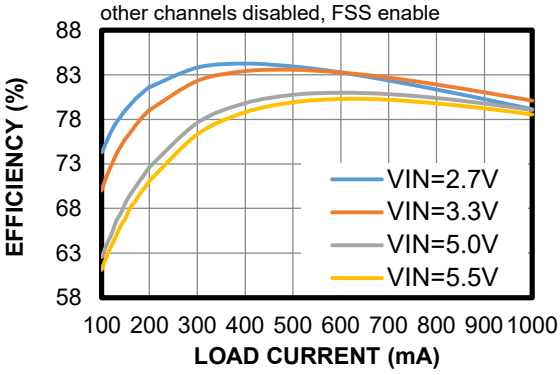
Efficiency vs. Load Current (Buck 6)

V_{OUT6} = 0.8V, f_{SW} = 2MHz, FCCM, L6 = 1µH, DCR = 35mΩ, VCTA25201B-1R0MS6-99, other channels disabled, FSS enable



Efficiency vs. Load Current (Buck 6)

V_{OUT6} = 0.8V, f_{SW} = 2MHz, FCCM, L6 = 1µH, DCR = 35mΩ, VCTA25201B-1R0MS6-99, other channels disabled, FSS enable



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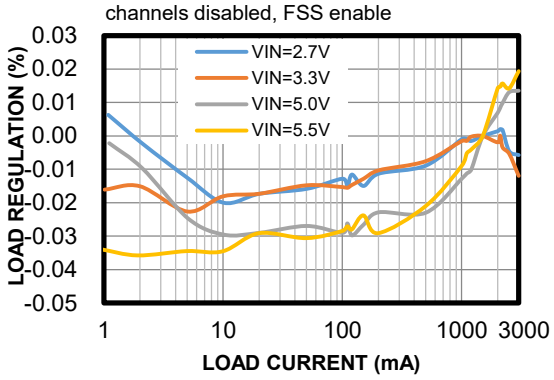
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

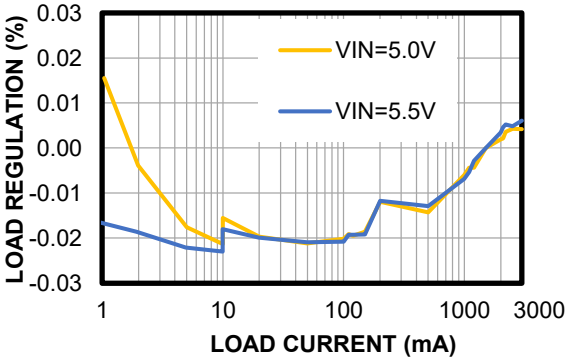
Load Regulation (Buck 1)

V_{OUT1} = 1.0V, f_{SW} = 2MHz, FCCM, L1 = 0.47µH, DCR = 6.3mΩ, VCHA042A-R47MS6, other channels disabled, FSS enable



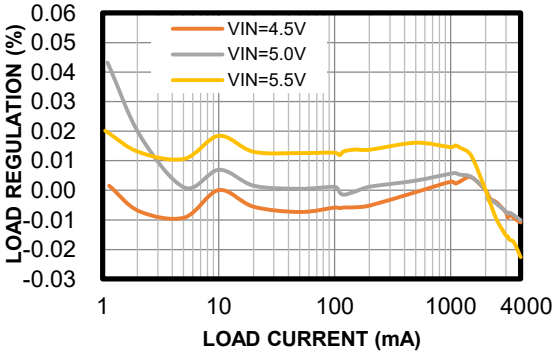
Load Regulation (Buck 2)

V_{OUT2} = 2.0V, f_{SW} = 2MHz, FCCM, L2 = 0.68µH, DCR = 8.6mΩ, VCHA042A-R68MS6, other channels disabled, FSS enable



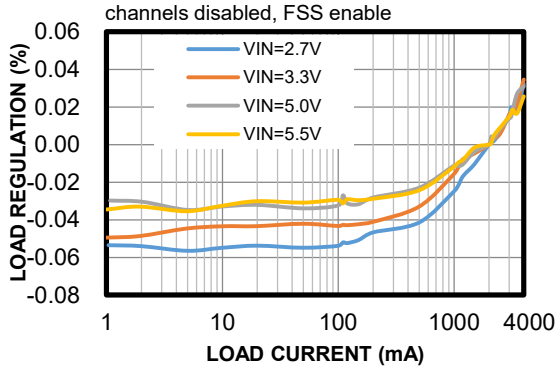
Load Regulation (Buck 3)

V_{OUT3} = 3.3V, f_{SW} = 2MHz, FCCM, L3 = 0.47µH, DCR = 6.3mΩ, VCHA042A-R47MS6, other channels disabled, FSS enable



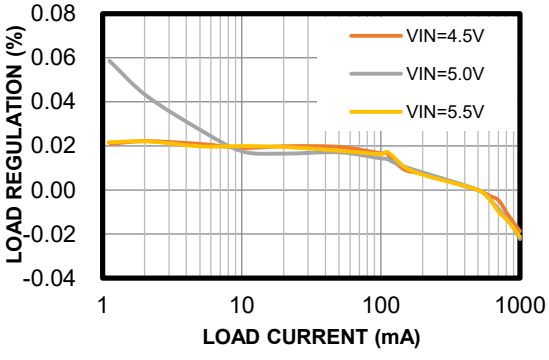
Load Regulation (Buck 4)

V_{OUT4} = 0.5V, f_{SW} = 2MHz, FCCM, L4 = 0.22µH, DCR=3.4mΩ, XGL4015-221MEB, other channels disabled, FSS enable



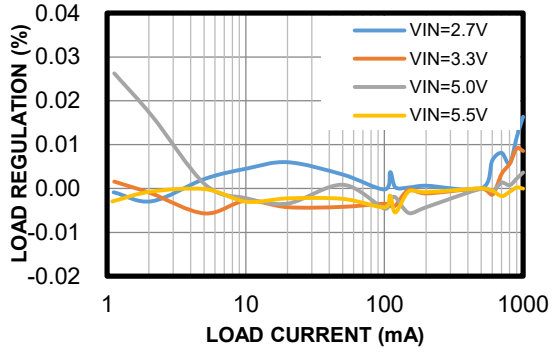
Load Regulation (Buck 5)

V_{OUT5} = 1.8V, f_{SW} = 2MHz, FCCM, L5 = 2.2µH, DCR = 70mΩ, VCTA25201B-2R2MS6, other channels disabled, FSS enable



Load Regulation (Buck 6)

V_{OUT6} = 0.8V, f_{SW} = 2MHz, FCCM, L6 = 1µH, DCR = 35mΩ, VCTA25201B-1R0MS6-99, other channels disabled, FSS enable



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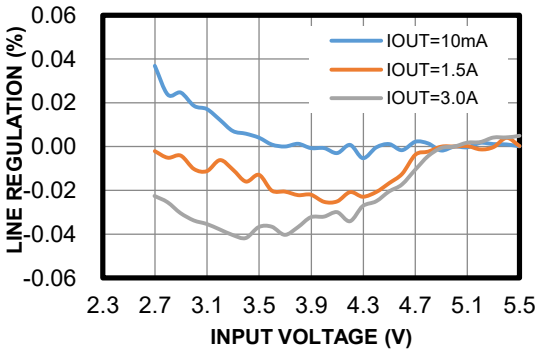
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

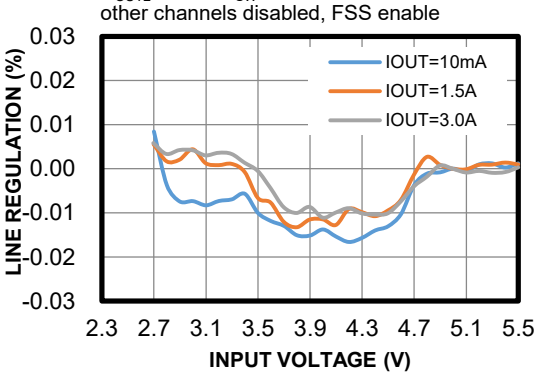
Line Regulation (Buck 1)

V_{OUT1} = 1.0V, f_{SW} = 2MHz, FCCM, other channels disabled, FSS enable



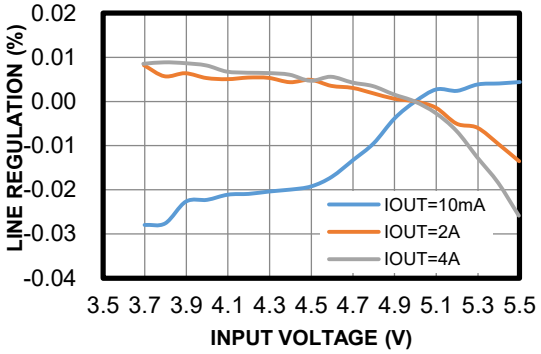
Line Regulation (Buck 2)

V_{OUT2} = 2.0V, f_{SW} = 2MHz, FCCM, other channels disabled, FSS enable



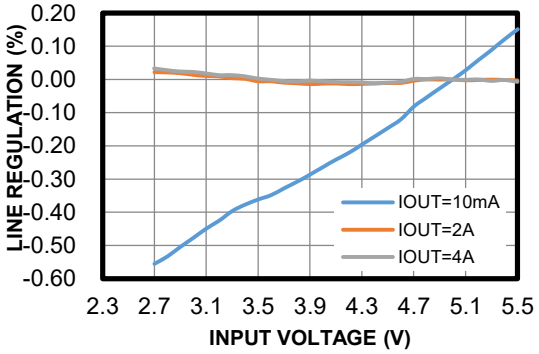
Line Regulation (Buck 3)

V_{OUT3} = 3.3V, f_{SW} = 2MHz, FCCM, other channels disabled, FSS enable



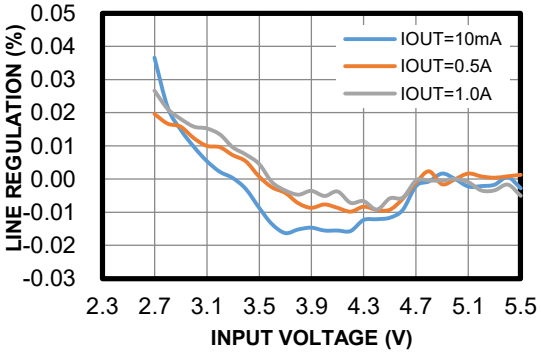
Line Regulation (Buck 4)

V_{OUT4} = 0.5V, f_{SW} = 2MHz, FCCM, other channels disabled, FSS enable



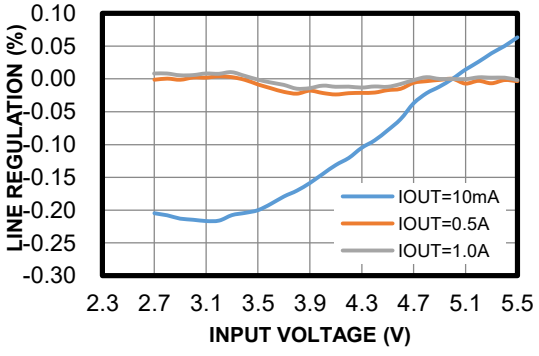
Line Regulation (Buck 5)

V_{OUT5} = 1.8V, f_{SW} = 2MHz, FCCM, other channels disabled, FSS enable



Line Regulation (Buck 6)

V_{OUT6} = 0.8V, f_{SW} = 2MHz, FCCM, other channels disabled, FSS enable



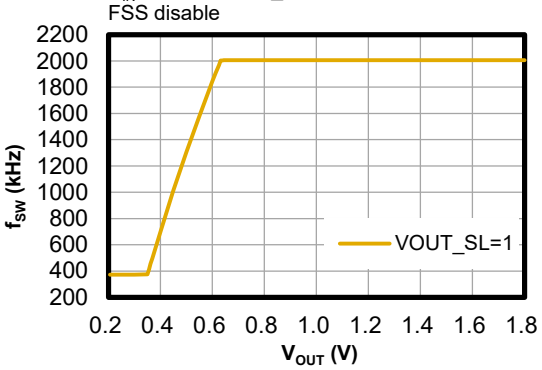
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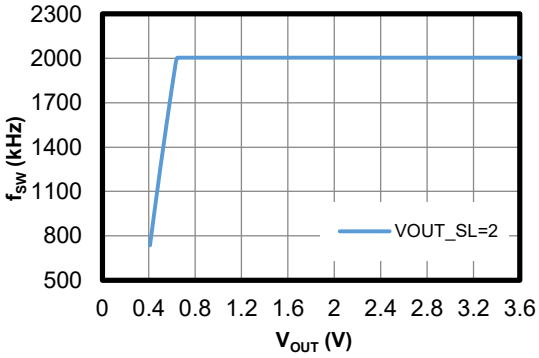
TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

f_{SW} vs. V_{OUT}
V_{IN} = 5.0V, VOUT_SL = 1, FCCM,
FSS disable



f_{SW} vs. V_{OUT}
V_{IN} = 5.0V, VOUT_SL = 2, FCCM,
FSS disable



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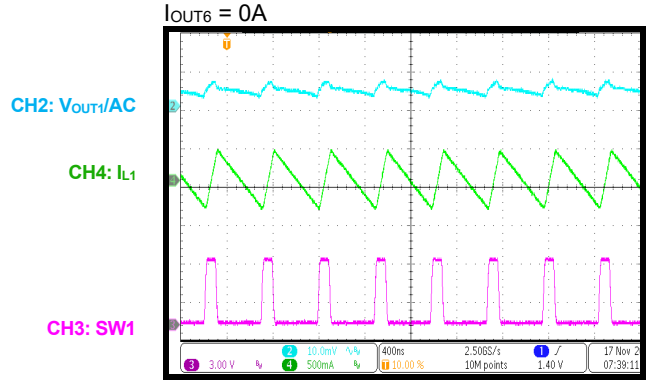
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

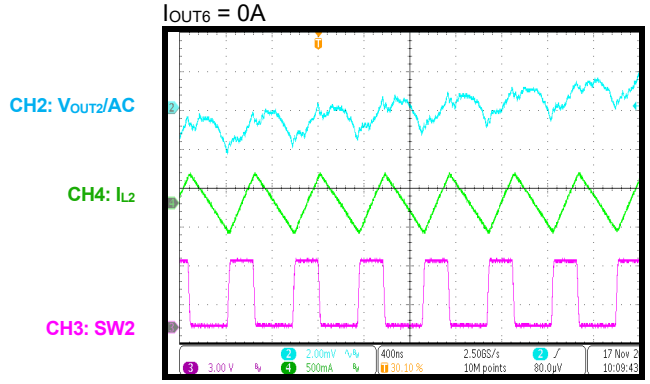
Steady State (Buck1)

FCCM, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



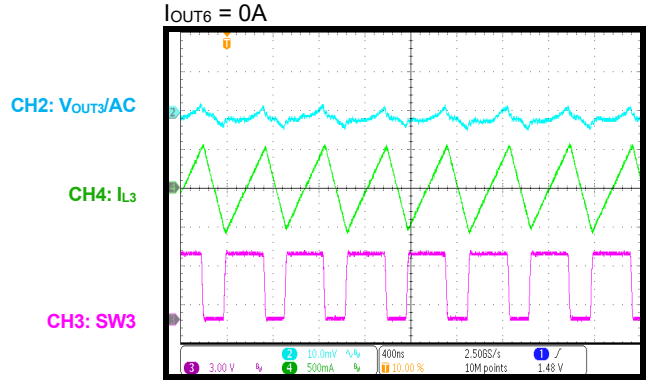
Steady State (Buck2)

FCCM, I_{OUT1} v= I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



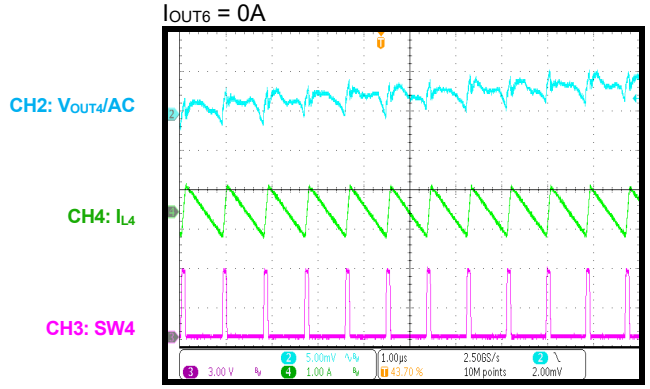
Steady State (Buck3)

FCCM, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



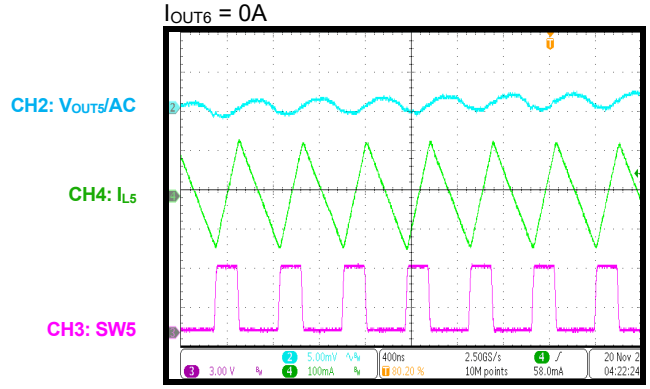
Steady State (Buck4)

FCCM, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



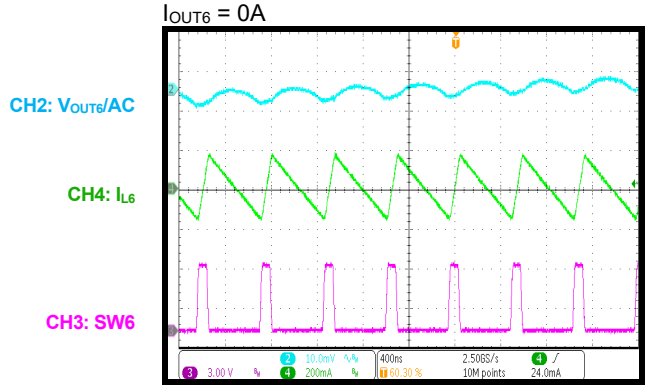
Steady State (Buck5)

FCCM, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



Steady State (Buck6)

FCCM, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

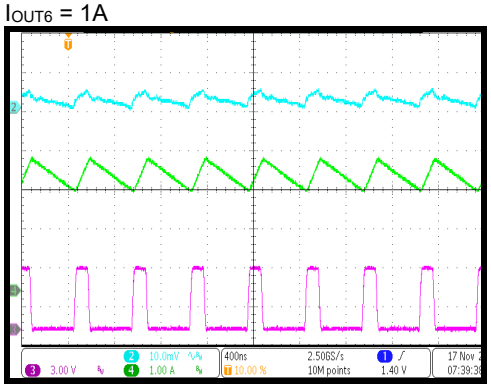
Steady State (Buck1)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A

CH2: V_{OUT1}/AC

CH4: I_{L1}

CH3: SW1



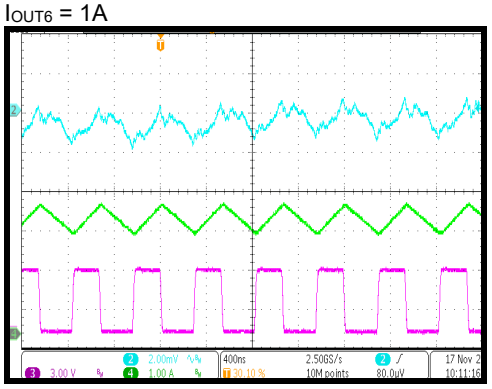
Steady State (Buck2)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A

CH2: V_{OUT2}/AC

CH4: I_{L2}

CH3: SW2



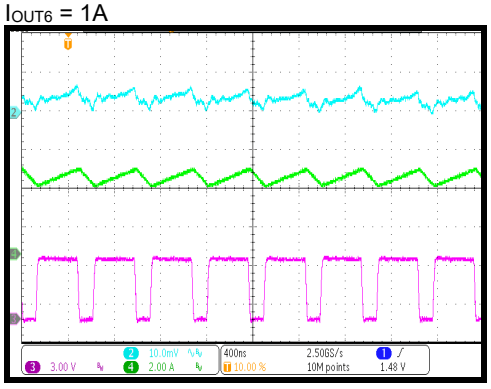
Steady State (Buck3)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A

CH2: V_{OUT3}/AC

CH4: I_{L3}

CH3: SW3



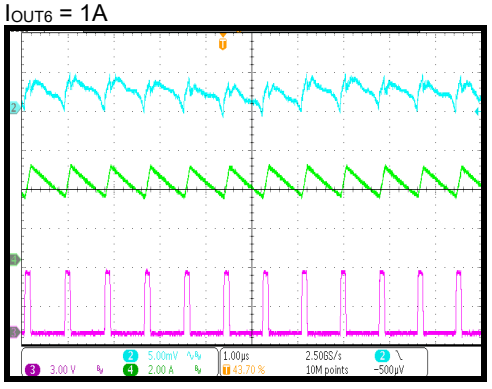
Steady State (Buck4)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A

CH2: V_{OUT4}/AC

CH4: I_{L4}

CH3: SW4



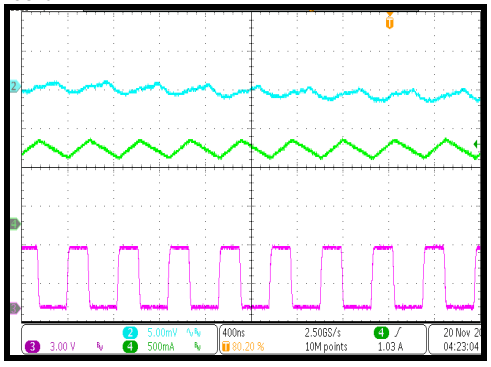
Steady State (Buck5)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A

CH2: V_{OUT5}/AC

CH4: I_{L5}

CH3: SW5



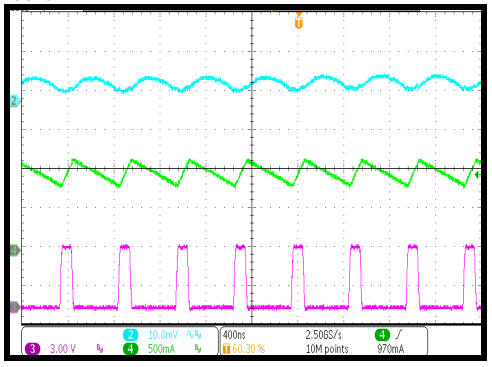
Steady State (Buck6)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A

CH2: V_{OUT6}/AC

CH4: I_{L6}

CH3: SW6



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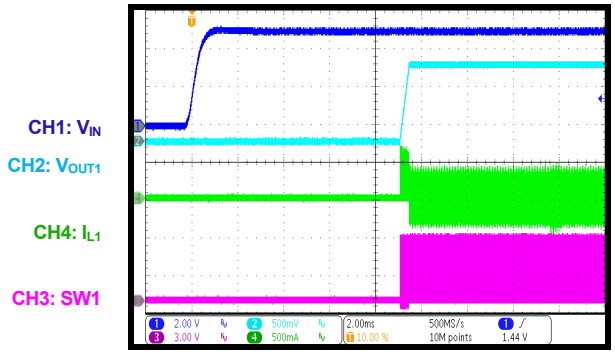
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

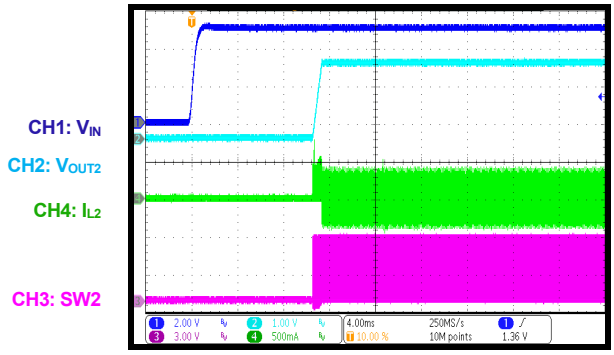
Start-Up through VIN (Buck1)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



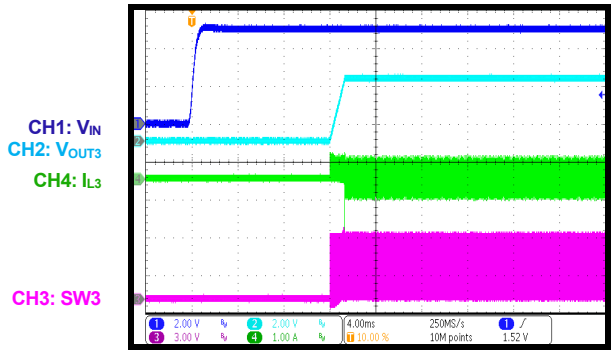
Start-Up through VIN (Buck2)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



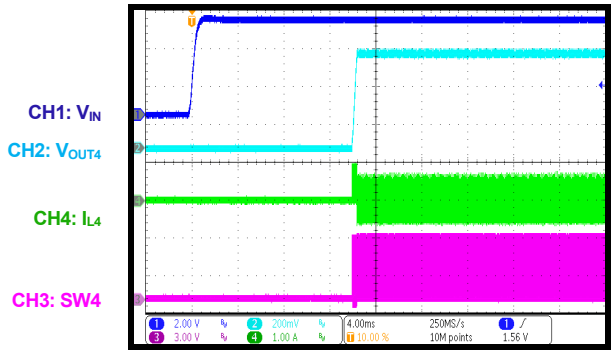
Start-Up through VIN (Buck3)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



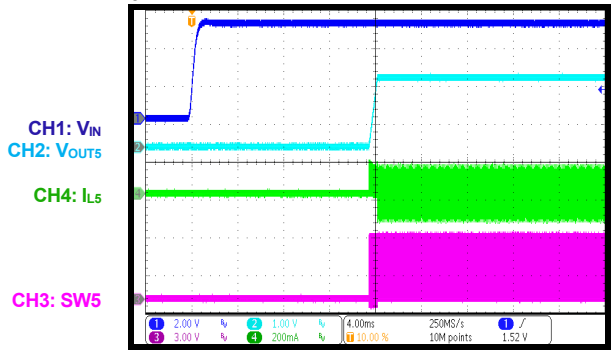
Start-Up through VIN (Buck4)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



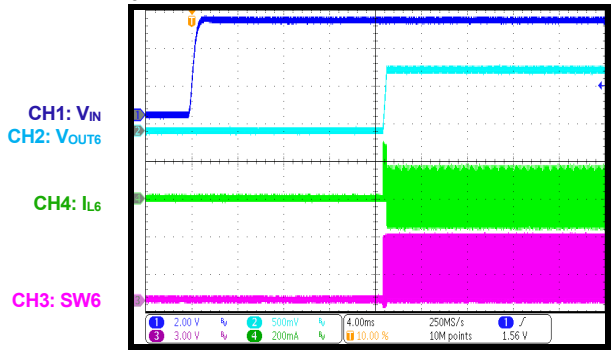
Start-Up through VIN (Buck5)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



Start-Up through VIN (Buck6)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



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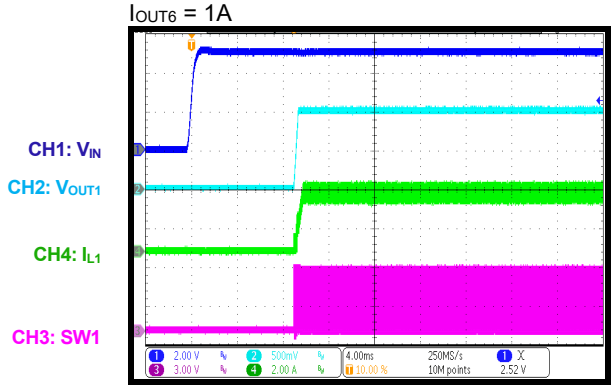
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

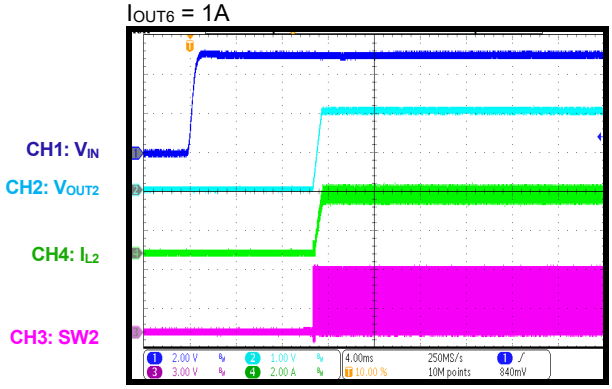
Start-Up through VIN (Buck1)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



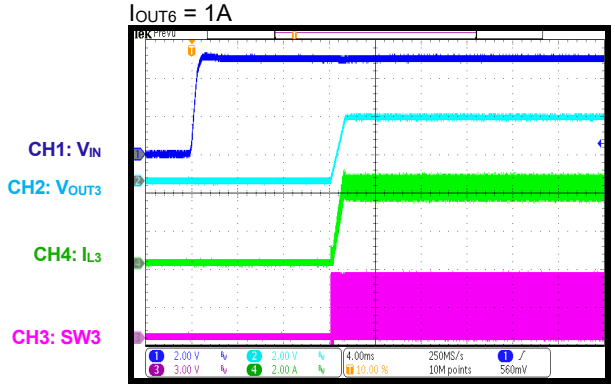
Start-Up through VIN (Buck2)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



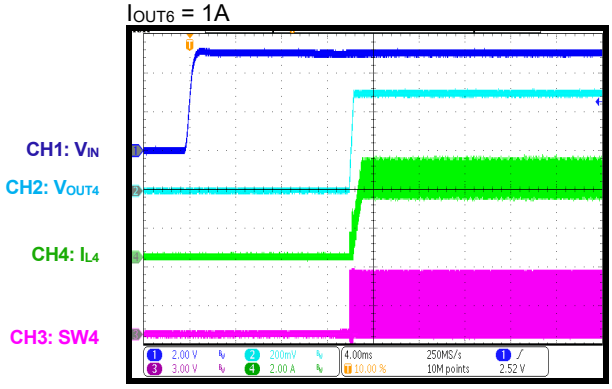
Start-Up through VIN (Buck3)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



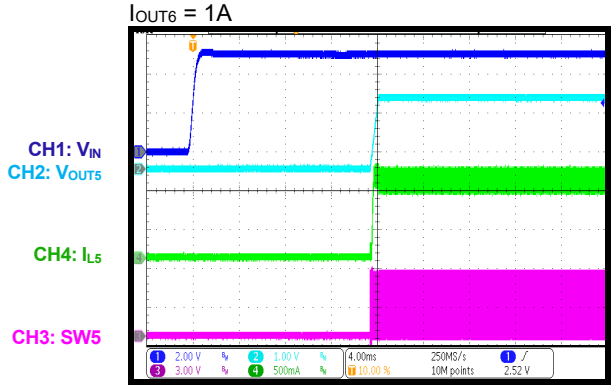
Start-Up through VIN (Buck4)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



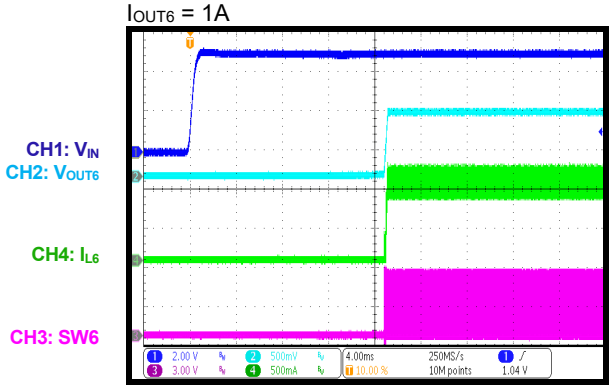
Start-Up through VIN (Buck5)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



Start-Up through VIN (Buck6)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



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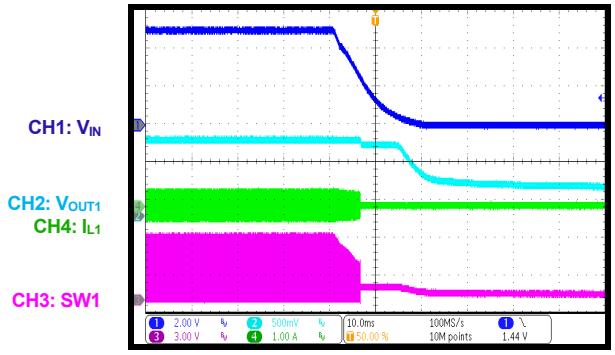
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

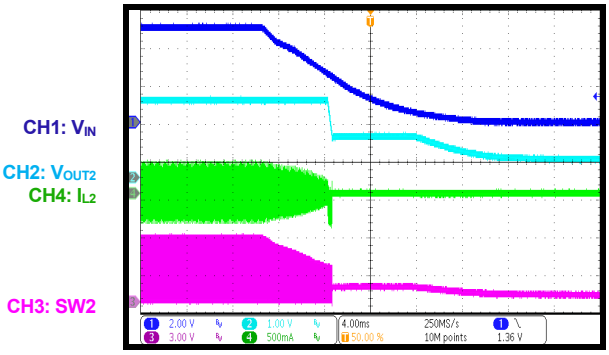
Shutdown through VIN (Buck1)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



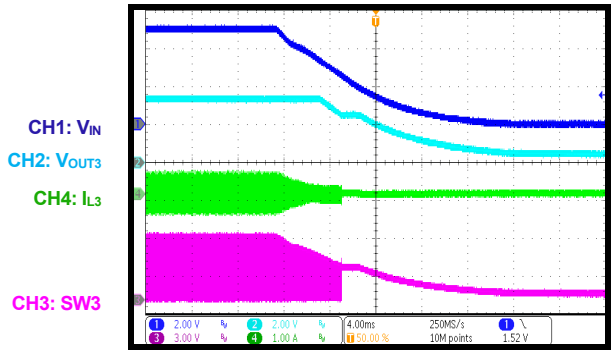
Shutdown through VIN (Buck2)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



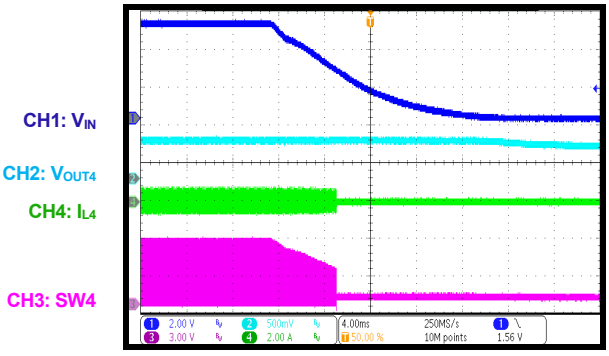
Shutdown through VIN (Buck3)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



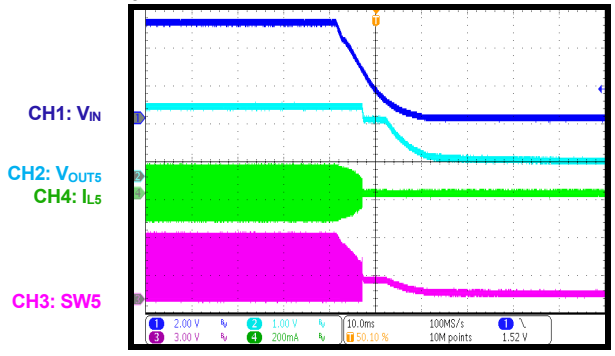
Shutdown through VIN (Buck4)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



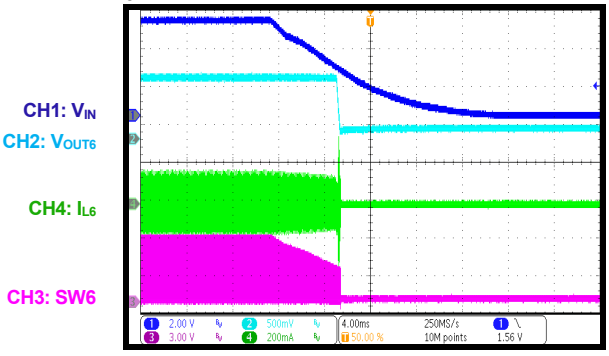
Shutdown through VIN (Buck5)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



Shutdown through VIN (Buck6)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



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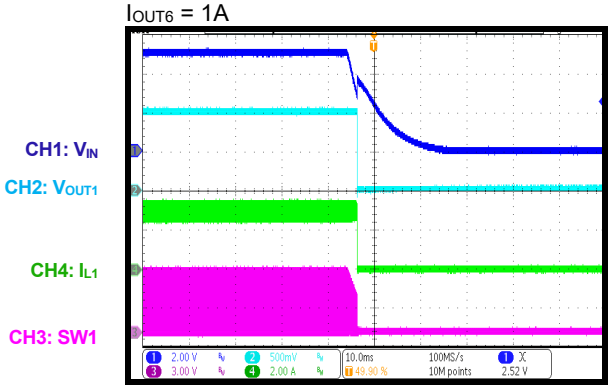
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

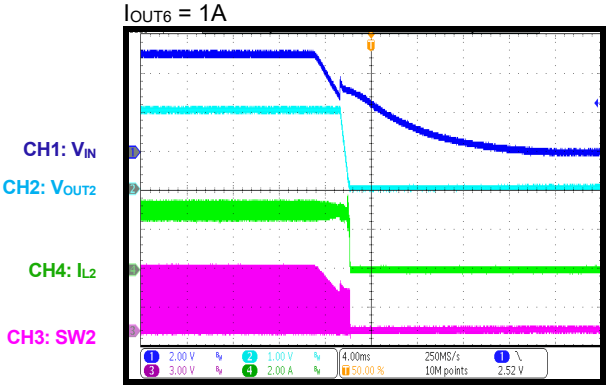
Shutdown through VIN (Buck1)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



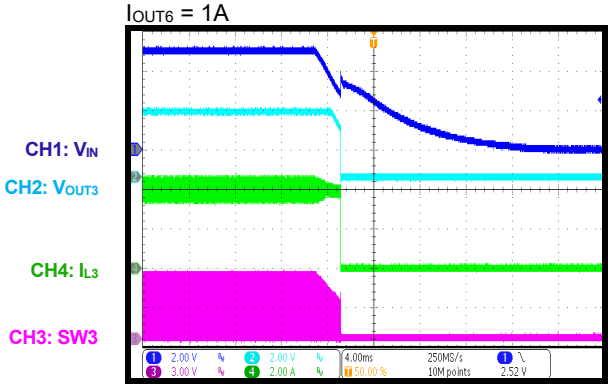
Shutdown through VIN (Buck2)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



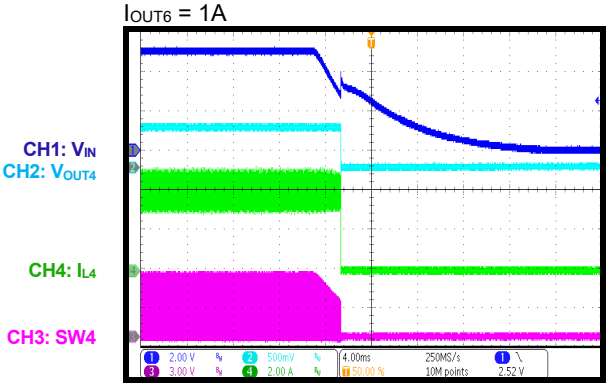
Shutdown through VIN (Buck3)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



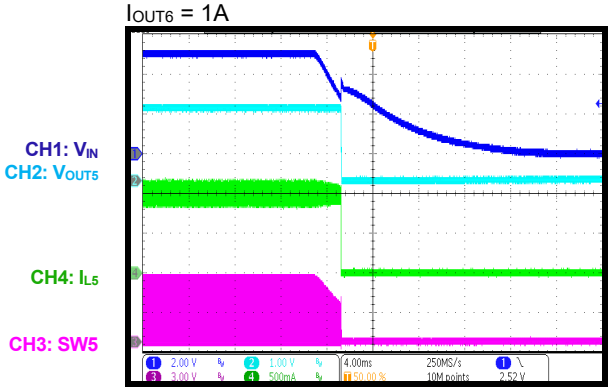
Shutdown through VIN (Buck4)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



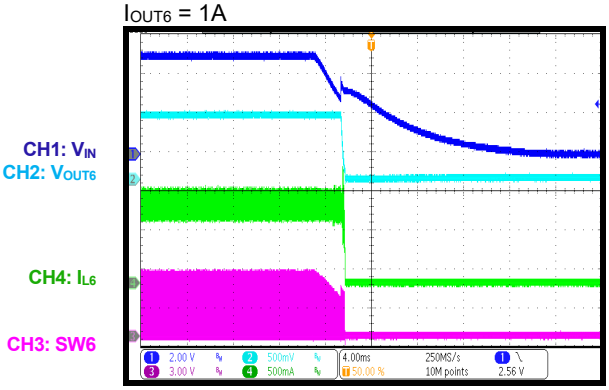
Shutdown through VIN (Buck5)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



Shutdown through VIN (Buck6)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



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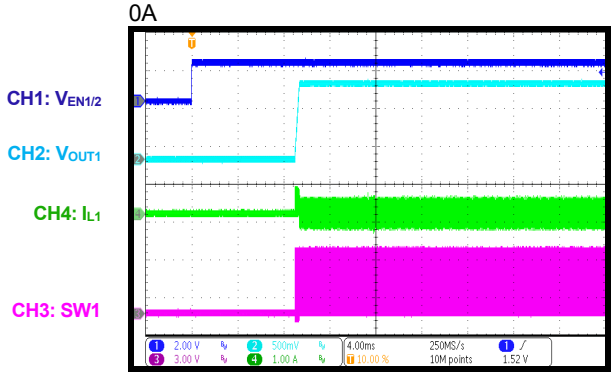
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

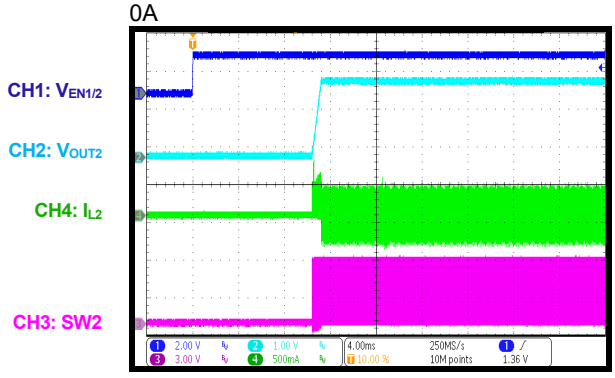
Start-Up through EN 0V to 2V (Buck1)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



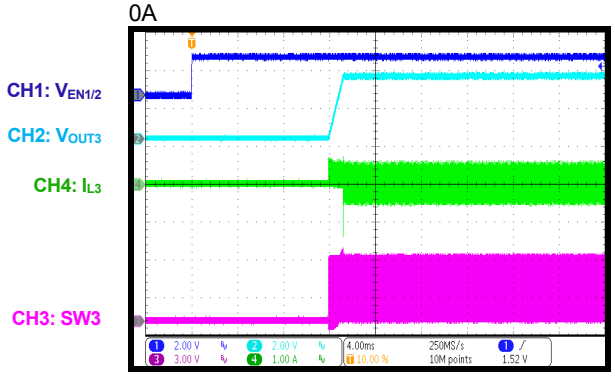
Start-Up through EN 0V to 2V (Buck2)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



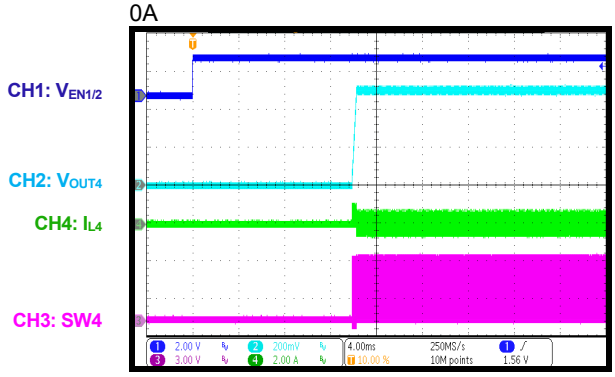
Start-Up through EN 0V to 2V (Buck3)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



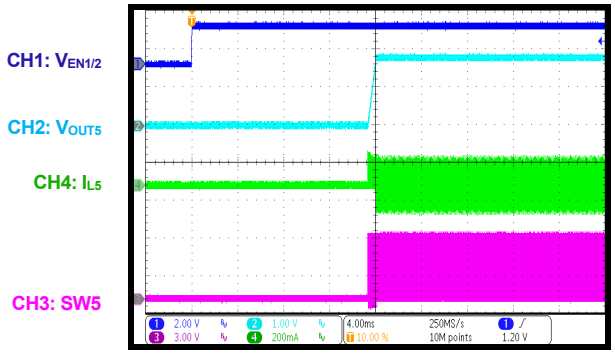
Start-Up through EN 0V to 2V (Buck4)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



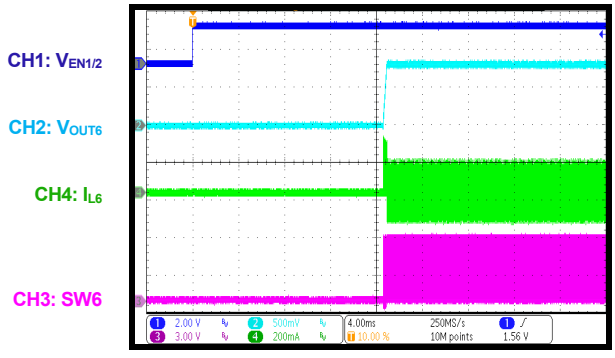
Start-Up through EN 0V to 2V (Buck5)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



Start-Up through EN 0V to 2V (Buck6)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



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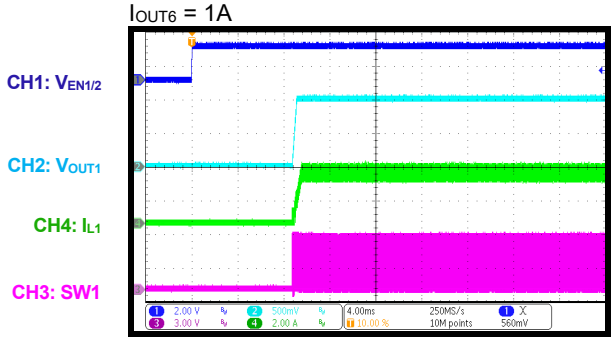
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

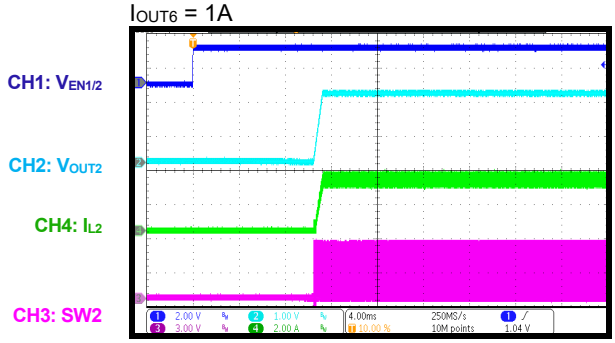
Start-Up through EN 0V to 2V (Buck1)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



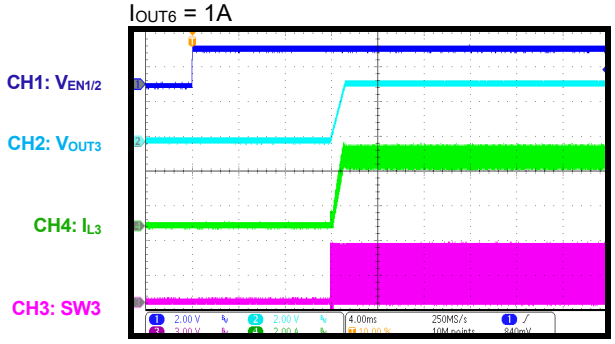
Start-Up through EN 0V to 2V (Buck2)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



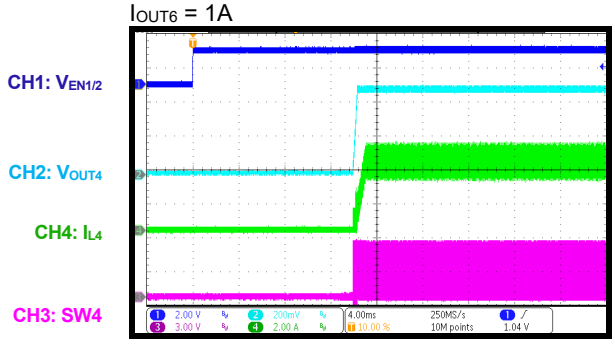
Start-Up through EN 0V to 2V (Buck3)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



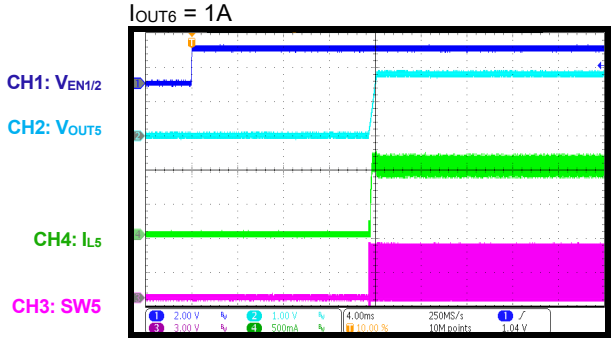
Start-Up through EN 0V to 2V (Buck4)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



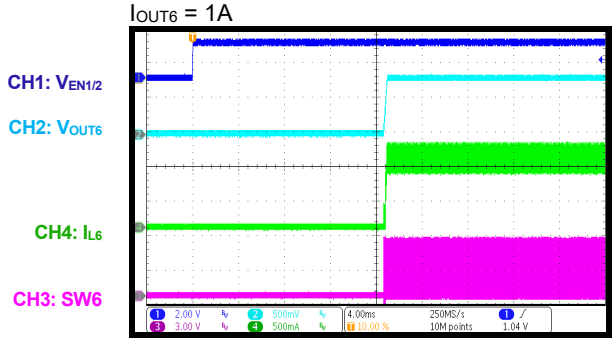
Start-Up through EN 0V to 2V (Buck5)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



Start-Up through EN 0V to 2V (Buck6)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



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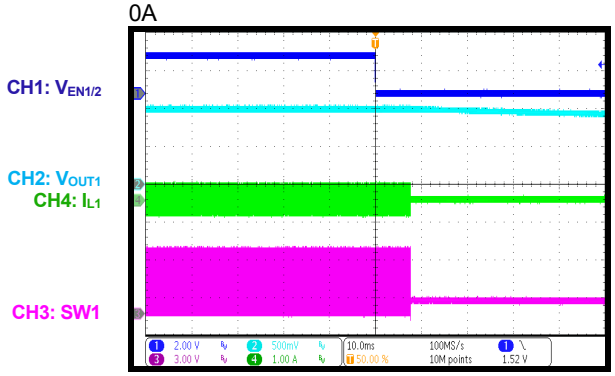
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

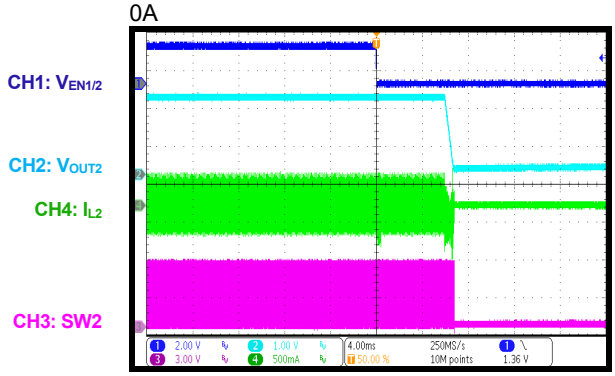
Shutdown through EN 2V to 0V (Buck1)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



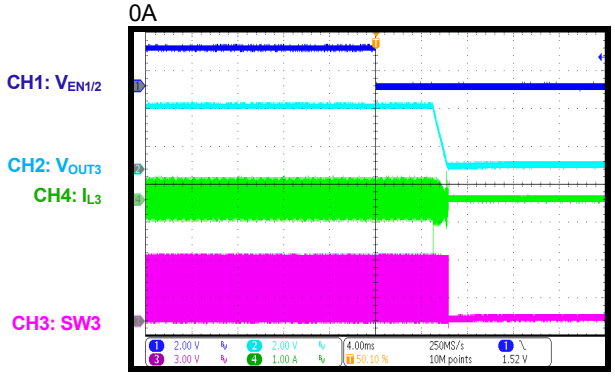
Shutdown through EN 2V to 0V (Buck2)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



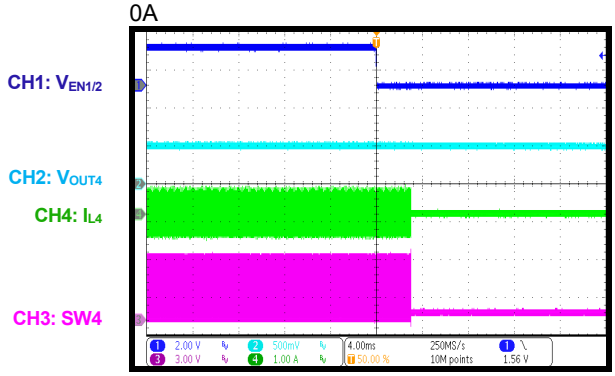
Shutdown through EN 2V to 0V (Buck3)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



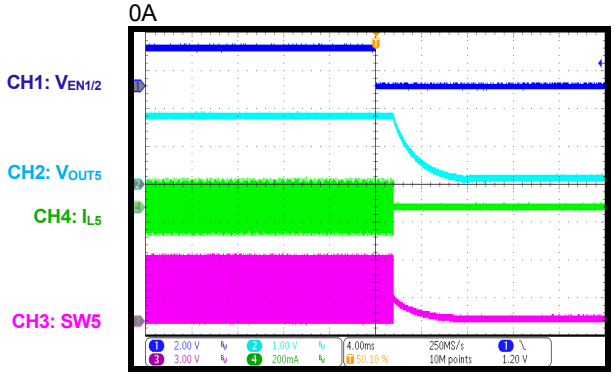
Shutdown through EN 2V to 0V (Buck4)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



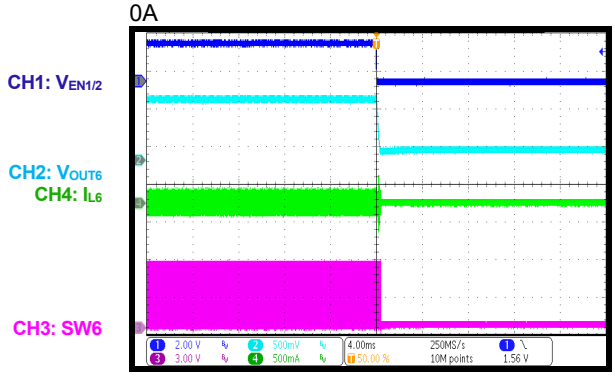
Shutdown through EN 2V to 0V (Buck5)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



Shutdown through EN 2V to 0V (Buck6)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



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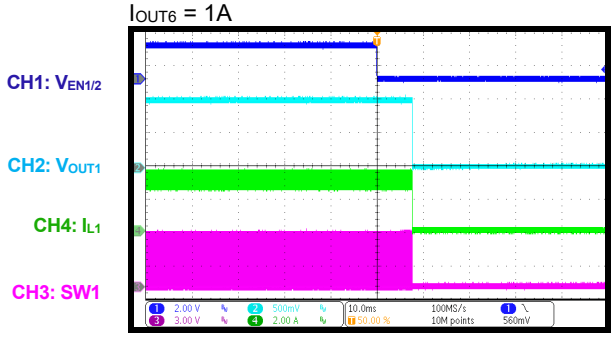
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

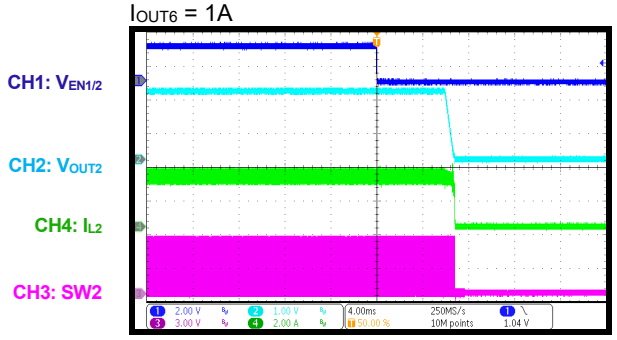
Shutdown through EN 2V to 0V (Buck1)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



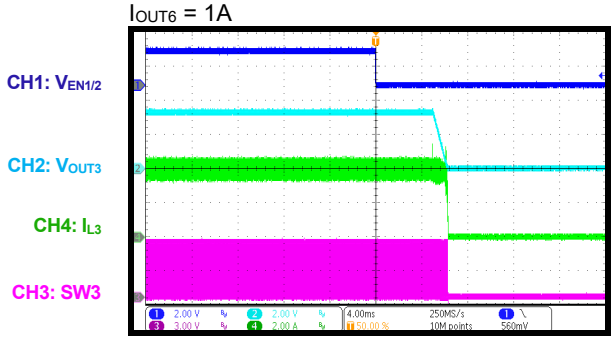
Shutdown through EN 2V to 0V (Buck2)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



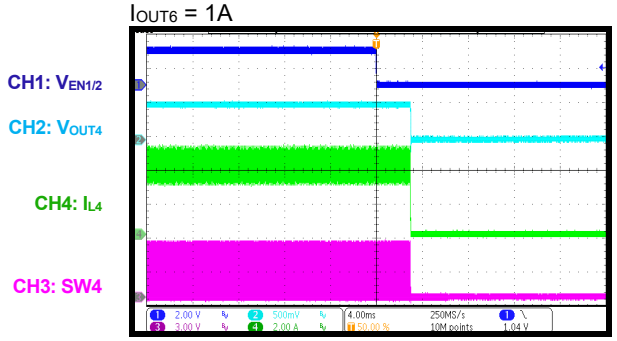
Shutdown through EN 2V to 0V (Buck3)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



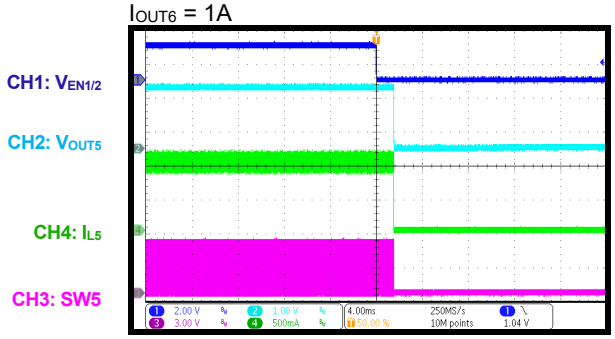
Shutdown through EN 2V to 0V (Buck4)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



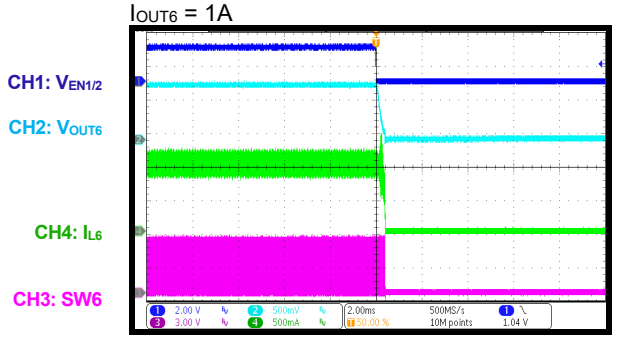
Shutdown through EN 2V to 0V (Buck5)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



Shutdown through EN 2V to 0V (Buck6)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



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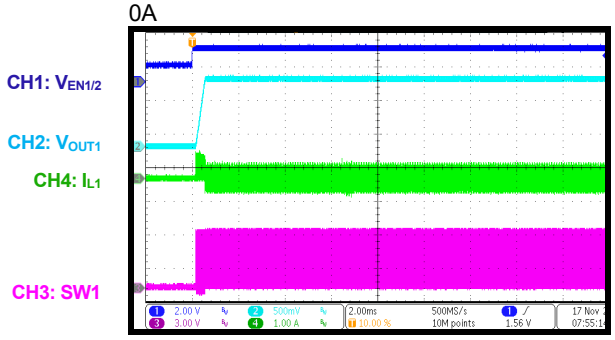
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

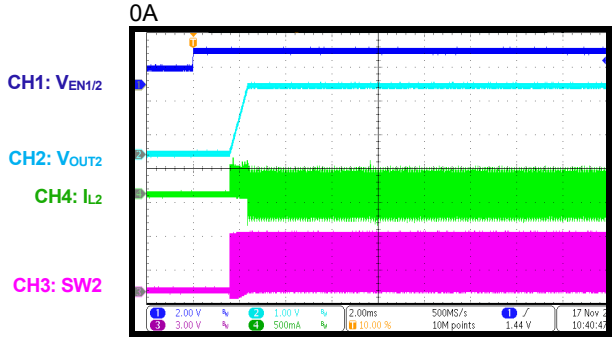
Start-Up through EN 1V to 2V (Buck1)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



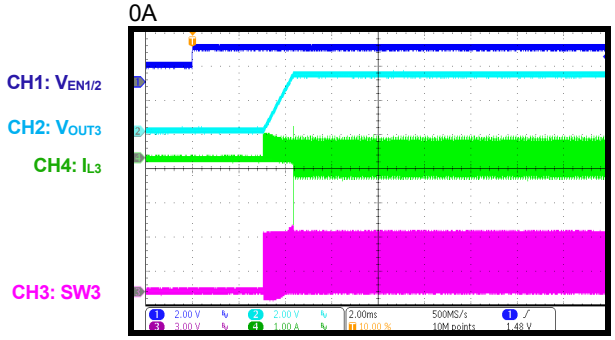
Start-Up through EN 1V to 2V (Buck2)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



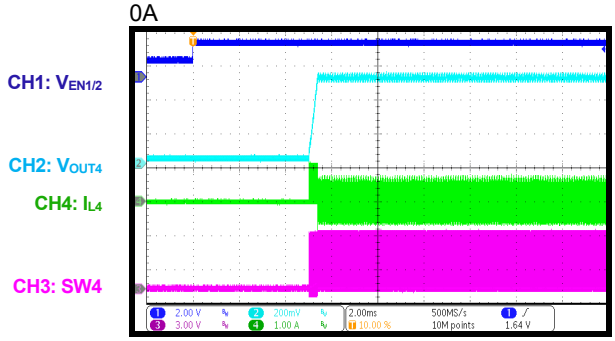
Start-Up through EN 1V to 2V (Buck3)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



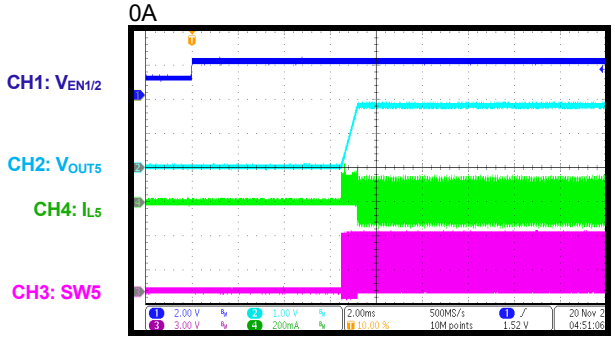
Start-Up through EN 1V to 2V (Buck4)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



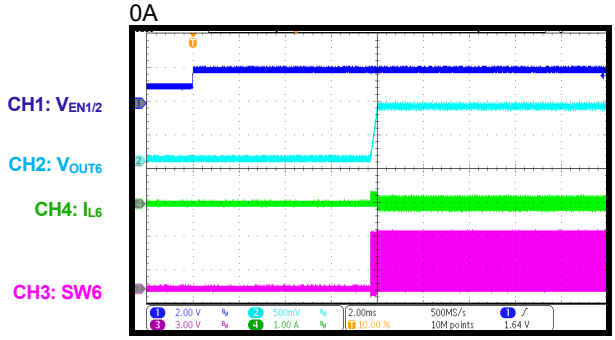
Start-Up through EN 1V to 2V (Buck5)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



Start-Up through EN 1V to 2V (Buck6)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



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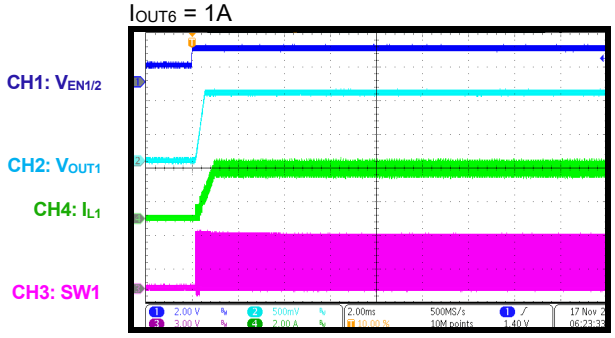
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

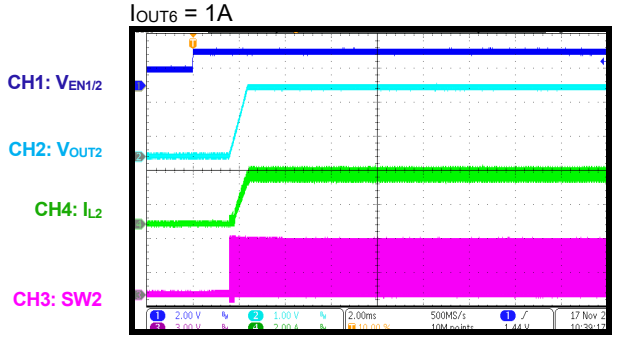
Start-Up through EN 1V to 2V (Buck1)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



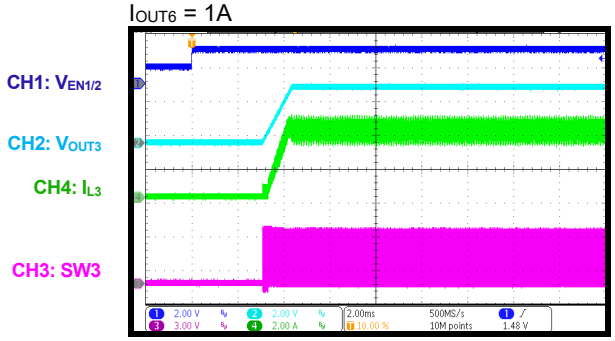
Start-Up through EN 1V to 2V (Buck2)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



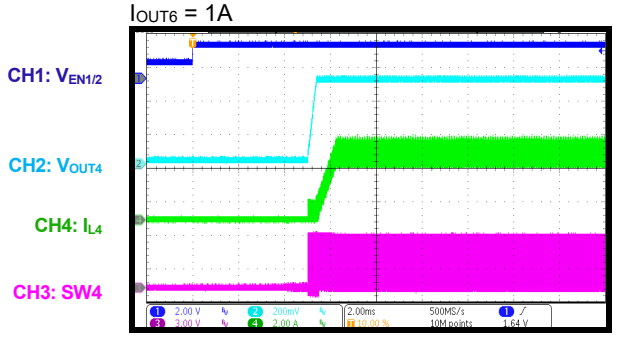
Start-Up through EN 1V to 2V (Buck3)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



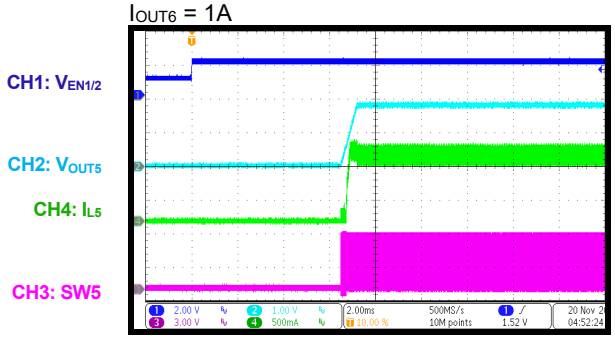
Start-Up through EN 1V to 2V (Buck4)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



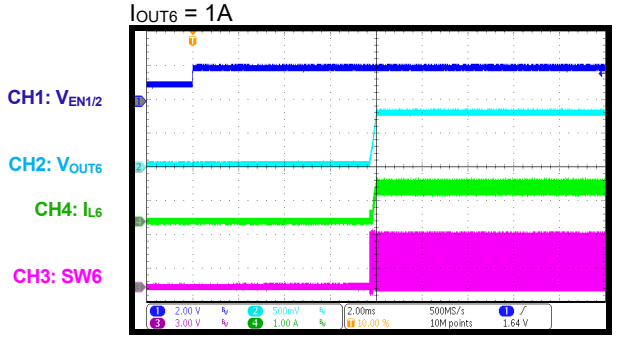
Start-Up through EN 1V to 2V (Buck5)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



Start-Up through EN 1V to 2V (Buck6)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



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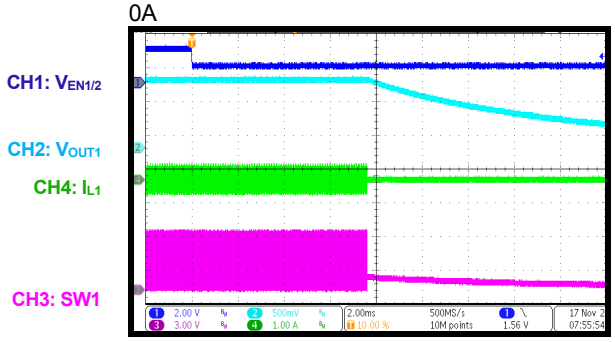
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

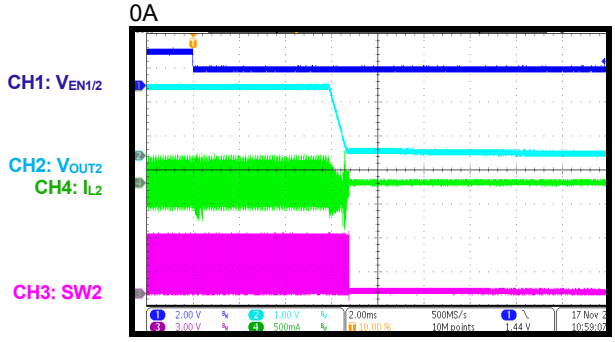
Shutdown through EN 2V to 1V (Buck1)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



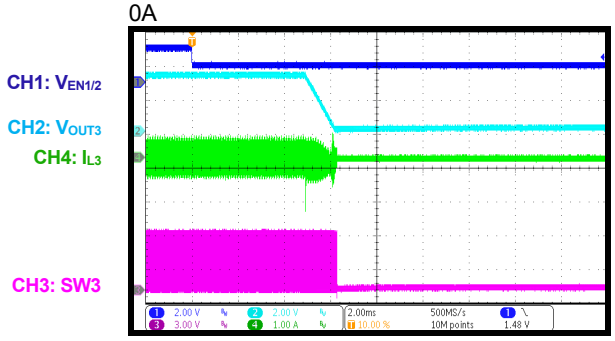
Shutdown through EN 2V to 1V (Buck2)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



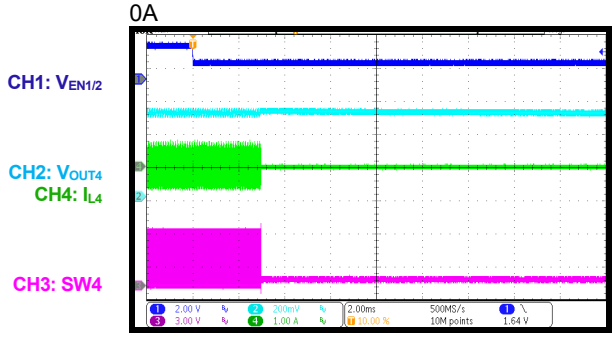
Shutdown through EN 2V to 1V (Buck3)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



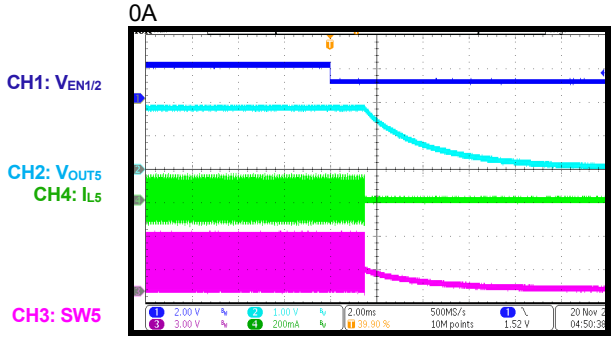
Shutdown through EN 2V to 1V (Buck4)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



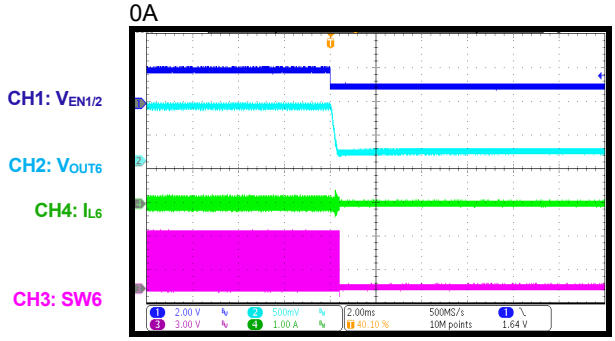
Shutdown through EN 2V to 1V (Buck5)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



Shutdown through EN 2V to 1V (Buck6)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



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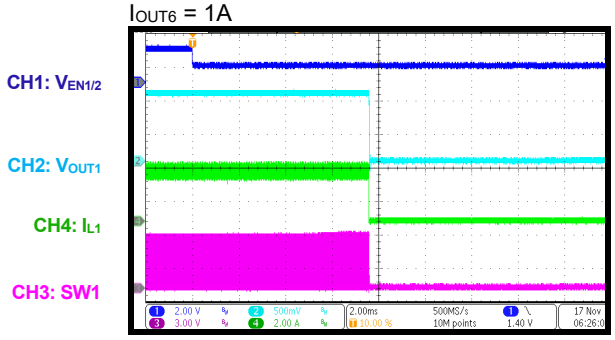
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

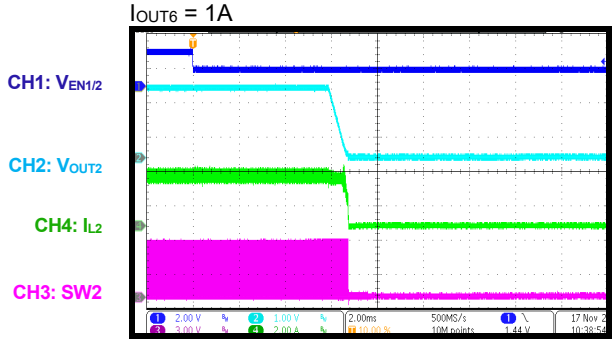
Shutdown through EN 2V to 1V (Buck1)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



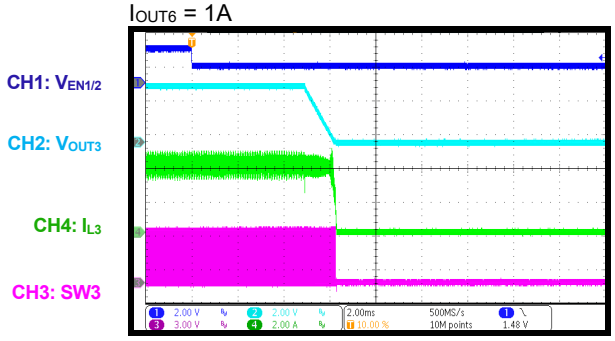
Shutdown through EN 2V to 1V (Buck2)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



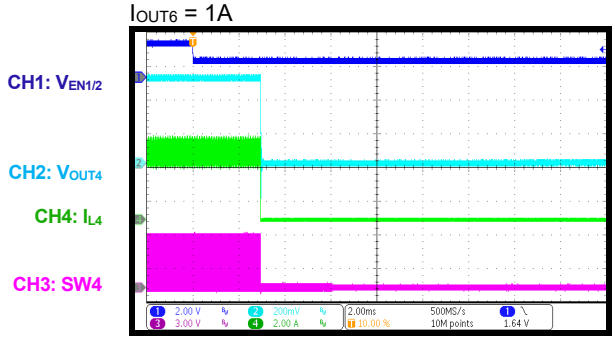
Shutdown through EN 2V to 1V (Buck3)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



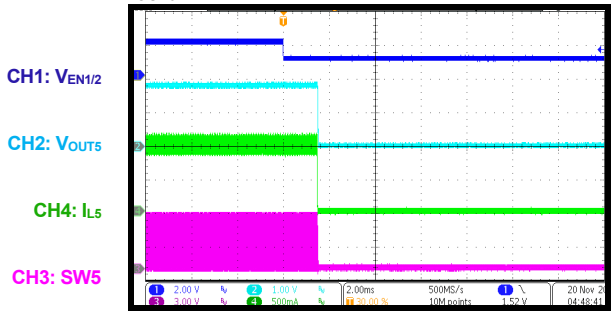
Shutdown through EN 2V to 1V (Buck4)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



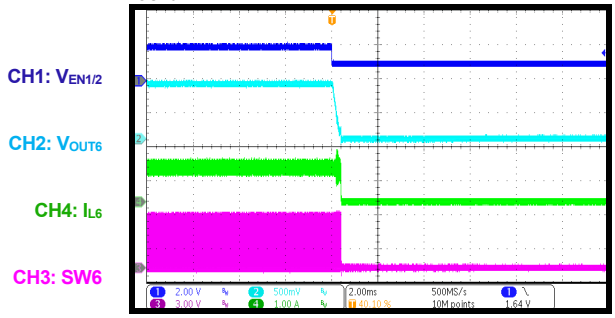
Shutdown through EN 2V to 1V (Buck5)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



Shutdown through EN 2V to 1V (Buck6)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



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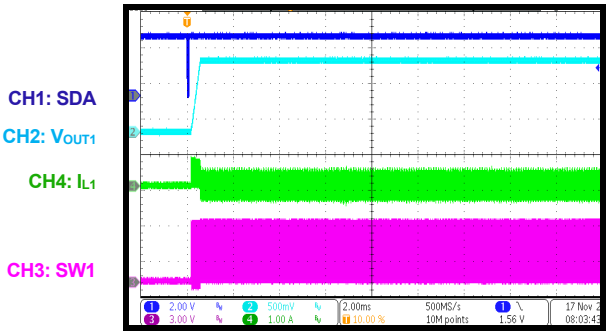
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

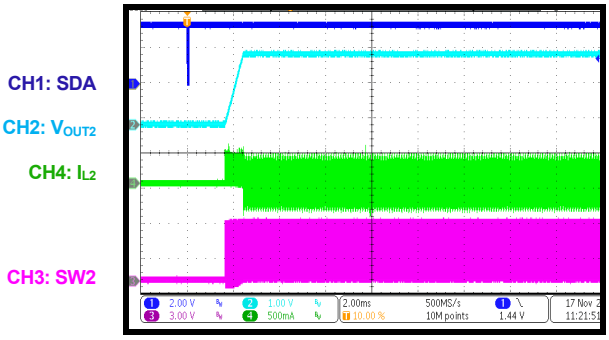
Start-Up through I²C (Buck1)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



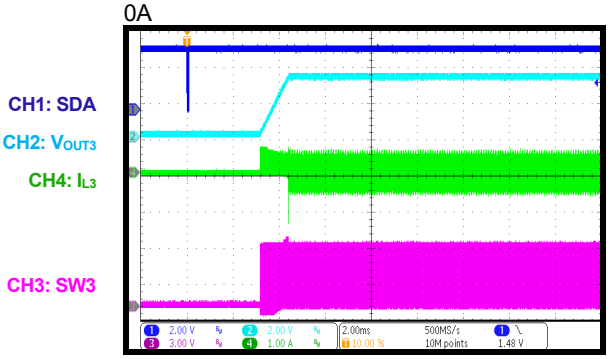
Start-Up through I²C (Buck2)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



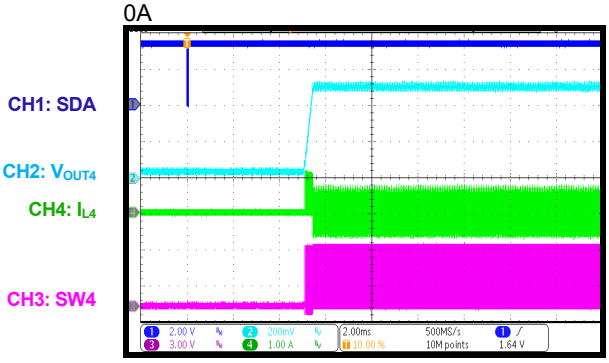
Start-Up through I²C (Buck3)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



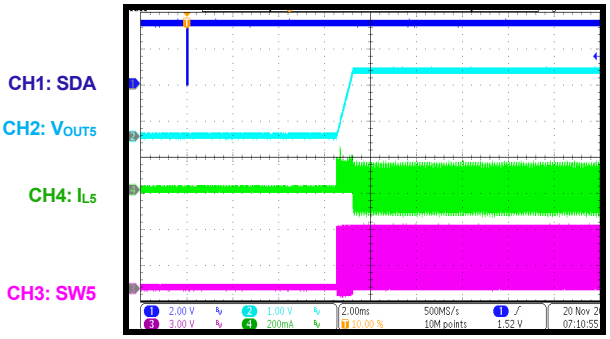
Start-Up through I²C (Buck4)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



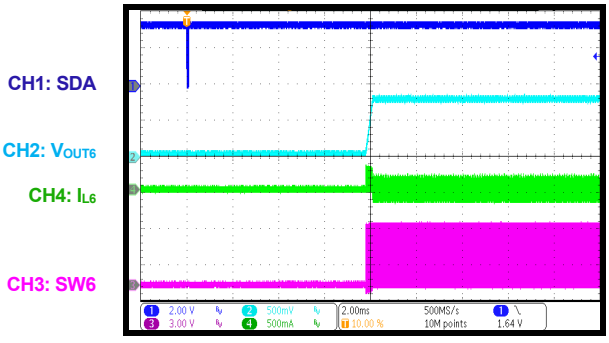
Start-Up through I²C (Buck5)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



Start-Up through I²C (Buck6)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



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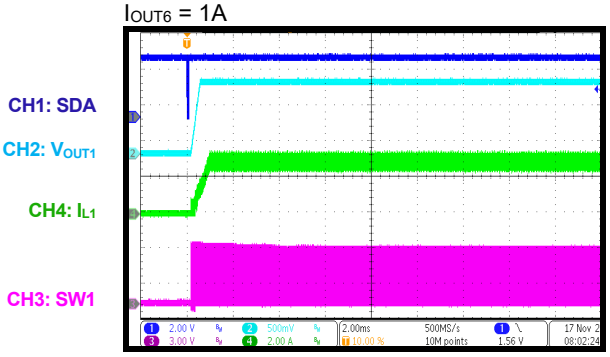
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

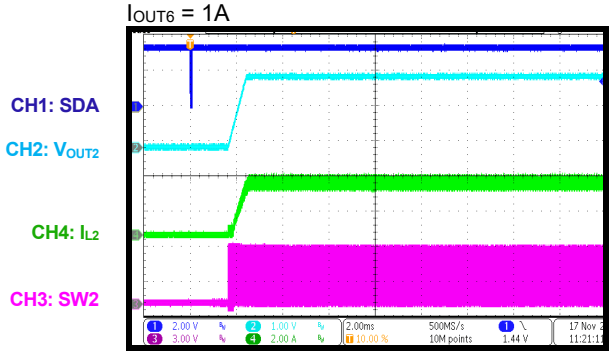
Start-Up through I²C (Buck1)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



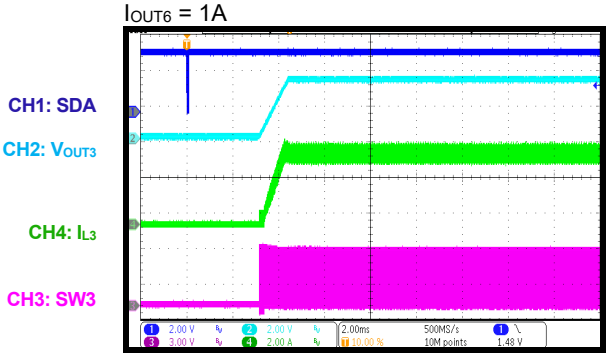
Start-Up through I²C (Buck2)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



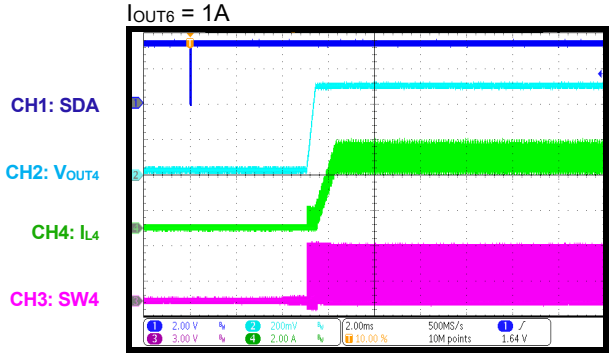
Start-Up through I²C (Buck3)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



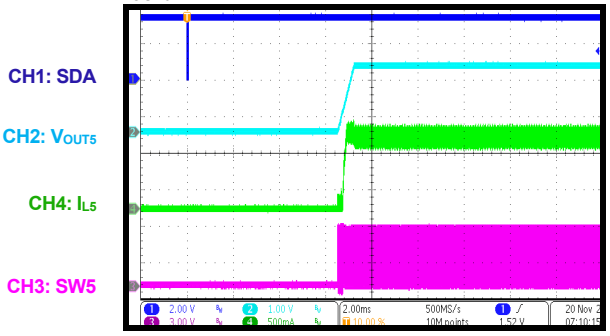
Start-Up through I²C (Buck4)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



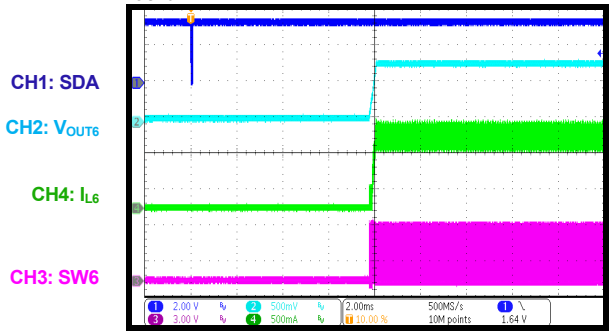
Start-Up through I²C (Buck5)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



Start-Up through I²C (Buck6)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



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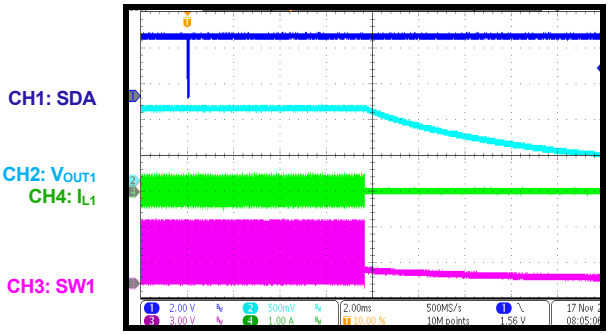
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

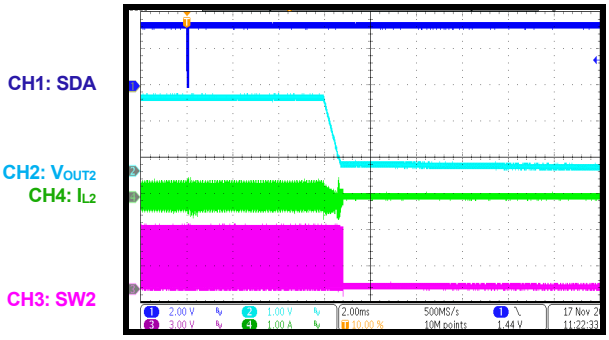
Shutdown through I²C (Buck1)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



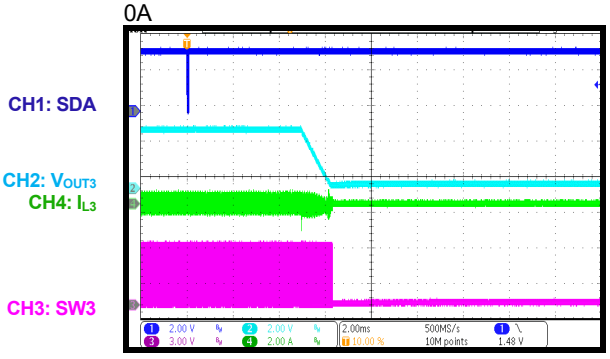
Shutdown through I²C (Buck2)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



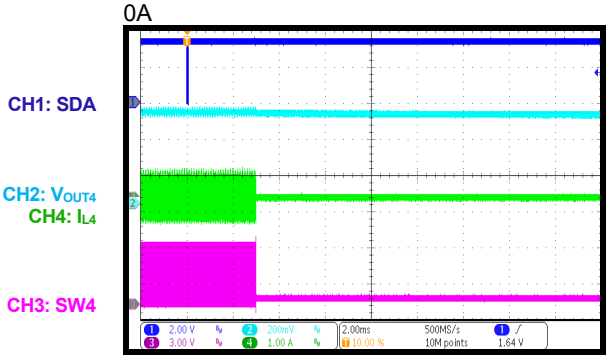
Shutdown through I²C (Buck3)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



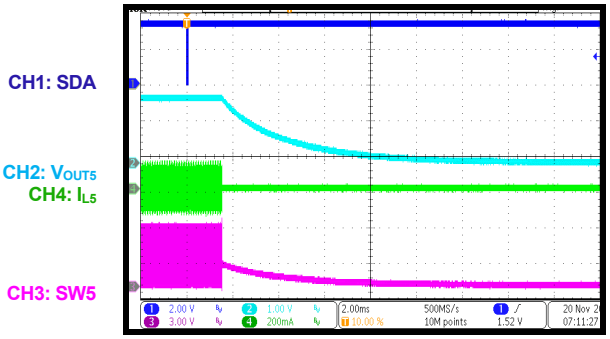
Shutdown through I²C (Buck4)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



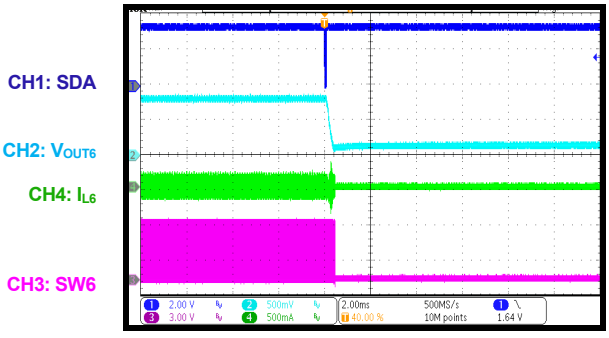
Shutdown through I²C (Buck5)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



Shutdown through I²C (Buck6)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



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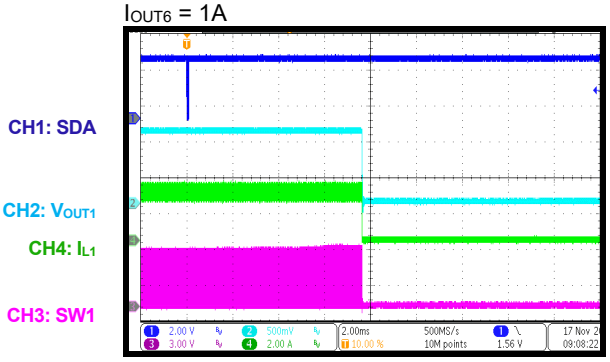
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

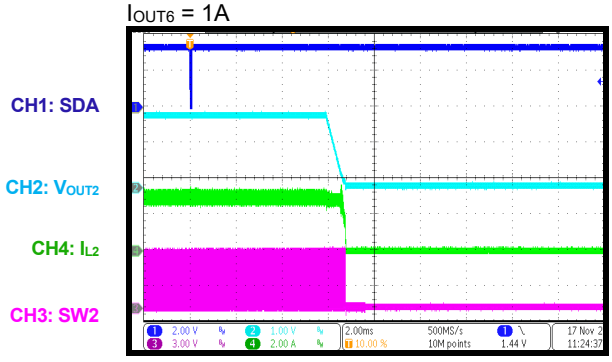
Start-Up through I²C (Buck1)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



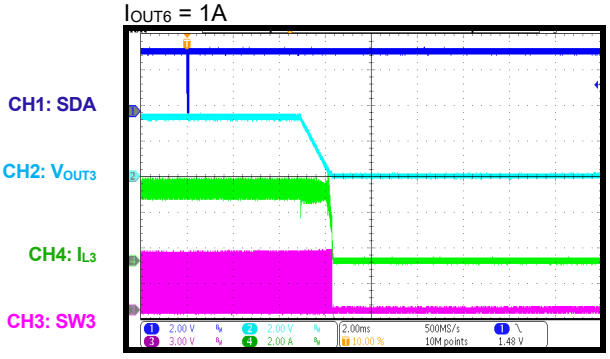
Start-Up through I²C (Buck2)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



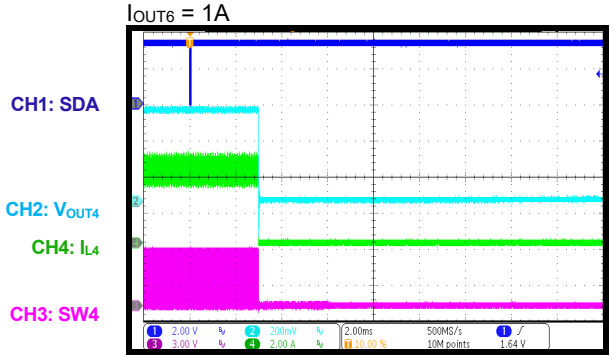
Start-Up through I²C (Buck3)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



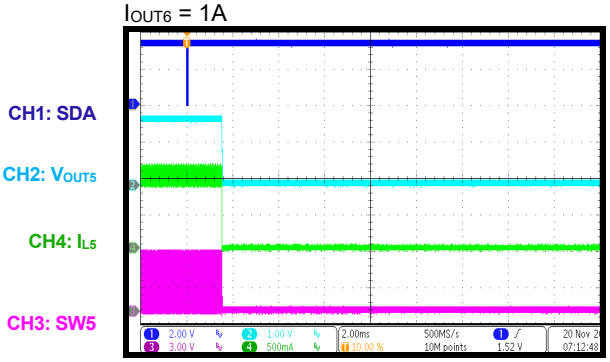
Start-Up through I²C (Buck4)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



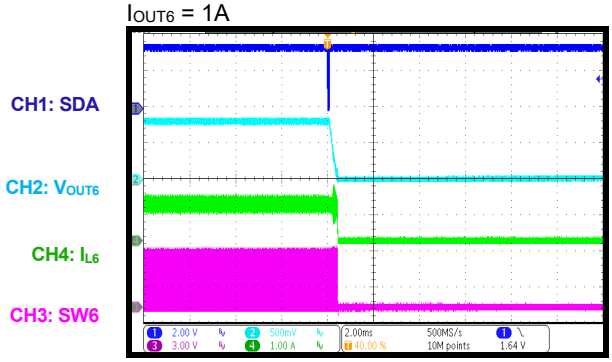
Start-Up through I²C (Buck5)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



Start-Up through I²C (Buck6)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



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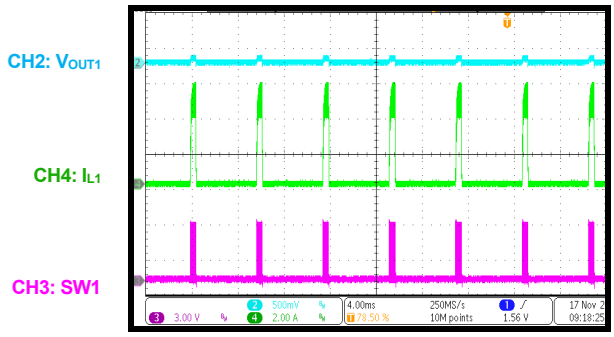
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

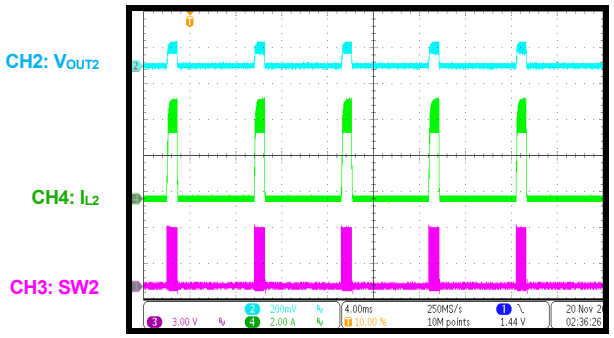
SCP Steady State (Buck1)

Buck1 SCP, other bucks no SCP



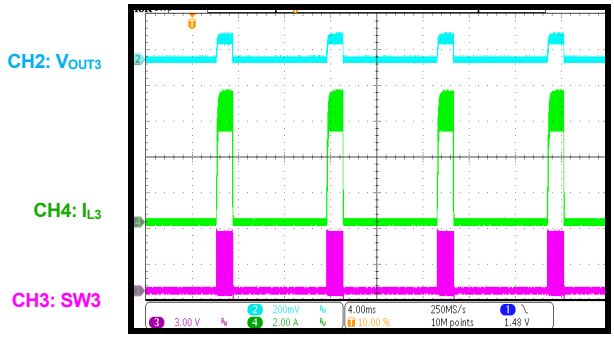
SCP Steady State (Buck2)

Buck2 SCP, other bucks no SCP



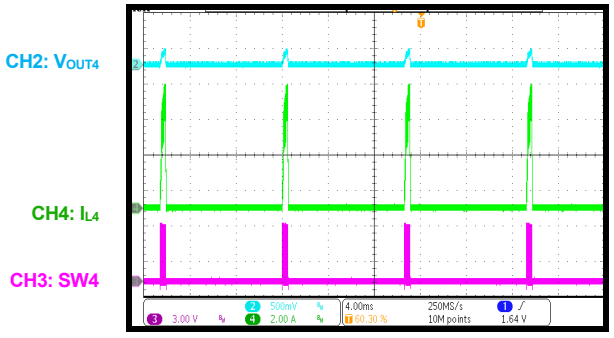
SCP Steady State (Buck3)

Buck3 SCP, other bucks no SCP



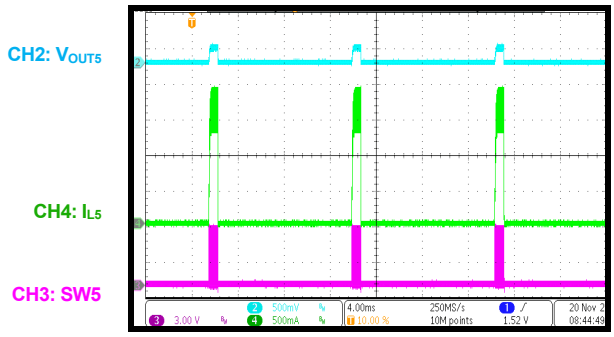
SCP Steady State (Buck4)

Buck4 SCP, other bucks no SCP



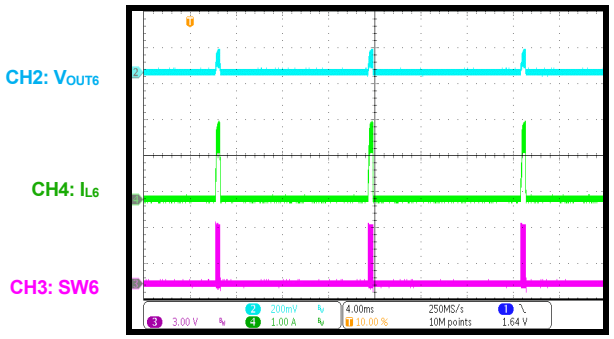
SCP Steady State (Buck5)

Buck5 SCP, other bucks no SCP



SCP Steady State (Buck6)

Buck6 SCP, other bucks no SCP



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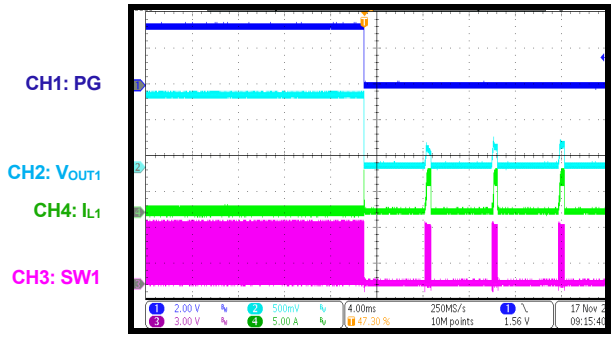
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

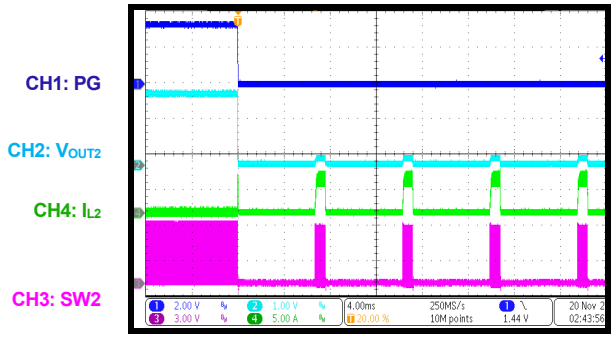
SCP Entry (Buck1)

Buck1 SCP entry, other bucks no SCP, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



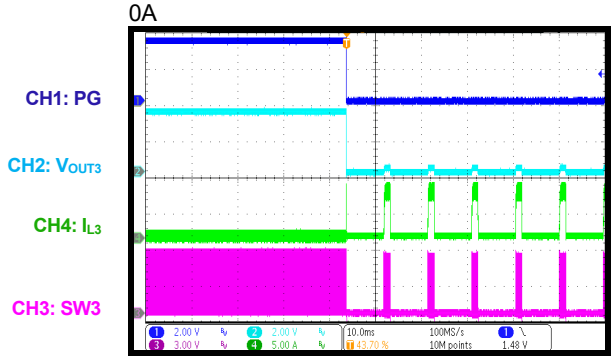
SCP Entry (Buck2)

Buck2 SCP entry, other bucks no SCP, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



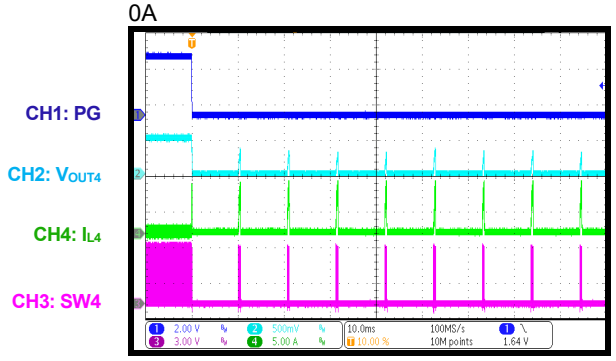
SCP Entry (Buck3)

Buck3 SCP entry, other bucks no SCP, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



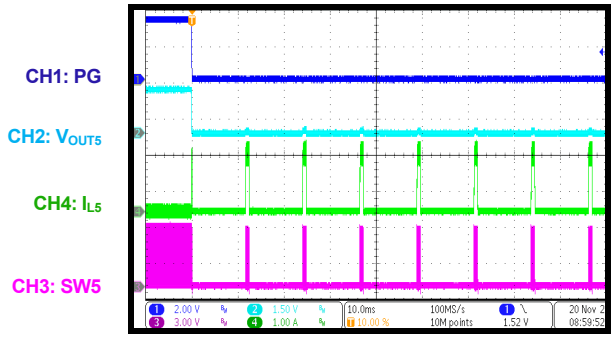
SCP Entry (Buck4)

Buck4 SCP entry, other bucks no SCP, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



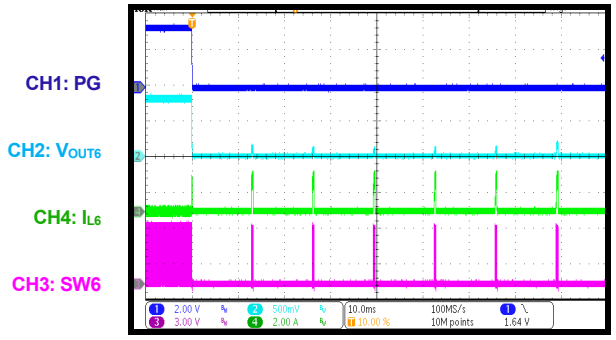
SCP Entry (Buck5)

Buck5 SCP entry, other bucks no SCP, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



SCP Entry (Buck6)

Buck6 SCP entry, other bucks no SCP, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



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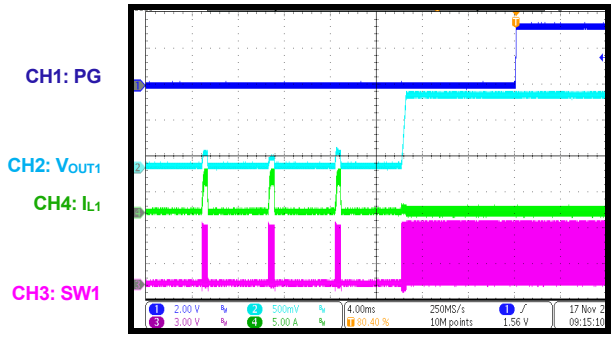
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

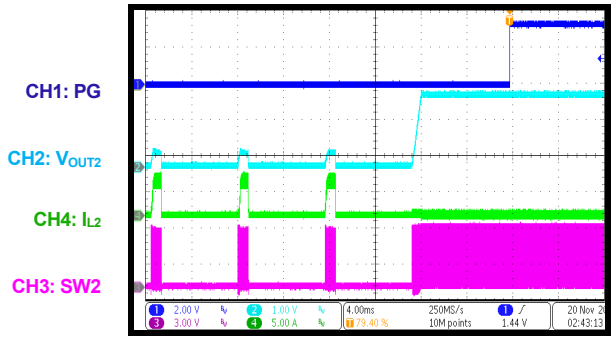
SCP Recovery (Buck1)

Buck1 SCP Recovery, other bucks no SCP, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



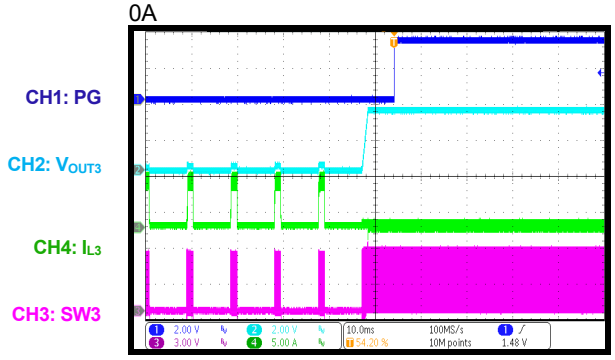
SCP Recovery (Buck2)

Buck2 SCP Recovery, other bucks no SCP, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



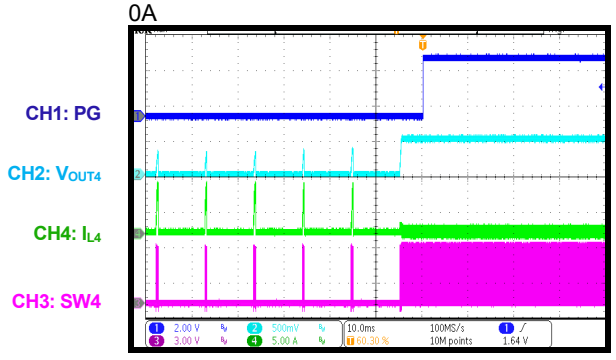
SCP Recovery (Buck3)

Buck3 SCP Recovery, other bucks no SCP, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



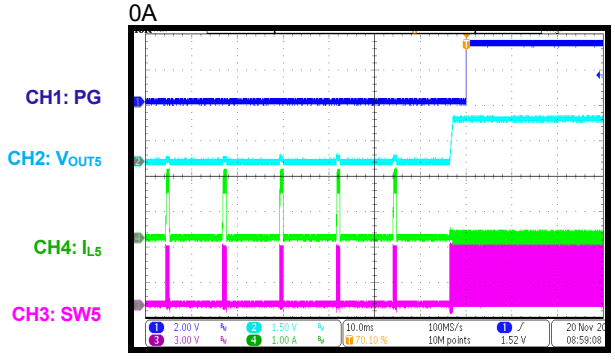
SCP Recovery (Buck4)

Buck4 SCP Recovery, other bucks no SCP, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



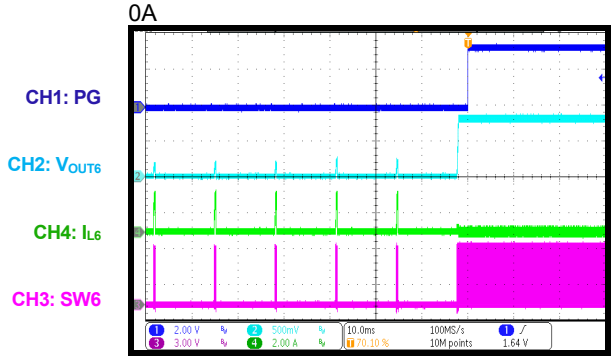
SCP Recovery (Buck5)

Buck5 SCP Recovery, other bucks no SCP, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



SCP Recovery (Buck6)

Buck6 SCP Recovery, other bucks no SCP, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



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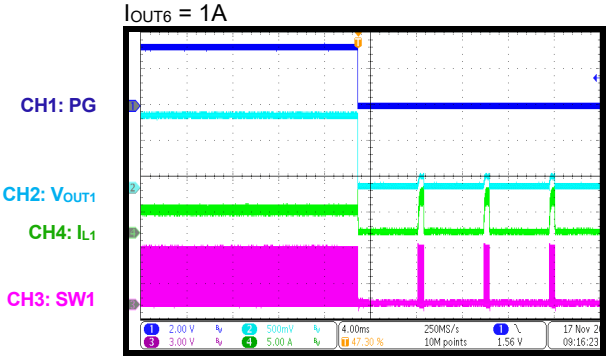
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

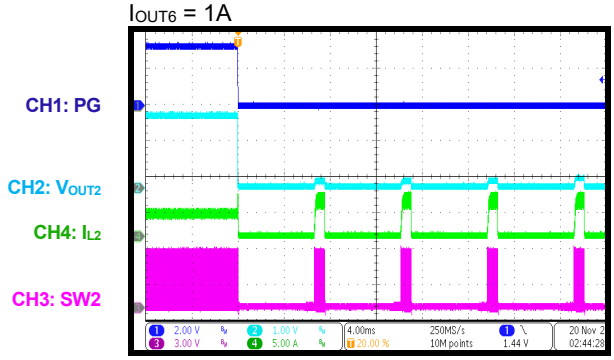
SCP Entry (Buck1)

Buck1 SCP entry, other bucks no SCP, I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



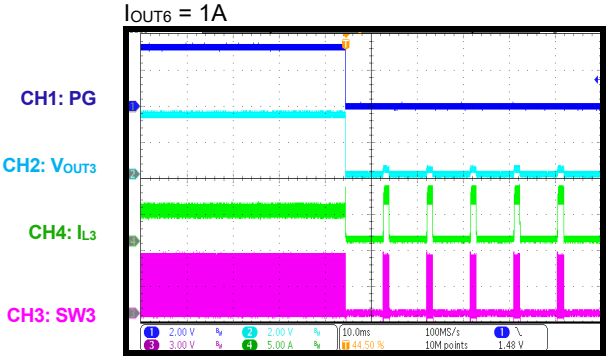
SCP Entry (Buck2)

Buck2 SCP entry, other bucks no SCP, I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



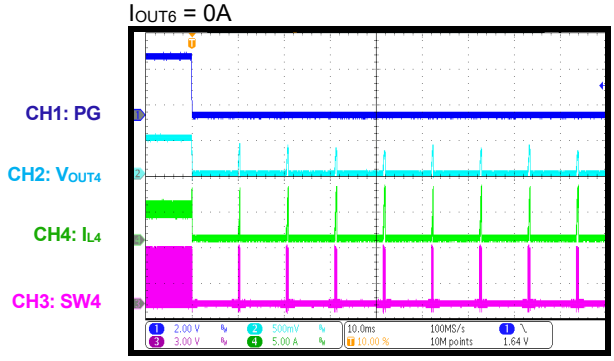
SCP Entry (Buck3)

Buck3 SCP entry, other bucks no SCP, I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



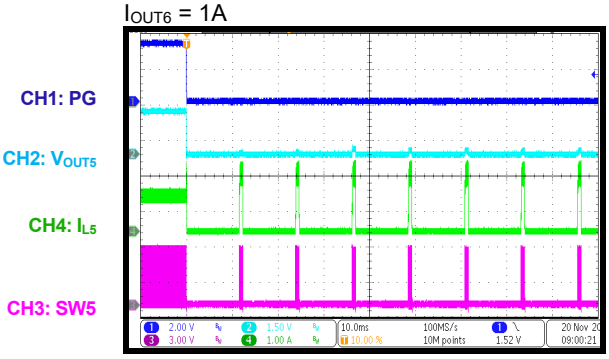
SCP Entry (Buck4)

Buck4 SCP entry, other bucks no SCP, FCCM, I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



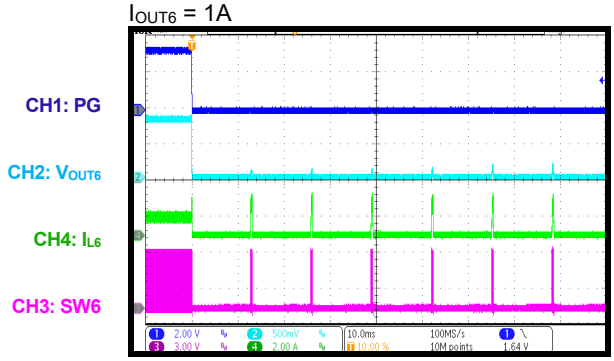
SCP Entry (Buck5)

Buck5 SCP entry, other bucks no SCP, I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



SCP Entry (Buck6)

Buck6 SCP entry, other bucks no SCP, I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



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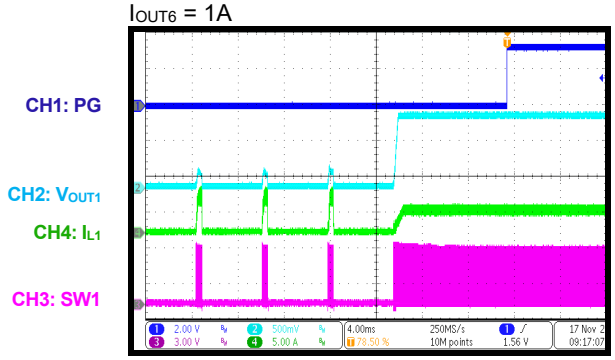
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

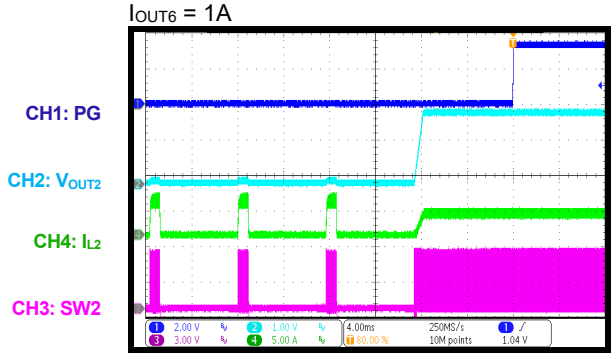
SCP Recovery (Buck1)

Buck1 SCP Recovery, other bucks no SCP, I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



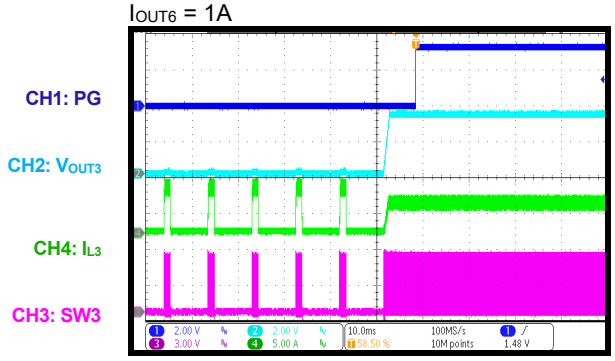
SCP Recovery (Buck2)

Buck2 SCP Recovery, other bucks no SCP, I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



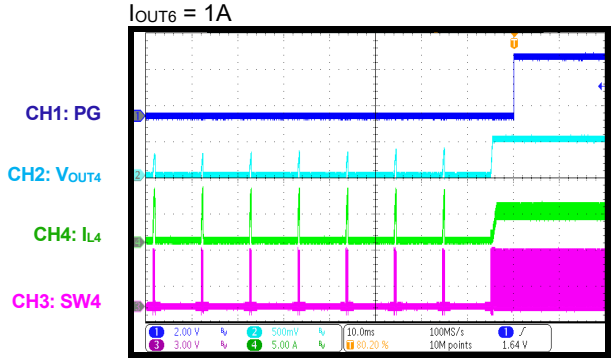
SCP Recovery (Buck3)

Buck3 SCP Recovery, other bucks no SCP, I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



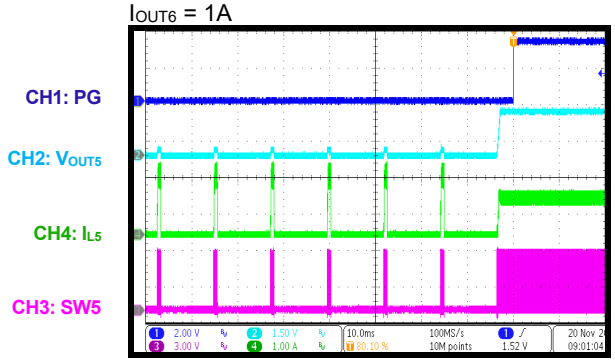
SCP Recovery (Buck4)

Buck4 SCP Recovery, other bucks no SCP, I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



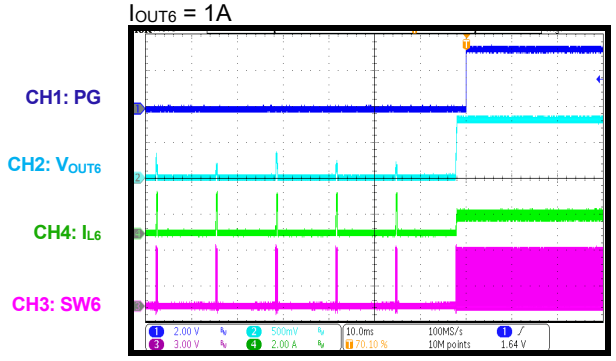
SCP Recovery (Buck5)

Buck5 SCP Recovery, other bucks no SCP, I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



SCP Recovery (Buck6)

Buck6 SCP Recovery, other bucks no SCP, I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



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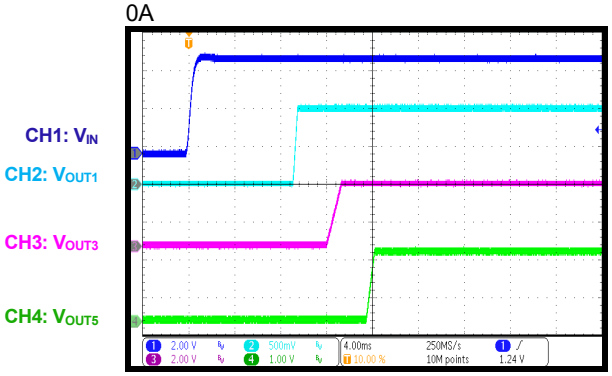
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

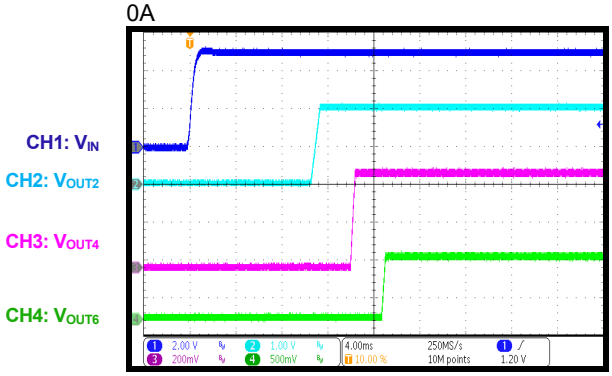
Start-Up through VIN (start-up sequence)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



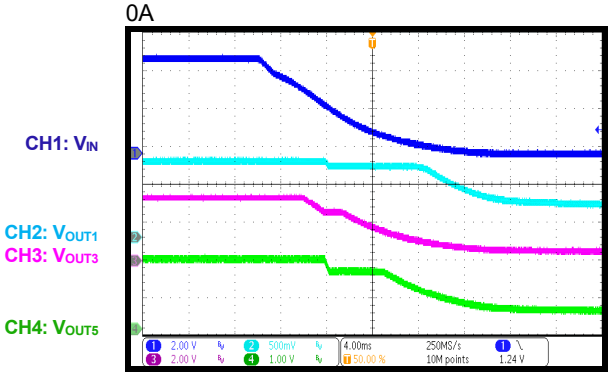
Start-Up through VIN (start-up sequence)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



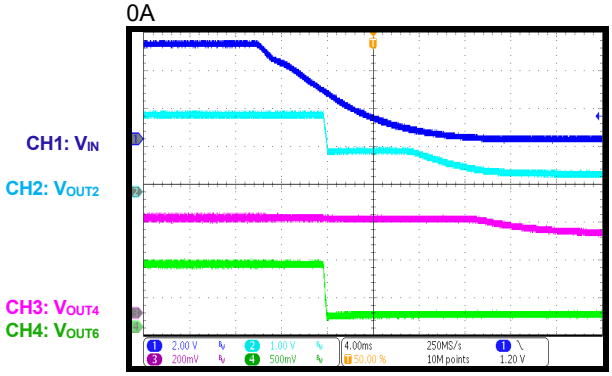
Shutdown through VIN (start-up sequence)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



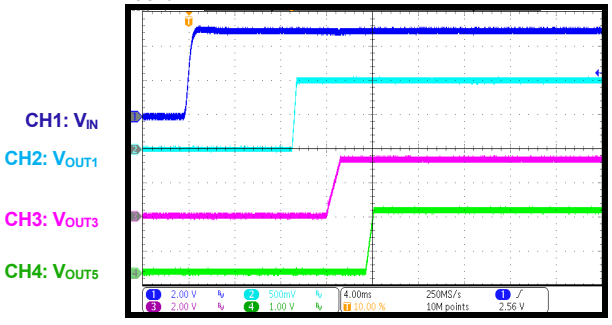
Shutdown through VIN (start-up sequence)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



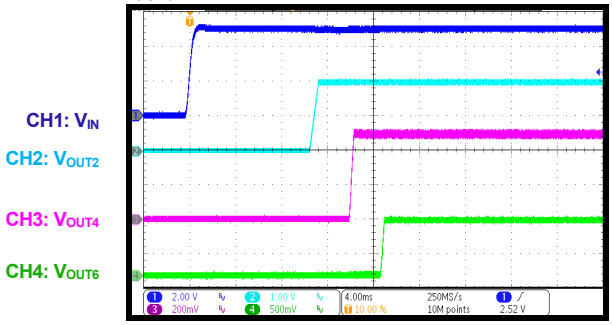
Start-Up through VIN (start-up sequence)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



Start-Up through VIN (start-up sequence)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



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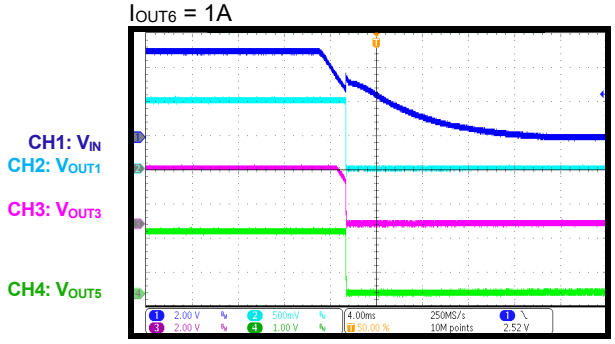
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

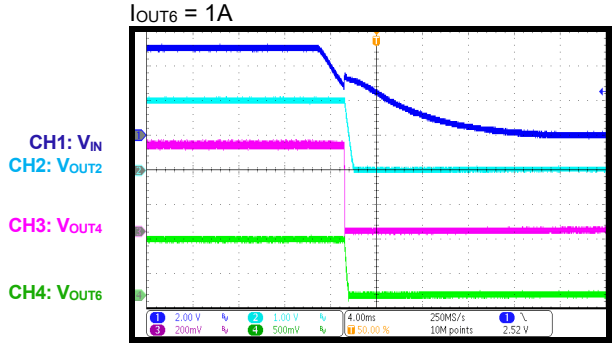
Shutdown through VIN (start-up sequence)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



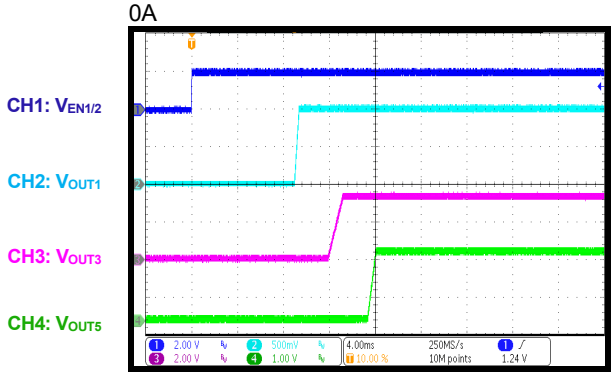
Shutdown through VIN (start-up sequence)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



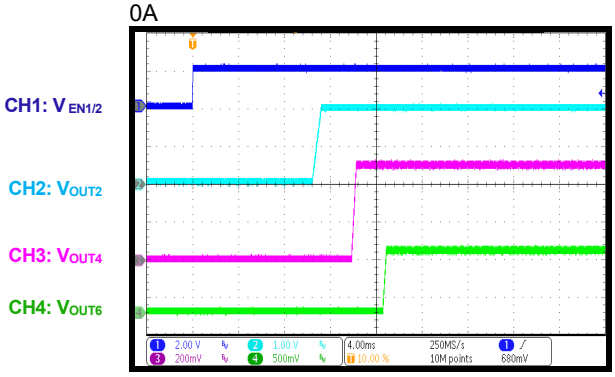
Start-Up through EN 0V to 2V (start-up sequence)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



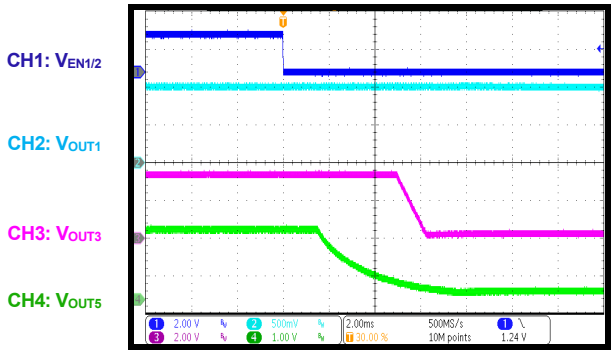
Start-Up through VEN 0V to 2V (start-up sequence)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



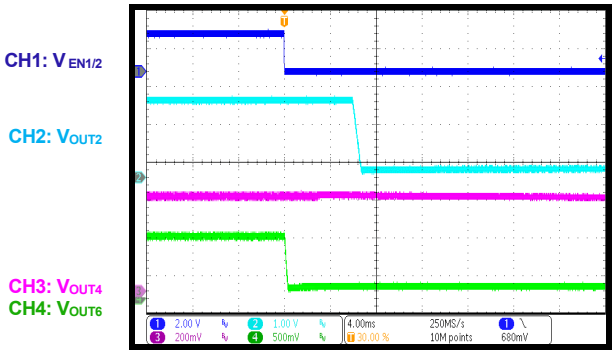
Shutdown through EN 0V to 2V (shutdown sequence)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



Shutdown through EN 0V to 2V (shutdown sequence)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



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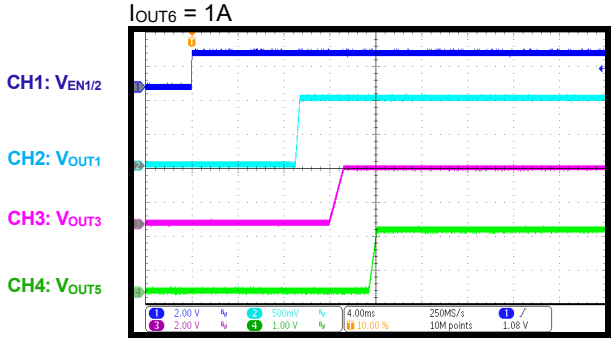
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

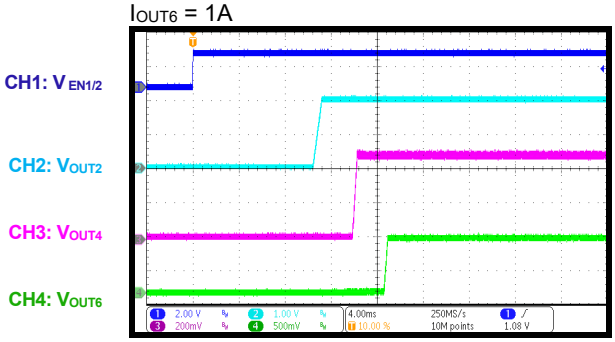
Start-Up through EN 0V to 2V (start-up sequence)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



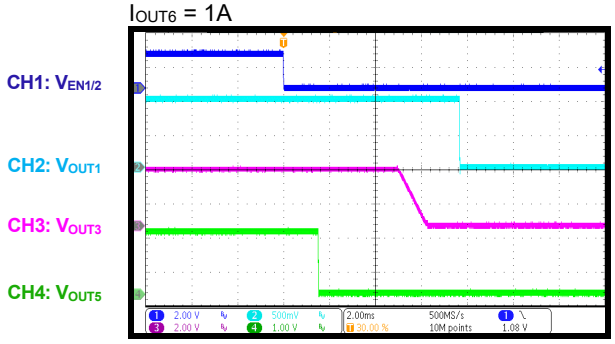
Start-Up through VEN 0V to 2V (start-up sequence)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



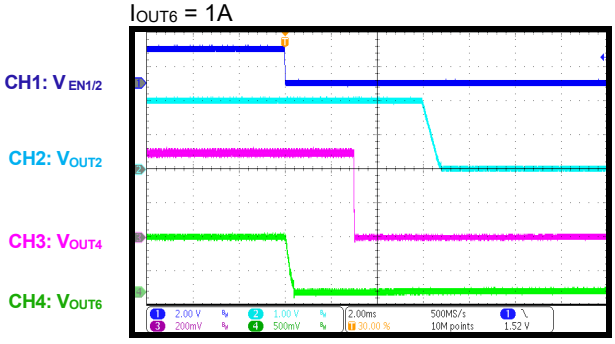
Shutdown through EN 0V to 2V (shutdown sequence)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



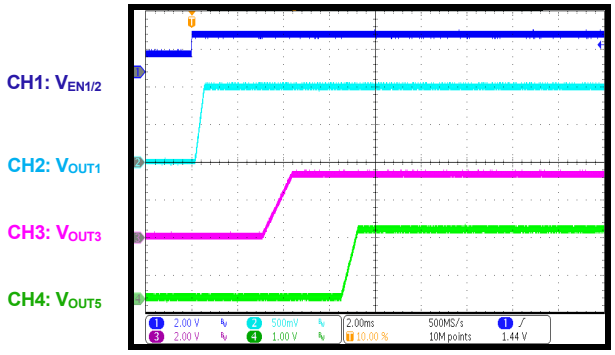
Shutdown through EN 0V to 2V (shutdown sequence)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



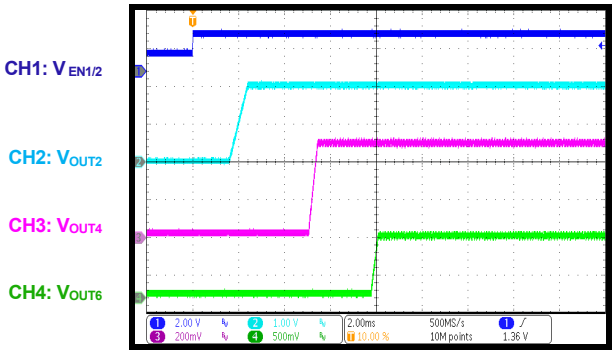
Start-Up through EN 1V to 2V (start-up sequence)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



Start-Up through VEN 1V to 2V (start-up sequence)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



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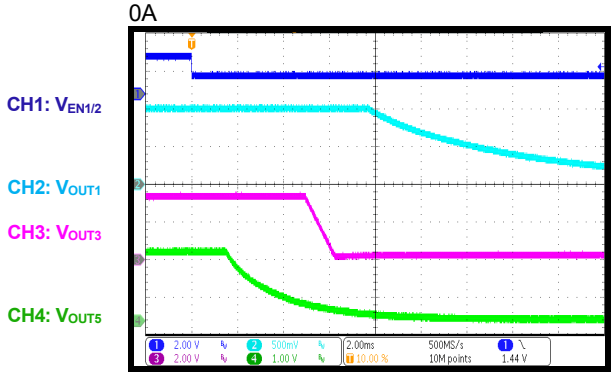
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

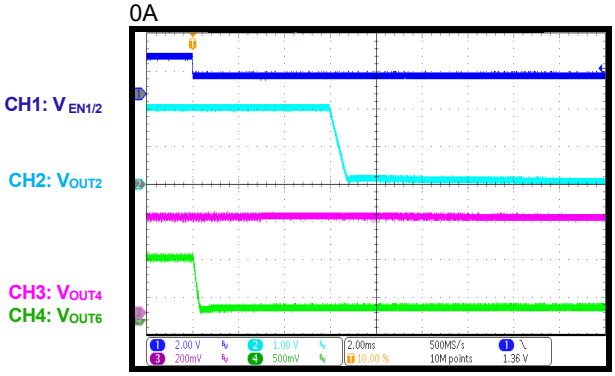
Shutdown through EN 1V to 2V (shutdown sequence)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



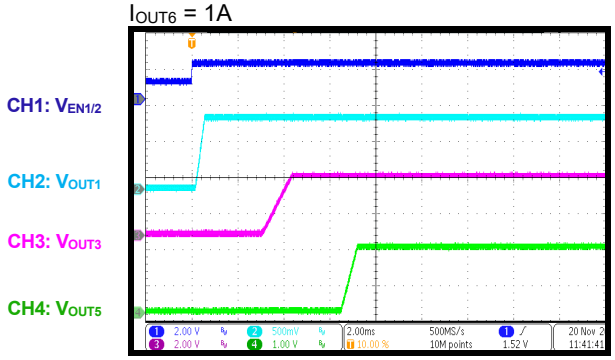
Shutdown through EN 1V to 2V (shutdown sequence)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



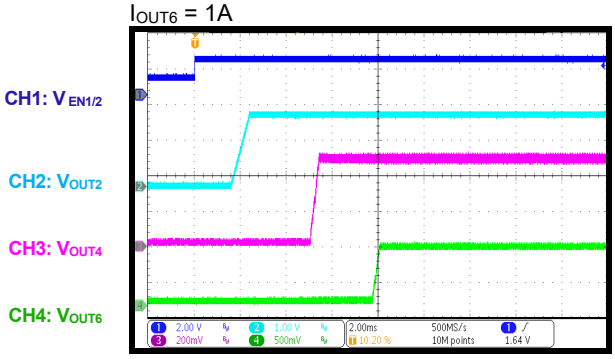
Start-Up through EN 1V to 2V (start-up sequence)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



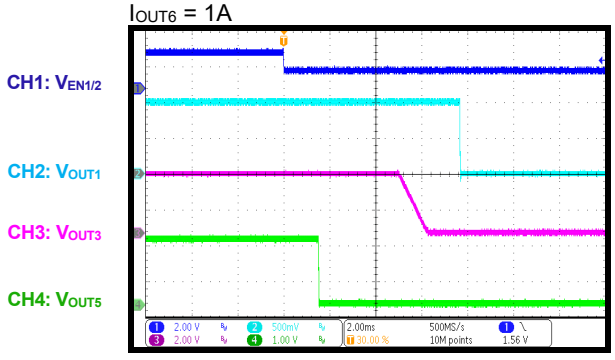
Start-Up through VEN 1V to 2V (start-up sequence)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



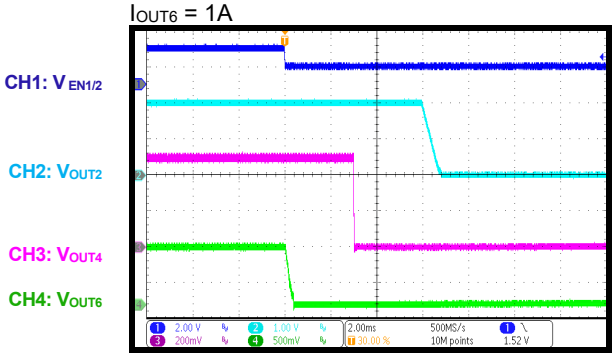
Shutdown through EN 1V to 2V (shutdown sequence)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



Shutdown through EN 1V to 2V (shutdown sequence)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



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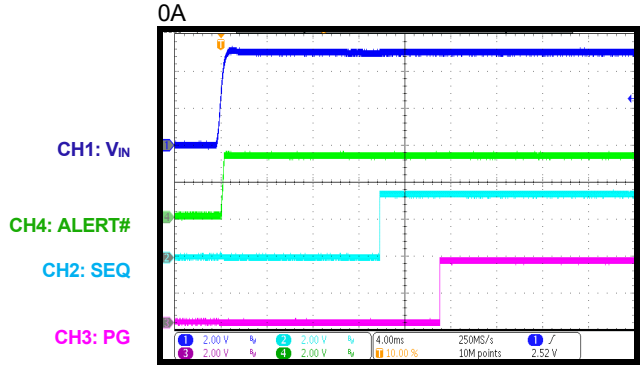
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

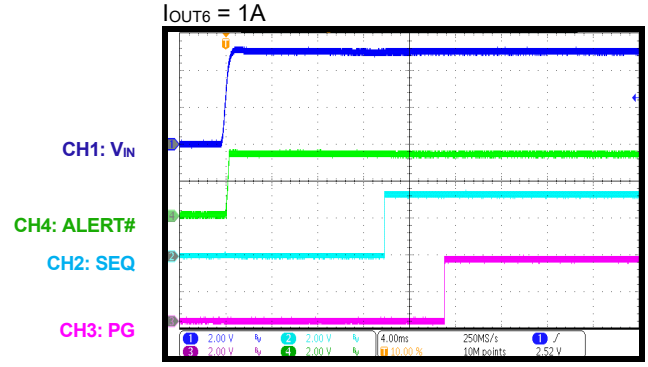
Start-Up through VIN (Other functional pins)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



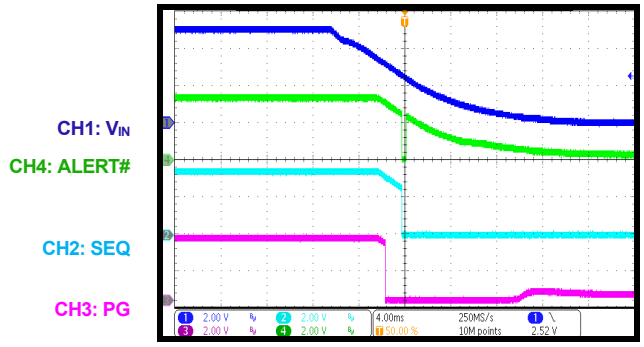
Start-Up through VIN (Other functional pins)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



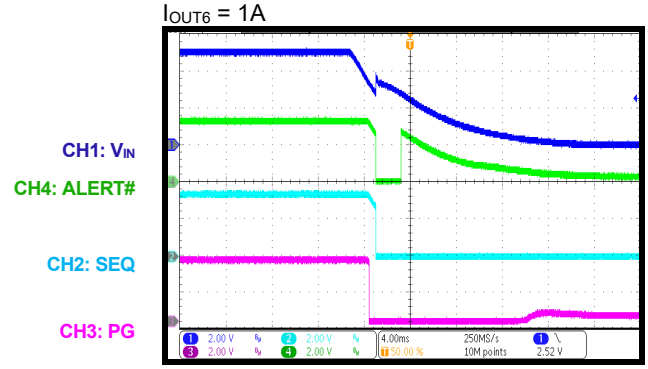
Shutdown through VIN (Other functional pins)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



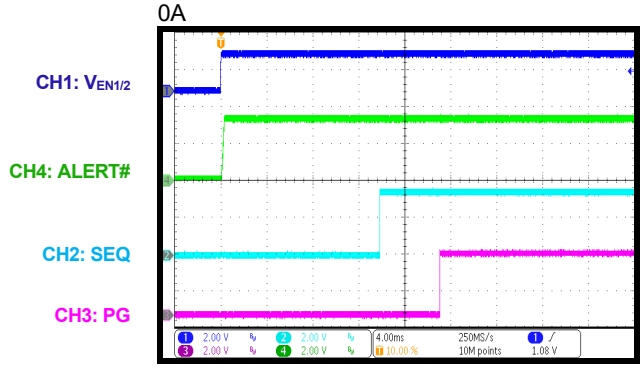
Shutdown through VIN (Other functional pins)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



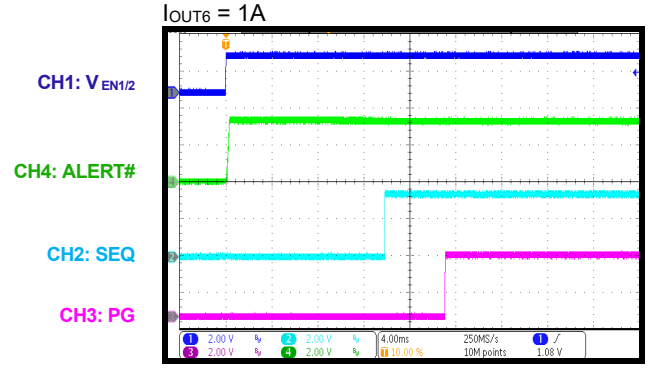
Start-Up through EN 0V to 2V (Other functional pins)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



Start-Up through EN 0V to 2V (Other functional pins)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



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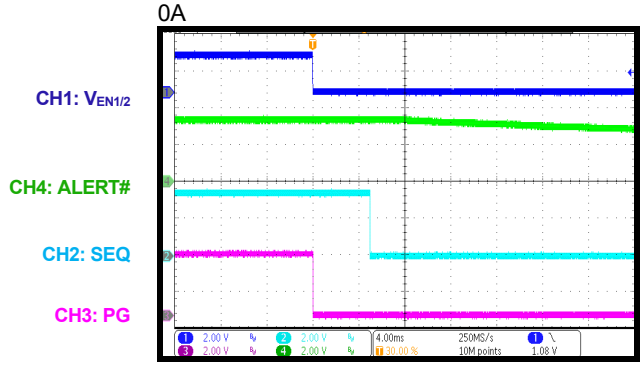
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TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

V_{IN} = 5V, V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, L1 = 470nH, L2 = 680nH, L3 = 470nH, L4 = 220nH, L5 = 2.2µH, L6 = 1.0µH, C_{OUT1} = 22+47µF, C_{OUT2} = 2x22µF, C_{OUT3} = 2x22µF, C_{OUT4} = 7x10µF, C_{OUT5} = 4.7+10µF, C_{OUT6} = 2x4.7µF, f_{SW} = 2MHz, T_A = +25°C, unless otherwise noted.

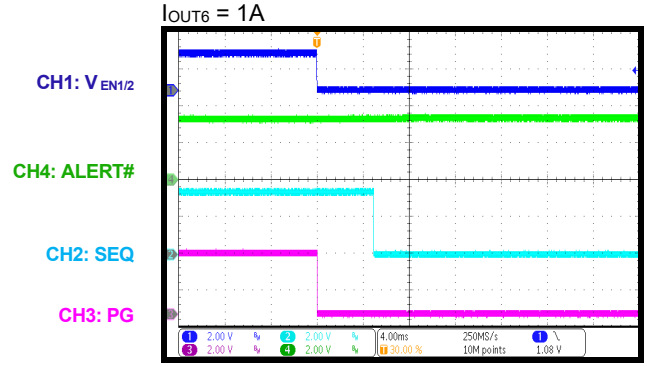
Shutdown through EN 2V to 0V (Other functional pins)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



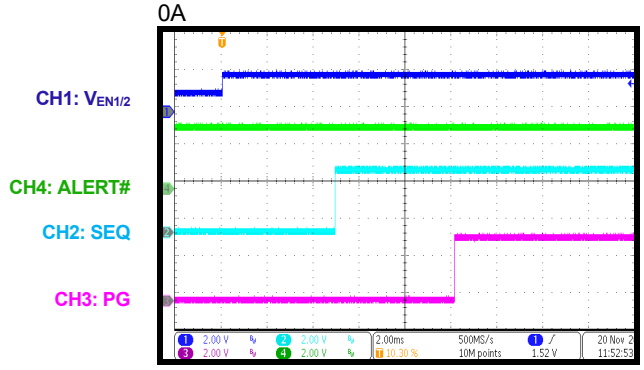
Shutdown through EN 2V to 0V (Other functional pins)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



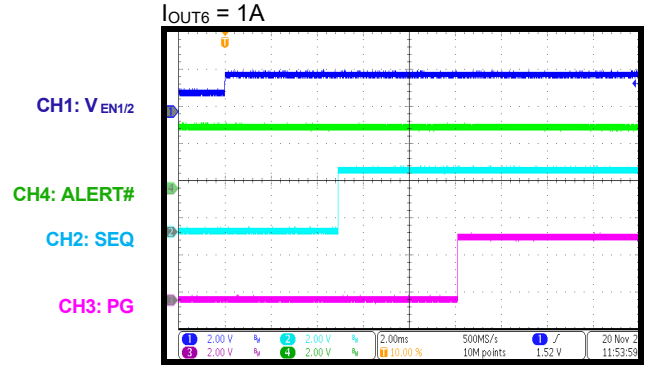
Start-Up through EN 1V to 2V (Other functional pins)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



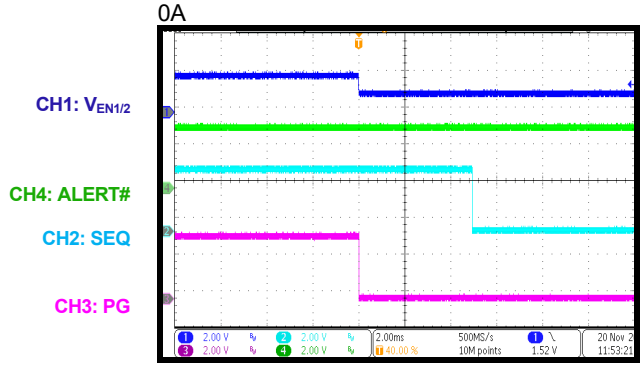
Start-Up through EN 1V to 2V (Other functional pins)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



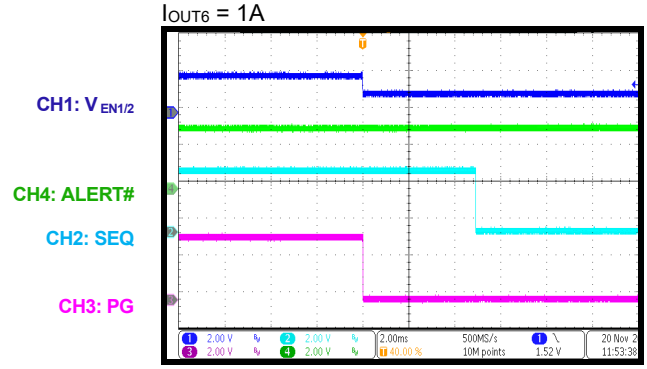
Shutdown through EN 2V to 1V (Other functional pins)

I_{OUT1} = I_{OUT2} = I_{OUT3} = I_{OUT4} = I_{OUT5} = I_{OUT6} = 0A



Shutdown through EN 2V to 1V (Other functional pins)

I_{OUT1} = I_{OUT2} =3A, I_{OUT3} = I_{OUT4} =4A, I_{OUT5} = I_{OUT6} = 1A



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FUNCTIONAL BLOCK DIAGRAM

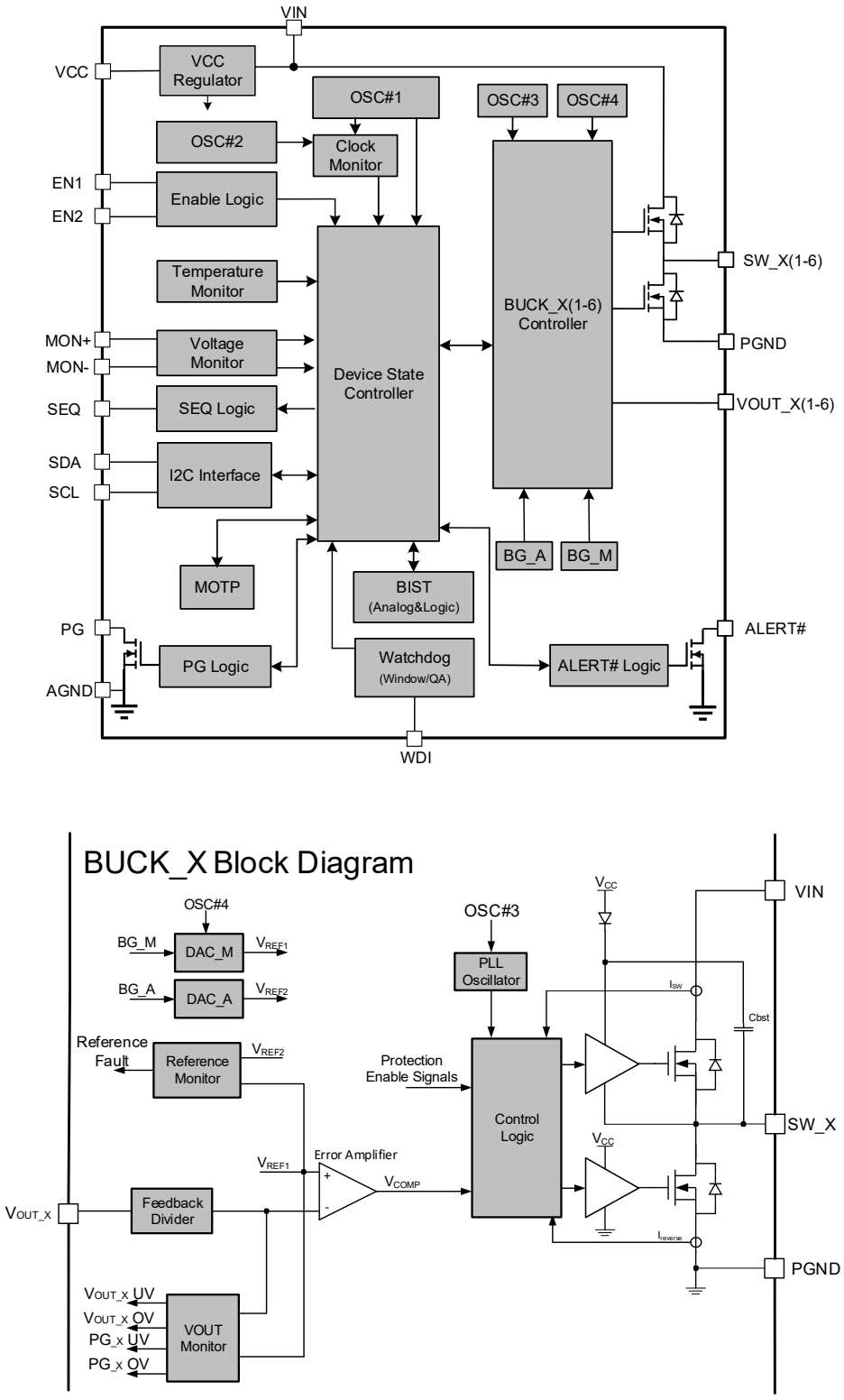


Figure 2: Functional Block Diagram

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POWER-UP and POWER-DOWN BEHAVIOR

The power-up behavior of MPQ70165FS is as shown in Figure 3. Figure 4 shows power-down behavior based on EN off action. The device may not be able to respond to MCU communication before PRE-DIAG, so if the MCU boots faster than 2rd Restore OTP, it should wait until 2rd Restore OTP is complete to use PMBus communication. If the Logic BIST fail, the device can't start machine. If the Analog BIST fail, the device transits to DEEP SAFE state.

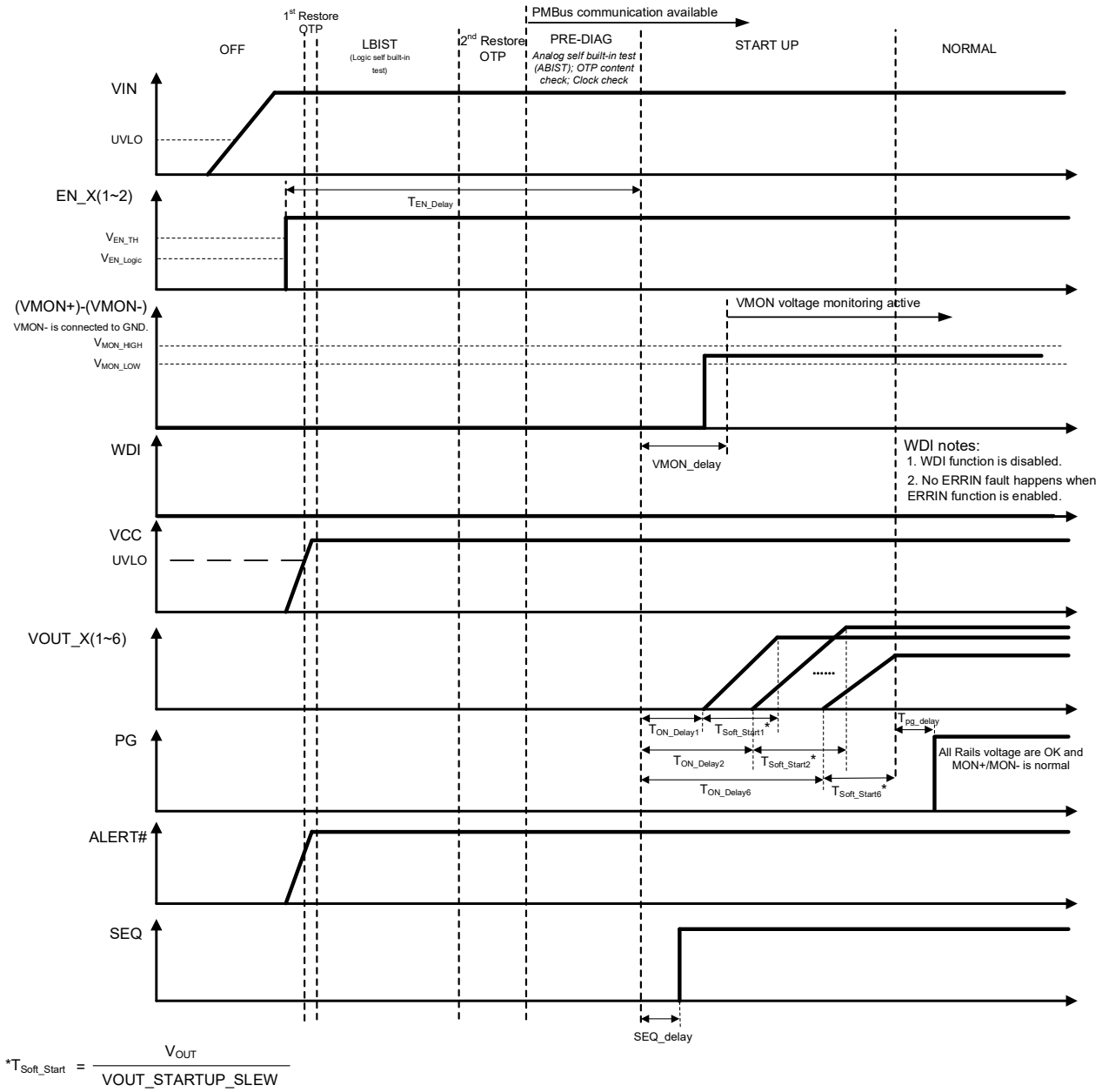


Figure 3: Power-up Behavior

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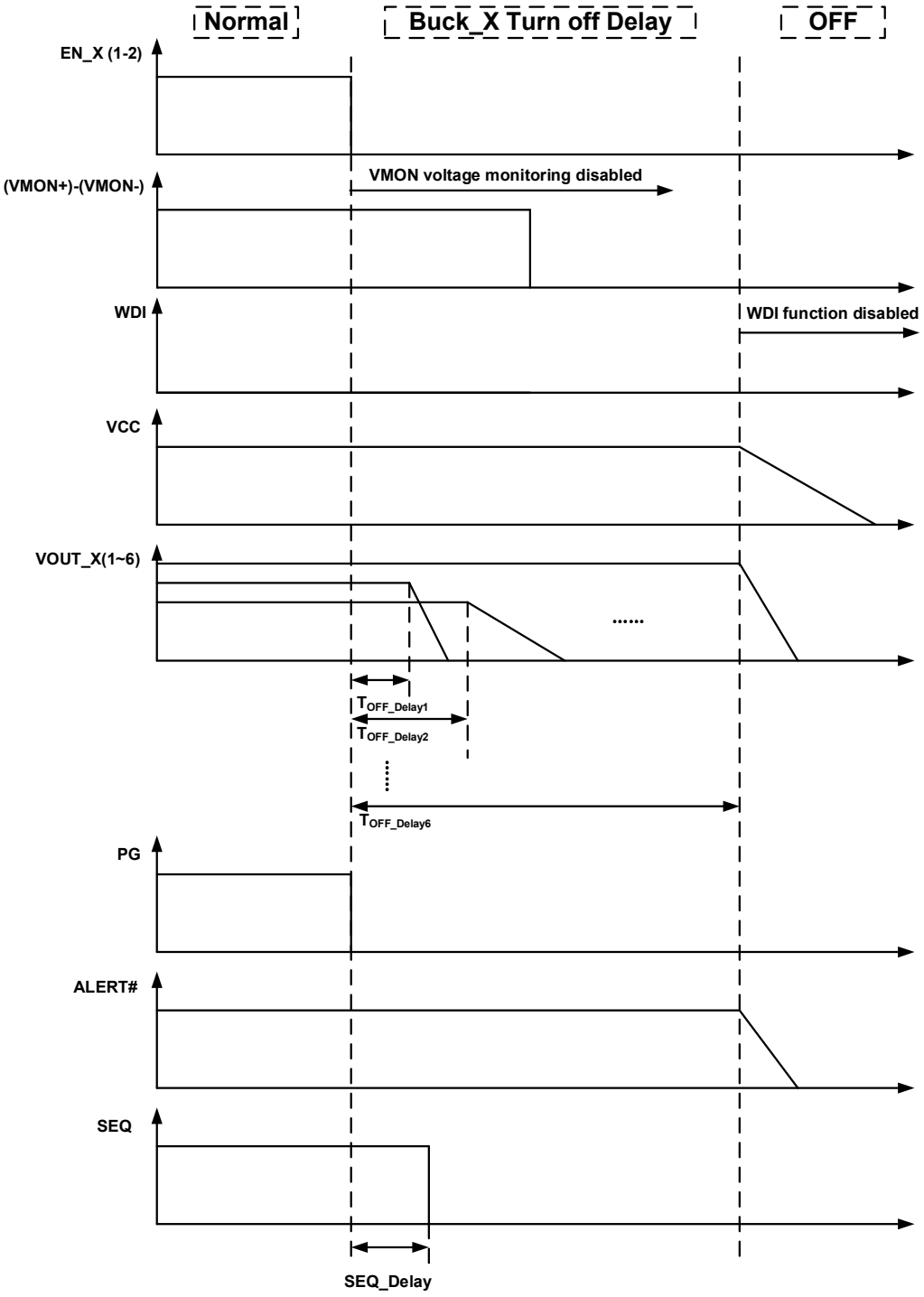


Figure 4: Power-down Behavior

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OPERATION

The MPQ70165FS is a highly flexible power solution for Advanced Driver-assistance Systems (ADAS) and System on Chip (SoC). It integrates six high frequency, synchronous rectified step-down converters. Integrated functional safety features improve system safety scores. An external auxiliary voltage monitor and sequencer allow the protection to be extended to additional supplies. PMBus and multi-page OTP allows significant configuration to meet the requirements of the target SoC.

High Efficiency Buck Regulator

Buck1 to Buck6 are synchronous step-down DC/DC with OVP/UVP, soft-start and configurable soft-stop.

These bucks operate with a fixed frequency, peak current mode control to regulate the output voltage (V_{OUT}). A pulse-width modulation (PWM) cycle is initiated by the internal clock. At the clock's rising edge, the high-side MOSFET (HS-FET) turns on and remains on until the control signal hits the COMP voltage (V_{COMP}), which is the output of the internal error amplifier. The output voltage of the error amplifier depends on the difference of the output feedback voltage and the internal reference voltage; it will determine how much energy should be transferred to the load. The higher the load current, the higher the COMP voltage.

When the HS-FET is off, the low-side MOSFET (LS-FET) turns on immediately and remains on until the next cycle starts. During this time, the inductor current will flow through the LS-FET. In order to avoid shoot-through, a dead time is inserted to avoid the HS-FET and LS-FET turning on at the same time. For each turn on/off period in a switching cycle, the HS-FET will remain on/off with a minimum on/off time limit.

If the current in the HS-FET cannot reach the value set by V_{COMP} within one PWM period, then the HS-FET remains on and skips a turn-off operation. The HS-FET is forced on until it reaches the value set by V_{COMP}, or its maximum on time (t_{ON_MAX}) (~3µs) is complete. When HS-FET is off, LS-FET turns on immediately and remains on minimum off time. This mode extends the duty cycle, which achieves low dropout when $V_{IN} \approx V_{OUT}$.

Multiphase Operation

Multiphase Operation is available, and can be configured with OTP. Multiphase pairs are Buck1 & Buck2, Buck3 & Buck4, and Buck5 & Buck6. Other combinations do not support parallel work. When two bucks are set to multiphase work, MPQ70165FS will cut V_{COMP} of secondary buck. Instead of, MPQ70165FS will connect V_{COPM} of primary buck to loop control logic of secondary buck. This operation can ensure that the V_{COMP} of the secondary buck is the same as that of the primary buck. The inductance, output and input voltages of the two Bucks are the same, so their output currents are basically the same.

Two bucks work in parallel, should configure same parameters on both PAGEs. For example, Buck1/2 are configured to multiphase operation. V_{OUT1} and V_{OUT2} shall be configured to same value by VOUT_COMMAND. See register map section for more information. If disable parallel buck pair through PMBus interface, only set OPERATION under PAGE 0 or PAGE 2 or PAGE 4 or PAGE FF. Other PAGEs will be invalid.

When set two Bucks worked in parallel, two Bucks' inductor can't be omitted and should be same inductance value. VOUT pins of the two bucks should be connected together. Please refer to the Figure 5 for parallel pair's peripheral circuit (as Buck1 & Buck2 for example).

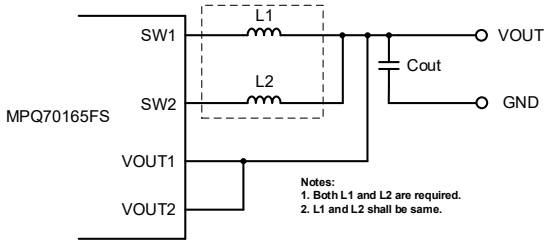


Figure 5: Parallel Pair's Peripheral Circuit

Frequency Foldback

MPQ70165FS provides a frequency foldback point to achieve low output voltage under high input voltage. When VOUT_COMMAND (21h) setting voltage exceeds 0.65V, Buck's switch frequency (f_{SW}) is fixed 2MHz frequency. When VOUT_COMMAND (21h) setting voltage drops below of 0.65V, Buck's f_{SW} will drop. The lowest

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f_{SW} is about 400kHz under normal operation.

In multiphase operation, secondary buck's f_{SW} is controlled by own VOUT_COMMAND (21h) setting voltage. So, it is recommended that secondary buck's VOUT_COMMAND (21h) setting voltage shall be same with primary buck.

Phase Shift

MPQ70165FS provide two phase clock signals. They have 180° phase shift.

Buck1 / Buck3 / Buck6 clocks share a phase, and it is 0°. Buck2 / Buck4 / Buck5 clocks share a phase, and it is 180°. Figure 6 shows their phase relation.

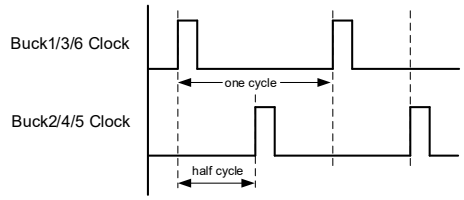


Figure 6: Clock Phase Shift

But if someone Buck's fSW happens frequency foldback, this Buck's phase will be not fixed than another Buck without frequency foldback.

Light Load Operation

The MPQ70165FS can support forced continuous conduction mode (FCCM) mode.

In FCCM, MPQ70165FS allows the inductor current to reach negative values, up to the negative current limit (see Figure 7).

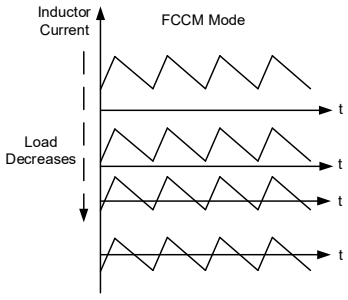


Figure 7: FCCM Mode

Forced CCM provides the best transient and frequency performance. The advantage of FCCM are the controllable frequency and lower output ripple and smaller peak inductor current at light load.

VIN/VCC Under-Voltage Lock Out

Input under voltage lock out is derived from VIN on pin 3. If VIN drops below of $V_{IN_UVLO_FALLING}$ threshold, but VCC is higher than $V_{CC_UVLO_FALLING}$ and enable signal is ready, part will shut down all output according to turn off delay sequence and state machine will enter deep safe state. If any buck enables soft-stop or 100 Ω discharge, V_{OUT} will have external discharge path. PMBus communication can still work normally. If VIN increase to exceed $V_{IN_UVLO_Rising}$ threshold, MPQ70165FS will start up again. MPQ70165FS don't support record VIN UVLO fault by register.

If VCC is lower than VCC_UVLO, all outputs will be shut down at the same time and without external discharge path although enable soft-stop or 100 Ω discharge. PMBus communication is unavailable and digital block is reset.

ERROR Input

MPQ70165FS provides ERROR input (ERRIN) function to receive system fault. If enable ERRIN function, watchdog mode shall be configured to Question-Answer watchdog mode by WATCHDOG_CONFIG (B2h). WDI pin will be reused to ERRIN pin.

When ERRIN level exceeds min WDI high threshold, MPQ70165FS can identify an ERRIN fault. All channels' output will be shut down and PG / ALERT# are also pulled low. State machine will enter deep safe state. MPQ70165FS will re-start until ERRIN level drops below of WDI low threshold. Figure 8 shows ERRIN reaction.

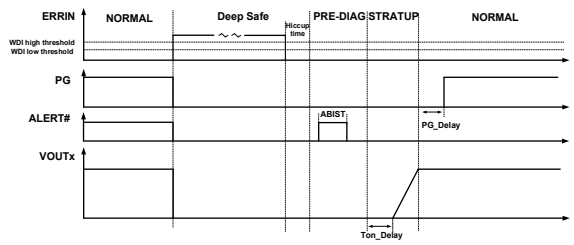


Figure 8: ERRIN Reaction

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Enable Group

MPQ70165FS provide two enable pin to turn on / off the part. Six bucks' on / off, VMON and SEQ on / off can be configured to different enable pin. That can be set by ENABLE_GROUP register.

While any enable pin voltage exceeds EN logic on threshold, V_{CC} will be generated by V_{IN}. The enable voltage rises further, and when the enable voltage exceeds the EN rising threshold, VMON, SEQ and six bucks will be active.

Enable On Sequencing

Each output can be assigned to either enable domain 1 or 2. After PRE DIAG passed, if the corresponding enable is driven high, each assigned output will rise after the programmed delay. For each Buck, their delay time can be independently and freely configured through the **TON_DELAY** register.

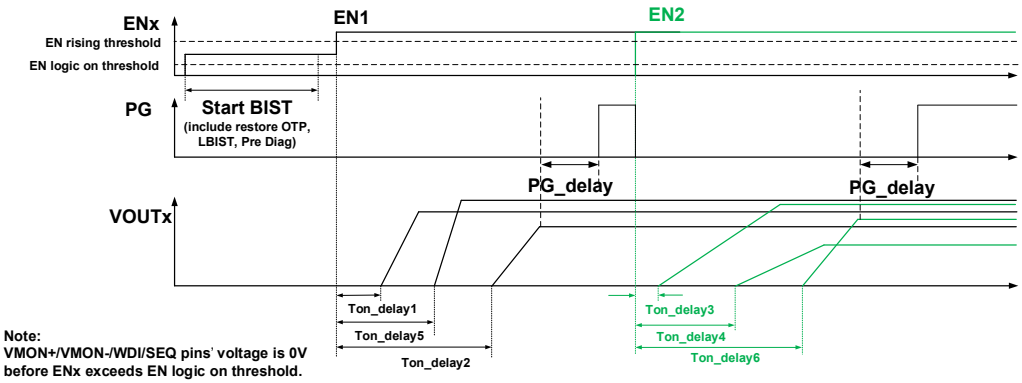


Figure 9: Turn On Timing Diagram with EN Group Configuration

Figure 9 shows an application example where EN1 controls Buck1/2/5, EN2 controls Buck3/4/6. VMON voltage is normal before EN1 exceeds EN rising threshold.

In Figure 9, when EN1 drive high, buck1/2/5 will start on sequence. After buck2 soft start ending, PG signal pulls high after TPG_DELAY time until EN2 pulled high. If EN2 is pulled high before PG pulled high, PG won't be pulled high after PG delay time until all outputs are normal.

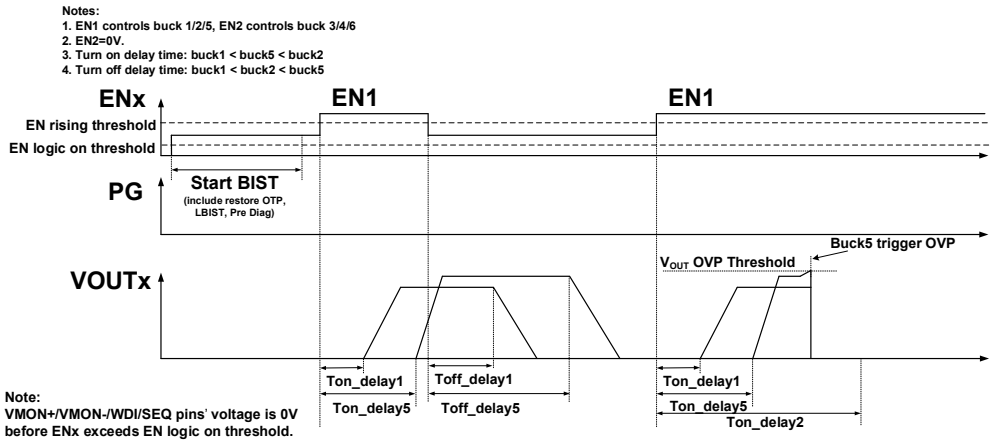


Figure 10: Timing Diagram with EN1 off or OVP Trigger

Figure 10 shown, EN1 pulled down during the soft-start of buck5, buck1/5 will enter turn off delay state and buck2 will not startup. If buck5 triggers V_{OUT} over voltage protection (OVP) before buck2 soft-start and after buck5 soft start ending, buck2 don't startup and buck1/5 will be shutdown at the same. MPQ70165 has

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same behavior when buck 5 triggers V_{OUT} under voltage protection (UVP) instead of OVP.

Enable Off Sequencing

In normal state, when the corresponding enable is driven low, each assigned output will shutdown after the programmed delay. For each Buck, their delay time can be independently and freely configured through the **TOFF_DELAY** register.

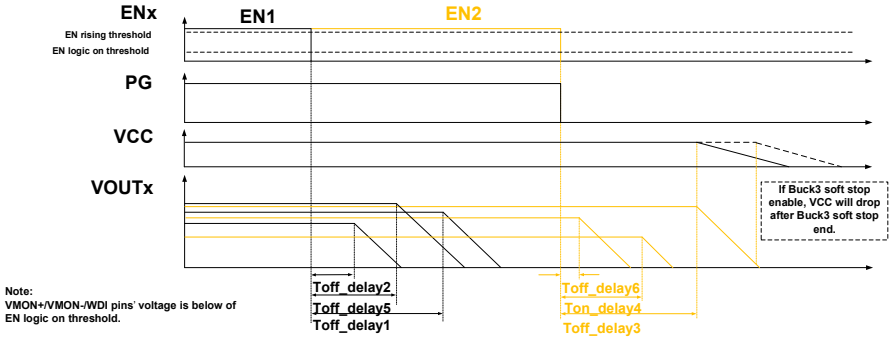


Figure 11: Turn Off Timing Diagram with EN Group Configuration

Figure 11 shows an application example where EN1 controls Buck1/2/5, EN2 controls Buck3/4/6. VMON voltage is normal during power off sequence. In the event that no other errors occur, PG will not be pulled down until the last EN is pulled down.

In Figure 11, VCC will drop after toff_delay3 timing ending. So buck3 don't have external 100 Ω discharge path although its 100 Ω discharger enabled. If buck3 enable soft-stop function, buck3's output will shut down according to its shutdown slew rate and VCC will drop after soft stop ending.

Output Discharge

The MPQ70165FS supports two output discharge modes: 100Ω discharge resistor; soft stop. This discharge mode can be set to enable or disabled independently for each buck and supports OTP.

The discharge resistor is connected between the VOUT pin and AGND. When VCC is ready and 100Ω discharge resistor enabled, the 100Ω discharge resistor will be connected to circuit after MPQ70165FS receives the shutdown signal. The energy on the output capacitor is discharged through the 100Ω resistor. During normal operation, the discharge resistor will be disconnected.

For soft stop function detail, please see Internal Soft-Stop section. If soft stop enabled, output drops with programmed shutdown slew rate.

When both discharge paths are enabled at the same time, the soft stop feature is dominant, but the 100Ω discharge resistance is also connected between VOUT and AGND

Internal Soft-Start

The soft-start function is implemented to prevent the PMIC output voltage from overshooting during start-up.

When MPQ70165FS starts up, the internal reference voltage (Vref) that ramps up from 0V. The Vref as error amplifier positive input voltage. After Vref reaches the rated voltage (the rated voltage will change with VOUT_COMMAND setting value), MPQ70165FS still needs to wait ~30us delay time, and then the soft start period is complete. The soft start time of each output can be adjusted by setting different VOUT startup slew rate (VOUT_STARTUP_SLEW). The slew rate can be OTP.

Internal Soft-Stop

The soft-stop function is implemented to discharge the output voltage according to programmed shutdown slew rate.

When MPQ70165FS starts turns off, the soft stop voltage ramps down from reference voltage to 0V. The output follows the soft stop voltage. This will cause charge to be transferred from the output to the input. Soft stop can be

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enabled and slew rate adjusted for each output with OTP.

Power Good and ALERT# Indicators

Power Good and ALERT# are open drain outputs. In normal state, Power Good will assert if the Buck voltage is not within the power good range or VMON voltage exceeds monitoring range. The power good range of each buck can be adjusted by register **PG_CONFIG**. The VMON voltage monitoring range can also be adjusted by related registers (**VMON_LOW_THRESHOLD**, **VMON_HIGH_THRESHOLD** and **MON_RANGE**). PG and ALERT# will assert to indicate a fault with the status of the device. See detail actions at Table 1. If PG_REACTION is set to 1, some faults will cause different action at Table 2. Some action can be masked by **ALTERT#_MAPPING** and **PG_MAPPING** registers. If MPQ70165FS happens deep safe events, PG will be pulled low immediately and enter deep safe state. Then, PMIC will restart according to normal power up sequence after hiccup off time done. When all Bucks voltage is within allowable window and VMON is normal, PG will be pulled high after PG_Delay.

Table 1: PG&ALERT# response for different fault type (PG_REACTION=0)

Fault Type	PG assert	ALERT# assert	Bucks' Output	State machine
Thermal shutdown	Y	Y	Off	DEEP SAFE
Thermal warning	Y	Y	On	NORMAL
PMBus CRC error	Y(10)(11)	Y(12)	On	NORMAL
PMBus communication error	Y(10)(11)	Y(12)	On	NORMAL
Busy	N	Y	On	NORMAL
Force PG assert by register	Y	N	On	NORMAL
Force ALERT# assert by register	N	Y	On	NORMAL
V_{OUT} Max / Min register warning	Y	Y	On	NORMAL
Over voltage protection	Y	Y	Off	DEEP SAFE
Under voltage protection	Y	Y	Off	DEEP SAFE
PG threshold over voltage	Y	N	On	NORMAL
PG threshold under voltage	Y	N	On	NORMAL
Register Write Check error	N	Y	On	NORMAL
OTP CRC check error	Y(10)(13)	Y(12)	Off(10)	DEEP SAFE (10)
ABIST fault	N	Y	Off	DEEP SAFE
Memory CRC Check error	Y(10)	Y(12)	Off(10)	DEEP SAFE (10)
State Machine Error	Y	Y	Off	DEEP SAFE
LBIST fault	N	N	Off	DEEP SAFE
Frequency check Fail	Y(13)	Y	Off	DEEP SAFE
Monitor over voltage	Y	N	On	NORMAL
Monitor under voltage	Y	N	On	NORMAL
Watchdog fault	N	N	On	NORMAL
Watchdog failure	Y(10)	Y(12)	On	NORMAL(14)
Reference fault	Y(10)	Y(12)	Off(10)	DEEP SAFE (10)
ERRIN fault	Y	Y	Off	DEEP SAFE

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Table 2: PG&ALERT# response for different fault type (PG_REACTION=1)

Fault Type	PG assert	ALERT# assert	Bucks' Output	State machine
PG threshold over voltage	Y	N	Off	DEEP SAFE
PG threshold under voltage	Y	N	Off	DEEP SAFE
Monitor over voltage	Y	N	Off	DEEP SAFE
Monitor under voltage	Y	N	Off	DEEP SAFE

Notes:

10) Behavior can be configured via PG_MAPPING (C6h).
11) If only the fault occurs, PG pulls low and maintains low status within PG_DELAY time. After that, if no other fault occurs, the PG will pull high.
12) Behavior can be configured via ALERT#_MAPPING (C4h).
13) Only in normal state, PG can be pulled low while fault happen.
14) Behavior can be configured via WD_FAULT_CFG (bit2) of WATCHDOG_CONFIG (B2h).

Output Over-Voltage Protection

MPQ70165FS provides a fixed, higher threshold to protect output, also known as the over-voltage protection (OVP) threshold. When any output exceeds the OVP threshold duration ~10us, all outputs shut off at the same time, and state machine will be transferred to DEEP SAFE state and timing hiccup time. If wake up signal is enabled, MPQ70165FS will perform PRE DIAG automatically. PRE DIAG passed, the normal power up sequence will begin again. Figure 12 shows an example of buck1 triggering OVP.

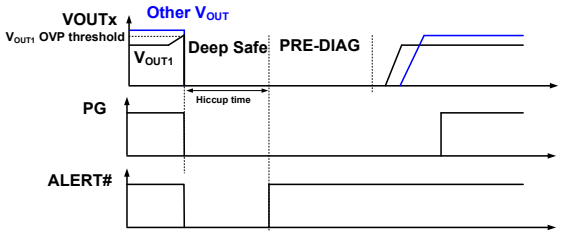


Figure 12: OVP Reaction

OVP is not always activated. During soft start period, MPQ70165FS masks OVP. So MPQ70165FS doesn't respond OVP until soft start ends After the soft start ends, if the channel doesn't trigger OVP, MPQ70165FS doesn't report OVP fault.

Channels' output discharge will also operate during the hiccup time except for the channel where OVP occurs.

Output Under-Voltage Protection

MPQ70165FS provides a fixed, lower threshold to protect output, also known as the under-voltage protection (UVP) threshold. When any output exceeds the UVP threshold duration ~10us, all outputs shut off at the same time, and state machine will be transferred to DEEP SAFE state and timing hiccup time. If wake up signal is enabled, MPQ70165FS will perform PRE DIAG automatically. PRE DIAG passed, the normal power up sequence will begin again. Figure 13 shows an example of buck1 triggering UVP.

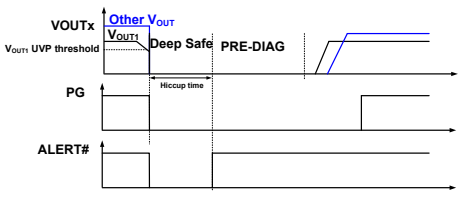


Figure 13: UVP Reaction

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UVP is not always activated. During soft start period, MPQ70165FS masks UVP. So MPQ70165FS doesn't respond UVP until soft start ends After the soft start ends, if the channel doesn't trigger UVP, MPQ70165FS doesn't report UVP fault.

Channels' output discharge will also operate during the hiccup time except for the channel where UVP occurs.

Output Current Protection

The power MOSFET current is accurately sensed via a current-sense MOSFET. It is then fed to the high-speed current comparator for current mode control. The current comparator takes this sensed current as one of its inputs.

When the HS-FET turns on, the comparator is blanked until the end of the turn-on transition to avoid noise. Then the comparator compares the power MOSFET current with V_{COMP}. When the sensed current exceeds V_{COMP}, the comparator outputs low to turn off the HS-FET. During LS-FET turns on, if I_L exceed valley current limit value at next clock arriving, LS-FET will continue turned on until I_L drops below of valley current limit value. The internal power MOSFET's maximum current is internally limited cycle by cycle.

Compensation

The MPQ70165FS integrates internal compensation for each buck output. The compensation is adjustable with OTP. If changes are made over OTP, they will be made when that output is disabled and then enabled.

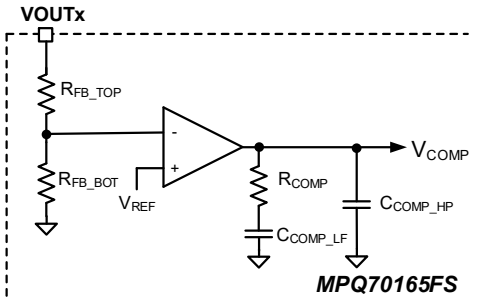


Figure 14: Compensation Network

Figure 14 shows a typical Type II compensation network is fully integrated in the chip. The components values are all programmable by PMBus interface. When VOUT_SL = 1, R_{FB_TOP} : R_{FB_BOT} = 1:3; When VOUT_SL = 2, R_{FB_TOP} : R_{FB_BOT} = 5:3.

Frequency Spread Spectrum (FSS)

Spread spectrum modulates the clock frequency with a triangle wave. The modulation range is typically 15% and the frequency is 9kHz, see Figure 15. This can be enabled or disabled over PMBus interface.

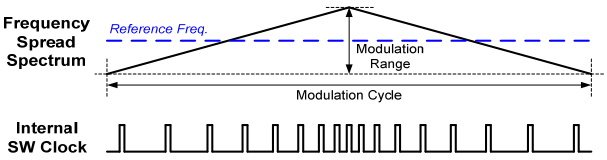


Figure 15: Frequency Spread Spectrum

When FSS is enabled, the operating frequency will decrease from 2MHz to the minimum value, and then the operating frequency will slowly increase. After reaching the maximum, slowly decrease. When FSS is disabled, the operating frequency will change from the current value to 2MHz.

External Voltage Monitor

MPQ70165FS provides an external voltage monitor function. This function can detect any input or output voltage on all system.

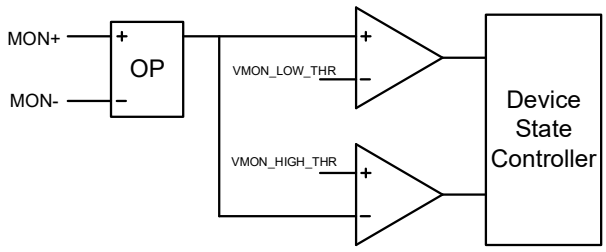


Figure 16: Monitor Voltage Detection

In Figure 16, VMON_LOW_THR and VMON_HIGH_THR are configurable through PMBus interface. For MON+ / MON- pins, in addition to being the input to VMON, it also serves as the enable signal for the internal VCC and some internal circuits. So, if EN voltage is 0V, but MON+ or MON- voltage exceeds EN logic on threshold, MPQ70165FS can also perform these steps (from 1st RESTORE OTP to PRE-DIAG) and then PMBus interface can work.

In normal state and no other fault happens, when external input voltage exceeds the monitor range, PG will be pulled down. If the VMON voltage recovers normal, PG will be pulled high after PG delay timing ending.

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In PG_REACTION = 1, if the VMON voltage exceeds the monitor range, MPQ70165FS doesn't only pull low PG, but also enter DEEP SAFE state.

Sequence Output

MPQ70165FS provides a sequence output with push - pull structure. After PRE-DIAG passed, the related enable pin is pulled high and then timing SEQ delay time, SEQ will be pulled high. If the related enable pin is pulled low, SEQ will be pulled low after SEQ delay timing ending. See Figure 17 for SEQ on / off timing diagram.

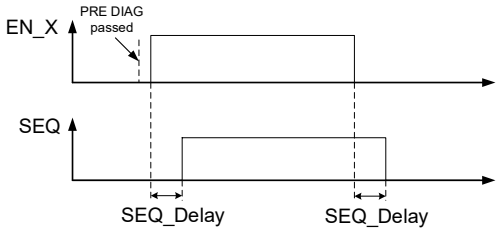


Figure 17: SEQ Timing Diagram

MPQ70165FS supports the configuration of which EN controls SEQ output, see ENBALE_GROUP register. SEQ delay time can be configured by SEQ_DELAY register.

For SEQ pin, in addition to being the sequence output, it also serves as the enable signal for the internal VCC and some internal circuits. So, if EN voltage is 0V, but SEQ voltage exceeds EN logic on threshold, MPQ70165FS can also perform these steps (from 1st RESTORE OTP to PRE-DIAG) and then PMBus interface can work.

VOUT Command Protection

MPQ70165FS provides a protection for VOUT COMMAND. When VOUT COMMAND value exceeds the range of VOUT_MIN (2Bh) / VOUT_MAX (24h), the actual V_{OUT} will follow V_{OUT_MAX} / V_{OUT_MIN}. This protection is not same with V_{OUT} OVP / UVP.

Multi-Page One-Time Programmable (MOTP) Memory

The MPQ70165FS provides an MOTP memory function to set custom default parameters.

MPS provides GUI (Virtual Bench Pro 4.0) and digital interface tool (EVKT-USBI2C-03) to configure the MPQ70165FS during the development process. Contact an MPS FAE to configure in application.

Only the MPQ70165FS-0000 code supports two OTP chances. Other codes only support once OTP. Before performing OTP, V_{IN} shall exceed 5V and state machine shall be NORMAL state. Then the host send a write command to STORE_USER_ALL register, but the command doesn't include any data. At OTP programmed period, the host shall not send any write command to MPQ70165FS. If the host send a write command, MPQ70165FS will report a busy fault. So, the host should stop PMBus write operation to MPQ70165FS within 1s after OTP programmed start.

Diagnostics

Each output is monitored for OV/UV. Each DAC output (reference for each buck) is monitored for VOUT_MIN (2Bh) / VOUT_MAX (24h). The digital clock speed is monitored. The state of PG and ALERT# are monitored when asserting. The digital logic of the PMBus registers, functional safety state machine, power good, ALERT# and sequencing should be check with LBIST during start up. The DAC comparators, power good comparators and monitor comparators are checked with ABIST during start up. The memory and OTP are protected by CRC. PMBus transactions are protected by PEC.

- **PMBus Protocol PEC**

  Data transfers through the PMBus have an attached packet error checking (PEC) code. The slave device rejects data unless the PEC matches the master device. PEC is an optional function related to SAFETY_CONFIG register. See the Packet Error Checking (PEC) section for more details.

  When MPQ70165FS trigger PMBus PEC fault, state machine has no reaction and all channels' output are normal. Only the ALERT# and PG will be pulled down and register can record fault flag. But PG will be pulled high automatically after PG_DELAY time.

- **Register Write Check**

  MPQ70165FS provides automatic read-back when the host send write command. If

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an error occurs, ALERT# will be pulled low and this write command is invalid.

- **Logic Self Built-in Test (LBIST)**

  The logic circuits are checked during LBIST. During LBIST, PMBus interface can't be used. If an error is detected, MPQ70165FS will not start-up.

- **Analog Self Built-in Test (ABIST)**

  The aim of ABIST is to verify correct function of diagnostic circuits (e.g., voltage supervisor) and increase the detection of latent faults.

  ABIST typically involves exercising diagnostic circuits into and out of fault scenarios by injecting currents or voltages into the diagnostic circuit to ensure the diagnostic circuit can switch to both faulted and non-faulted states. In other words, it is an analog circuit controlled by a digital circuit that verifies correct functionality of analog safety mechanisms.

  While ABIST is performing:

  - No assertion of ALERT#: The assertion of ALERT# due to the occurrence of any event which is supposed to trigger the relationship action and is part of the logic which is affected by the ABIST control is blocked.
  - No transition of state: The state machine will not change its state according to the reaction on detected emulated faults.

  Test items are:

  - Reference comparators
  - PG comparators
  - Auxiliary monitor voltage comparators

  If the ABIST failed, the device transits to DEEP SAFE state. ABIST fault will be reported by STATUS_MFR_SPECIFIC register. After wait hiccup time (configured via HICCUP_TIMER register), if wake up signal is available, state machine will re-enter PRE-DIAG. Write blank data to CLEAR_FAULTS register can clear ABIST fault register flag.

- **Clock Monitoring**

  Two digital clocks are implemented as Clk1, Clk2. Clk1 is primary digital clock and Clk2 is auxiliary monitor clock. Clk1 or Clk2 can start count at the same time. When counter in Clk1 arrive 1000, MPQ70165FS will check count number in Clk2. If it lower/higher than our threshold, the clock error counter will be increased 1. If Clk1 or Clk2 counts to 2048, the clock error counter is also increased to 1. If it is within the normal range, the clock error counter will be decreased 1.

  In NORMAL state, if the clock error counter exceeds 7, the part will pull low ALERT# and PG. Meanwhile, all channels shutdown at the same time and state machine enters DEEP SAFE state.

  Clock check fault will be reported by STATUS_FREQ register. After waiting for a hiccup time (configured via HICCUP_TIMER), if wake up signal is available, state machine will re-enter PRE-DIAG. If enable signal is high, MPQ70165FS will perform on sequence after PRE DIAG passed. Write blank data to 03h register can clear Clock check fault register (E7h bit0) flag.

  Besides, MPQ70165FS has other two clocks. Clk3 is used as a reference for Buck switching frequency. Clk4 provides a counting function for reference voltage rising / falling during soft start / stop. These two clocks are not monitoring by Clk2.

- **Reference Monitoring**

  MPQ70165FS has two 1.5V Bandgap voltages inside, one of which is used to generate reference voltage 1 (VREF1) to stabilize the output voltage in the control loop. It is also a reference value for monitoring the output voltage depth OVP / UVP. Another Bandgap voltage generates an auxiliary reference voltage 2 (VREF2) to monitor VREF1. If the difference between VREF1 and VREF2 exceeds a certain threshold of VREF2, MPQ70165FS will pull PG and ALERT# and all output will shut off at the same time.

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- **Thermal Warning and Shutdown**

  Thermal Warning is indicated by asserting a bit in the STATUS_TEMPERATURE register and asserting the PG and ALERT# pin.

  Thermal shutdown disables the device until the temperature drops below the hysteresis point. PG and ALERT# will also be asserted while in thermal shutdown.

  While MPQ70165FS recovery from TSD or TWN, PG will be pulled high automatically. But ALERT# is still low. Only through writing CLEAR_FAULTS register to clear TSD or TWN fault, ALERT# can be pulled high again.

- **OTP CRC Check**

  When MPQ70165FS has used one OTP chance, this CRC check shall be performed during 2nd Restore OTP and PRE-DIAG. MPQ70165FS use CRC-16 (Polynomial: 0xC002) to check 464 bits that include user page and trim page.

  Meanwhile, we also support continuous or once OTP CRC check by register at Normal State.

  This CRC check can be configured as:

  - Detect rise-edge (default): MFR_OTP_MEM_CHK 0xDA bit 1 transfer 0 to 1; based on bit 2=0.
  - Continuous: set MFR_OTP_MEM_CHK 0xDA bit 2=1, bit 1 = 1.

  If the OTP CRC check failed, the device transits to DEEP SAFE state and ALERT# / PG are pulled down. OTP CRC check fault will be reported. After waiting for a hiccup time (configured via HICCUP_TIMER register), if wake up signal is available, state machine will re-enter PRE-DIAG. Write blank data to CLEAR_FAULTS register can clear OTP CRC check fault register flag. MPQ70165FS detects OTP CRC check failure at different state, the part provides different register fault bit to report this fault.

  In 2nd RESTORE OTP state, if the OTP CRC check failed, the MFR_OTP_MEM_STATUS bit5 will be set to 1. In PRE-DIAG or NORMAL state, the MFR_OTP_MEM_STATUS bit1 will be set to 1 while OTP CRC check failed.

- **Memory (Register) CRC check**

  In NORMAL State, MPQ70165FS supports continuous or once Memory CRC check. It uses CRC-16 (Polynomial: 0xC002) to check 464 bits that include user page and trim page. It can detect at least 2 bits of failure of the registers. The reaction can be triggered when one bit error occurs.

  This CRC check can be configured as:

  - Detect rise-edge of 0xDA bit 0 (default): MFR_OTP_MEM_CHK 0xDA bit 0 transfer 0 to 1; based on bit 2=0.
  - Continuous: set MFR_OTP_MEM_CHK 0xDA bit 2=1, bit 0 = 1

  If the Memory CRC check failed, the device transits to DEEP SAFE state. Memory CRC check fault will be reported by MFR_OTP_MEM_STATUS bit2. After waiting for a hiccup time (configured via HICCUP_TIMER register), if wake up signal is available, state machine will re-enter PRE-DIAG. Write blank data to CLEAR_FAULTS register can clear Memory CRC check fault register (MFR_OTP_MEM_STATUS bit2) flag.

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PMBUS INTERFACE

PMBus Serial Interface Description

The Power Management Bus (PMBus) is an open-standard, power-management protocol that defines a means of communication with power conversion and other devices. The PMBus is a two-wire, bidirectional serial interface, consisting of a serial data line (SDA) and a serial clock line (SCL). The lines are externally pulled to a bus voltage when they are “idle”. Connecting to the line, a host device generates the SCL signal and device address and arranges the communication sequence. The MPQ70165FS works as a slave-only device which supports both the standard mode (100kbit/s), fast mode (400kbit/s) and fast mode plus (1Mbit/s) bidirectional data transfer, adding flexibility to the power supply solution. The output voltage, transition slew rate or other interesting parameters of the converter can be instantaneously controlled by PMBus interface.

Data Validity

One clock pulse is generated for each data bit transferred. The data on the SDA line must be stable during the high period of the clock. The HIGH or LOW state of the SDA line can only change when the clock signal on the SCL line is LOW (see Figure 18).

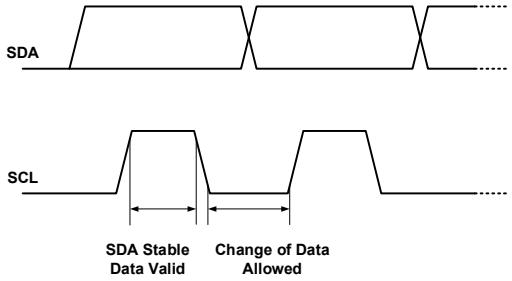


Figure 18: Bit Transfer on the PMBus

START and STOP Conditions

The START and STOP conditions are signaled by the host device, which signifies the beginning and the end of the PMBus transfer. The START condition is defined as the SDA signal transitioning from HIGH to LOW while the SCL line is HIGH. The STOP condition is defined as the SDA signal transitioning from LOW to HIGH while the SCL is HIGH as shown in Figure 19.

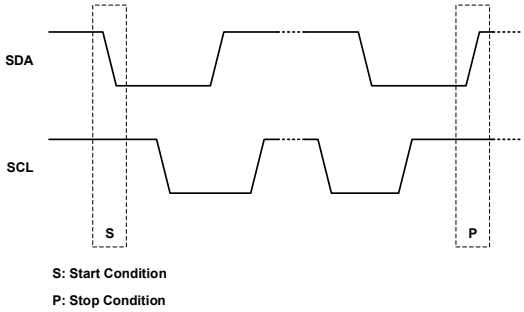


Figure 19: START & STOP Conditions

START and STOP conditions are generated always by the host. The bus is considered active after the START condition. The bus is considered idle again after the last ack data for write command. If it’s a read command, the bus is considered idle again during the last non-ack data. The bus stays active if a repeated START is generated instead of a STOP condition. The START and repeated START conditions are functionally identical.

Transfer Data

Every byte puts on the SDA line must be 8-bit long. Each byte must be followed by an Acknowledge bit. The acknowledge-related clock pulse is generated by the host. The transmitter releases the SDA line (HIGH) during the acknowledge clock pulse. The receiver must pull down the SDA line during the acknowledge clock pulse, so that it remains stable LOW during the HIGH period of the clock pulse.

Data transfers follow the format shown in Figure 20. After the START condition, a slave address is sent. This address is 7-bits long followed by an eighth bit, which is a data direction bit (R/W). A bit “0”’ indicates a transmission (WRITE), and a bit “1” indicates a request for data (READ). A data transfer is terminated always by a STOP condition generated by the host. However, if a host still wishes to communicate on the bus, it can generate a repeated START condition and address another slave without first generating a STOP condition.

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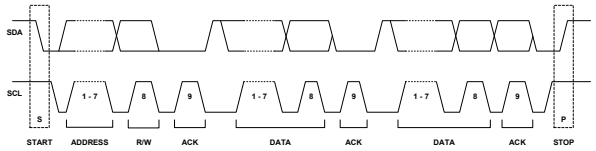


Figure 20: A Complete Data Transfer

Packet Error Checking (PEC)

The Packet Error Checking mechanism is employed to improve the communication reliability and robustness. Packet Error Checking, whenever applicable, is implemented by appending a Packet Error Code (PEC) after the data of each message transfer.

The PEC is a CRC-8 (abbreviated Polynomial: 0x83, which is ignored the last bit for full Polynomial: 0x107) error-checking byte, calculated on all the message bytes (including addresses, read/write bits, command codes, data). The PEC is appended to the message by the device that supplied the last data byte.

If an incorrect PEC is received, a communication fault is triggered. The power good flag and ALERET# will assert.

PMBus Communication Failure

A data transmission fault occurs when the data is not properly transferred between the devices.

There are several data transmission faults. All sorts of faults are listed below.

- Sending too few data
- Reading too few data
- The host sends too many bytes
- The host reads too many bytes
- Improperly set read bit in the address byte
- Unsupported command code

The communication failure will be recorded in the register **STATUS_CML (7Eh)**.

Write/Read Sequence

All PMBus commands are supported by MPQ70165FS following the write/read sequence shown in Figure 21 and Figure 22. Five kinds of commands are implemented with or without PEC:

1) Send command only
2) Write byte
3) Write word
4) Read byte
5) Read word

If the host write command to a read-only register, the MPQ70165FS will report invalid data fault in INVALID_DATA (7Eh bit6). If the host read command from a write-only register, the MPQ70165FS will return 0xFF to host. If host send a write command with data to a register that is defined no data for this command, MPQ70165FS will not perform this write command.

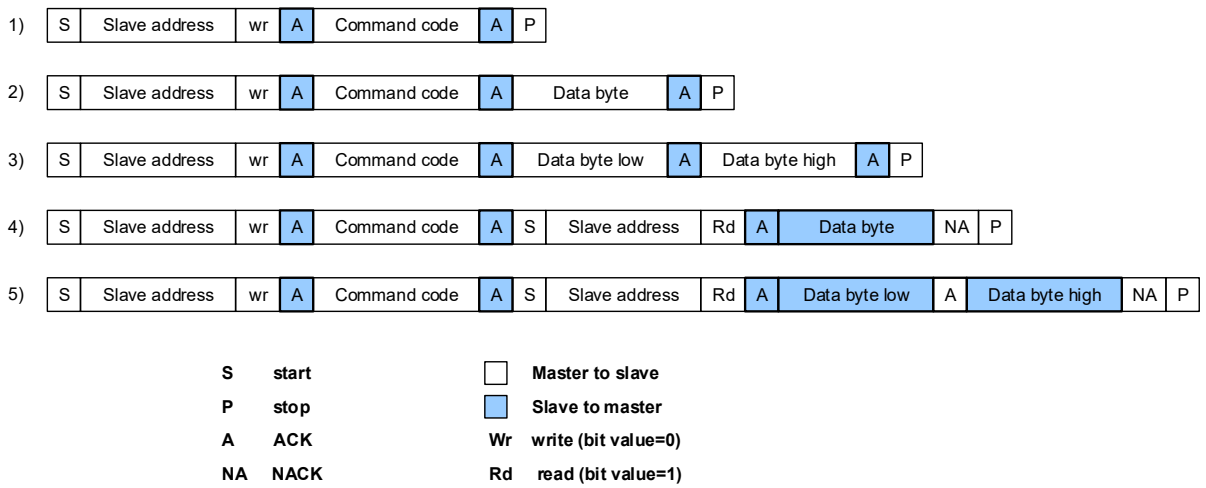


Figure 21: PMBus Write/Read Sequence without PEC

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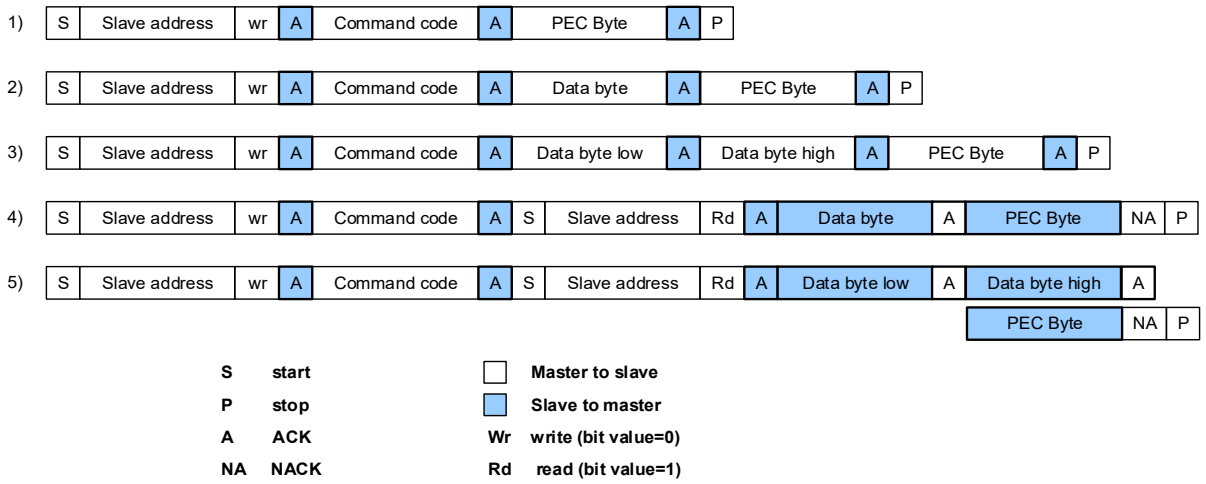


Figure 22: PMBus Write/Read Sequence with PEC

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STATE MACHINE DESCRIPTION

The state machine describes the different states of operation. There are 9 states in the device, POWER OFF, 1st RESTORE OTP, LBIST, 2nd RESTORE OTP, DEEP SAFE, PRE-DIAG, STARTUP, NORMAL, STORE_USER, RESTORE_USER.

Each state is described below and represented in the state diagram shown in Figure 23.

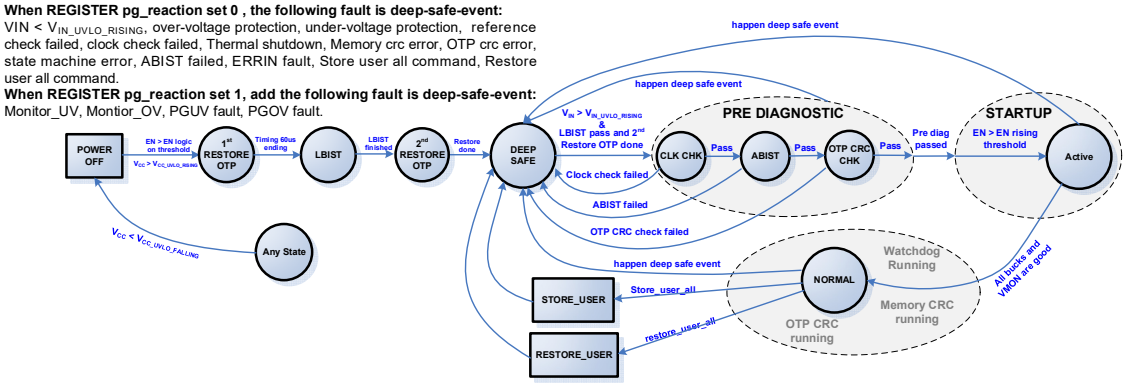


Figure 23: State Diagram of the Operating States

POWER OFF STATE:

MPQ70165FS is in POWER OFF state if V_{CC} is lower than $V_{CC_UVLO_RISING}$. All functions are disable during this state.

1st RESTORE_OTP State:

When V_{CC} exceeds $V_{CC_UVLO_RISING}$ and EN exceeds V_{EN_LOG}, MPQ70165FS will begin to do 1st restore OTP. The 1st RESTORE OTP provides a LBIST trigger signal to inform MPQ70165FS begin to perform Logic Build-In Self-Test (LBIST). During this state, PMBus interface is not available and all channels have not output.

LBIST State:

After the 1st RESTORE OTP starts a certain time (~60us), the part enters the LBIST state. In this state, the MPQ70165FS will test digital logic. Once LBIST finished, state machine will enter the next state. During this state, PMBus interface is not available, and all channels have not output.

2nd RESTORE OTP State:

LBIST can mess up the internal configuration, so 2nd RESTORE OTP is used to ensure that the register is restored to the preset value before startup. In addition, 2nd restore OTP does an OTP CRC check during the load process. During this state, PMBus interface is not available, all channels have not output and ALERT# pin can indicate fault.

DEEP SAFE State:

After 2nd RESTORE OTP finished, state machine will be transferred to DEEP SAFE State automatically. All output rails are disabled in DEEP SAFE state. There are multiple reasons to enter the DEEP SAFE state. These trigger events are listed in the state diagram and are listed in the following contents:

- $V_{IN} < V_{IN_UVLO_FALLING}$
- State machine error
- Over temperature shutdown
- Clock check failed
- Memory CRC check error
- OTP CRC check error
- $V_{OUTX} > V_{OUT_OV}$
- $V_{OUTX} < V_{OUT_UV}$
- Reference check failed
- ABIST failed
- ERRIN fault
- STORE USER ALL
- RESTORE USER ALL

Setting bit PG_REACTION in the SAFETY_CONFIG register will extend to additional deep safe events:

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- Auxiliary voltage monitor UV/OV
- PG UV/OV

Below deep safe event can be masked through write PG_MAPPING register.

- Memory CRC check error
- OTP CRC check error (only in Normal state)
- Reference check failure

DEEP SAFE state brings the system to a known safe status after a critical failure by turning off all the output rails. And the ALERT# and PG pins are automatically asserted low to indicate the status (ALERT# will be high when auxiliary voltage monitor UV/OV and PG UV/OV deep safe event occurs). The DEEP SAFE state will be maintained for a configurable hiccup time (2ms by default) to avoid any fast toggling behavior. During this state, PMBus interface is available, all channels have not output and ALERT# pin can indicate fault.

PRE-DIAG State:

When LBIST and 2nd RESTORE OTP are passed, MPQ70165FS will generate a wake-up signal after V_{IN} exceeds $V_{IN_UVLO_RISING}$. As soon as a valid wake-up signal occurs, the device moves to the PRE-DIAG state from the DEEP SAFE state. In the PRE-DIAG state, power-on diagnosis is performed before output rails ramp-up.

The pre diagnosis includes the following items:

- Clock check
- ABIST
- OTP CRC check

This clock check is different from Clock Monitoring. During this clock check process, MPQ70165FS increase CLK1's frequency, causing a clock check error. If MPQ70165FS can detect clock check error, this clock check will be passed.

During PRE-DIAG state, PMBus interface is available, all channels have not output and ALERT# pin can indicate fault. ERRIN function is active and can detect error input signal.

STARTUP State:

Once the PRE-DIAG is passed, the device enters the STARTUP state. When EN group signals exceed V_{EN_RISING}, SEQ, VMON and all channels' output will be active.

During this state, PMBus interface is available, all channels start, SEQ is available, VMON is available and ALERT# pin can indicate fault.

NORMAL State:

MPQ70165FS enters the NORMAL state after all output voltage that controlled EN's level exceeds V_{EN_RISING} soft-start process is ready.

The continuously OTP CRC or Memory CRC only allowed to run in NORMAL state. The watchdog only allowed starts to run in NORMAL state. The NORMAL state is the standard operating state for MPQ70165FS. During this state, PMBus interface is available, all output converters are running if corresponding EN level is high; SEQ and VMON function are active if corresponding EN level is high; PG / ALERT# pins can indicate fault.

STORE_USER State:

The STORE_USER state writes the present data from registers to the internal OTP content. Then, MPQ70165FS transits to the DEEP SAFE state and restart. Only writing to **STORE_USER_ALL** register from NORMAL state can enter STORE_USER state. Refer to REGISTER MAP for OTP available registers.

During this state, PMBus interface is not available, all channels are off, and ALERT# pins can indicate fault.

RESTORE_USER State:

The RESTORE_USER state copies the entire contents of OTP values to the matching locations in the registers. The values in the registers are overwritten by the value retrieved from the OTP. Any items in OTP that do not have matching locations in the operating memory are ignored. Then, MPQ70165FS transits to the DEEP SAFE state and restart. Only writing to **RESTORE_USER_ALL** register from NORMAL state can enter RESTORE_USER state. Refer to REGISTER MAP for OTP available registers.

During this state, PMBus interface is not available, all channels are off, and ALERT# pins can indicate fault.

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WATCHDOG OPERATION

The watchdog (WD) is used to monitor the software execution of an external microcontroller. It can reset registers' value of MPQ70165FS (configurable via WATCHDOG_CONFIG register) if the microcontroller stops serving the watchdog due to a lock up in the software. The watchdog is only running among the NORMAL states and remains non-actively in other state. The respective watchdog configurations can be selected online by PMBus interface.

There are two different watchdog configurations in MPQ70165FS: Window watchdog and Question-Answer (Q-A) watchdog, which can be selected via the bit WD_MODE in the WATCHDOG_CONFIG register. If WD_MODE is set to Window watchdog, WDI pin will be used to receive watchdog signal. If WD_MODE is set to Q-A watchdog, WDI pin will be re-used to receive external error signal.

Window Watchdog Operation

Window Watchdog is the default operating configuration of the MPQ70165FS watchdog. The watchdog refreshment is defined as follow and the timing diagram is as shown in Figure 24 and 25.

- The microcontroller that is being monitored shall send a periodic pulse signal to the watchdog input pin WDI within the "open window" and this pulse width shall exceed WDI deglitch time (t_{WD_DG}), which is called a good WD refreshment.
- If a pulse is sent during the "closed window", which is called a bad WD refreshment, then the watchdog error counter increases by 1 signaling a potential microcontroller fault has occurred.
- If a pulse is not sent or the pulse width is smaller than t_{WD_DG}, which is called a WD time-out, WD_TO_ERR bit in the WATCHDOG_STATUS_1 register will be set and then the watchdog error counter increases by 1.

A good WD refreshment, bad WD refreshment or WD time-out immediately result in starting a new window watchdog sequence by a new "closed window" followed by an "open window".

Moreover, the watchdog uses a deglitch filter with t_{WD_DG} filter time and an internal system clock to create the internally generated watchdog pulse. The rising edge of the trigger on the WDI pin shall occur at least the t_{WD_DG} time before the end of "open window" to generate a good WD refreshment.

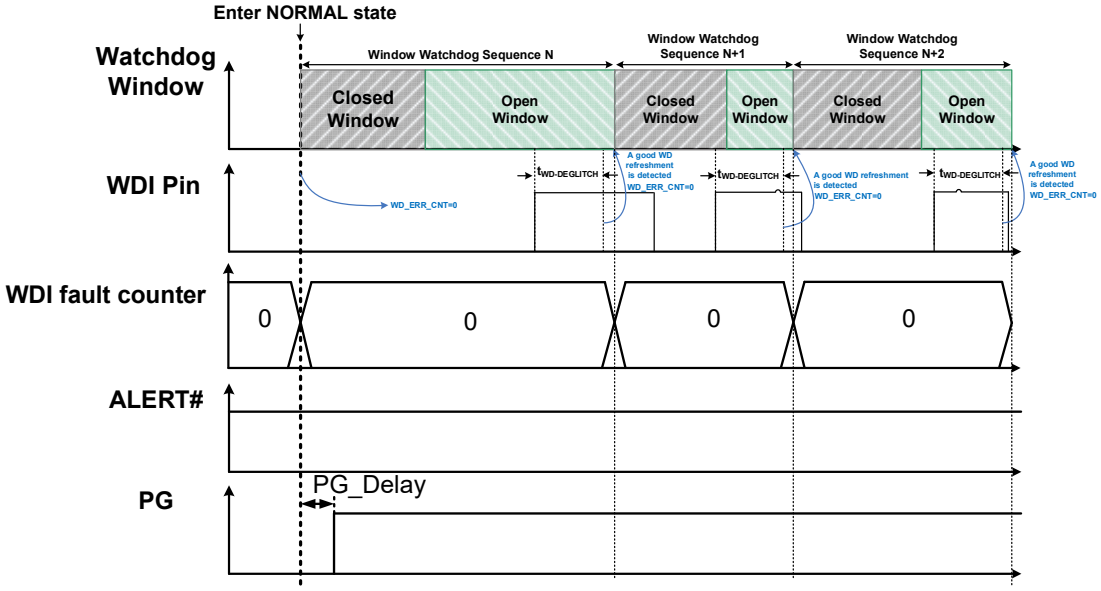


Figure 24: Timing Diagram of Window Watchdog- Good WD Refreshments

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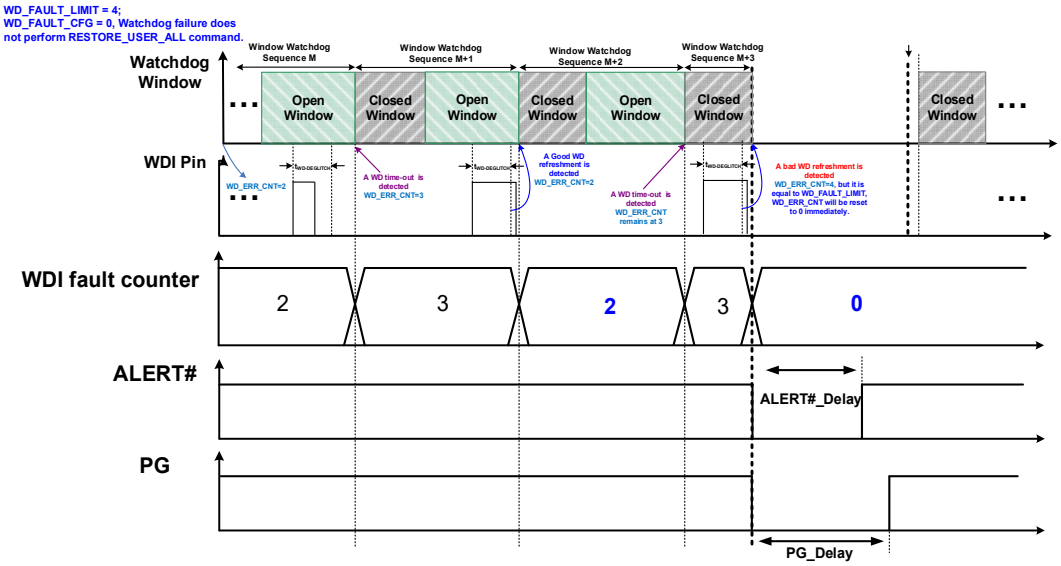


Figure 25: Timing Diagram of Window Watchdog- Bad WD Refreshments and WD Time-out

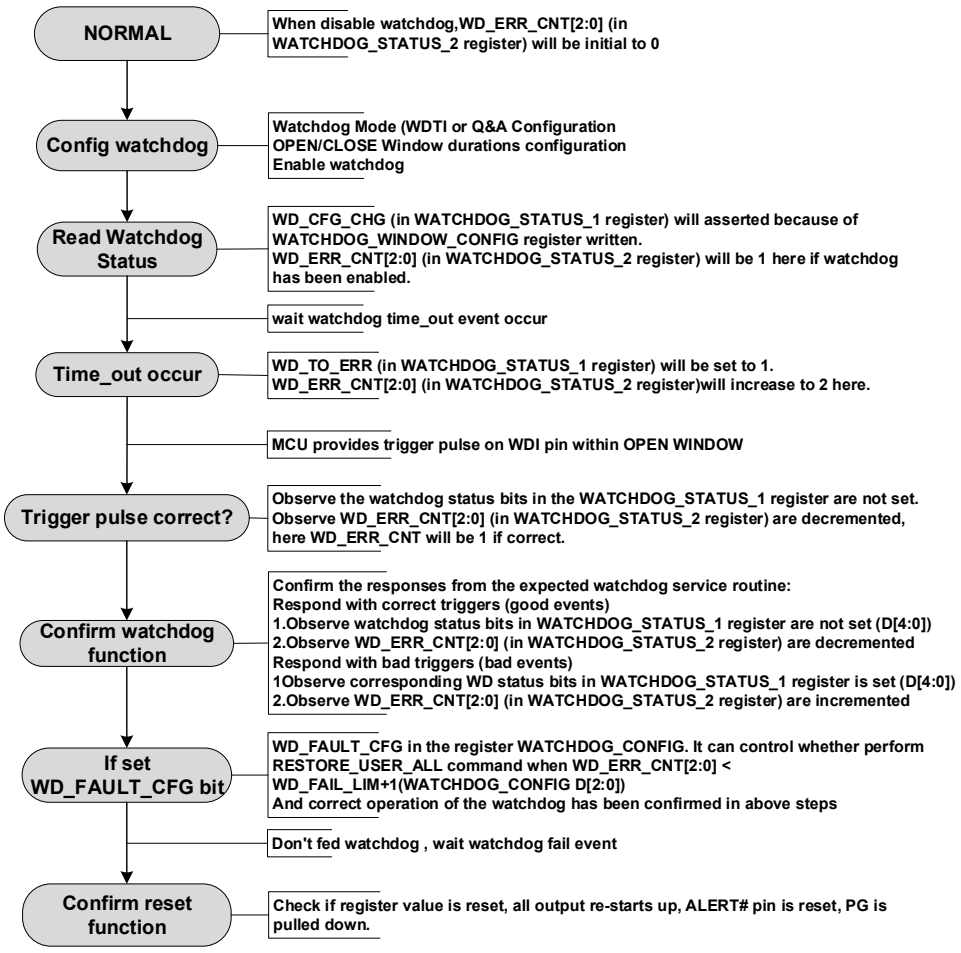


Figure 26: Window Watchdog Check Flow

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Question-Answer Watchdog Operation

Question-Answer watchdog is another operation mode to monitor a microcontroller. Setting bit WD_MODE = 1 in the WATCHDOG_CONFIG register configures the watchdog for Q-A watchdog. The watchdog mode shall be configured before enabling watchdog function.

In Q-A watchdog, the device provides a question (TOKEN) for the microcontroller in the WATCHDOG_TOKEN_VALUE register. The microcontroller performs a fixed series of arithmetic operations on the question to calculate the required 32-bit answer. This answer is split into 4 answer bytes. The microcontroller writes these answer bytes through PMBus, 1 byte at a time, into the WATCHDOG_ANSWER register. The device verifies that the microcontroller returned the answer bytes within the specified timing windows, and that the answer bytes are correct. A new question is generated only if a good WD refreshment occurred in the previous Q-A watchdog sequence causing the Token Counter (internal counter) to increment and generate a new question.

The microcontroller sends the correct answer bytes calculated for the current question within the correct Q-A watchdog window and in the correct order, which is called a good WD refreshment. Then the watchdog error counter, WD_ERR_CNT decrements by 1 and the internal token counter increments by 1, causing a new question to be generated.

If the microcontroller sends these situations, an event is called a bad WD refreshment:

- The correct answer bytes, but the last answer isn't in the correct Q-A watchdog window.
- The incorrectly calculated answer bytes.
- The correct answer bytes in the wrong order.

If MPQ70165FS detects a bad WD refreshment, the watchdog error counter WD_ERR_CNT increases by 1. That signals a potential microcontroller fault has occurred. Meanwhile, the internal Token Counter does not change. Thus the question does not change. The WD_SEQU_ERR bit in the register of WATCHDOG_STATUS_1 sets to 1 and the value of the WD_TOKEN_ERR bit in the same register is determined by whether all 4 answer bytes are returned in the "closed window".

If the microcontroller stops sending answer bytes, which is called a WD time-out, WD_TO_ERR bit will be set and then the watchdog error counter WD_ERR_CNT increases by 1.

A good WD refreshment, bad WD refreshment or WD time-out immediately result in starting a new Q-A watchdog sequence. Only good WD refreshment can generate a new question in the next WD sequence. When a bad WD refreshment or WD time-out is detected, the watchdog question of next WD sequence is same with previous sequence. See Table 3 for more about Good or Bad Q-A WD refreshment.

Table 3: Good or Bad Q-A WD Refreshment

Style	Number of WD Answers		Actions	WD Status bits in WATCHDOG_STATUS_1 (BAh)				Comments
	Close Window	Open Window		WD_ANS_ERR	WD_SEQU_ERR	WD_TO_ERR	WD_TOKEN_ERR	
Bad	0 Answer	0 Answer	-New WD cycle starts after the end of open window. -Watchdog fault count increase 1 -New WD cycle starts with same WD token value.	0	1	1	0	No answers.
Bad	0 Answer	4 INCORRECT Answers	-New WD cycle starts after the 4th WD answer -Watchdog fault count increase 1 -New WD cycle starts with same WD token value.	1	1	0	0	Four incorrect answers are sent during Open Window
Good	0 Answer	4 CORRECT Answers	-New WD cycle starts after the 4th WD answer -Watchdog fault count decrease 1 -New WD cycle starts with a new WD token value.	0	0	0	0	Four CORRECT answers are sent during Open Window

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Table 3: Good or Bad Q-A WD Refreshment *(continued)*

Style	Number of WD Answers		Actions	WD Status bits in WATCHDOG_STATUS_1 (BAh)				Comments
	Close Window	Open Window		WD_ANS_ERR	WD_SEQU_ERR	WD_TO_ERR	WD_TOKEN_ERR	
Bad	0 Answer	1 CORRECT Answer	-New WD cycle starts after the end of open window -Watchdog fault count increase 1 -New WD cycle starts with same WD token value.	0	1	1	0	Less than 3 CORRECT Answers in close window and 1 CORRECT Answer in open window (WD_ANSW_CNT[1:0] < 3)
	1 CORRECT Answer	1 CORRECT Answer		0	1	1	0	
	2 CORRECT Answers	1 CORRECT Answer		0	1	1	0	
Bad	0 Answer	1 INCORRECT Answer	-New WD cycle starts after the end of open window -Watchdog fault count increase 1 -New WD cycle starts with same WD token value.	1	1	1	0	Less than 3 CORRECT Answers in close window and 1 INCORRECT Answer in open window (WD_ANSW_CNT[1:0] < 3)
	1 CORRECT Answer	1 INCORRECT Answer		1	1	1	0	
	2 CORRECT Answers	1 INCORRECT Answer		1	1	1	0	
Good	0 Answer	4 CORRECT Answers	-New WD cycle starts after 4th WD answer -Watchdog fault count decrease 1 -New WD cycle starts with a new WD token value.	0	0	0	0	Less than 3 CORRECT Answers in close window and more than 1 CORRECT Answer in open window (WD_ANSW_CNT[1:0] = 3)
	1 CORRECT Answer	3 CORRECT Answers		0	0	0	0	
	2 CORRECT Answers	2 CORRECT Answers		0	0	0	0	
Bad	0 Answer	4 INCORRECT Answers	-New WD cycle starts after 4th WD answer -Watchdog fault count increase 1 -New WD cycle starts with same WD token value.	1	1	0	0	Less than 3 CORRECT Answers in close window and more than 1 INCORRECT Answer in open window (WD_ANSW_CNT[1:0] = 3)
	1 CORRECT Answer	3 INCORRECT Answers		1	1	0	0	
	2 CORRECT Answers	2 INCORRECT Answers		1	1	0	0	
Bad	0 Answer	3 CORRECT Answers	-New WD cycle starts after the end of open window. -Watchdog fault count increase 1 -New WD cycle starts with same WD token value.	0	1	1	0	Less than 3 INCORRECT Answers in close window and more than 1 CORRECT Answer in open window (WD_ANSW_CNT[1:0] < 3)
	1 INCORRECT Answer	2 CORRECT Answers		1	1	1	0	
	2 INCORRECT Answers	1 CORRECT Answer		1	1	1	0	
Bad	0 Answer	3 INCORRECT Answers	-New WD cycle starts after the end of open window. -Watchdog fault count increase 1 -New WD cycle starts with same WD token value.	1	1	1	0	Less than 3 INCORRECT Answers in close window and more than 1 INCORRECT Answer in open window (WD_ANSW_CNT[1:0] < 3)
	1 INCORRECT Answers	2 INCORRECT Answers		1	1	1	0	
	2 INCORRECT Answers	1 INCORRECT Answer		1	1	1	0	
Good	0 Answer	4 CORRECT Answers	-New WD cycle starts after 4th WD answer -Watchdog fault count decrease 1 -New WD cycle starts with a new WD token value.	0	0	0	0	Less than 3 INCORRECT Answers in close window and more than 1 CORRECT Answer in open window (WD_ANSW_CNT[1:0] = 3)
Bad	1 INCORRECT Answer	3 CORRECT Answers	-New WD cycle starts after 4th WD answer -Watchdog fault count increase 1 -New WD cycle starts with same WD token value.	1	1	0	0	
	2 INCORRECT Answers	2 CORRECT Answers		1	1	0	0	
Bad	0 Answer	4 INCORRECT Answers	-New WD cycle starts after 4th WD answer -Watchdog fault count increase 1 -New WD cycle starts with same WD token value.	1	1	0	0	Less than 3 INCORRECT Answers in close window and more than 1 INCORRECT Answer in open window (WD_ANSW_CNT[1:0] = 3)
	1 INCORRECT Answer	3 INCORRECT Answers		1	1	0	0	
	2 INCORRECT Answers	2 INCORRECT Answers		1	1	0	0	

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Table 3: Good or Bad Q-A WD Refreshment *(continued)*

Style	Number of WD Answers		Actions	WD Status bits in WATCHDOG_STATUS_1 (BAh)				Comments
	Close Window	Open Window		WD_ANS_ERR	WD_SEQU_ERR	WD_TO_ERR	WD_TOKEN_ERR	
Bad	3 CORRECT Answers	0 Answer	-New WD cycle starts after the end of open window -Watchdog fault count increase 1 -New WD cycle starts with same WD token value.	0	1	1	0	Less than 4 CORRECT Answers in close window and 0 Answer in open window (WD_ANSW_CNT[1:0] < 3)
	2 CORRECT Answers	0 Answer		0	1	1	0	
	1 CORRECT Answer	0 Answer		0	1	1	0	
Good	3 CORRECT Answers	1 CORRECT Answer	-New WD cycle starts after the 4th WD answer -Watchdog fault count decrease 1 -New WD cycle starts with a new WD token value.	0	0	0	0	3 CORRECT Answers in close window and 1 CORRECT Answer in open window. (WD_ANSW_CNT[1:0] = 3)
Bad	3 CORRECT Answers	1 INCORRECT Answer	-New WD cycle starts after the 4th WD answer -Watchdog fault count increase 1 -New WD cycle starts with same WD token value.	1	1	0	0	3 CORRECT Answers in close window and 1 INCORRECT Answer in open window. (WD_ANSW_CNT[1:0] = 3)
Bad	3 INCORRECT Answers	0 Answer	-New WD cycle starts after the end of open window -Watchdog fault count increase 1 -New WD cycle starts with same WD token value.	1	1	1	0	3 INCORRECT Answers in close window and 0 Answer in open window. (WD_ANSW_CNT[1:0] < 3)
Bad	3 INCORRECT Answers	1 CORRECT Answer	-New WD cycle starts after the 4th WD answer -Watchdog fault count increase 1 -New WD cycle starts with same WD token value.	1	1	0	0	3 INCORRECT Answers in close window and 1 CORRECT Answer in open window. (WD_ANSW_CNT[1:0] = 3)
Bad	3 INCORRECT Answers	1 INCORRECT Answer	-New WD cycle starts after the 4th WD answer -Watchdog fault count increase 1 -New WD cycle starts with same WD token value.	1	1	0	0	3 INCORRECT Answers in close window and 1 INCORRECT Answer in open window. (WD_ANSW_CNT[1:0] = 3)
Bad	4 CORRECT Answers	0 Answer	-New WD cycle starts after the 4th WD answer -Watchdog fault count increase 1 -New WD cycle starts with same WD token value.	0	1	0	1	4 CORRECT or INCORRECT Answers in close window. (WD_ANSW_CNT[1:0] = 3)
	3 CORREC Answers + 1 INCORRECT Answer	0 Answer		1	1	0	1	
	2 CORREC Answers + 2 INCORRECT Answers	0 Answer		1	1	0	1	
	1 CORREC Answer + 3 INCORRECT Answers	0 Answer		1	1	0	1	

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The Q-A watchdog is performed step-by-step as follow.

Step 1: Set adjustable close and open window time. If watchdog is disabled, configuring close / open window time sets the WD_CFG_CHG status bit to 1. If watchdog is enabled, configuring close / open window time not only sets the WD_CFG_CHG status bit to 1, but also causes the watchdog fault count increment by 1. Meanwhile, window will be refreshed to a new close window.

Step 2: Set WD_MODE to QA watchdog. The watchdog mode shall be configured before enabling watchdog function. If change watchdog mode during watchdog operation, the microcontroller shall re-configure close / open window time to refresh watchdog window to close window.

Step 3: Set WD_FAULT_CFG and WD_FAULT_LIMIT. WD_FAULT_CFG can control whether MPQ70165FS preforms once RESTORE_USR_ALL command when watchdog failure happens. If the watchdog fault counter is equal to watchdog fault limit, MPQ70165FS will report watchdog failure at STATUS_MFR_SPECIFIC (80h) register.

Step 4: Set WD_EN to enable watchdog. Once MPQ70165FS receives the command of watchdog enable, MPQ70165FS will start the first watchdog sequence. In every watchdog sequence, the close window is timed first. Microcontroller should read WATCHDOG_STATUS_1 bit4:0 and WATCHDOG_STATUS_2 bit2:0 to get watchdog fault status and count.

Step 5: The microcontroller reads the question (WD_TOKEN_VALUE [4:0]) from register WATCHDOG_TOKEN_VALUE. A 5-bit seed in the WD_TOKEN_SEED [4:0] (either stored in MPQ70165FS or from microcontroller) is applied to a Linear Feedback Shift Register (LFSR) inside MPQ70165FS to generate a 5-bit question (token). In every watchdog sequence, the close window is timed first. At the beginning of each watchdog sequence, the question is read first. Within a watchdog sequence, the watchdog question can be read only once. If microcontroller reads the question again in the process of replying to the answer, the answer count will be refreshed to default value.

Step 6: The microcontroller reads the answer count in WD_ANS_CNT [4:3]. According to answer count order of 2’b11, 2’b10, 2b’01, 2b00, these answers are called Answer-3, Answer-2, Answer-1, and Answer-0. The default value is 2’b11 to indicate that MPQ70165FS expects the microcontroller to send the Answer–3 in WD_ANS [7:0]. In a watchdog sequence, when MPQ70165FS receives an answer, WD_ANS_CNT [4:3] is reduced by 1. And WD_ANS_CNT[4:3] will always change in the order of 2'b11, 2'b10, 2'b01, 2'b00. If microcontroller reads the question again in the process of replying answers, the answer count will be refreshed to default value.

Step 7: In a watchdog sequence, microcontroller shall perform a pre-defined (Figure 27) calculation to calculate the corresponding 8-bit answer based on watchdog question and answer count. So, these answers are only applicable to this watchdog sequence. According to answer count order of 2’b11, 2’b10, 2b’01, 2b00, these answers are called Answer-3, Answer-2, Answer-1, and Answer-0.

Step 8: Microcontroller shall send these answers to WD_ANS [7:0] one by one. If MPQ70165FS receives less than four answers in a watchdog sequence, MPQ70165FS will report time-out error at the ending of open window.

Step 9: The answer sent by the microcontroller is verified by MPQ70165FS. For each question, MPQ70165FS requires 4 correct answer bytes from the microcontroller with the correct order. Answer-3, Answer-2, and Answer-1 may be in "closed window" or "open window" in the correct order, and Answer-0 shall be in "open window" to be detected as a good WD refreshment.

Step 10: The new Q-A watchdog sequence results from a good WD refreshment or a bad WD refreshment or a time-out error. Only the good WD refreshment can make MPQ70165FS generate a new question for the microcontroller. In cases of bad WD refreshment or WD time-out, a new question will not be generated.

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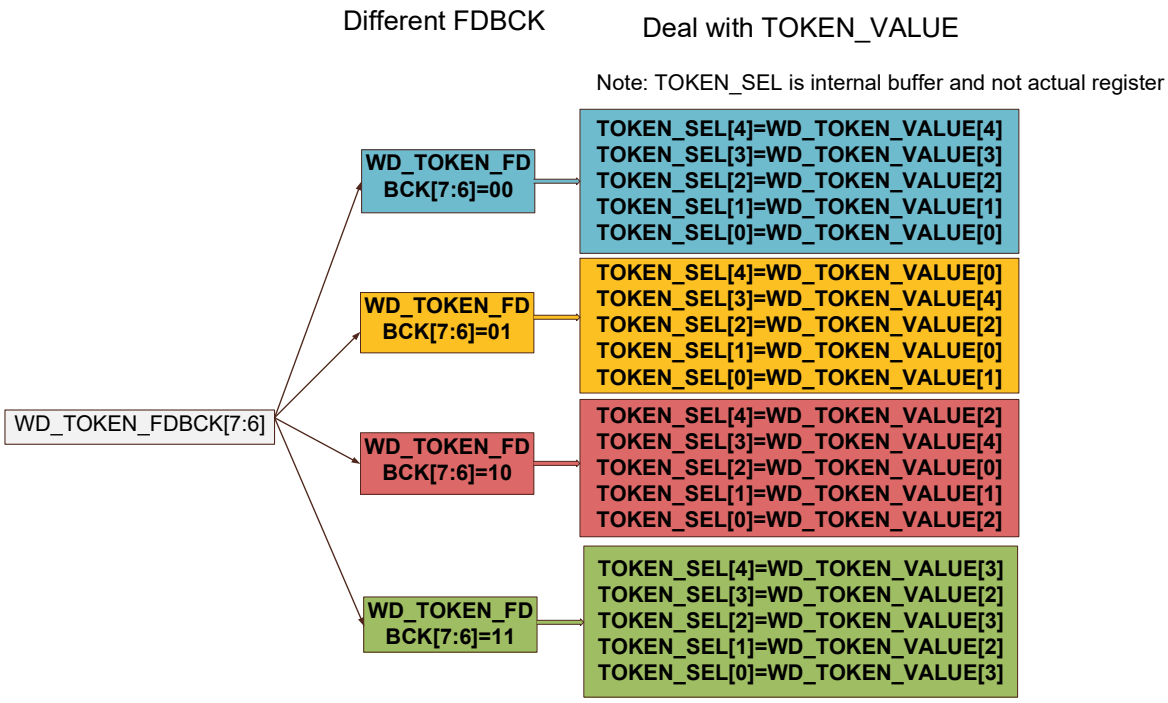


Figure 27: Answer Calculation with Different FDBCK

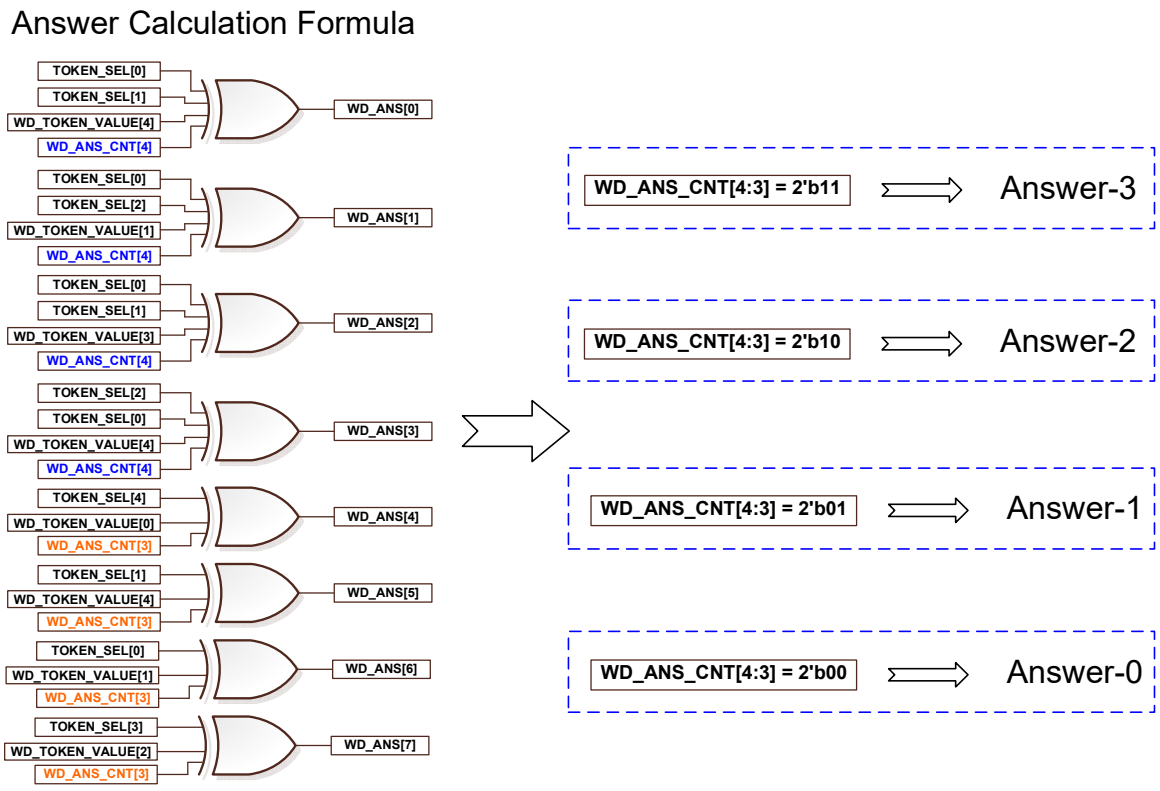


Figure 28: Answer Calculation Formula

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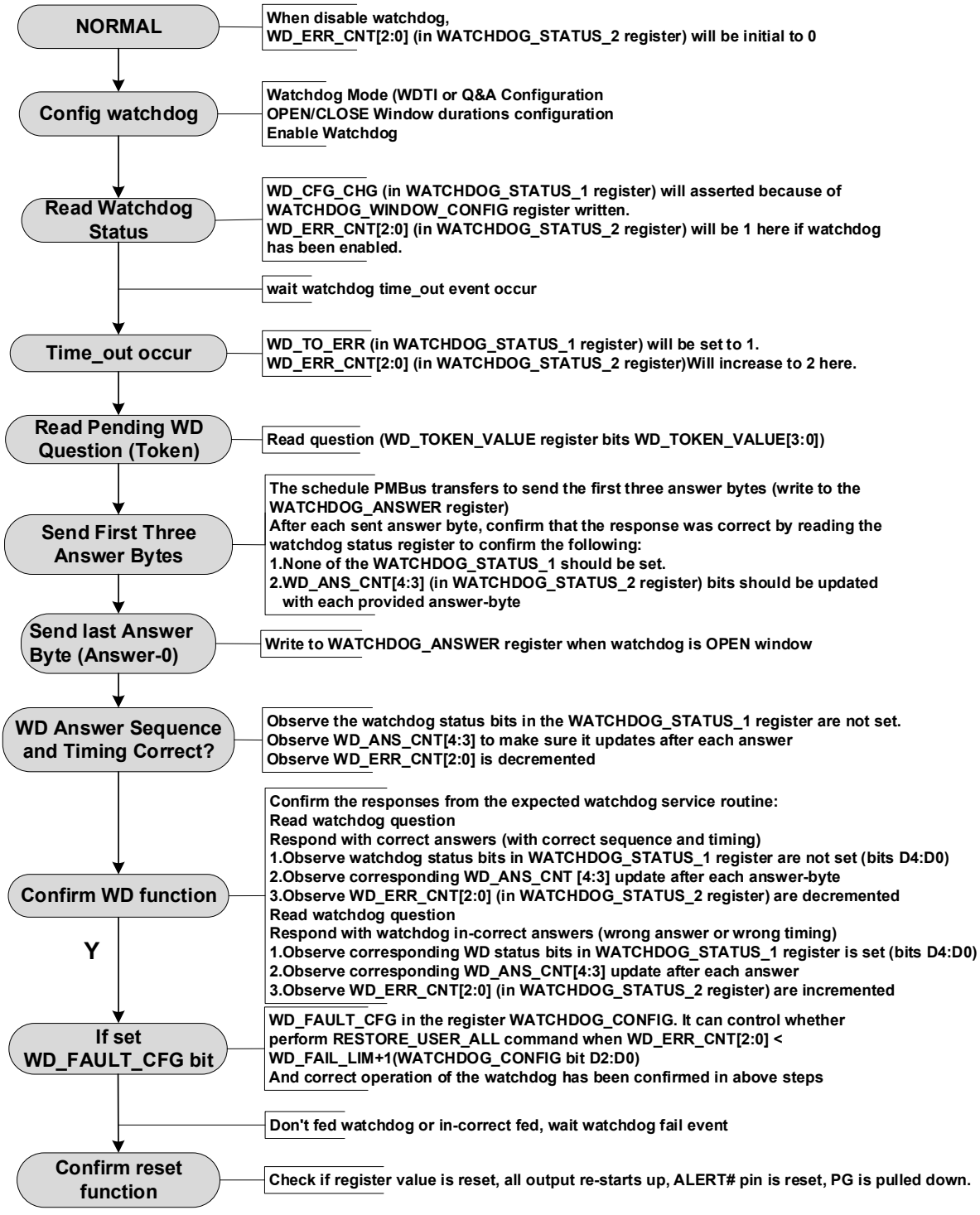


Figure 29: QA Watchdog Check Flow

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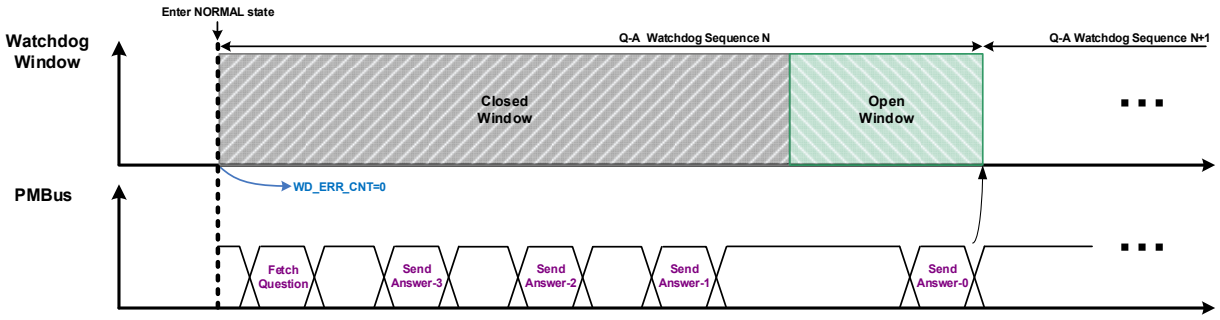


Figure 30: Timing Diagram of Q-A Watchdog WD Refreshments

Watchdog Error Counter and Watchdog Fail Flag

MPQ70165FS provides a watchdog error counter WD_ERR_CNT in the register of WATCHEDOG_STATUS_2 which is increment or decrement after every watchdog refreshment.

The watchdog error counter ranges from 0 to 7, it is initialized to 0 after the power-on- reset. A good WD refreshment decreases the WD_ERR_CNT by 1, down to the minimum of 0. A bad WD refreshment and WD time-out increase the WD_ERR_CNT by 1, up to the maximum of 7, moreover, WD time-out sets the WD_TO_ERR bit in the WATCHEDOG_STATUS_1 register. Once Watchdog Failure Limit minus the watchdog error counter equals 1, a coming new bad WD refreshment or WD time-out leads to a watchdog fail.

If a watchdog fail event occurs, the watchdog fail flag WD_FAULT bit in the register of STATUS_MFR_SPECIFIC is set to 1. Then all registers are reset or not, which can be configured by WD_FAULT_CFG.

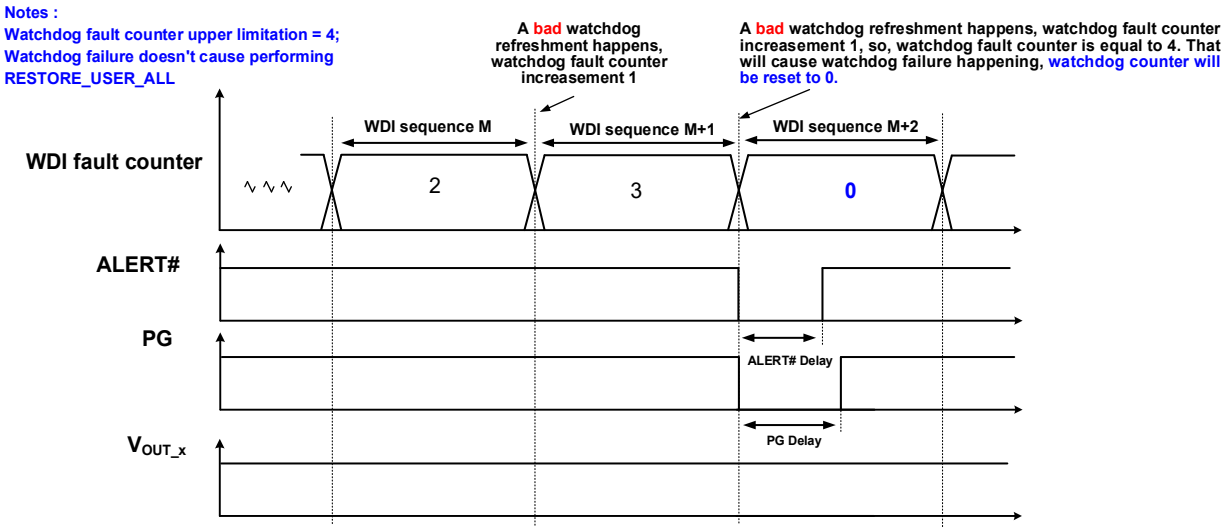


Figure 31: Watchdog Failure with Non-perform RESTORE_USER_ALL

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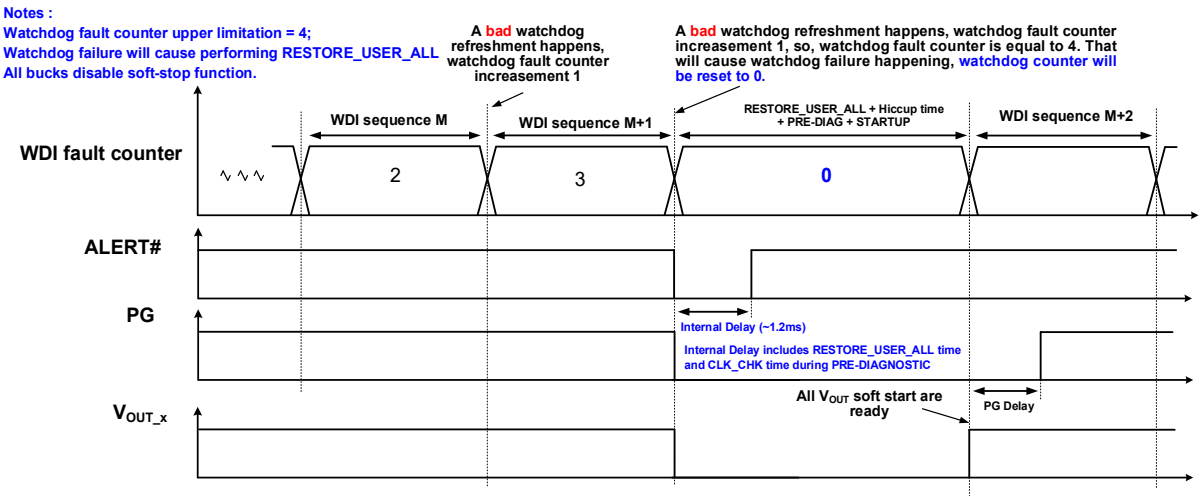


Figure 32: Watchdog Failure with Perform RESTORE_USER_ALL

Watchdog Sequence

One window watchdog sequence in MPQ70165FS consists of a "closed window" followed by an "open window". Both the window and Q-A watchdog configurations shares the same watchdog sequence settings.

The watchdog window sequence is derived from the values in WD_OPEN_WINDOW and WD_CLOSE_WINDOW bits in the register of WATCHDOG_WINDOW_CONFIG. The MCU can program the time period of "closed window" (T_{WD_OPEN}) and "open window" (T_{WD_CLS}) by writing the above register.

The watchdog uses MPQ70165FS's internal system clock as a time reference and suffers ±10% tolerances of the internal oscillator, microcontroller should allow a reasonable timing margin while programming.

Watchdog Synchronization

When MPQ70165FS enters the NORMAL state, the external microcontroller should have been powered properly and started its own initialization. The watchdog can be reconfigured by the external microcontroller in the NORMAL state.

In watchdog enabled, the microcontroller can write to WATCHDOG_WINDOW_CONFIG register to start a new watchdog sequence. That can synchronize the microcontroller timer with the watchdog timer. After a write access to the register by microcontroller, the WD_CFG_CHG bit in the register of WATCHDOG_STATUS_1 is set to 1 and the watchdog error counter WD_ERR_CNT increases by 1, the device immediately starts a new watchdog sequence. In the new watchdog sequence, the close / open window time will follow the new written value.

When the microcontroller timer is synchronized with the watchdog timer, a good WD refreshment from the microcontroller immediately starts a new watchdog sequence. In this way, the microcontroller stays synchronized with the watchdog sequence.

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REGISTER MAP

Command Code	Command Name	Type	Bytes	Page 0 (Rail 1)	Page 1 (Rail 2)	Page 2 (Rail 3)	Page 3 (Rail 4)	Page 4 (Rail 5)	Page 5 (Rail 6)	Page FF (All rails)
00h	PAGE	R/W	1	✓	✓	✓	✓	✓	✓	✓
01h	OPERATION	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
03h(16)	CLEAR_FAULTS	W	0(17)	✓	✓	✓	✓	✓	✓	✓
10h(16)	WRITE_PROTECT	R/W	1	✓						✓
15h(16)	STORE_USER_ALL	W	0(17)	✓						✓
16h(16)	RESTORE_USER_ALL	W	0(17)	✓						✓
19h(16)	CAPABILITY	R	1	✓						✓
20h(16)	VOUT_MODE	R	1	✓						✓
21h	VOUT_COMMAND	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
24h	VOUT_MAX	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
29h	VOUT_SCALE_LOOP	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
2Bh	VOUT_MIN	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
60h	TON_DELAY	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
64h	TOFF_DELAY	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
78h(16)	STATUS_BYTE	R	1	✓	✓	✓	✓	✓	✓	✓(15)
79h(16)	STATUS_WORD	R	2	✓	✓	✓	✓	✓	✓	✓(15)
7Ah(16)	STATUS_VOUT	R	1	✓	✓	✓	✓	✓	✓	✓(15)
7Dh(16)	STATUS_TEMPERATURE	R	1	✓						✓
7Eh(16)	STATUS_CML	R	1	✓						✓
80h(16)	STATUS_MFR_SPECIFIC	R	1	✓						✓
9Dh(16)	SECURE_LOCKOUT	R/W	1	✓						✓
9Eh	MULTIPHASE_CONFIG	R/W	1	✓						✓
A0h	HICCUP_TIMER	R/W	1	✓						✓
A1h(16)	STATUS_PIN_STATE	R	1	✓						✓
A3h	VOUT_STARTUP_SLEW	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
A5h	VOUT_SHUTDOWN_SLEW	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
A7h	VOUT_SLEW	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
A9h	OUTPUT_DISCHARGE	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
ABh	OUTPUT_SHUTDOWN_MODE	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
ADh	FREQUENCY_DITHER	R/W	1	✓						✓
B0h	COMPENSATION_CONFIG_1	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
B1h	COMPENSATION_CONFIG_2	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
B2h	WATCHDOG_CONFIG	R/W	1	✓						✓
B4h	WATCHDOG_WINDOW_CONFIG	R/W	1	✓						✓
B6h	WATCHDOG_TOKEN_FDBCK	R/W	1	✓						✓

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REGISTER MAP *(continued)*

Command Code	Command Name	Type	Bytes	Page 0 (Rail 1)	Page 1 (Rail 2)	Page 2 (Rail 3)	Page 3 (Rail 4)	Page 4 (Rail 5)	Page 5 (Rail 6)	Page FF (All rails)
B8h(16)	WATCHDOG_TOKEN_VALUE	R	1	✓						✓
BAh(16)	WATCHDOG_STATUS_1	R	1	✓						✓
BCh(16)	WATCHDOG_STATUS_2	R	1	✓						✓
BEh	WATCHDOG_ANSWER	R/W	1	✓						✓
C0h	ALERT#_PG_FORCE_ASSERT	R/W	1	✓						✓
C2h	SAFETY_CONFIG	R/W	1	✓						✓
C4h	ALERT#_MAPPING	R/W	1	✓						✓
C6h	PG_MAPPING	R/W	1	✓						✓
C8h	PG_ALERT#_DELAY	R/W	1	✓						✓
CAh	PG_CONFIG	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
CCh	LIGHT_LOAD	R/W	1	✓						✓
CEh	ILIM_SCALE	R/W	1	✓	✓	✓	✓	✓	✓	✓(15)
CFh	ENABLE_GROUP	R/W	1	✓						✓
D0h	SEQ_DELAY	R/W	1	✓						✓
D1h	VMON_LOW_THRESHOLD	R/W	1	✓						✓
D2h	VMON_HIGH_THRESHOLD	R/W	1	✓						✓
D3h	VMON _DELAY	R/W	1	✓						✓
D4h	MON_RANGE	R/W	1	✓						✓
D5h	ADDRESS	R/W	1	✓						✓
D6h	CONFIG_CODE	R/W	1	✓						✓
D7h	OTP_CRC	R	2	✓						✓
D9h	MEMORY_CRC	R	2	✓						✓
DAh	MFR_OTP_MEM_CHK	R/W	1	✓						✓
DBh	MFR_OTP_MEM_STATUS	R	1	✓						✓
E7h	STATUS_FREQ	R	1	✓						✓

Notes:

15) When reading this data, the data being read is actually the value of Page 0. When this data is written, all Pages are written to the same data in this register.
16) These registers don't participate in Memory CRC check.
17) MPQ70165FS will not respond to write command with data.

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PAGE 0 REGISTER MAP

PAGE (00h)

Format: Unsigned binary

The PAGE command provides the ability to configure, control and monitor all outputs through only one physical address. All subsequent commands (pages is “Y” at REGISTER MAP section) that allow multipage action are applied to the corresponding regulators selected by the PAGE command.

Bits	Access	Bit Name	Description
7:0	R/W	PAGE	In Page FF, write command to other registers can change all pages’ configuration, including page 0-5. In Page 0 – 5, write command to other registers can only change the current page’s configuration. So, the PAGE command shall be written before modifying other registers. 8’b00000000: VOUT1 8’b00000001: VOUT2 8’b00000010: VOUT3 8’b00000011: VOUT4 8’b00000100: VOUT5 8’b00000101: VOUT6 8’b11111111: All Others: Invalid commands

OPERATION (01h)

Format: Unsigned binary

The OPERATION command is used to configure the output on/off state of Buck 1, in conjunction with input from the EN_X pin.

Bits	Access	Bit Name	Description
7	R/W	OPERATION	Enable signal is available, MPQ70165FS support interface to turn on / off channels through PMBus. 1’b1: Output on (if enable high) 1’b0: Output off
6:0	R/W	RESERVED	Reserved.

CLEAR_FAULTS (03h)

This command is used to clear any fault bit in all status registers: STATUS_BYTE (78h), STATUS_WORD (79h), STATUS_VOUT (7Ah), STATUS_TEMPERATURE (7Dh), STATUS_CML (7Eh), STATUS_MFR_SPECIFIC (80h), WATCHDOG_STATUS_1 (BAh), MFR_OTP_MEM_STATUS (DBh) and STATUS_FREQ (E7h)

This command is written only. There is no data byte for this command.

WRITE_PROTECT (10h)

Format: Unsigned binary

The WRITE_PROTECT command is used to control writing to the converter. The intent of this command is to provide protection against accident changes. It’s not intended to provide protection against deliberate or malicious changes to the converter’s configuration or operation. All the supported commands may have their parameters read, regardless of the WRITE_PROTECT settings.

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Bits	Access	Bit Name	Description
7:0	R/W	WRITE_PROTECT	When PMBus interface is available, the register can be written. If host try to write a protected command, MPQ70165FS will report invalid data fault and ALERT# will be pulled down. CLEAR FAULT, STORE USER ALL, RESTORE USER ALL are still available, regardless of the WRITE_PROTECT settings. 8'b10000000: Disable all writes except to the WRITE_PROTECT command 8'b01000000: Disable all writes except to the WRITE_PROTECT, OPERATION and PAGE commands 8'b00100000: Disable all writes except to the WRITE_PROTECT, OPERATION, PAGE and VOUT_COMMAND commands 8'b00000000: Enable writes to all commands. Others: Invalid commands

STORE_USER_ALL (15h)

This command instructs the MPQ70165FS to copy the contents of the operating memory to the matching locations in the OTP, except the internal trim registers. This process will operate when MPQ70165FS receives a STORE_USER_ALL command from the PMBus interface.

This command is written only. There is no data byte for this command.

RESTORE_USER_ALL (16h)

This command instructs the MPQ70165FS to copy the contents from the OTP and over-write the matching locations in the operating memory. Trim registers are not over-written by this process. Any items in OTP that do not have matching locations in the operating memory are ignored.

It's permitted to use the RESTORE_USER_ALL command while the MPQ70165FS is operating. However, the MPQ70165FS may be unresponsive during the operation with unpredictable, undesirable or even catastrophic results.

This command is written only. There is no data byte for this command.

CAPABILITY (19h)

Format: Unsigned binary

This command provides one byte to return key PMBus features that MPQ70165FS can support.

Bits	Access	Bit Name	Description
7	R	PACKET_ERR_CHECKING	1'b1: Packet Error Checking is supported Others: Invalid commands
6:5	R	MAX_BUS_SPEED	2'b10: Maximum supported bus speed is 1 MHz Others: Invalid commands
4	R	SMBALERT#	1'b0: The device does not have a SMBALERT# pin and does not support the SMBus ALERT# Response protocol Others: Invalid commands
3	R	NUMERIC_FORMAT	1'b0: Numeric data is in LINEAR11, ULINEAR16, SLINEAR16, or DIRECT format Others: Invalid commands
2	R	AVSBUS_SUPPORT	1'b0: AVSBus Not supported Others: Invalid commands
1:0	R	RESERVED	Reserved.

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VOUT_MODE (20h)

Format: Unsigned binary

The VOUT_MODE command is used to command and read the output voltage mode. The 3 MSB bits are used to determine the data format (only direct format is supported in MPQ70165FS).

Bits	Access	Bit Name	Description
7:0	R	VOUT_MODE	8'b01000000: Direct mode. Coefficients are m=160, R=0, b=-33.

VOUT_COMMAND (21h)

Format: Unsigned binary

The VOUT_COMMAND command sets the output voltage of Buck1.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_CMD	The command is used to set output voltage. Steps as below: 1. Select page x by PAGE (00h). 2. Change V_{OUT} by VOUT_COMMAND. V_{OUT} is influenced by VOUT_SCALE_LOOP, see the below description for detail V_{OUT} range. 1. When VOUT_SL = 1, V_{OUT} can be set from 0.20625V to 1.8V, 6.25mV/step. 2. When VOUT_SL = 2, V_{OUT} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT} = (VOUT_CMD x 6.25mV + 206.25mV) x VOUT_SL

VOUT_MAX (24h)

Format: Unsigned binary

The VOUT_MAX command sets an upper limit on the output voltage that Buck 1 can command regardless of any other commands or combinations. The intent of this command is to provide a safeguard against a user accidental setting the output voltage to a possible destructive level rather than to be the primary output overvoltage protection.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_MAX	When VOUT_SL = 1, V_{OUT_MAX} can be set from 0.20625V to 1.8V, 6.25mV/step. When VOUT_SL = 2, V_{OUT_MAX} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT_MAX} = (VOUT_MAX x 6.25mV + 206.25mV) x VOUT_SL Attempting to write a higher value to VOUT_COMMAND causes VOUT_COMMAND to be set to VOUT_MAX and asserts a VOUT_MAX_MIN warning.

VOUT_SCALE_LOOP (29h)

Format: Unsigned binary

The VOUT_SCALE_LOOP command sets the scale for the V_{OUT}, V_{OUT_MAX}, V_{OUT_MIN}, V_{OUT} start-up slew rate, V_{OUT} shutdown slew rate and V_{OUT} slew rate on Page 0.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	VOUT_SC_L	According to application requirement, set adjust scale. The device should be off via the OPERATINO command before change the scale. 1'b0: VOUT_SL= 1 1'b1: VOUT_SL= 2

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VOUT_MIN (2Bh)

Format: Unsigned binary

The VOUT_MIN command sets a lower limit on the output voltage that Buck 1 can command regardless of any other commands or combinations. The intent of this command is to provide a safeguard against a user accidentally setting the output voltage to a possible destructive level rather than to be the primary output under-voltage protection.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_MIN	When VOUT_SL = 1, V_{OUT_MIN} can be set from 0.20625V to 1.8V, 6.25mV/step. When VOUT_SL = 2, V_{OUT_MIN} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT_MIN} = (VOUT_MIN x 6.25mV + 206.25mV) x VOUT_SL Attempting to write a lower value to VOUT_COMMAND causes VOUT_COMMAND to be set to VOUT_MIN, and asserts a VOUT_MAX_MIN warning

TON_DELAY (60h)

Format: Unsigned binary

The TON_DELAY command sets the turn-on delay time for Buck 1.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	TON_DELAY	Every buck's turn on delay time can be set independently. Please change page vis PAGE command, and then set the register to configure turn on delay time. It can be calculated with the following equation: Startup Delay = TON_DELAY x 0.25ms. (0-7.75ms)

TOFF_DELAY (64h)

Format: Unsigned binary

The TOFF_DELAY command sets the turn-off delay time for Buck1.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	TOFF_DELAY	Every buck's turn off delay time can be set independently. Please change page vis PAGE command, and then set the register to configure turn off delay time. Only when enable is low or PMBus off command or V_{IN} < UVLO, the device can perform the turn off sequence. It can be calculated with the following equation: Shutdown Delay = TOFF_DELAY x 0.25ms. (0-7.75ms)

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STATUS_BYTE (78h)

Format: Unsigned binary

The STATUS_BYTE command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	BUSY	While MPQ70165FS is performing OTP or restore OTP, host send PMBus write command. MPQ70165FS will report the Busy fault and ALERT# will be pulled down (see Table 1 at page 39). The bit is set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No busy fault 1'b1: A fault was declared because the device was busy and unable to respond
6	R	OFF	The bit can indicate channels' current status. For different channels' status, only change page via PAGE command before reading this bit. When the channel's status in between t_{ON_DELAY} ending and t_{OFF_DELAY} ending, the bit will be set to 0. Besides, this bit will be set to 1. 1'b0: Device is on 1'b1: Device is off
5	R	VOUT_OV_FAULT	When one channel's V_{OUT} exceeds V_{OUT} OV threshold duration ~10us, the channel will happen V_{OUT} OV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 7Ah bit7. 1'b0: No output over voltage fault has occurred 1'b1: An output over voltage fault has occurred
4	R	VOUT_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 79h bit6 and 7Ah bit4. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
3	R	RESERVED	RESERVED
2	R	TEMPERATURE	When MPQ70165FS's junction temperature exceeds TWN threshold or TSD threshold, the bit will be set to 1 at all pages. But the bit cannot accurately determine which thermal fault occurs. Please read 7Dh to ger the detail thermal fault information. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No TSD fault and TWN fault occurred 1'b1: A TSD fault or TWN fault has occurred
1	R	CML	When MPQ70165FS happens PMBus fault or OTP CRC (during 2nd restore OTP) fault, this bit will be set to 1 at all pages. PMBus fault include unsupported command fault, unsupported data fault, PEC fault and communication fault (include start error, stop error and long error). Please read 7Eh to ger the detail fault information. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No PMBus or OTP CRC (during 2nd restore OTP) fault has occurred. 1'b1: A PMBus or OTP CRC (during 2nd restore OTP) fault has occurred.

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0	R	FAULT_OTHER	In a channel, when VOUT COMMAND exceeds VOUT_MAX COMMAND or VOUT COMMAND lower than VOUT_MIN COMMAND, the bit will be set to1 under corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No VOUT MAXMIN fault has occurred. 1'b1: VOUT MAXMIN fault has occurred.

STATUS_WORD (79h)

Format: Unsigned binary

This command returns two bytes of information with a summary of the device fault/warning condition. The higher byte gives more detailed information of the fault conditions. The higher byte shares with register STATUS_BYTE.

Bits	Access	Bit Name	Description
15:8	R	STATUS_BYTE	The same as STATUS BYTE.
7	R	VOUT_FAULT	When one channel happens V_{OUT} related fault, this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No output voltage fault has occurred 1'b1: An output voltage fault has occurred. (including OV fault, UV fault, VOUT MAXMIN fault, PG_OV fault and PG_UV fault)
6	R	VO_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4 and 7Ah bit4. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
5	R	RESERVED	Reserved.
4	R	MANUFAC_FAULT	When MPQ70165FS happens reference fault, ABIST fault or watchdog fail fault, this bit will be set to 1 at all pages. Please read 80h to ger the detail fault information. This fault flag will be cleared once the device performs ABIST if only reference fault happened. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No manufacturer specific fault has occurred 1'b1: A manufacturer specific fault has occurred. (Including Reference fault, ABIST fault and Watchdog fail event)
3	R	PGOOD	The bit can indicate if channels' output exceeds the PGOOD range. For different channels' PGOOD status, only change page via PAGE command before reading this bit. When channel's output is not within PGOOD range, the bit will be set to 1 at corresponding page. Other page doesn't set this bit to 1. When channel's output is within PGOOD range, the bit will be set to 0 automatically at corresponding page. 1'b0: Power Good is high 1'b1: Power Good is low
2:0	R	RESERVED	Reserved.

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STATUS_VOUT (7Ah)

Format: Unsigned binary

The STATUS_VOUT command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	VOUT_OV_FAULT	When one channel's V_{OUT} exceeds V_{OUT} OV threshold duration ~10us, the channel will happen V_{OUT} OV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4. 1'b0: No output over voltage fault has occurred 1'b1: An output over voltage fault has occurred
6:5	R	RESERVED	Reserved.
4	R	VOUT_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4 and 79h bit6. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
3	R	RESERVED	Reserved.
2	R	PG_OV	When one channel's V_{OUT} exceeds PG upper threshold, the channel will happen PGOV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No pgova fault has occurred 1'b1: A pgova fault has occurred
1	R	PG_UV	When one channel's V_{OUT} is lower than PG lower threshold, the channel will happen PGUV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No pguva fault has occurred 1'b1: A pguva fault has occurred
0	R	VOUT_MAX_MIN	In a channel, when VOUT COMMAND exceeds VOUT_MAX COMMAND or VOUT COMMAND lower than VOUT_MIN COMMAND, the bit will be set to1 under corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No V_{OUT_CMD} maxmin fault has occurred 1'b1: An V_{OUT_CMD} maxmin fault has occurred

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STATUS_ TEMPERATURE (7Dh)

Format: Unsigned binary

The STATUS_TEMPERATURE command returns the value of a number of flags indicating the state of the MPQ70165FS junction temperature.

Bits	Access	Bit Name	Description
7	R	TEMP_OT_FAULT	When MPQ70165FS's junction temperature exceeds TSD threshold, the bit will be set to 1 at all pages. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No over temperature fault has occurred 1'b1: An over temperature fault has occurred
6	R	TEMP_OT_WARNING	When MPQ70165FS's junction temperature exceeds TWN threshold, the bit will be set to 1 at all pages. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No over temperature warning has occurred 1'b1: An over temperature warning has occurred
5:0	R	RESERVED	Reserved.

STATUS_CML (7Eh)

Format: Unsigned binary

The STATUS_CML command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	INVALID_CMD	If host sends any write/read command to a register that MPQ70165FS does not define, the device will report invalid command fault and this bit set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No invalid or unsupported commands received 1'b1: An invalid or unsupported command has been received
6	R	INVALID_DATA	If host send write command to an only read register or a protected register by WRITE PROTETCION or a lock register by SECURE_LOCKOUT, the device will report invalid data fault and this bit set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No invalid data received 1'b1: Invalid data has been received
5	R	PEC_ERROR	Under PEC enabled, only when host send a write command with incorrect PEC value, MPQ70165FS will report PEC fault and this bit is set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: Packet Error Check not failed 1'b1: Packet Error Check failed
4	R	CRC_ERROR	In 2nd RESTORE OTP, if OTP CRC check value is incorrect, MPQ70165FS will report start up load OTP CRC fail, and this bit will be set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: CRC did not fail 1'b1: CRC failed (OTP CRC check fail)
3:1	R	RESERVED	Reserved.

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0	R	COM_FAULT	If MPQ70165FS happens start error or stop error or long error, this bit will be set to 1. Long error: When host send a read command with two bytes to an MPQ70165FS register with one byte definition, MPQ70165FS will report long error. Start error: During the process of slave address or command address or data or PEC value, if host send a start signal, MPQ70165FS will report start error. Stop error: During the process of slave address or command address or data or PEC value, if host send a stop signal, MPQ70165FS will report stop error. 1'b0: No communication fault received 1'b1: A communication fault has occurred (START err, STOP err, LONG err)

STATUS_ MFR_SPECIFIC (80h)

Format: Unsigned binary

The STATUS_MFR_SPECIFIC command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	RESERVED	Reserved.
6	R	VMON_OV	After VMON function is active, if VMON voltage exceeds VMON_HIGH_THRESHOLD (D2h), MPQ70165FS will report VMON over-voltage fault. This bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No VMON_OV fault has occurred 1'b1: A VMON_OV fault has occurred
5	R	VMON_UV	After VMON function is active, if VMON voltage drop below of VMON_LOW_THRESHOLD (D1h), MPQ70165FS will report VMON under-voltage fault. This bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No VMON_UV fault has occurred 1'b1: A VMON_UV fault has occurred
4	R	ERRIN_FAULT	When WDI pin is reused to error input pin, if this pin voltage level is high, MPQ70165FS will report ERRIN fault. This bit will be set to 1. 1'b0: No ERRIN fault has occurred 1'b1: An ERRIN fault has occurred
3	R	LBIST_FAULT	During LBIST, if any logic circuits have error, MPQ70165FS will report LBIST fault and this bit will be set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No LBIST fault has occurred 1'b1: A LBIST fault has occurred
2	R	REFERENCE_FAULT	When the difference between the main reference is more than normal range (97% ~ 103% x auxiliary reference), MPQ70165FS will report reference fault and this bit will be set to 1 at all pages. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No reference fault has occurred 1'b1: A reference fault has occurred
1	R	ABIST_FAULT	If the output of any of the comparators checked by ABIST is detected as not pulling up or down properly, MPQ70165FS will report an ABIST fault and this bit will be set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No ABIST fault has occurred 1'b1: An ABIST fault has occurred

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0	R	WD_FAULT	If microcontroller doesn't provide good watchdog event to MPQ70165FS cause watchdog fault count being equal to watchdog fault limit, MPQ70165FS will report watchdog fail and this bit will be set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No watchdog fail has occurred 1'b1: A watchdog fail has occurred

SECURE_LOCKOUT (9Dh)

Format: Unsigned binary

The SECURE_LOCKOUT command to lockout write access to most registers.

Bits	Access	Bit Name	Description
7:0	R/W	SECURE_LOCKOUT	Write 01011010 to lockout write access to most registers. Registers then can still be written are listed as N under "Lockout?" This can only be cleared by setting all enables to low.

MULTIPHASE_CONFIG (9Eh)

Format: Unsigned binary

The MULTIPHASE_CONFIG command to set channels work in parallel or independent.

Bits	Access	Bit Name	Description
7:3	R	RESERVED	Reserved.
2	R/W	MULTIPHASE_B5B6	Set parallel or independent work between buck5 and buck6. Please see Multiphase Operation section for detail information. 1'b0: Independent 1'b1: Multiphase, Buck5 runs as the primary.
1	R/W	MULTIPHASE_B3B4	Set parallel or independent work between buck3 and buck4. Please see Multiphase Operation section for detail information. 1'b0: Independent 1'b1: Multiphase, Buck3 runs as the primary.
0	R/W	MULTIPHASE_B1B2	Set parallel or independent work between buck1 and buck2. Please see Multiphase Operation section for detail information. 1'b0: Independent 1'b1: Multiphase, Buck1 runs as the primary.

HICCUP_TIMER (A0h)

Format: Unsigned binary

The HICCUP_TIMER command to set the hiccup time during hiccup state.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	HICCUP_TIMER	When MPQ70165FS happens deep safe event causing state machine to enter deep safe state and maintain hiccup time. Therefore, hiccup time can be regarded as the time when a deep safe state is maintained. 2'b00: 2ms 2'b01: 4ms 2'b10: 6ms 2'b11: 8ms

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STATUS_PIN_STATE (A1h)

Format: Unsigned binary

The STATUS_PIN_STATE command to indicate the pin voltage status.

Bits	Access	Bit Name	Description
7:4	R	RESERVED	Reserved.
3	R	EN_2_STATE	Indicate EN2 pin voltage status is high or low. When EN2 pin voltage exceeds EN rising threshold, this bit will be set to 1. When EN2 pin voltage is lower than EN falling threshold, this bit will be set to 0. 1'b0: Low 1'b1: High
2	R	EN_1_STATE	Indicate EN1 pin voltage status is high or low. When EN1 pin voltage exceeds EN rising threshold, this bit will be set to 1. When EN1 pin voltage is lower than EN falling threshold, this bit will be set to 0. 1'b0: Low 1'b1: High
1	R	PG_MEAS_STATE	Indicate PG pin voltage status is high or low. When PG pin voltage exceeds EN logic on threshold, this bit will be set to 1. When PG pin voltage is lower than EN logic on threshold, this bit will be set to 0. 1'b0: Low 1'b1: High
0	R	ALERT#_MEAS_STATE	Indicate ALERT# pin voltage status is high or low. When ALERT# pin voltage exceeds EN logic on threshold, this bit will be set to 1. When ALERT# pin voltage is lower than EN logic on threshold, this bit will be set to 0. 1'b0: Low 1'b1: High

VOUT_STARTUP_SLEW (A3h)

Format: Unsigned binary

The VOUT_STARTUP_SLEW command to set the rising rate of Buck 1 output during soft start.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_STARTUP_SLEW	These bits can control the rising rate of channels' output voltage. Please change page vis PAGE command, and then set the register to configure rising rate. To make the output voltage more linear during the soft start period, it is recommended to choose 1.25mV/us or 2.5mV/us. 2'b00: VOUT_SL x 1.25mV/us 2'b01: VOUT_SL x 2.5mV/us 2'b10: VOUT_SL x 5mV/us 2'b11: VOUT_SL x 10mV/us

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VOUT_ SHUTDOWN _SLEW (A5h)

Format: Unsigned binary

The VOUT_SHUTDOWN_SLEW command to set the rising rate of Buck 1 output during soft stop.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_SHUTDOWN_SLEW	These bits can control the rising rate of channels' output voltage. Please change page vis PAGE command, and then set the register to configure rising rate. To make the output voltage more linear during the soft stop period, it is recommended to choose 1.25mV/us or 2.5mV/us. 2'b00: VOUT_SL x 1.25mV/us 2'b01: VOUT_SL x 2.5mV/us 2'b10: VOUT_SL x 5mV/us 2'b11: VOUT_SL x 10mV/us

VOUT_SLEW (A7h)

Format: Unsigned binary

The VOUT_ SLEW command to set the rising rate of Buck 1 output during changing output voltage online.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_SLEW	These bits can control the rising rate of channels' output voltage during changing output voltage online via PMBus. Please change page vis PAGE command, and then set the register to configure rising rate. 2'b00: VOUT_SL x 2.5mV/us 2'b01: VOUT_SL x 5mV/us 2'b10: VOUT_SL x 10mV/us 2'b11: VOUT_SL x 20mV/us

OUTPUT_DISCHARGE (A9h)

Format: Unsigned binary

The OUTPUT_DISCHARGE command to set the discharge path of Buck 1 enabled or disabled.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1	R/W	SS_EN	This bit can control soft stop function to enabled or disabled. Please change page vis PAGE command, and then set the register to configure rising rate. For detail information, please see Output Discharge and Internal Soft Stop sections. 1'b0: Soft Stop Disable 1'b1: Soft Stop Enable
0	R/W	DISCHARGE_EN	This bit can control 100 Ω discharge resistance to be or not to be connected between VOUT and AGND. Please change page vis PAGE command, and then set the register to configure rising rate. For detail information, please see Output Discharge section. 1'b0: 100 Ω Output Discharge Disable 1'b1: 100 Ω Output Discharge Enable

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OUTPUT_SHUTDOWN_MODE (ABh)

Format: Unsigned binary

The OUTPUT_SHUTDOWN_MODE command to set the shutdown mode of Buck 1.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	OUTPUT_SHUTDOWN_MODE	MPQ70165FS only support the Time-Based Shutdown Mode. When EN off or PMBus off, the channel will shut down after the corresponding turn off delay time timing ending. 1'b0: Time Based Shutdown Mode

FREQUENCY_DITHER (ADh)

Format: Unsigned binary

The FREQUENCY_DITHER command to enable or disable frequency spread spectrum in MPQ70165FS.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	FREQUENCY_DITHER	Frequency spread spectrum Control. Please see Frequency Spread Spectrum (FSS) section. 1'b0: Disable 1'b1: Enable

COMPENSATION_CONFIG_1 (B0h)

Format: Unsigned binary

The COMPENSATION_CONFIG_1 command configures the compensation network parameters of Buck 1.

Bits	Access	Bit Name	Description
7:4	R/W	RCOMP	Sets R_{COMP}. The default selection in 1000. 4'b0000: 550K 4'b0001: 600K 4'b0010: 450K 4'b0011: 500K 4'b0100: 750K 4'b0101: 800K 4'b0110: 650K 4'b0111: 700K 4'b1000: 150K 4'b1001: 200K 4'b1010: 50K 4'b1011: 100K 4'b1100: 350K 4'b1101: 400K 4'b1110: 250K 4'b1111: 300K
3	R	RESERVED	Reserved.
2:0	R/W	CCOMP_HP	Sets C_{COMP_HP}. The default selection is 011. 100fF/LSB. 3'b000: 0 x LSB … 3'b111: +7 x LSB

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COMPENSATION_CONFIG_2 (B1h)

Format: Unsigned binary

The COMPENSATION_CONFIG_2 command configures the compensation network parameters of Buck 1.

Bits	Access	Bit Name	Description
7	R	RESERVED	Reserved.
6:4	R/W	CCOMP_LF	Sets C_{COMP_LF}. The default selection is 011. 3'b000: 44.5pF 3'b001: 49.5pF 3'b010: 54.5pF 3'b011: 59.5pF 3'b100: 24.5pF 3'b101: 29.5pF 3'b110: 34.5pF 3'b111: 39.5pF
3:2	R	RESERVED	RESERVED
1:0	R/W	SLOPE_COMP	Sets SLOPE_COMP. The default selection is 10. 2'b00: I_{SLOPE} 2'b01: I_{SLOPE} x 2 2'b10: I_{SLOPE} x 3 2'b11: I_{SLOPE} x 4

WATCHDOG_CONFIG (B2h)

Format: Unsigned binary

The WATCHDOG_CONFIG command configures watchdog parameters.

Bits	Access	Bit Name	Description
7:6	R	RESERVED	Reserved.
5	R/W	WD_MODE	Set the watchdog mode to Window or Question – Answer mode. If the microcontroller changes the watchdog mode while the watchdog is working, the close/open window time must be reconfigured after the watchdog mode is changed. Make sure the window starts timing from close window in the new mode. 1'b0: Window Watchdog, WDI pin as watchdog input pin. 1'b1: QA Watchdog, WDI pin as external error input pin.
4	R/W	WD_EN	This bit can control watchdog to be enabled or disabled. Configuring other watchdog parameters before enabling watchdog. 1'b0: Watchdog disabled 1: Watchdog enabled
3	R/W	WD_FAULT_CFG	When MPQ70165FS detect a watchdog failure, this bit can control whether MPQ70165FS will perform RESTORE_USER_ALL command automatically or not. 1'b0: Watchdog failure does not perform RESTORE_USER_ALL command. 1'b1: Watchdog failure cause MPQ70165FS to perform RESTORE_USER_ALL command automatically.
2:0	R/W	WD_FAULT_LIMIT	Set an upper limit to indicate watchdog fail. When watchdog fault count is equal to the upper limit, MPQ70165FS will report watchdog fail and 80h bit0 will be set to 1. Watchdog Fault Limit= WD_FAULT_LIMIT [2:0] + 1

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WATCHDOG_WINDOW_CONFIG (B4h)

Format: Unsigned binary

The WATCHDOG_WINDOW_CONFIG command configures watchdog Close / Open window time.

Bits	Access	Bit Name	Description
7:6	R	RESERVED	Reserved.
5:3	R/W	WD_CLOSE_WINDOW	Set Close Window time during watchdog operation. 3'b000: 1ms 3'b001: 2ms 3'b010: 3ms 3'b011: 4ms 3'b100: 6ms 3'b101: 8ms 3'b110: 10ms 3'b111: 12ms
2:0	R/W	WD_OPEN_WINDOW	Set Open Window time during watchdog operation. 3'b000: 1ms 3'b001: 2ms 3'b010: 3ms 3'b011: 4ms 3'b100: 6ms 3'b101: 8ms 3'b110: 10ms 3'b111: 12ms

WATCHDOG_TOKEN_FDBCK (B6h)

Format: Unsigned binary

The WATCHDOG_TOKEN_FDBCK command configures watchdog parameters.

Bits	Access	Bit Name	Description
7:5	R/W	WD_TOKEN_FDBCK	These bits are only available for Q-A watchdog mode. They can control the sequence of generated questions and respective answers.
4:0	R/W	WD_TOKEN_SEED	These bits are only available for Q-A watchdog mode. They are used to generate a set of new questions.

WATCHDOG_TOKEN_VALUE (B8h)

Format: Unsigned binary

The WATCHDOG_TOKEN_VALUE command indicates watchdog question during one Q-A watchdog sequence.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R	WD_TOKEN_VALUE	These bits are only available for Q-A watchdog mode. They return to watchdog question. It is recommended to read the question when a new watchdog sequence just begins.

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WATCHDOG_STATUS_1 (BAh)

Format: Unsigned binary

The WATCHDOG_STATUS_1 command indicates some watchdog faults.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4	R	WD_ANS_ERR	This is only available in Q-A watchdog mode. If any of the answers received by MPQ70165FS is incorrect, this bit is set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: Answer Correct 1'b1: Answer Incorrect
3	R	WD_CFG_CHG	This is available in Window and Q-A watchdog mode. If the microcontroller writes to the B4h register of the MPQ70165FS, this bit will be set to 1 regardless of the value written. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: B4h register is not written 1'b1: B4h register is written
2	R	WD_SEQU_ERR	This is only available in Q-A watchdog mode. If any of the answers received by MPQ70165FS is incorrect or correct Answer-0 is sent in close window or happens time-out, this bit is set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No error 1'b1: Token early or wrong answer or time out
1	R	WD_TO_ERR	This is available in Window and Q-A watchdog mode. If MPQ70165FS don't receive valid trigger signal during window mode or MPQ70165FS receives less than four answers during Q-A mode, this bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No watchdog time-out event occurred 1'b1: Watchdog time-out event occurred
0	R	WD_TOKEN_ERR	This is available in Window and Q-A watchdog mode. In Q-A mode, if the Answer-0 is sent in close window, this bit is set to 1 regardless of incorrect or correct answer. In Window mode, if WDI valid pulse is identified in close window, this bit is set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No early watchdog error 1'b1: An early watchdog error occurs

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WATCHDOG_STATUS_2 (BCh)

Format: Unsigned binary

The WATCHDOG_STATUS_2 command indicates watchdog answer order and watchdog fault count.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:3	R	WD_ANS_CNT	This is only available in Q-A watchdog mode. These two bits indicate the current answer order that needs to be written. It is recommended to read this bit before the microcontroller writes the answer to ensure that the answer being written is required by MPQ70165FS. Default value is 2'b11. Each time MPQ70165FS receives an answer, the two bits are updated. The order of update is always 2'b11 -> 2'b10 -> 2'b01 -> 2'b00. 2'b00: 1st watchdog answer needs to be written 2'b01: 2nd watchdog answer needs to be written 2'b10: 3rd watchdog answer needs to be written 2'b11: 4th watchdog answer needs to be written
2:0	R	WD_ERR_CNT	This is available in Window and Q-A watchdog mode. These bits record the number of watchdog fault. But when MPQ70165FS perform a good watchdog event, the counter will be decreased 1. Therefore, the value of counter does not represent the number of bad watchdog events detected by MPQ70165FS in the normal state. The maximum value of this counter is 7. Therefore, with the upper limit set to 8, when the counter is 7, MPQ70165FS then detects a bad event or time-out, and the counter does not increase. Watchdog Fault Counter = WD_ERR_CNT [2:0]

WATCHDOG_ANSWER (BEh)

Format: Unsigned binary

The WATCHDOG_ANSWER command receives watchdog answers.

Bits	Access	Bit Name	Description
7:0	R/W	WD_ANS	This is only available in Q-A watchdog mode. Microcontroller send Watchdog Answer to MPQ70165FS. Please refer to Figure 27 and 28 for the Answer calculation formula.

ALERT#_PG_FORCE_ASSERT (C0h)

Format: Unsigned binary

The ALERT#_PG_FORCE_ASSERT command forces PG and ALERT# to assert.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1	R/W	ALERT#_FORCE_ASSERT	Write 1'b1 to this bit, ALERT# will be low. Write 1'b0 to this bit, ALERT# will be pulled high if no other fault happens. 1'b0: Do not force ALERT# to be asserted 1'b1: Force ALERT# to be asserted
0	R/W	PG_FORCE_ASSERT	Write 1'b1 to this bit, PG will be low. Write 1'b0 to this bit, PG will be pulled high if no other fault happens. 1'b0: Do not force PG to be asserted 1'b1: Force PG to be asserted

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SAFETY_CONFIG (C2h)

Format: Unsigned binary

The SAFETY_CONFIG command enables certain functions.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4	R/W	OVP	The MPQ70165FS always provides over-voltage protection for the output voltage. 1'b1: Over Voltage Protection Enable (don't support disable OVP)
3	R/W	UVP	The MPQ70165FS always provides over-voltage protection for the output voltage. 1'b1: Under voltage Protection Enable (don't support disable UVP)
2	R/W	BIST_EN	The bit can control to enable or disable BIST function. BIST here includes PRE-DIANOSTIC and Memory CRC check. 1'b0: BIST Disabled 1'b1: BIST Enabled (Default)
1	R/W	PG_REACTION	Set PG reaction to 1, MPQ70165FS will support these faults causing state machine to enter deep safe state except pulling-down PG signal. If PG reaction is set to 0, these faults only pull-down PG signal. These faults include PG OV, PG UV, VMON OV and VMON UV. 1'b0: PG low fault (only for PGUV/OV, VMON OV/UV fault) does not move to safe state (Default) 1'b1: PG low fault (only for PGUV/OV, VMON OV/UV fault) causes safe state (all outputs off)
0	R/W	PEC_REQ	This bit can enable or disable PMBus interface communication CRC check, here we call it the PEC function. If PEC is disable, the host don't need send PEC value during write command and the MPQ70165FS don't turn PEC value during read command. If PEC is optional, the host can choose to send or not send PEC value during write command and the MPQ70165FS will definitely send PEC value during read command. 1'b0: PEC Disable (Default) 1'b1: PEC Optional

ALTERT#_MAPPING (C4h)

Format: Unsigned binary

The ALERT#_MAPPING command can configure ALERT# reaction when some faults happen.

Bits	Access	Bit Name	Description
7:6	R	RESERVED	Reserved.
5	R/W	WD_ALERT#_MAPPING	This bit can control whether watchdog failure resets ALERT# signal. When the bit is set to 1, ALERT# isn't also reset and only the register report fault information. 1'b0: Watchdog Failure causes ALERT# reset & register assertion (default) 1'b1: Watchdog Failure cause register assertion only
4	R/W	CRC_ALERT#_MAPPING	This bit can control whether OTP CRC check (during 2nd restore OTP) fail pulls down ALERT# signal. When the bit is set to 1, although MPQ70165FS happens OTP CRC check fail during 2nd restore OTP, ALERT# isn't also pulled down and only the register report fault information. 1'b0: OTP CRC check (during 2nd restore OTP) Failure causes deep safe state and ALERT# low & register assertion (default) 1'b1: OTP CRC check (during 2nd restore OTP) Failure cause deep safe state and register assertion

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3	R/W	PMBus _CML_ALERT#_MAPPING	This bit can control whether ALERT# is pulled down when MPQ70165FS detect unsupported command fault or unsupported data fault or communication fault (include start error, stop error, long error). 1'b0: PMBus Communication Failure causes ALERT# low & register assertion (default) 1'b1: PMBus Communication Failure cause register assertion only
2	R/W	PMBus _CRC_ALERT#_MAPPING	This bit can control whether ALERT# is pulled down when MPQ70165FS detect PEC fault. 1'b0: PMBus CRC Failure causes ALERT# low & register assertion(default) 1'b1: PMBus CRC Failure cause register assertion only
1	R/W	REF_FAULT_ALERT #_MAPPING	This bit can control whether ALERT# is pulled down when MPQ70165FS detect reference fault. 1'b0: Reference check Failure causes ALERT# low & register assertion (default) 1'b1: Reference check Failure only causes register assertion
0	R/W	BIST_FAULT_ALERT #_MAPPING	This bit can control whether ALERT# is pulled down when MPQ70165FS detect BIST failure. BITS failure includes ABIST fault, Memory CRC (in Normal state) fault, OTP CRC (in 2nd Restore OTP, Pre-Diagnostic and Normal state) fault and Clock (in Pre-Diagnostic state) fault. 1'b0: BIST Failure causes ALERT# low & register assertion (default) 1'b1: BIST Failure only causes register assertion

PG_MAPPING (C6h)

Format: Unsigned binary

The PG_MAPPING command can configure PG reaction and state machine reaction when some faults happen.

Bits	Access	Bit Name	Description
7:6	R	RESERVED	Reserved.
5	R/W	WD_PG_MAPPING	This bit can control whether watchdog fail pulls down PG signal. When the bit is set to 1, although MPQ70165FS happens watchdog fail because watchdog fault count is equal to watchdog fault limit, PG isn't also pulled down and only the register report fault information. 1'b0: Watchdog Fail causes PG low & register assertion (default) 1'b1: Watchdog Fail cause register assertion only
4	R/W	CRC_PG_MAPPING	This bit can control whether MPQ70165FS masks the OTP CRC check failure during 2nd Restore OTP. If this bit is set to1, state machine doesn't stuck in DEEP SAFE state with hiccup time even though MPQ70165FS detect OTP CRC (during 2nd Restore OTP) check failure. 1'b0: OTP CRC Failure causes deep safe state & register assertion (default) 1'b1: OTP CRC Failure cause register assertion only
3	R/W	PMBus_CML_PG_MAPPING	This bit can control whether PG is pulled down when MPQ70165FS detect unsupported command fault or unsupported data fault or communication fault (include start error, stop error, long error). 1'b0: PMBus Communication Failure causes PG low & register assertion (default) 1'b1: PMBus Communication Failure cause register assertion only
2	R/W	PMBus_CRC_PG_MAPPING	This bit can control whether PG is pulled down when MPQ70165FS detect PEC failure. 1'b0: PMBus CRC Failure causes PG low & register assertion(default) 1'b1: PMBus CRC Failure cause register assertion only

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1	R/W	REF_FAULT_PG_MAPPING	This bit can control whether PG is pulled down and state machine enters deep safe state when MPQ70165FS detect reference failure. 1'b0: Reference Failure causes deep safe state and PG low & register assertion (default) 1'b1: Reference Failure cause register assertion only
0	R/W	BIST_FAULT_PG_MAPPING	This bit can control whether PG is pulled down and state machine enters deep safe state when MPQ70165FS detect BIST failure. BITS failure includes ABIST fault, Memory CRC (in Normal state) fault, OTP CRC (in 2nd Restore OTP, Pre-Diagnostic and Normal state) fault and Clock (in Pre-Diagnostic state) fault. But for the process of pre-diagnostic, once ABIST fault or Clock check fault or OTP CRC check fault is detected, MPQ70165FS will report pre-diagnostic failure at pre-diagnostic ending. The pre-diagnostic failure will cause state machine entering deep safe state when this bit is set to 1. 1'b0: BIST Failure causes deep safe state and PG low & register assertion (default) 1'b1: BIST Failure cause register assertion only

PG_ALERT#_DELAY (C8h)

Format: Unsigned binary

The PG_ALERT#_DELAY command can configure PG and ALERT# delay time.

Bits	Access	Bit Name	Description
7:4	R	RESERVED	Reserved.
3:2	R/W	ALERT#_DELAY	When all faults disappear, ALERT# can be pulled high again through write clear fault command. After writing clear fault command, timing ALERT# delay time, ALERT# will be pulled high. 2'b00: Immediate 2'b01: 2ms 2'b10: 5ms 2'b11: 10ms
1:0	R/W	PG_DELAY	When all faults disappear and state enter normal state, PG can be pulled high again automatically after waiting PG delay time. 2'b00: Immediate 2'b01: 2ms 2'b10: 5ms 2'b11: 10ms

PG_CONFIG (CAh)

Format: Unsigned binary

The PG_CONFIG command can configure PG OV / UV threshold of Buck 1.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	PG_THRESHOLD	This bit can set different PG OV rising threshold and PG UV falling threshold for different channel. Please change page vis PAGE command, and then set the register to configure PG threshold. 1'b1: +4.5% / - 3.5% 1'b0: +6.5% / - 5.5%

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LIGHT_LOAD (CCh)

Format: Unsigned binary

The LIGHT_LOAD command can configure the work mode to FCCM at light load.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	MODE	The bit can set the work mode is FCCM at light load. 1'b0: Invalid command. 1'b1: FCCM

ILIM_SCALE (CEh)

Format: Unsigned binary

The ILIM_SCALE command can configure the current limit scale of Buck 1.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	ILIM_SCALE	These bits can set the scale of peak / valley / reverse current limit, default value is 100% (2'b00). Please change page vis PAGE command, and then set the register to configure current limit scale. 2'b00: 100% 2'b01: 75% 2'b10: 50%

ENABLE_GROUP (CFh)

Format: Unsigned binary

The ENABLE_GROUP command can configure these functions are controlled to EN1 or EN2.

Bits	Access	Bit Name	Description
7	R/W	MON_EN_GUP	Setting EN1 or EN2 controls whether the VMON function is enabled or disabled. 1'b0: Monitor in Enable Group 1 1'b1: Monitor in Enable Group 2
6	R/W	SEQ_EN_GUP	Setting EN1 or EN2 controls whether the SEQ function is enabled or disabled. 1'b0: Sequencer in Enable Group 1 1'b1: Sequencer in Enable Group 2
5	R/W	BK6_EN_GUP	Setting EN1 or EN2 controls whether the buck6 is enabled or disabled. 1'b0: Buck 6 in Enable Group 1 1'b1: Buck 6 in Enable Group 2
4	R/W	BK5_EN_GUP	Setting EN1 or EN2 controls whether the buck5 is enabled or disabled. 1'b0: Buck 5 in Enable Group 1 1'b1: Buck 5 in Enable Group 2
3	R/W	BK4_EN_GUP	Setting EN1 or EN2 controls whether the buck4 is enabled or disabled. 1'b0: Buck 4 in Enable Group 1 1'b1: Buck 4 in Enable Group 2
2	R/W	BK3_EN_GUP	Setting EN1 or EN2 controls whether the buck3 is enabled or disabled. 1'b0: Buck 3 in Enable Group 1 1'b1: Buck 3 in Enable Group 2

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1	R/W	BK2_EN_GUP	Setting EN1 or EN2 controls whether the buck2 is enabled or disabled. 1'b0: Buck 2 in Enable Group 1 1'b1: Buck 2 in Enable Group 2
0	R/W	BK1_EN_GUP	Setting EN1 or EN2 controls whether the buck1 is enabled or disabled. 1'b0: Buck 1 in Enable Group 1 1'b1: Buck 1 in Enable Group 2

SEQ_DELAY (D0h)

Format: Unsigned binary

The SEQ_DELAY command can configure SEQ output delay time.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	SEQ_DELAY	After PRE-DIAG state, when related enable pin voltage exceeds EN rising threshold, MPQ70165FS begin to time the SEQ delay time. Once the delay timing ending, SEQ will be pulled high. When related enable pin voltage ramp down to the below of EN falling threshold, MPQ70165FS begin to time the SEQ delay time. Once the delay timing ending, SEQ will be pulled low. Sequence Output Startup Delay = SEQ_DELAY [4:0] x 0.25ms. (0 – 7.75ms)

VMON_LOW_THRESHOLD (D1h)

Format: Unsigned binary

The VMON_LOW_THRESHOLD command can set the lower limit of voltage monitor range.

Bits	Access	Bit Name	Description
7:0	R/W	VMON_LOW_THR	MON_RANGE[0] = 0: V_{MON_LOW} setting, from 0.4125V to 3.6V V_{MON_LOW} = VMON_LOW_THR [7:0] x 12.5mV + 412.5mV MON_RANGE[0] = 1: V_{MON_LOW} setting, from 0.2V to 3.3875V V_{MON_LOW} = VMON_LOW_THR [7:0] x 12.5mV + 200mV

VMON_ HIGH _THRESHOLD (D2h)

Format: Unsigned binary

The VMON_HIGH_THRESHOLD command can set the upper limit of voltage monitor range.

Bits	Access	Bit Name	Description
7:0	R/W	VMON_HIGH_THR	MON_RANGE[0] = 0: V_{MON_HIGH} setting, from 0.4125V to 3.6V V_{MON_HIGH} = VMON_HIGH_THR [7:0] x 12.5mV + 412.5mV MON_RANGE[0] = 1: V_{MON_HIGH} setting, from 0.2V to 3.3875V V_{MON_HIGH} = VMON_HIGH_THR [7:0] x 12.5mV + 200mV

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VMON_DELAY (D3h)

Format: Unsigned binary

The VMON_DELAY command can set the delay time of VMON available.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	VMON_DELAY	After PRE-DIAG state, when related enable pin voltage exceeds EN rising threshold, MPQ70165FS begin to time the VMON delay time. Once the delay timing ending, VMON function will be active. MPQ70165FS only detect VMON voltage during VMON active state. MON Block on/off Delay = VMON_DELAY[4:0] x 0.25ms. (0 – 7.75ms)

MON_RANGE (D4h)

Format: Unsigned binary

The MON_RANGE command can set the voltage monitor range.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1	R/W	MON_ENABLE	This bit is used to enable or disable VMON function when the corresponding enable pin is high level. 1'b0: Disable 1'b1: Enable
0	R/W	MON_RANGE	This bit can set the VMON range. Please see the below formula. 1'b0: $V_{MON_REF_HIGH}$ = (VMON_HIGH_THR [7:0] + 33) x 12.5mV $V_{MON_REF_LOW}$ = (VMON_LOW_THR [7:0] + 33) x 12.5mV 1'b1: $V_{MON_REF_HIGH}$ = (VMON_HIGH_THR [7:0] + 16) x12.5mV $V_{MON_REF_LOW}$ = (VMON_LOW_THR [7:0] + 16) x 12.5mV

ADDRESS (D5h)

Format: Unsigned binary

The ADDRESS command can set the PMBus address.

Bits	Access	Bit Name	Description
7	R	RESERVED	Reserved.
6:0	R/W	ADDRESS	MPQ70165FS support to set different PMBus address. (0x00 ~ 0x7F)

CONFIG_CODE (D6h)

Format: Unsigned binary

The CONFIG_CODE command can indicate whether user page 1 / 2 are OTP programmed.

Bits	Access	Bit Name	Description
7:0	R/W	CONFIG_CODE	For MPS provided MPQ70165FS, if the register value is 0x00, MPQ70165FS has two OTP chance. If the register value is 0x01, MPQ70165FS has one OTP chance. If the register value is 0x03, MPQ70165FS has no OTP chance.

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OTP_CRC (D7h)

Format: Unsigned binary

The OTP_CRC command can return OTP CRC value.

Bits	Access	Bit Name	Description
15:0	R	OTP_CRC	The register stores the CRC value of user OTP data. Only when user page 1 is programmed or user page 1 and page 2 are programmed, the register will return to a valid value. Otherwise, the register value is always 0x0000.

MEMORY _CRC (D9h)

Format: Unsigned binary

The MEMORY_CRC command can return Memory CRC value.

Bits	Access	Bit Name	Description
15:0	R	MEMORY_CRC	After Memory CRC check is activated, the register stores the CRC value of the current registers' value. Please refer to register map, which registers participate in memory CRC check.

MFR_OTP_MEM_CHK (DAh)

Format: Unsigned binary

The MFR_OTP_MEM_CHK command can configure the work mode of OTP CRC check and Memory CRC check.

Bits	Access	Bit Name	Description
7:3	R	RESERVED	Reserved.
2	R/W	MFR_OTP_MEM_CHK	The bit can set whether OTP CRC check and Memory CRC check are continuous check at Normal state. OTP CRC check: If the bit is set to 1, the OTP CRC check is always working when DAh bit1 = 1'b1. If the bit is set to 0, the OTP CRC check will be performed when DAh bit1 is changed from 1'b0 to 1'b1. Memory CRC check: If the bit is set to 1, the Memory CRC check is always working when DAh bit0 = 1'b1. If the bit is set to 0, the Memory CRC check will perform when DAh bit0 is changed from 1'b0 to 1'b1. 1'b1: continuous check after OTP CRC check and Memory CRC check config enable. 1'b0: check once after OTP CRC check and Memory CRC check changed from disable to enable.
1	R/W	MFR_OTP_CRC_CHK_EN	The bit can control whether OTP CRC check is enabled or disabled. 1'b0: Disable 1: Enable
0	R/W	MFR_MEM_CRC_CHK_EN	The bit can control whether Memory CRC check is enabled or disabled. 1'b0: Disable 1'b1: Enable

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MFR_OTP_MEM_STATUS (DBh)

Format: Unsigned binary

The MFR_OTP_MEM_STATUS command can indicate some related faults of OTP CRC and Memory CRC.

Bits	Access	Bit Name	Description
7:6	R	RESERVED	Reserved.
5	R	STRUP_OTP_CRC_ERR	When MPQ70165FS detect OTP CRC error during 2nd Restore OTP, the bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No start up load OTP CRC error 1'b1: Start up load OTP CRC error
4	R	STRUP_OTP_IND_ERR	There are 3 bits inside the MPQ70165FS to indicate the usage of OTP. 3 'b000 means no page is programmed. 3' b001 means internal trim page is programmed. 3 'b011 means trim page and 1st user page are programmed. 3 'b111 for trim page, 1st user page and 2nd user page are both burned. On 2nd Restore OTP, if MPQ70165FS detects anything other than the four case above, the bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No start up load OTP indicator error 1'b1: Start up load OTP indicator error
3	R	STORE_OTP_ERR	When MPQ70165FS has one or two OTP chance and then host send Store User All completed, if the 3 bits OTP indicator don't be changed, the bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No store OTP fail 1'b1: Store OTP fail
2	R	BIST_CHK_MEM_CRC_ERR	When MPQ70165FS detect Memory CRC error during Normal state, the bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No BIST check memory CRC error 1'b1: BIST check memory CRC error
1	R	BIST_CHK_OTP_CRC_ERR	When MPQ70165FS detect OTP CRC error during Pre-Diagnostic and Normal state, the bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No BIST check OTP CRC error 1'b1: BIST check OTP CRC error
0	R	BIST_CHK_OTP_IND_ERR	There are 3 bits inside the MPQ70165FS to indicate the usage of OTP. 3 'b000 means no page is programmed. 3' b001 means internal trim page is programmed. 3 'b011 means trim page and 1st user page are programmed. 3 'b111 for trim page, 1st user page and 2nd user page are both burned. On OTP CRC check (during Pre-diagnostic and Normal state), if MPQ70165FS detects anything other than the four case above, the bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No BIST check OTP indicator error 1'b1: BIST check OTP indicator error

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STATUS_FREQ (E7h)

Format: Unsigned binary

The STATUS_FREQ command can indicate clock check fault.

Bits	Access	Bit Name	Description
7:3	R	RESERVED	Reserved.
2	R	REG_W_CHK_ERR	When a written data of register isn't same as the intended written value, MPQ70165FS will report register write check error. ALERT# is pulled down. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No register write check error occurred 1'b1: A register write check error occurred
1	R	RESERVED	Reserved.
0	R	CLK_CHK_ERR	When MPQ70165FS internal clock error counter is equal to 8, the device will report clock check fault and the bit is set to 1. PG and ALERT# are pulled down, all channels are shutdown. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No clock check error occurred 1'b1: A clock check error occurred

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PAGE 1 REGISTER MAP

PAGE (00h)

Format: Unsigned binary

The PAGE command provides the ability to configure, control and monitor all outputs through only one physical address. All subsequent commands (pages is “Y” at REGISTER MAP section) that allow multipage action are applied to the corresponding regulators selected by the PAGE command.

Bits	Access	Bit Name	Description
7:0	R/W	PAGE	In Page FF, write command to other registers can change all pages’ configuration, including page 0-5. In Page 0 – 5, write command to other registers can only change the current page’s configuration. So, the PAGE command shall be written before modifying other registers. 8’b00000000: VOUT1 8’b00000001: VOUT2 8’b00000010: VOUT3 8’b00000011: VOUT4 8’b00000100: VOUT5 8’b00000101: VOUT6 8’b11111111: All Others: Invalid commands

OPERATION (01h)

Format: Unsigned binary

The OPERATION command is used to configure the output on/off state of the Buck 2, in conjunction with input from the EN_X pin.

Bits	Access	Bit Name	Description
7	R/W	OPERATION	Enable signal is available, MPQ70165FS support interface to turn on / off channels through PMBus. 1’b1: Output on (if enable high) 1’b0: Output off
6:0	R/W	RESERVED	Reserved.

CLEAR_FAULTS (03h)

This command is used to clear any fault bit in all status registers: STATUS_BYTE (78h), STATUS_WORD (79h), STATUS_VOUT (7Ah), STATUS_TEMPERATURE (7Dh), STATUS_CML (7Eh), STATUS_MFR_SPECIFIC (80h), WATCHDOG_STATUS_1 (BAh), MFR_OTP_MEM_STATUS (DBh) and STATUS_FREQ (E7h)

This command is written only. There is no data byte for this command.

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VOUT_COMMAND (21h)

Format: Unsigned binary

The VOUT_COMMAND command sets the output voltage of Buck 2.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_CMD	The command is used to set output voltage. Steps as below: 3. Select page x by PAGE (00h). 4. Change V_{OUT} by VOUT_COMMAND. V_{OUT} is influenced by VOUT_SCALE_LOOP, see the below description for detail V_{OUT} range. 3. When VOUT_SL = 1, V_{OUT} can be set from 0.20625V to 1.8V, 6.25mV/step. 4. When VOUT_SL = 2, V_{OUT} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT} = (VOUT_CMD x 6.25mV + 206.25mV) x VOUT_SL

VOUT_MAX (24h)

Format: Unsigned binary

The VOUT_MAX command sets an upper limit on the output voltage that Buck 2 can command regardless of any other commands or combinations. The intent of this command is to provide a safeguard against a user accidental setting the output voltage to a possible destructive level rather than to be the primary output overvoltage protection.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_MAX	When VOUT_SL = 1, V_{OUT_MAX} can be set from 0.20625V to 1.8V, 6.25mV/step. When VOUT_SL = 2, V_{OUT_MAX} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT_MAX} = (VOUT_MAX x 6.25mV + 206.25mV) x VOUT_SL Attempting to write a higher value to VOUT_COMMAND causes VOUT_COMMAND to be set to VOUT_MAX and asserts a VOUT_MAX_MIN warning.

VOUT_SCALE_LOOP (29h)

Format: Unsigned binary

The VOUT_SCALE_LOOP command sets the scale for the V_{OUT}, V_{OUT_MAX}, V_{OUT_MIN}, V_{OUT} start-up slew rate, V_{OUT} shutdown slew rate and V_{OUT} slew rate on Page 1.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	VOUT_SC_L	According to application requirement, set adjust scale. The device should be off via the OPERATINO command before change the scale. 1'b0: VOUT_SL= 1 1'b1: VOUT_SL= 2

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VOUT_MIN (2Bh)

Format: Unsigned binary

The VOUT_MIN command sets a lower limit on the output voltage that Buck 2 can command regardless of any other commands or combinations. The intent of this command is to provide a safeguard against a user accidentally setting the output voltage to a possible destructive level rather than to be the primary output under-voltage protection.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_MIN	When VOUT_SL = 1, V_{OUT_MIN} can be set from 0.20625V to 1.8V, 6.25mV/step. When VOUT_SL = 2, V_{OUT_MIN} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT_MIN} = (VOUT_MIN x 6.25mV + 206.25mV) x VOUT_SL Attempting to write a lower value to VOUT_COMMAND causes VOUT_COMMAND to be set to VOUT_MIN, and asserts a VOUT_MAX_MIN warning

TON_DELAY (60h)

Format: Unsigned binary

The TON_DELAY command sets the turn-on delay time for Buck 2.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	TON_DELAY	Every buck's turn on delay time can be set independently. Please change page vis PAGE command, and then set the register to configure turn on delay time. It can be calculated with the following equation: Startup Delay = TON_DELAY x 0.25ms. (0-7.75ms)

TOFF_DELAY (64h)

Format: Unsigned binary

The TOFF_DELAY command sets the turn-off delay time for Buck 2.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	TOFF_DELAY	Every buck's turn off delay time can be set independently. Please change page vis PAGE command, and then set the register to configure turn off delay time. Only when enable is low or PMBus off command or V_{IN} < UVLO, the device can perform the turn off sequence. It can be calculated with the following equation: Shutdown Delay = TOFF_DELAY x 0.25ms. (0-7.75ms)

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STATUS_BYTE (78h)

Format: Unsigned binary

The STATUS_BYTE command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	BUSY	While MPQ70165FS is performing OTP or restore OTP, host send PMBus write command. MPQ70165FS will report the Busy fault and ALERT# will be pulled down (see Table 1 at page 39). The bit is set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No busy fault 1'b1: A fault was declared because the device was busy and unable to respond
6	R	OFF	The bit can indicate channels' current status. For different channels' status, only change page via PAGE command before reading this bit. When the channel's status in between t_{ON_DELAY} ending and t_{OFF_DELAY} ending, the bit will be set to 0. Besides, this bit will be set to 1. 1'b0: Device is on 1'b1: Device is off
5	R	VOUT_OV_FAULT	When one channel's V_{OUT} exceeds V_{OUT} OV threshold duration ~10us, the channel will happen V_{OUT} OV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 7Ah bit7. 1'b0: No output over voltage fault has occurred 1'b1: An output over voltage fault has occurred
4	R	VOUT_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 79h bit6 and 7Ah bit4. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
3	R	RESERVED	Reserved.
2	R	TEMPERATURE	When MPQ70165FS's junction temperature exceeds TWN threshold or TSD threshold, the bit will be set to 1 at all pages. But the bit cannot accurately determine which thermal fault occurs. Please read 7Dh to ger the detail thermal fault information. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No TSD fault and TWN fault occurred 1'b1: A TSD fault or TWN fault has occurred
1	R	CML	When MPQ70165FS happens PMBus fault or OTP CRC (during 2nd restore OTP) fault, this bit will be set to 1 at all pages. PMBus fault include unsupported command fault, unsupported data fault, PEC fault and communication fault (include start error, stop error and long error). Please read 7Eh to ger the detail fault information. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No PMBus or OTP CRC (during 2nd restore OTP) fault has occurred. 1'b1: A PMBus or OTP CRC (during 2nd restore OTP) fault has occurred.

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0	R	FAULT_OTHER	In a channel, when VOUT COMMAND exceeds VOUT_MAX COMMAND or VOUT COMMAND lower than VOUT_MIN COMMAND, the bit will be set to1 under corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No VOUT MAXMIN fault has occurred. 1'b1: VOUT MAXMIN fault has occurred.

STATUS_WORD (79h)

Format: Unsigned binary

This command returns two bytes of information with a summary of the device fault/warning condition. The higher byte gives more detailed information of the fault conditions. The higher byte shares with register STATUS_BYTE.

Bits	Access	Bit Name	Description
15:8	R	STATUS_BYTE	The same as STATUS BYTE.
7	R	VOUT_FAULT	When one channel happens V_{OUT} related fault, this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No output voltage fault has occurred 1'b1: An output voltage fault has occurred. (including OV fault, UV fault, VOUT MAXMIN fault, PG_OV fault and PG_UV fault)
6	R	VO_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4 and 7Ah bit4. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
5	R	RESERVED	Reserved.
4	R	MANUFAC_FAULT	When MPQ70165FS happens reference fault, ABIST fault or watchdog fail fault, this bit will be set to 1 at all pages. Please read 80h to ger the detail fault information. This fault flag will be cleared once the device performs ABIST if only reference fault happened. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No manufacturer specific fault has occurred 1'b1: A manufacturer specific fault has occurred. (Including Reference fault, ABIST fault and Watchdog fail event)
3	R	PGOOD	The bit can indicate if channels' output exceeds the PGOOD range. For different channels' PGOOD status, only change page via PAGE command before reading this bit. When channel's output is not within PGOOD range, the bit will be set to 1 at corresponding page. Other page doesn't set this bit to 1. When channel's output is within PGOOD range, the bit will be set to 0 automatically at corresponding page. 1'b0: Power Good is high 1'b1: Power Good is low
2:0	R	RESERVED	Reserved.

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STATUS_VOUT (7Ah)

Format: Unsigned binary

The STATUS_VOUT command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	VOUT_OV_FAULT	When one channel's V_{OUT} exceeds V_{OUT} OV threshold duration ~10us, the channel will happen V_{OUT} OV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4. 1'b0: No output over voltage fault has occurred 1'b1: An output over voltage fault has occurred
6:5	R	RESERVED	Reserved.
4	R	VOUT_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4 and 79h bit6. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
3	R	RESERVED	Reserved.
2	R	PG_OV	When one channel's V_{OUT} exceeds PG upper threshold, the channel will happen PGOV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No pgova fault has occurred 1'b1: A pgova fault has occurred
1	R	PG_UV	When one channel's V_{OUT} is lower than PG lower threshold, the channel will happen PGUV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No pguva fault has occurred 1'b1: A pguva fault has occurred
0	R	VOUT_MAX_MIN	In a channel, when VOUT COMMAND exceeds VOUT_MAX COMMAND or VOUT COMMAND lower than VOUT_MIN COMMAND, the bit will be set to1 under corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No V_{OUT_CMD} maxmin fault has occurred 1'b1: An V_{OUT_CMD} maxmin fault has occurred

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VOUT_STARTUP_SLEW (A3h)

Format: Unsigned binary

The VOUT_STARTUP_SLEW command to set the rising rate of Buck 2 output during soft start.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_STARTUP_SLEW	These bits can control the rising rate of channels' output voltage. Please change page vis PAGE command, and then set the register to configure rising rate. To make the output voltage more linear during the soft start period, it is recommended to choose 1.25mV/us or 2.5mV/us. 2'b00: VOUT_SL x 1.25mV/μs 2'b01: VOUT_SL x 2.5mV/μs 2'b10: VOUT_SL x 5mV/μs 2'b11: VOUT_SL x 10mV/μs

VOUT_ SHUTDOWN _SLEW (A5h)

Format: Unsigned binary

The VOUT_SHUTDOWN_SLEW command to set the rising rate of Buck 2 output during soft stop.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_SHUTDOWN_SLEW	These bits can control the rising rate of channels' output voltage. Please change page vis PAGE command, and then set the register to configure rising rate. To make the output voltage more linear during the soft stop period, it is recommended to choose 1.25mV/us or 2.5mV/us. 2'b00: VOUT_SL x 1.25mV/μs 2'b01: VOUT_SL x 2.5mV/μs 2'b10: VOUT_SL x 5mV/μs 2'b11: VOUT_SL x 10mV/μs

VOUT_SLEW (A7h)

Format: Unsigned binary

The VOUT_ SLEW command to set the rising rate of Buck 2 output during changing output voltage online.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_SLEW	These bits can control the rising rate of channels' output voltage during changing output voltage online via PMBus. Please change page vis PAGE command, and then set the register to configure rising rate. 2'b00: VOUT_SL x 2.5mV/μs 2'b01: VOUT_SL x 5mV/μs 2'b10: VOUT_SL x 10mV/μs 2'b11: VOUT_SL x 20mV/μs

OUTPUT_DISCHARGE (A9h)

Format: Unsigned binary

The OUTPUT_DISCHARGE command to set the discharge path of Buck 2 enabled or disabled.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.

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1	R/W	SS_EN	This bit can control soft stop function to enabled or disabled. Please change page vis PAGE command, and then set the register to configure rising rate. For detail information, please see Output Discharge and Internal Soft Stop sections. 1'b0: Soft Stop Disable 1'b1: Soft Stop Enable
0	R/W	DISCHARGE_EN	This bit can control 100 Ω discharge resistance to be or not to be connected between VOUT and AGND. Please change page vis PAGE command, and then set the register to configure rising rate. For detail information, please see Output Discharge section. 1'b0: 100 Ω Output Discharge Disable 1'b1: 100 Ω Output Discharge Enable

OUTPUT_SHUTDOWN_MODE (ABh)

Format: Unsigned binary

The OUTPUT_SHUTDOWN_MODE command to set the shutdown mode of Buck 2.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	OUTPUT_SHUTDOWN_MODE	MPQ70165FS only support the Time-Based Shutdown Mode. When EN off or PMBus off, the channel will shut down after the corresponding turn off delay time timing ending. 1'b0: Time Based Shutdown Mode

COMPENSATION_CONFIG_1 (B0h)

Format: Unsigned binary

The COMPENSATION_CONFIG_1 command configures the compensation network parameters of Buck 2.

Bits	Access	Bit Name	Description
7:4	R/W	RCOMP	Sets R_{COMP}. The default selection in 1000. 4'b0000: 550K 4'b0001: 600K 4'b0010: 450K 4'b0011: 500K 4'b0100: 750K 4'b0101: 800K 4'b0110: 650K 4'b0111: 700K 4'b1000: 150K 4'b1001: 200K 4'b1010: 50K 4'b1011: 100K 4'b1100: 350K 4'b1101: 400K 4'b1110: 250K 4'b1111: 300K
3	R	RESERVED	Reserved.
2:0	R/W	CCOMP_HP	Sets C_{COMP_HP}. The default selection is 011. 100fF/LSB. 3'b000: 0 x LSB … 3'b111: +7 x LSB

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COMPENSATION_CONFIG_2 (B1h)

Format: Unsigned binary

The COMPENSATION_CONFIG_2 command configures the compensation network parameters of Buck 2.

Bits	Access	Bit Name	Description
7	R	RESERVED	Reserved.
6:4	R/W	CCOMP_LF	Sets C_{COMP_LF}. The default selection is 011. 3'b000: 44.5pF 3'b001: 49.5pF 3'b010: 54.5pF 3'b011: 59.5pF 3'b100: 24.5pF 3'b101: 29.5pF 3'b110: 34.5pF 3'b111: 39.5pF
3:2	R	RESERVED	Reserved.
1:0	R/W	SLOPE_COMP	Sets SLOPE_COMP. The default selection is 10. 2'b00: I_{SLOPE} 2'b01: I_{SLOPE} x 2 2'b10: I_{SLOPE} x 3 2'b11: I_{SLOPE} x 4

PG_CONFIG (CAh)

Format: Unsigned binary

The PG_CONFIG command can configure PG OV / UV threshold of Buck 2.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	PG_THRESHOLD	This bit can set different PG OV rising threshold and PG UV falling threshold for different channel. Please change page vis PAGE command, and then set the register to configure PG threshold. 1'b1: +4.5% / - 3.5% 1'b0: +6.5% / - 5.5%

ILIM_SCALE (CEh)

Format: Unsigned binary

The ILIM_SCALE command can configure the current limit scale of Buck 2.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	ILIM_SCALE	These bits can set the scale of peak / valley / reverse current limit, default value is 100% (2'b00). Please change page vis PAGE command, and then set the register to configure current limit scale. 2'b00: 100% 2'b01: 75% 2'b10: 50%

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PAGE 2 REGISTER MAP

PAGE (00h)

Format: Unsigned binary

The PAGE command provides the ability to configure, control and monitor all outputs through only one physical address. All subsequent commands (pages is “Y” at REGISTER MAP section) that allow multipage action are applied to the corresponding regulators selected by the PAGE command.

Bits	Access	Bit Name	Description
7:0	R/W	PAGE	In Page FF, write command to other registers can change all pages’ configuration, including page 0-5. In Page 0 – 5, write command to other registers can only change the current page’s configuration. So, the PAGE command shall be written before modifying other registers. 8’b00000000: VOUT1 8’b00000001: VOUT2 8’b00000010: VOUT3 8’b00000011: VOUT4 8’b00000100: VOUT5 8’b00000101: VOUT6 8’b11111111: All Others: Invalid commands

OPERATION (01h)

Format: Unsigned binary

The OPERATION command is used to configure the output on/off state of the Buck 3, in conjunction with input from the EN_X pin.

Bits	Access	Bit Name	Description
7	R/W	OPERATION	Enable signal is available, MPQ70165FS support interface to turn on / off channels through PMBus. 1’b1: Output on (if enable high) 1’b0: Output off
6:0	R/W	RESERVED	Reserved.

CLEAR_FAULTS (03h)

This command is used to clear any fault bit in all status registers: STATUS_BYTE (78h), STATUS_WORD (79h), STATUS_VOUT (7Ah), STATUS_TEMPERATURE (7Dh), STATUS_CML (7Eh), STATUS_MFR_SPECIFIC (80h), WATCHDOG_STATUS_1 (BAh), MFR_OTP_MEM_STATUS (DBh) and STATUS_FREQ (E7h)

This command is written only. There is no data byte for this command.

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VOUT_COMMAND (21h)

Format: Unsigned binary

The VOUT_COMMAND command sets the output voltage of Buck 3.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_CMD	The command is used to set output voltage. Steps as below: 5. Select page x by PAGE (00h). 6. Change V_{OUT} by VOUT_COMMAND. V_{OUT} is influenced by VOUT_SCALE_LOOP, see the below description for detail V_{OUT} range. 5. When VOUT_SL = 1, V_{OUT} can be set from 0.20625V to 1.8V, 6.25mV/step. 6. When VOUT_SL = 2, V_{OUT} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT} = (VOUT_CMD x 6.25mV + 206.25mV) x VOUT_SL

VOUT_MAX (24h)

Format: Unsigned binary

The VOUT_MAX command sets an upper limit on the output voltage that Buck 3 can command regardless of any other commands or combinations. The intent of this command is to provide a safeguard against a user accidental setting the output voltage to a possible destructive level rather than to be the primary output overvoltage protection.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_MAX	When VOUT_SL = 1, V_{OUT_MAX} can be set from 0.20625V to 1.8V, 6.25mV/step. When VOUT_SL = 2, V_{OUT_MAX} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT_MAX} = (VOUT_MAX x 6.25mV + 206.25mV) x VOUT_SL Attempting to write a higher value to VOUT_COMMAND causes VOUT_COMMAND to be set to VOUT_MAX and asserts a VOUT_MAX_MIN warning.

VOUT_SCALE_LOOP (29h)

Format: Unsigned binary

The VOUT_SCALE_LOOP command sets the scale for the V_{OUT}, V_{OUT_MAX}, V_{OUT_MIN}, V_{OUT} start-up slew rate, V_{OUT} shutdown slew rate and V_{OUT} slew rate on Page 2.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	VOUT_SC_L	According to application requirement, set adjust scale. The device should be off via the OPERATINO command before change the scale. 1'b0: VOUT_SL= 1 1'b1: VOUT_SL= 2

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VOUT_MIN (2Bh)

Format: Unsigned binary

The VOUT_MIN command sets a lower limit on the output voltage that Buck 3 can command regardless of any other commands or combinations. The intent of this command is to provide a safeguard against a user accidentally setting the output voltage to a possible destructive level rather than to be the primary output under-voltage protection.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_MIN	When VOUT_SL = 1, V_{OUT_MIN} can be set from 0.20625V to 1.8V, 6.25mV/step. When VOUT_SL = 2, V_{OUT_MIN} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT_MIN} = (VOUT_MIN x 6.25mV + 206.25mV) x VOUT_SL Attempting to write a lower value to VOUT_COMMAND causes VOUT_COMMAND to be set to VOUT_MIN, and asserts a VOUT_MAX_MIN warning

TON_DELAY (60h)

Format: Unsigned binary

The TON_DELAY command sets the turn-on delay time for Buck 3.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	TON_DELAY	Every buck's turn on delay time can be set independently. Please change page vis PAGE command, and then set the register to configure turn on delay time. It can be calculated with the following equation: Startup Delay = TON_DELAY x 0.25ms. (0-7.75ms)

TOFF_DELAY (64h)

Format: Unsigned binary

The TOFF_DELAY command sets the turn-off delay time for Buck 3.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	TOFF_DELAY	Every buck's turn off delay time can be set independently. Please change page vis PAGE command, and then set the register to configure turn off delay time. Only when enable is low or PMBus off command or V_{IN} < UVLO, the device can perform the turn off sequence. It can be calculated with the following equation: Shutdown Delay = TOFF_DELAY x 0.25ms. (0-7.75ms)

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STATUS_BYTE (78h)

Format: Unsigned binary

The STATUS_BYTE command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	BUSY	While MPQ70165FS is performing OTP or restore OTP, host send PMBus write command. MPQ70165FS will report the Busy fault and ALERT# will be pulled down (see Table 1 at page 39). The bit is set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No busy fault 1'b1: A fault was declared because the device was busy and unable to respond
6	R	OFF	The bit can indicate channels' current status. For different channels' status, only change page via PAGE command before reading this bit. When the channel's status in between t_{ON_DELAY} ending and t_{OFF_DELAY} ending, the bit will be set to 0. Besides, this bit will be set to 1. 1'b0: Device is on 1'b1: Device is off
5	R	VOUT_OV_FAULT	When one channel's V_{OUT} exceeds V_{OUT} OV threshold duration ~10us, the channel will happen V_{OUT} OV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 7Ah bit7. 1'b0: No output over voltage fault has occurred 1'b1: An output over voltage fault has occurred
4	R	VOUT_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 79h bit6 and 7Ah bit4. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
3	R	RESERVED	Reserved.
2	R	TEMPERATURE	When MPQ70165FS's junction temperature exceeds TWN threshold or TSD threshold, the bit will be set to 1 at all pages. But the bit cannot accurately determine which thermal fault occurs. Please read 7Dh to ger the detail thermal fault information. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No TSD fault and TWN fault occurred 1'b1: A TSD fault or TWN fault has occurred
1	R	CML	When MPQ70165FS happens PMBus fault or OTP CRC (during 2nd restore OTP) fault, this bit will be set to 1 at all pages. PMBus fault include unsupported command fault, unsupported data fault, PEC fault and communication fault (include start error, stop error and long error). Please read 7Eh to ger the detail fault information. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No PMBus or OTP CRC (during 2nd restore OTP) fault has occurred. 1'b1: A PMBus or OTP CRC (during 2nd restore OTP) fault has occurred.

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0	R	FAULT_OTHER	In a channel, when VOUT COMMAND exceeds VOUT_MAX COMMAND or VOUT COMMAND lower than VOUT_MIN COMMAND, the bit will be set to1 under corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No VOUT MAXMIN fault has occurred. 1'b1: VOUT MAXMIN fault has occurred.

STATUS_WORD (79h)

Format: Unsigned binary

This command returns two bytes of information with a summary of the device fault/warning condition. The higher byte gives more detailed information of the fault conditions. The higher byte shares with register STATUS_BYTE.

Bits	Access	Bit Name	Description
15:8	R	STATUS_BYTE	The same as STATUS BYTE.
7	R	VOUT_FAULT	When one channel happens V_{OUT} related fault, this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No output voltage fault has occurred 1'b1: An output voltage fault has occurred. (including OV fault, UV fault, VOUT MAXMIN fault, PG_OV fault and PG_UV fault)
6	R	VO_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4 and 7Ah bit4. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
5	R	RESERVED	Reserved.
4	R	MANUFAC_FAULT	When MPQ70165FS happens reference fault, ABIST fault or watchdog fail fault, this bit will be set to 1 at all pages. Please read 80h to ger the detail fault information. This fault flag will be cleared once the device performs ABIST if only reference fault happened. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No manufacturer specific fault has occurred 1'b1: A manufacturer specific fault has occurred. (Including Reference fault, ABIST fault and Watchdog fail event)
3	R	PGOOD	The bit can indicate if channels' output exceeds the PGOOD range. For different channels' PGOOD status, only change page via PAGE command before reading this bit. When channel's output is not within PGOOD range, the bit will be set to 1 at corresponding page. Other page doesn't set this bit to 1. When channel's output is within PGOOD range, the bit will be set to 0 automatically at corresponding page. 1'b0: Power Good is high 1'b1: Power Good is low
2:0	R	RESERVED	Reserved.

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STATUS_VOUT (7Ah)

Format: Unsigned binary

The STATUS_VOUT command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	VOUT_OV_FAULT	When one channel's V_{OUT} exceeds V_{OUT} OV threshold duration ~10us, the channel will happen V_{OUT} OV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4. 1'b0: No output over voltage fault has occurred 1'b1: An output over voltage fault has occurred
6:5	R	RESERVED	Reserved.
4	R	VOUT_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4 and 79h bit6. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
3	R	RESERVED	Reserved.
2	R	PG_OV	When one channel's V_{OUT} exceeds PG upper threshold, the channel will happen PGOV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No pgova fault has occurred 1'b1: A pgova fault has occurred
1	R	PG_UV	When one channel's V_{OUT} is lower than PG lower threshold, the channel will happen PGUV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No pguva fault has occurred 1'b1: A pguva fault has occurred
0	R	VOUT_MAX_MIN	In a channel, when VOUT COMMAND exceeds VOUT_MAX COMMAND or VOUT COMMAND lower than VOUT_MIN COMMAND, the bit will be set to1 under corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No V_{OUT_CMD} maxmin fault has occurred 1'b1: An V_{OUT_CMD} maxmin fault has occurred

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VOUT_STARTUP_SLEW (A3h)

Format: Unsigned binary

The VOUT_STARTUP_SLEW command to set the rising rate of Buck 3 output during soft start.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_STARTUP_SLEW	These bits can control the rising rate of channels' output voltage. Please change page vis PAGE command, and then set the register to configure rising rate. To make the output voltage more linear during the soft start period, it is recommended to choose 1.25mV/us or 2.5mV/us. 2'b00: VOUT_SL x 1.25mV/us 2'b01: VOUT_SL x 2.5mV/us 2'b10: VOUT_SL x 5mV/us 2'b11: VOUT_SL x 10mV/us

VOUT_ SHUTDOWN _SLEW (A5h)

Format: Unsigned binary

The VOUT_SHUTDOWN_SLEW command to set the rising rate of Buck 3 output during soft stop.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_SHUTDOWN_SLEW	These bits can control the rising rate of channels' output voltage. Please change page vis PAGE command, and then set the register to configure rising rate. To make the output voltage more linear during the soft stop period, it is recommended to choose 1.25mV/us or 2.5mV/us. 2'b00: VOUT_SL x 1.25mV/us 2'b01: VOUT_SL x 2.5mV/us 2'b10: VOUT_SL x 5mV/us 2'b11: VOUT_SL x 10mV/us

VOUT_SLEW (A7h)

Format: Unsigned binary

The VOUT_ SLEW command to set the rising rate of Buck 3 output during changing output voltage online.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_SLEW	These bits can control the rising rate of channels' output voltage during changing output voltage online via PMBus. Please change page vis PAGE command, and then set the register to configure rising rate. 2'b00: VOUT_SL x 2.5mV/us 2'b01: VOUT_SL x 5mV/us 2'b10: VOUT_SL x 10mV/us 2'b11: VOUT_SL x 20mV/us

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OUTPUT_DISCHARGE (A9h)

Format: Unsigned binary

The OUTPUT_DISCHARGE command to set the discharge path of Buck 3 enabled or disabled.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1	R/W	SS_EN	This bit can control soft stop function to enabled or disabled. Please change page vis PAGE command, and then set the register to configure rising rate. For detail information, please see Output Discharge and Internal Soft Stop sections. 1'b0: Soft Stop Disable 1'b1: Soft Stop Enable
0	R/W	DISCHARGE_EN	This bit can control 100 Ω discharge resistance to be or not to be connected between VOUT and AGND. Please change page vis PAGE command, and then set the register to configure rising rate. For detail information, please see Output Discharge section. 1'b0: 100 Ω Output Discharge Disable 1'b1: 100 Ω Output Discharge Enable

OUTPUT_SHUTDOWN_MODE (ABh)

Format: Unsigned binary

The OUTPUT_SHUTDOWN_MODE command to set the shutdown mode of Buck 3.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	OUTPUT_SHUTDOWN_MODE	MPQ70165FS only support the Time-Based Shutdown Mode. When EN off or PMBus off, the channel will shut down after the corresponding turn off delay time timing ending. 1'b0: Time Based Shutdown Mode

COMPENSATION_CONFIG_1 (B0h)

Format: Unsigned binary

The COMPENSATION_CONFIG_1 command configures the compensation network parameters of Buck 3.

Bits	Access	Bit Name	Description
7:4	R/W	RCOMP	Sets R_{COMP}. The default selection in 1000. 4'b0000: 550K 4'b0001: 600K 4'b0010: 450K 4'b0011: 500K 4'b0100: 750K 4'b0101: 800K 4'b0110: 650K 4'b0111: 700K 4'b1000: 150K 4'b1001: 200K 4'b1010: 50K 4'b1011: 100K 4'b1100: 350K 4'b1101: 400K 4'b1110: 250K 4'b1111: 300K
3	R	RESERVED	Reserved.

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2:0	R/W	CCOMP_HP	Sets C_{COMP_HP}. The default selection is 011. 100fF/LSB. 3'b000: 0 x LSB … 3'b111: +7 x LSB

COMPENSATION_CONFIG_2 (B1h)

Format: Unsigned binary

The COMPENSATION_CONFIG_2 command configures the compensation network parameters of Buck 3.

Bits	Access	Bit Name	Description
7	R	RESERVED	Reserved.
6:4	R/W	CCOMP_LF	Sets C_{COMP_LF}. The default selection is 011. 3'b000: 44.5pF 3'b001: 49.5pF 3'b010: 54.5pF 3'b011: 59.5pF 3'b100: 24.5pF 3'b101: 29.5pF 3'b110: 34.5pF 3'b111: 39.5pF
3:2	R	RESERVED	Reserved.
1:0	R/W	SLOPE_COMP	Sets SLOPE_COMP. The default selection is 10. 2'b00: I_{SLOPE} 2'b01: I_{SLOPE} x 2 2'b10: I_{SLOPE} x 3 2'b11: I_{SLOPE} x 4

PG_CONFIG (CAh)

Format: Unsigned binary

The PG_CONFIG command can configure PG OV / UV threshold of Buck 3.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	PG_THRESHOLD	This bit can set different PG OV rising threshold and PG UV falling threshold for different channel. Please change page vis PAGE command, and then set the register to configure PG threshold. 1'b1: +4.5% / - 3.5% 1'b0: +6.5% / - 5.5%

ILIM_SCALE (CEh)

Format: Unsigned binary

The ILIM_SCALE command can configure the current limit scale of Buck 3.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	ILIM_SCALE	These bits can set the scale of peak / valley / reverse current limit, default value is 100% (2'b00). Please change page vis PAGE command, and then set the register to configure current limit scale. 2'b00: 100% 2'b01: 75% 2'b10: 50%

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PAGE 3 REGISTER MAP

PAGE (00h)

Format: Unsigned binary

The PAGE command provides the ability to configure, control and monitor all outputs through only one physical address. All subsequent commands (pages is “Y” at REGISTER MAP section) that allow multipage action are applied to the corresponding regulators selected by the PAGE command.

Bits	Access	Bit Name	Description
7:0	R/W	PAGE	In Page FF, write command to other registers can change all pages’ configuration, including page 0-5. In Page 0 – 5, write command to other registers can only change the current page’s configuration. So, the PAGE command shall be written before modifying other registers. 8’b00000000: VOUT1 8’b00000001: VOUT2 8’b00000010: VOUT3 8’b00000011: VOUT4 8’b00000100: VOUT5 8’b00000101: VOUT6 8’b11111111: All Others: Invalid commands

OPERATION (01h)

Format: Unsigned binary

The OPERATION command is used to configure the output on/off state of the Buck 4, in conjunction with input from the EN_X pin.

Bits	Access	Bit Name	Description
7	R/W	OPERATION	Enable signal is available, MPQ70165FS support interface to turn on / off channels through PMBus. 1’b1: Output on (if enable high) 1’b0: Output off
6:0	R/W	RESERVED	Reserved.

CLEAR_FAULTS (03h)

This command is used to clear any fault bit in all status registers: STATUS_BYTE (78h), STATUS_WORD (79h), STATUS_VOUT (7Ah), STATUS_TEMPERATURE (7Dh), STATUS_CML (7Eh), STATUS_MFR_SPECIFIC (80h), WATCHDOG_STATUS_1 (BAh), MFR_OTP_MEM_STATUS (DBh) and STATUS_FREQ (E7h)

This command is written only. There is no data byte for this command.

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VOUT_COMMAND (21h)

Format: Unsigned binary

The VOUT_COMMAND command sets the output voltage of Buck 4.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_CMD	The command is used to set output voltage. Steps as below: 7. Select page x by PAGE (00h). 8. Change V_{OUT} by VOUT_COMMAND. V_{OUT} is influenced by VOUT_SCALE_LOOP, see the below description for detail V_{OUT} range. 7. When VOUT_SL = 1, V_{OUT} can be set from 0.20625V to 1.8V, 6.25mV/step. 8. When VOUT_SL = 2, V_{OUT} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT} = (VOUT_CMD x 6.25mV + 206.25mV) x VOUT_SL

VOUT_MAX (24h)

Format: Unsigned binary

The VOUT_MAX command sets an upper limit on the output voltage that Buck 4 can command regardless of any other commands or combinations. The intent of this command is to provide a safeguard against a user accidental setting the output voltage to a possible destructive level rather than to be the primary output overvoltage protection.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_MAX	When VOUT_SL = 1, V_{OUT_MAX} can be set from 0.20625V to 1.8V, 6.25mV/step. When VOUT_SL = 2, V_{OUT_MAX} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT_MAX} = (VOUT_MAX x 6.25mV + 206.25mV) x VOUT_SL Attempting to write a higher value to VOUT_COMMAND causes VOUT_COMMAND to be set to VOUT_MAX and asserts a VOUT_MAX_MIN warning.

VOUT_SCALE_LOOP (29h)

Format: Unsigned binary

The VOUT_SCALE_LOOP command sets the scale for the V_{OUT}, V_{OUT_MAX}, V_{OUT_MIN}, V_{OUT} start-up slew rate, V_{OUT} shutdown slew rate and V_{OUT} slew rate on Page 3.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	VOUT_SC_L	According to application requirement, set adjust scale. The device should be off via the OPERATINO command before change the scale. 1'b0: VOUT_SL= 1 1'b1: VOUT_SL= 2

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VOUT_MIN (2Bh)

Format: Unsigned binary

The VOUT_MIN command sets a lower limit on the output voltage that Buck 4 can command regardless of any other commands or combinations. The intent of this command is to provide a safeguard against a user accidentally setting the output voltage to a possible destructive level rather than to be the primary output under-voltage protection.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_MIN	When VOUT_SL = 1, V_{OUT_MIN} can be set from 0.20625V to 1.8V, 6.25mV/step. When VOUT_SL = 2, V_{OUT_MIN} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT_MIN} = (VOUT_MIN x 6.25mV + 206.25mV) x VOUT_SL Attempting to write a lower value to VOUT_COMMAND causes VOUT_COMMAND to be set to VOUT_MIN, and asserts a VOUT_MAX_MIN warning

TON_DELAY (60h)

Format: Unsigned binary

The TON_DELAY command sets the turn-on delay time for Buck 4.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	TON_DELAY	Every buck's turn on delay time can be set independently. Please change page vis PAGE command, and then set the register to configure turn on delay time. It can be calculated with the following equation: Startup Delay = TON_DELAY x 0.25ms. (0-7.75ms)

TOFF_DELAY (64h)

Format: Unsigned binary

The TOFF_DELAY command sets the turn-off delay time for Buck 4.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	TOFF_DELAY	Every buck's turn off delay time can be set independently. Please change page vis PAGE command, and then set the register to configure turn off delay time. Only when enable is low or PMBus off command or V_{IN} < UVLO, the device can perform the turn off sequence. It can be calculated with the following equation: Shutdown Delay = TOFF_DELAY x 0.25ms. (0-7.75ms)

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STATUS_BYTE (78h)

Format: Unsigned binary

The STATUS_BYTE command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	BUSY	While MPQ70165FS is performing OTP or restore OTP, host send PMBus write command. MPQ70165FS will report the Busy fault and ALERT# will be pulled down (see Table 1 at page 39). The bit is set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No busy fault 1'b1: A fault was declared because the device was busy and unable to respond
6	R	OFF	The bit can indicate channels' current status. For different channels' status, only change page via PAGE command before reading this bit. When the channel's status in between t_{ON_DELAY} ending and t_{OFF_DELAY} ending, the bit will be set to 0. Besides, this bit will be set to 1. 1'b0: Device is on 1'b1: Device is off
5	R	VOUT_OV_FAULT	When one channel's V_{OUT} exceeds V_{OUT} OV threshold duration ~10us, the channel will happen V_{OUT} OV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 7Ah bit7. 1'b0: No output over voltage fault has occurred 1'b1: An output over voltage fault has occurred
4	R	VOUT_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 79h bit6 and 7Ah bit4. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
3	R	RESERVED	Reserved.
2	R	TEMPERATURE	When MPQ70165FS's junction temperature exceeds TWN threshold or TSD threshold, the bit will be set to 1 at all pages. But the bit cannot accurately determine which thermal fault occurs. Please read 7Dh to ger the detail thermal fault information. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No TSD fault and TWN fault occurred 1'b1: A TSD fault or TWN fault has occurred
1	R	CML	When MPQ70165FS happens PMBus fault or OTP CRC (during 2nd restore OTP) fault, this bit will be set to 1 at all pages. PMBus fault include unsupported command fault, unsupported data fault, PEC fault and communication fault (include start error, stop error and long error). Please read 7Eh to ger the detail fault information. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No PMBus or OTP CRC (during 2nd restore OTP) fault has occurred. 1'b1: A PMBus or OTP CRC (during 2nd restore OTP) fault has occurred.

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0	R	FAULT_OTHER	In a channel, when VOUT COMMAND exceeds VOUT_MAX COMMAND or VOUT COMMAND lower than VOUT_MIN COMMAND, the bit will be set to1 under corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No VOUT MAXMIN fault has occurred. 1'b1: VOUT MAXMIN fault has occurred.

STATUS_WORD (79h)

Format: Unsigned binary

This command returns two bytes of information with a summary of the device fault/warning condition. The higher byte gives more detailed information of the fault conditions. The higher byte shares with register STATUS_BYTE.

Bits	Access	Bit Name	Description
15:8	R	STATUS_BYTE	The same as STATUS BYTE.
7	R	VOUT_FAULT	When one channel happens V_{OUT} related fault, this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No output voltage fault has occurred 1'b1: An output voltage fault has occurred. (including OV fault, UV fault, VOUT MAXMIN fault, PG_OV fault and PG_UV fault)
6	R	VO_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4 and 7Ah bit4. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
5	R	RESERVED	Reserved.
4	R	MANUFAC_FAULT	When MPQ70165FS happens reference fault, ABIST fault or watchdog fail fault, this bit will be set to 1 at all pages. Please read 80h to ger the detail fault information. This fault flag will be cleared once the device performs ABIST if only reference fault happened. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No manufacturer specific fault has occurred 1'b1: A manufacturer specific fault has occurred. (Including Reference fault, ABIST fault and Watchdog fail event)
3	R	PGOOD	The bit can indicate if channels' output exceeds the PGOOD range. For different channels' PGOOD status, only change page via PAGE command before reading this bit. When channel's output is not within PGOOD range, the bit will be set to 1 at corresponding page. Other page doesn't set this bit to 1. When channel's output is within PGOOD range, the bit will be set to 0 automatically at corresponding page. 1'b0: Power Good is high 1'b1: Power Good is low
2:0	R	RESERVED	Reserved.

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STATUS_VOUT (7Ah)

Format: Unsigned binary

The STATUS_VOUT command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	VOUT_OV_FAULT	When one channel's V_{OUT} exceeds V_{OUT} OV threshold duration ~10us, the channel will happen V_{OUT} OV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4. 1'b0: No output over voltage fault has occurred 1'b1: An output over voltage fault has occurred
6:5	R	RESERVED	Reserved.
4	R	VOUT_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4 and 79h bit6. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
3	R	RESERVED	Reserved.
2	R	PG_OV	When one channel's V_{OUT} exceeds PG upper threshold, the channel will happen PGOV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No pgova fault has occurred 1'b1: A pgova fault has occurred
1	R	PG_UV	When one channel's V_{OUT} is lower than PG lower threshold, the channel will happen PGUV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No pguva fault has occurred 1'b1: A pguva fault has occurred
0	R	VOUT_MAX_MIN	In a channel, when VOUT COMMAND exceeds VOUT_MAX COMMAND or VOUT COMMAND lower than VOUT_MIN COMMAND, the bit will be set to1 under corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No V_{OUT_CMD} maxmin fault has occurred 1'b1: An V_{OUT_CMD} maxmin fault has occurred

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VOUT_STARTUP_SLEW (A3h)

Format: Unsigned binary

The VOUT_STARTUP_SLEW command to set the rising rate of Buck 4 output during soft start.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_STARTUP_SLEW	These bits can control the rising rate of channels' output voltage. Please change page vis PAGE command, and then set the register to configure rising rate. To make the output voltage more linear during the soft start period, it is recommended to choose 1.25mV/us or 2.5mV/us. 2'b00: VOUT_SL x 1.25mV/us 2'b01: VOUT_SL x 2.5mV/us 2'b10: VOUT_SL x 5mV/us 2'b11: VOUT_SL x 10mV/us

VOUT_ SHUTDOWN _SLEW (A5h)

Format: Unsigned binary

The VOUT_SHUTDOWN_SLEW command to set the rising rate of Buck 4 output during soft stop.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_SHUTDOWN_SLEW	These bits can control the rising rate of channels' output voltage. Please change page vis PAGE command, and then set the register to configure rising rate. To make the output voltage more linear during the soft stop period, it is recommended to choose 1.25mV/us or 2.5mV/us. 2'b00: VOUT_SL x 1.25mV/us 2'b01: VOUT_SL x 2.5mV/us 2'b10: VOUT_SL x 5mV/us 2'b11: VOUT_SL x 10mV/us

VOUT_SLEW (A7h)

Format: Unsigned binary

The VOUT_ SLEW command to set the rising rate of Buck 4 output during changing output voltage online.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_SLEW	These bits can control the rising rate of channels' output voltage during changing output voltage online via PMBus. Please change page vis PAGE command, and then set the register to configure rising rate. 2'b00: VOUT_SL x 2.5mV/us 2'b01: VOUT_SL x 5mV/us 2'b10: VOUT_SL x 10mV/us 2'b11: VOUT_SL x 20mV/us

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OUTPUT_DISCHARGE (A9h)

Format: Unsigned binary

The OUTPUT_DISCHARGE command to set the discharge path of Buck 4 enabled or disabled.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1	R/W	SS_EN	This bit can control soft stop function to enabled or disabled. Please change page vis PAGE command, and then set the register to configure rising rate. For detail information, please see Output Discharge and Internal Soft Stop sections. 1'b0: Soft Stop Disable 1'b1: Soft Stop Enable
0	R/W	DISCHARGE_EN	This bit can control 100 Ω discharge resistance to be or not to be connected between VOUT and AGND. Please change page vis PAGE command, and then set the register to configure rising rate. For detail information, please see Output Discharge section. 1'b0: 100 Ω Output Discharge Disable 1'b1: 100 Ω Output Discharge Enable

OUTPUT_SHUTDOWN_MODE (ABh)

Format: Unsigned binary

The OUTPUT_SHUTDOWN_MODE command to set the shutdown mode of Buck 4.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	OUTPUT_SHUTDOWN_MODE	MPQ70165FS only support the Time-Based Shutdown Mode. When EN off or PMBus off, the channel will shut down after the corresponding turn off delay time timing ending. 1'b0: Time Based Shutdown Mode

COMPENSATION_CONFIG_1 (B0h)

Format: Unsigned binary

The COMPENSATION_CONFIG_1 command configures the compensation network parameters of Buck 4.

Bits	Access	Bit Name	Description
7:4	R/W	RCOMP	Sets R_{COMP}. The default selection in 1000. 4'b0000: 550K 4'b0001: 600K 4'b0010: 450K 4'b0011: 500K 4'b0100: 750K 4'b0101: 800K 4'b0110: 650K 4'b0111: 700K 4'b1000: 150K 4'b1001: 200K 4'b1010: 50K 4'b1011: 100K 4'b1100: 350K 4'b1101: 400K 4'b1110: 250K 4'b1111: 300K
3	R	RESERVED	Reserved.

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2:0	R/W	CCOMP_HP	Sets C_{COMP_HP}. The default selection is 011. 100fF/LSB. 3'b000: 0 x LSB … 3'b111: +7 x LSB

COMPENSATION_CONFIG_2 (B1h)

Format: Unsigned binary

The COMPENSATION_CONFIG_2 command configures the compensation network parameters of Buck 4.

Bits	Access	Bit Name	Description
7	R	RESERVED	Reserved.
6:4	R/W	CCOMP_LF	Sets C_{COMP_LF}. The default selection is 011. 3'b000: 44.5pF 3'b001: 49.5pF 3'b010: 54.5pF 3'b011: 59.5pF 3'b100: 24.5pF 3'b101: 29.5pF 3'b110: 34.5pF 3'b111: 39.5pF
3:2	R	RESERVED	Reserved.
1:0	R/W	SLOPE_COMP	Sets SLOPE_COMP. The default selection is 10. 2'b00: I_{SLOPE} 2'b01: I_{SLOPE} x 2 2'b10: I_{SLOPE} x 3 2'b11: I_{SLOPE} x 4

PG_CONFIG (CAh)

Format: Unsigned binary

The PG_CONFIG command can configure PG OV / UV threshold of Buck 4.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	PG_THRESHOLD	This bit can set different PG OV rising threshold and PG UV falling threshold for different channel. Please change page vis PAGE command, and then set the register to configure PG threshold. 1'b1: +4.5% / - 3.5% 1'b0: +6.5% / - 5.5%

ILIM_SCALE (CEh)

Format: Unsigned binary

The ILIM_SCALE command can configure the current limit scale of Buck 4.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	ILIM_SCALE	These bits can set the scale of peak / valley / reverse current limit, default value is 100% (2'b00). Please change page vis PAGE command, and then set the register to configure current limit scale. 2'b00: 100% 2'b01: 75% 2'b10: 50%

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PAGE 4 REGISTER MAP

PAGE (00h)

Format: Unsigned binary

The PAGE command provides the ability to configure, control and monitor all outputs through only one physical address. All subsequent commands (pages is “Y” at REGISTER MAP section) that allow multipage action are applied to the corresponding regulators selected by the PAGE command.

Bits	Access	Bit Name	Description
7:0	R/W	PAGE	In Page FF, write command to other registers can change all pages’ configuration, including page 0-5. In Page 0 – 5, write command to other registers can only change the current page’s configuration. So, the PAGE command shall be written before modifying other registers. 8’b00000000: VOUT1 8’b00000001: VOUT2 8’b00000010: VOUT3 8’b00000011: VOUT4 8’b00000100: VOUT5 8’b00000101: VOUT6 8’b11111111: All Others: Invalid commands

OPERATION (01h)

Format: Unsigned binary

The OPERATION command is used to configure the output on/off state of the Buck 5, in conjunction with input from the EN_X pin.

Bits	Access	Bit Name	Description
7	R/W	OPERATION	Enable signal is available, MPQ70165FS support interface to turn on / off channels through PMBus. 1’b1: Output on (if enable high) 1’b0: Output off
6:0	R/W	RESERVED	Reserved.

CLEAR_FAULTS (03h)

This command is used to clear any fault bit in all status registers: STATUS_BYTE (78h), STATUS_WORD (79h), STATUS_VOUT (7Ah), STATUS_TEMPERATURE (7Dh), STATUS_CML (7Eh), STATUS_MFR_SPECIFIC (80h), WATCHDOG_STATUS_1 (BAh), MFR_OTP_MEM_STATUS (DBh) and STATUS_FREQ (E7h)

This command is written only. There is no data byte for this command.

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VOUT_COMMAND (21h)

Format: Unsigned binary

The VOUT_COMMAND command sets the output voltage of Buck 5.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_CMD	The command is used to set output voltage. Steps as below: 9. Select page x by PAGE (00h). 10. Change V_{OUT} by VOUT_COMMAND. V_{OUT} is influenced by VOUT_SCALE_LOOP, see the below description for detail V_{OUT} range. 9. When VOUT_SL = 1, V_{OUT} can be set from 0.20625V to 1.8V, 6.25mV/step. 10. When VOUT_SL = 2, V_{OUT} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT} = (VOUT_CMD x 6.25mV + 206.25mV) x VOUT_SL

VOUT_MAX (24h)

Format: Unsigned binary

The VOUT_MAX command sets an upper limit on the output voltage that Buck 5 can command regardless of any other commands or combinations. The intent of this command is to provide a safeguard against a user accidental setting the output voltage to a possible destructive level rather than to be the primary output overvoltage protection.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_MAX	When VOUT_SL = 1, V_{OUT_MAX} can be set from 0.20625V to 1.8V, 6.25mV/step. When VOUT_SL = 2, V_{OUT_MAX} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT_MAX} = (VOUT_MAX x 6.25mV + 206.25mV) x VOUT_SL Attempting to write a higher value to VOUT_COMMAND causes VOUT_COMMAND to be set to VOUT_MAX and asserts a VOUT_MAX_MIN warning.

VOUT_SCALE_LOOP (29h)

Format: Unsigned binary

The VOUT_SCALE_LOOP command sets the scale for the V_{OUT}, V_{OUT_MAX}, V_{OUT_MIN}, V_{OUT} start-up slew rate, V_{OUT} shutdown slew rate and V_{OUT} slew rate on Page 4.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	VOUT_SC_L	According to application requirement, set adjust scale. The device should be off via the OPERATINO command before change the scale. 1'b0: VOUT_SL= 1 1'b1: VOUT_SL= 2

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VOUT_MIN (2Bh)

Format: Unsigned binary

The VOUT_MIN command sets a lower limit on the output voltage that Buck 5 can command regardless of any other commands or combinations. The intent of this command is to provide a safeguard against a user accidentally setting the output voltage to a possible destructive level rather than to be the primary output under-voltage protection.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_MIN	When VOUT_SL = 1, V_{OUT_MIN} can be set from 0.20625V to 1.8V, 6.25mV/step. When VOUT_SL = 2, V_{OUT_MIN} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT_MIN} = (VOUT_MIN x 6.25mV + 206.25mV) x VOUT_SL Attempting to write a lower value to VOUT_COMMAND causes VOUT_COMMAND to be set to VOUT_MIN, and asserts a VOUT_MAX_MIN warning

TON_DELAY (60h)

Format: Unsigned binary

The TON_DELAY command sets the turn-on delay time for Buck 5.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	TON_DELAY	Every buck's turn on delay time can be set independently. Please change page vis PAGE command, and then set the register to configure turn on delay time. It can be calculated with the following equation: Startup Delay = TON_DELAY x 0.25ms. (0-7.75ms)

TOFF_DELAY (64h)

Format: Unsigned binary

The TOFF_DELAY command sets the turn-off delay time for Buck 5.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	TOFF_DELAY	Every buck's turn off delay time can be set independently. Please change page vis PAGE command, and then set the register to configure turn off delay time. Only when enable is low or PMBus off command or V_{IN} < UVLO, the device can perform the turn off sequence. It can be calculated with the following equation: Shutdown Delay = TOFF_DELAY x 0.25ms. (0-7.75ms)

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STATUS_BYTE (78h)

Format: Unsigned binary

The STATUS_BYTE command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	BUSY	While MPQ70165FS is performing OTP or restore OTP, host send PMBus write command. MPQ70165FS will report the Busy fault and ALERT# will be pulled down (see Table 1 at page 39). The bit is set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No busy fault 1'b1: A fault was declared because the device was busy and unable to respond
6	R	OFF	The bit can indicate channels' current status. For different channels' status, only change page via PAGE command before reading this bit. When the channel's status in between t_{ON_DELAY} ending and t_{OFF_DELAY} ending, the bit will be set to 0. Besides, this bit will be set to 1. 1'b0: Device is on 1'b1: Device is off
5	R	VOUT_OV_FAULT	When one channel's V_{OUT} exceeds V_{OUT} OV threshold duration ~10us, the channel will happen V_{OUT} OV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 7Ah bit7. 1'b0: No output over voltage fault has occurred 1'b1: An output over voltage fault has occurred
4	R	VOUT_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 79h bit6 and 7Ah bit4. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
3	R	RESERVED	Reserved.
2	R	TEMPERATURE	When MPQ70165FS's junction temperature exceeds TWN threshold or TSD threshold, the bit will be set to 1 at all pages. But the bit cannot accurately determine which thermal fault occurs. Please read 7Dh to ger the detail thermal fault information. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No TSD fault and TWN fault occurred 1'b1: A TSD fault or TWN fault has occurred
1	R	CML	When MPQ70165FS happens PMBus fault or OTP CRC (during 2nd restore OTP) fault, this bit will be set to 1 at all pages. PMBus fault include unsupported command fault, unsupported data fault, PEC fault and communication fault (include start error, stop error and long error). Please read 7Eh to ger the detail fault information. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No PMBus or OTP CRC (during 2nd restore OTP) fault has occurred. 1'b1: A PMBus or OTP CRC (during 2nd restore OTP) fault has occurred.

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0	R	FAULT_OTHER	In a channel, when VOUT COMMAND exceeds VOUT_MAX COMMAND or VOUT COMMAND lower than VOUT_MIN COMMAND, the bit will be set to1 under corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No VOUT MAXMIN fault has occurred. 1'b1: VOUT MAXMIN fault has occurred.

STATUS_WORD (79h)

Format: Unsigned binary

This command returns two bytes of information with a summary of the device fault/warning condition. The higher byte gives more detailed information of the fault conditions. The higher byte shares with register STATUS_BYTE.

Bits	Access	Bit Name	Description
15:8	R	STATUS_BYTE	The same as STATUS BYTE.
7	R	VOUT_FAULT	When one channel happens V_{OUT} related fault, this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No output voltage fault has occurred 1'b1: An output voltage fault has occurred. (including OV fault, UV fault, VOUT MAXMIN fault, PG_OV fault and PG_UV fault)
6	R	VO_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4 and 7Ah bit4. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
5	R	RESERVED	Reserved.
4	R	MANUFAC_FAULT	When MPQ70165FS happens reference fault, ABIST fault or watchdog fail fault, this bit will be set to 1 at all pages. Please read 80h to ger the detail fault information. This fault flag will be cleared once the device performs ABIST if only reference fault happened. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No manufacturer specific fault has occurred 1'b1: A manufacturer specific fault has occurred. (Including Reference fault, ABIST fault and Watchdog fail event)
3	R	PGOOD	The bit can indicate if channels' output exceeds the PGOOD range. For different channels' PGOOD status, only change page via PAGE command before reading this bit. When channel's output is not within PGOOD range, the bit will be set to 1 at corresponding page. Other page doesn't set this bit to 1. When channel's output is within PGOOD range, the bit will be set to 0 automatically at corresponding page. 1'b0: Power Good is high 1'b1: Power Good is low
2:0	R	RESERVED	Reserved.

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STATUS_VOUT (7Ah)

Format: Unsigned binary

The STATUS_VOUT command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	VOUT_OV_FAULT	When one channel's V_{OUT} exceeds V_{OUT} OV threshold duration ~10us, the channel will happen V_{OUT} OV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4. 1'b0: No output over voltage fault has occurred 1'b1: An output over voltage fault has occurred
6:5	R	RESERVED	Reserved.
4	R	VOUT_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4 and 79h bit6. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
3	R	RESERVED	Reserved.
2	R	PG_OV	When one channel's V_{OUT} exceeds PG upper threshold, the channel will happen PGOV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No pgova fault has occurred 1'b1: A pgova fault has occurred
1	R	PG_UV	When one channel's V_{OUT} is lower than PG lower threshold, the channel will happen PGUV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No pguva fault has occurred 1'b1: A pguva fault has occurred
0	R	VOUT_MAX_MIN	In a channel, when VOUT COMMAND exceeds VOUT_MAX COMMAND or VOUT COMMAND lower than VOUT_MIN COMMAND, the bit will be set to1 under corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No V_{OUT_CMD} maxmin fault has occurred 1'b1: An V_{OUT_CMD} maxmin fault has occurred

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VOUT_STARTUP_SLEW (A3h)

Format: Unsigned binary

The VOUT_STARTUP_SLEW command to set the rising rate of Buck 5 output during soft start.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_STARTUP_SLEW	These bits can control the rising rate of channels' output voltage. Please change page vis PAGE command, and then set the register to configure rising rate. To make the output voltage more linear during the soft start period, it is recommended to choose 1.25mV/us or 2.5mV/us. 2'b00: VOUT_SL x 1.25mV/us 2'b01: VOUT_SL x 2.5mV/us 2'b10: VOUT_SL x 5mV/us 2'b11: VOUT_SL x 10mV/us

VOUT_ SHUTDOWN _SLEW (A5h)

Format: Unsigned binary

The VOUT_SHUTDOWN_SLEW command to set the rising rate of Buck 5 output during soft stop.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_SHUTDOWN_SLEW	These bits can control the rising rate of channels' output voltage. Please change page vis PAGE command, and then set the register to configure rising rate. To make the output voltage more linear during the soft stop period, it is recommended to choose 1.25mV/us or 2.5mV/us. 2'b00: VOUT_SL x 1.25mV/us 2'b01: VOUT_SL x 2.5mV/us 2'b10: VOUT_SL x 5mV/us 2'b11: VOUT_SL x 10mV/us

VOUT_SLEW (A7h)

Format: Unsigned binary

The VOUT_ SLEW command to set the rising rate of Buck 5 output during changing output voltage online.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_SLEW	These bits can control the rising rate of channels' output voltage during changing output voltage online via PMBus. Please change page vis PAGE command, and then set the register to configure rising rate. 2'b00: VOUT_SL x 2.5mV/us 2'b01: VOUT_SL x 5mV/us 2'b10: VOUT_SL x 10mV/us 2'b11: VOUT_SL x 20mV/us

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OUTPUT_DISCHARGE (A9h)

Format: Unsigned binary

The OUTPUT_DISCHARGE command to set the discharge path of Buck 5 enabled or disabled.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1	R/W	SS_EN	This bit can control soft stop function to enabled or disabled. Please change page vis PAGE command, and then set the register to configure rising rate. For detail information, please see Output Discharge and Internal Soft Stop sections. 1'b0: Soft Stop Disable 1'b1: Soft Stop Enable
0	R/W	DISCHARGE_EN	This bit can control 100 Ω discharge resistance to be or not to be connected between VOUT and AGND. Please change page vis PAGE command, and then set the register to configure rising rate. For detail information, please see Output Discharge section. 1'b0: 100 Ω Output Discharge Disable 1'b1: 100 Ω Output Discharge Enable

OUTPUT_SHUTDOWN_MODE (ABh)

Format: Unsigned binary

The OUTPUT_SHUTDOWN_MODE command to set the shutdown mode of Buck 5.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	OUTPUT_SHUTDOWN_MODE	MPQ70165FS only support the Time-Based Shutdown Mode. When EN off or PMBus off, the channel will shut down after the corresponding turn off delay time timing ending. 1'b0: Time Based Shutdown Mode

COMPENSATION_CONFIG_1 (B0h)

Format: Unsigned binary

The COMPENSATION_CONFIG_1 command configures the compensation network parameters of Buck 5.

Bits	Access	Bit Name	Description
7:4	R/W	RCOMP	Sets R_{COMP}. The default selection in 1000. 4'b0000: 550K 4'b0001: 600K 4'b0010: 450K 4'b0011: 500K 4'b0100: 750K 4'b0101: 800K 4'b0110: 650K 4'b0111: 700K 4'b1000: 150K 4'b1001: 200K 4'b1010: 50K 4'b1011: 100K 4'b1100: 350K 4'b1101: 400K 4'b1110: 250K 4'b1111: 300K
3	R	RESERVED	Reserved.

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2:0	R/W	CCOMP_HP	Sets C_{COMP_HP}. The default selection is 011. 100fF/LSB. 3'b000: 0 x LSB … 3'b111: +7 x LSB

COMPENSATION_CONFIG_2 (B1h)

Format: Unsigned binary

The COMPENSATION_CONFIG_2 command configures the compensation network parameters of Buck 5.

Bits	Access	Bit Name	Description
7	R	RESERVED	Reserved.
6:4	R/W	CCOMP_LF	Sets C_{COMP_LF}. The default selection is 011. 3'b000: 44.5pF 3'b001: 49.5pF 3'b010: 54.5pF 3'b011: 59.5pF 3'b100: 24.5pF 3'b101: 29.5pF 3'b110: 34.5pF 3'b111: 39.5pF
3:2	R	RESERVED	Reserved.
1:0	R/W	SLOPE_COMP	Sets SLOPE_COMP. The default selection is 10. 2'b00: I_{SLOPE} 2'b01: I_{SLOPE} x 2 2'b10: I_{SLOPE} x 3 2'b11: I_{SLOPE} x 4

PG_CONFIG (CAh)

Format: Unsigned binary

The PG_CONFIG command can configure PG OV / UV threshold of Buck 5.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	PG_THRESHOLD	This bit can set different PG OV rising threshold and PG UV falling threshold for different channel. Please change page vis PAGE command, and then set the register to configure PG threshold. 1'b1: +4.5% / - 3.5% 1'b0: +6.5% / - 5.5%

ILIM_SCALE (CEh)

Format: Unsigned binary

The ILIM_SCALE command can configure the current limit scale of Buck 5.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	ILIM_SCALE	These bits can set the scale of peak / valley / reverse current limit, default value is 100% (2'b00). Please change page vis PAGE command, and then set the register to configure current limit scale. 2'b00: 100% 2'b01: 75% 2'b10: 50%

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PAGE 5 REGISTER MAP

PAGE (00h)

Format: Unsigned binary

The PAGE command provides the ability to configure, control and monitor all outputs through only one physical address. All subsequent commands (pages is “Y” at REGISTER MAP section) that allow multipage action are applied to the corresponding regulators selected by the PAGE command.

Bits	Access	Bit Name	Description
7:0	R/W	PAGE	In Page FF, write command to other registers can change all pages’ configuration, including page 0-5. In Page 0 – 5, write command to other registers can only change the current page’s configuration. So, the PAGE command shall be written before modifying other registers. 8’b00000000: VOUT1 8’b00000001: VOUT2 8’b00000010: VOUT3 8’b00000011: VOUT4 8’b00000100: VOUT5 8’b00000101: VOUT6 8’b11111111: All Others: Invalid commands

OPERATION (01h)

Format: Unsigned binary

The OPERATION command is used to configure the output on/off state of the Buck 6, in conjunction with input from the EN_X pin.

Bits	Access	Bit Name	Description
7	R/W	OPERATION	Enable signal is available, MPQ70165FS support interface to turn on / off channels through PMBus. 1’b1: Output on (if enable high) 1’b0: Output off
6:0	R/W	RESERVED	Reserved.

CLEAR_FAULTS (03h)

This command is used to clear any fault bit in all status registers: STATUS_BYTE (78h), STATUS_WORD (79h), STATUS_VOUT (7Ah), STATUS_TEMPERATURE (7Dh), STATUS_CML (7Eh), STATUS_MFR_SPECIFIC (80h), WATCHDOG_STATUS_1 (BAh), MFR_OTP_MEM_STATUS (DBh) and STATUS_FREQ (E7h)

This command is written only. There is no data byte for this command.

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VOUT_COMMAND (21h)

Format: Unsigned binary

The VOUT_COMMAND command sets the output voltage of Buck 6.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_CMD	The command is used to set output voltage. Steps as below: 11. Select page x by PAGE (00h). 12. Change V_{OUT} by VOUT_COMMAND. V_{OUT} is influenced by VOUT_SCALE_LOOP, see the below description for detail V_{OUT} range. 11. When VOUT_SL = 1, V_{OUT} can be set from 0.20625V to 1.8V, 6.25mV/step. 12. When VOUT_SL = 2, V_{OUT} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT} = (VOUT_CMD x 6.25mV + 206.25mV) x VOUT_SL

VOUT_MAX (24h)

Format: Unsigned binary

The VOUT_MAX command sets an upper limit on the output voltage that Buck 6 can command regardless of any other commands or combinations. The intent of this command is to provide a safeguard against a user accidental setting the output voltage to a possible destructive level rather than to be the primary output overvoltage protection.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_MAX	When VOUT_SL = 1, V_{OUT_MAX} can be set from 0.20625V to 1.8V, 6.25mV/step. When VOUT_SL = 2, V_{OUT_MAX} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT_MAX} = (VOUT_MAX x 6.25mV + 206.25mV) x VOUT_SL Attempting to write a higher value to VOUT_COMMAND causes VOUT_COMMAND to be set to VOUT_MAX and asserts a VOUT_MAX_MIN warning.

VOUT_SCALE_LOOP (29h)

Format: Unsigned binary

The VOUT_SCALE_LOOP command sets the scale for the V_{OUT}, V_{OUT_MAX}, V_{OUT_MIN}, V_{OUT} start-up slew rate, V_{OUT} shutdown slew rate and V_{OUT} slew rate on Page 5.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	VOUT_SC_L	According to application requirement, set adjust scale. The device should be off via the OPERATINO command before change the scale. 1'b0: VOUT_SL= 1 1'b1: VOUT_SL= 2

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VOUT_MIN (2Bh)

Format: Unsigned binary

The VOUT_MIN command sets a lower limit on the output voltage that Buck 6 can command regardless of any other commands or combinations. The intent of this command is to provide a safeguard against a user accidentally setting the output voltage to a possible destructive level rather than to be the primary output under-voltage protection.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_MIN	When VOUT_SL = 1, V_{OUT_MIN} can be set from 0.20625V to 1.8V, 6.25mV/step. When VOUT_SL = 2, V_{OUT_MIN} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT_MIN} = (VOUT_MIN x 6.25mV + 206.25mV) x VOUT_SL Attempting to write a lower value to VOUT_COMMAND causes VOUT_COMMAND to be set to VOUT_MIN, and asserts a VOUT_MAX_MIN warning

TON_DELAY (60h)

Format: Unsigned binary

The TON_DELAY command sets the turn-on delay time for Buck 6.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	TON_DELAY	Every buck's turn on delay time can be set independently. Please change page vis PAGE command, and then set the register to configure turn on delay time. It can be calculated with the following equation: Startup Delay = TON_DELAY x 0.25ms. (0-7.75ms)

TOFF_DELAY (64h)

Format: Unsigned binary

The TOFF_DELAY command sets the turn-off delay time for Buck 6.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	TOFF_DELAY	Every buck's turn off delay time can be set independently. Please change page vis PAGE command, and then set the register to configure turn off delay time. Only when enable is low or PMBus off command or V_{IN} < UVLO, the device can perform the turn off sequence. It can be calculated with the following equation: Shutdown Delay = TOFF_DELAY x 0.25ms. (0-7.75ms)

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STATUS_BYTE (78h)

Format: Unsigned binary

The STATUS_BYTE command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	BUSY	While MPQ70165FS is performing OTP or restore OTP, host send PMBus write command. MPQ70165FS will report the Busy fault and ALERT# will be pulled down (see Table 1 at page 39). The bit is set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No busy fault 1'b1: A fault was declared because the device was busy and unable to respond
6	R	OFF	The bit can indicate channels' current status. For different channels' status, only change page via PAGE command before reading this bit. When the channel's status in between t_{ON_DELAY} ending and t_{OFF_DELAY} ending, the bit will be set to 0. Besides, this bit will be set to 1. 1'b0: Device is on 1'b1: Device is off
5	R	VOUT_OV_FAULT	When one channel's V_{OUT} exceeds V_{OUT} OV threshold duration ~10us, the channel will happen V_{OUT} OV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 7Ah bit7. 1'b0: No output over voltage fault has occurred 1'b1: An output over voltage fault has occurred
4	R	VOUT_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 79h bit6 and 7Ah bit4. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
3	R	RESERVED	Reserved.
2	R	TEMPERATURE	When MPQ70165FS's junction temperature exceeds TWN threshold or TSD threshold, the bit will be set to 1 at all pages. But the bit cannot accurately determine which thermal fault occurs. Please read 7Dh to ger the detail thermal fault information. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No TSD fault and TWN fault occurred 1'b1: A TSD fault or TWN fault has occurred
1	R	CML	When MPQ70165FS happens PMBus fault or OTP CRC (during 2nd restore OTP) fault, this bit will be set to 1 at all pages. PMBus fault include unsupported command fault, unsupported data fault, PEC fault and communication fault (include start error, stop error and long error). Please read 7Eh to ger the detail fault information. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No PMBus or OTP CRC (during 2nd restore OTP) fault has occurred. 1'b1: A PMBus or OTP CRC (during 2nd restore OTP) fault has occurred.

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0	R	FAULT_OTHER	In a channel, when VOUT COMMAND exceeds VOUT_MAX COMMAND or VOUT COMMAND lower than VOUT_MIN COMMAND, the bit will be set to1 under corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No VOUT MAXMIN fault has occurred. 1'b1: VOUT MAXMIN fault has occurred.

STATUS_WORD (79h)

Format: Unsigned binary

This command returns two bytes of information with a summary of the device fault/warning condition. The higher byte gives more detailed information of the fault conditions. The higher byte shares with register STATUS_BYTE.

Bits	Access	Bit Name	Description
15:8	R	STATUS_BYTE	The same as STATUS BYTE.
7	R	VOUT_FAULT	When one channel happens V_{OUT} related fault, this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No output voltage fault has occurred 1'b1: An output voltage fault has occurred. (including OV fault, UV fault, VOUT MAXMIN fault, PG_OV fault and PG_UV fault)
6	R	VO_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4 and 7Ah bit4. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
5	R	RESERVED	Reserved.
4	R	MANUFAC_FAULT	When MPQ70165FS happens reference fault, ABIST fault or watchdog fail fault, this bit will be set to 1 at all pages. Please read 80h to ger the detail fault information. This fault flag will be cleared once the device performs ABIST if only reference fault happened. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No manufacturer specific fault has occurred 1'b1: A manufacturer specific fault has occurred. (Including Reference fault, ABIST fault and Watchdog fail event)
3	R	PGOOD	The bit can indicate if channels' output exceeds the PGOOD range. For different channels' PGOOD status, only change page via PAGE command before reading this bit. When channel's output is not within PGOOD range, the bit will be set to 1 at corresponding page. Other page doesn't set this bit to 1. When channel's output is within PGOOD range, the bit will be set to 0 automatically at corresponding page. 1'b0: Power Good is high 1'b1: Power Good is low
2:0	R	RESERVED	Reserved.

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STATUS_VOUT (7Ah)

Format: Unsigned binary

The STATUS_VOUT command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	VOUT_OV_FAULT	When one channel's V_{OUT} exceeds V_{OUT} OV threshold duration ~10us, the channel will happen V_{OUT} OV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4. 1'b0: No output over voltage fault has occurred 1'b1: An output over voltage fault has occurred
6:5	R	RESERVED	Reserved.
4	R	VOUT_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4 and 79h bit6. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
3	R	RESERVED	Reserved.
2	R	PG_OV	When one channel's V_{OUT} exceeds PG upper threshold, the channel will happen PGOV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No pgova fault has occurred 1'b1: A pgova fault has occurred
1	R	PG_UV	When one channel's V_{OUT} is lower than PG lower threshold, the channel will happen PGUV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No pguva fault has occurred 1'b1: A pguva fault has occurred
0	R	VOUT_MAX_MIN	In a channel, when VOUT COMMAND exceeds VOUT_MAX COMMAND or VOUT COMMAND lower than VOUT_MIN COMMAND, the bit will be set to1 under corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No V_{OUT_CMD} maxmin fault has occurred 1'b1: An V_{OUT_CMD} maxmin fault has occurred

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VOUT_STARTUP_SLEW (A3h)

Format: Unsigned binary

The VOUT_STARTUP_SLEW command to set the rising rate of Buck 6 output during soft start.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_STARTUP_SLEW	These bits can control the rising rate of channels' output voltage. Please change page vis PAGE command, and then set the register to configure rising rate. To make the output voltage more linear during the soft start period, it is recommended to choose 1.25mV/us or 2.5mV/us. 2'b00: VOUT_SL x 1.25mV/us 2'b01: VOUT_SL x 2.5mV/us 2'b10: VOUT_SL x 5mV/us 2'b11: VOUT_SL x 10mV/us

VOUT_ SHUTDOWN _SLEW (A5h)

Format: Unsigned binary

The VOUT_SHUTDOWN_SLEW command to set the rising rate of Buck 6 output during soft stop.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_SHUTDOWN_SLEW	These bits can control the rising rate of channels' output voltage. Please change page vis PAGE command, and then set the register to configure rising rate. To make the output voltage more linear during the soft stop period, it is recommended to choose 1.25mV/us or 2.5mV/us. 2'b00: VOUT_SL x 1.25mV/us 2'b01: VOUT_SL x 2.5mV/us 2'b10: VOUT_SL x 5mV/us 2'b11: VOUT_SL x 10mV/us

VOUT_SLEW (A7h)

Format: Unsigned binary

The VOUT_ SLEW command to set the rising rate of Buck 6 output during changing output voltage online.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_SLEW	These bits can control the rising rate of channels' output voltage during changing output voltage online via PMBus. Please change page vis PAGE command, and then set the register to configure rising rate. 2'b00: VOUT_SL x 2.5mV/us 2'b01: VOUT_SL x 5mV/us 2'b10: VOUT_SL x 10mV/us 2'b11: VOUT_SL x 20mV/us

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OUTPUT_DISCHARGE (A9h)

Format: Unsigned binary

The OUTPUT_DISCHARGE command to set the discharge path of Buck 6 enabled or disabled.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1	R/W	SS_EN	This bit can control soft stop function to enabled or disabled. Please change page vis PAGE command, and then set the register to configure rising rate. For detail information, please see Output Discharge and Internal Soft Stop sections. 1'b0: Soft Stop Disable 1'b1: Soft Stop Enable
0	R/W	DISCHARGE_EN	This bit can control 100 Ω discharge resistance to be or not to be connected between VOUT and AGND. Please change page vis PAGE command, and then set the register to configure rising rate. For detail information, please see Output Discharge section. 1'b0: 100 Ω Output Discharge Disable 1'b1: 100 Ω Output Discharge Enable

OUTPUT_SHUTDOWN_MODE (ABh)

Format: Unsigned binary

The OUTPUT_SHUTDOWN_MODE command to set the shutdown mode of Buck 6.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	OUTPUT_SHUTDOWN_MODE	MPQ70165FS only support the Time-Based Shutdown Mode. When EN off or PMBus off, the channel will shut down after the corresponding turn off delay time timing ending. 1'b0: Time Based Shutdown Mode

COMPENSATION_CONFIG_1 (B0h)

Format: Unsigned binary

The COMPENSATION_CONFIG_1 command configures the compensation network parameters of Buck 6.

Bits	Access	Bit Name	Description
7:4	R/W	RCOMP	Sets R_{COMP}. The default selection in 1000. 4'b0000: 550K 4'b0001: 600K 4'b0010: 450K 4'b0011: 500K 4'b0100: 750K 4'b0101: 800K 4'b0110: 650K 4'b0111: 700K 4'b1000: 150K 4'b1001: 200K 4'b1010: 50K 4'b1011: 100K 4'b1100: 350K 4'b1101: 400K 4'b1110: 250K 4'b1111: 300K
3	R	RESERVED	Reserved.

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2:0	R/W	CCOMP_HP	Sets C_{COMP_HP}. The default selection is 011. 100fF/LSB. 3'b000: 0 x LSB … 3'b111: +7 x LSB

COMPENSATION_CONFIG_2 (B1h)

Format: Unsigned binary

The COMPENSATION_CONFIG_2 command configures the compensation network parameters of Buck 6.

Bits	Access	Bit Name	Description
7	R	RESERVED	Reserved.
6:4	R/W	CCOMP_LF	Sets C_{COMP_LF}. The default selection is 011. 3'b000: 44.5pF 3'b001: 49.5pF 3'b010: 54.5pF 3'b011: 59.5pF 3'b100: 24.5pF 3'b101: 29.5pF 3'b110: 34.5pF 3'b111: 39.5pF
3:2	R	RESERVED	Reserved.
1:0	R/W	SLOPE_COMP	Sets SLOPE_COMP. The default selection is 10. 2'b00: I_{SLOPE} 2'b01: I_{SLOPE} x 2 2'b10: I_{SLOPE} x 3 2'b11: I_{SLOPE} x 4

PG_CONFIG (CAh)

Format: Unsigned binary

The PG_CONFIG command can configure PG OV / UV threshold of Buck 6.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	PG_THRESHOLD	This bit can set different PG OV rising threshold and PG UV falling threshold for different channel. Please change page vis PAGE command, and then set the register to configure PG threshold. 1'b1: +4.5% / - 3.5% 1'b0: +6.5% / - 5.5%

ILIM_SCALE (CEh)

Format: Unsigned binary

The ILIM_SCALE command can configure the current limit scale of Buck 6.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	ILIM_SCALE	These bits can set the scale of peak / valley / reverse current limit, default value is 100% (2'b00). Please change page vis PAGE command, and then set the register to configure current limit scale. 2'b00: 100% 2'b01: 75% 2'b10: 50%

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PAGE FF REGISTER MAP

PAGE (00h)

Format: Unsigned binary

The PAGE command provides the ability to configure, control and monitor all outputs through only one physical address. All subsequent commands (pages is “Y” at REGISTER MAP section) that allow multipage action are applied to the corresponding regulators selected by the PAGE command.

Bits	Access	Bit Name	Description
7:0	R/W	PAGE	In Page FF, write command to other registers can change all pages’ configuration, including page 0-5. In Page 0 – 5, write command to other registers can only change the current page’s configuration. So, the PAGE command shall be written before modifying other registers. 8’b00000000: VOUT1 8’b00000001: VOUT2 8’b00000010: VOUT3 8’b00000011: VOUT4 8’b00000100: VOUT5 8’b00000101: VOUT6 8’b11111111: All Others: Invalid commands

OPERATION (01h)

Format: Unsigned binary

The OPERATION command is used to configure the output on/off state of all channels, in conjunction with input from the EN_X pin.

Bits	Access	Bit Name	Description
7	R/W	OPERATION	Enable signal is available, MPQ70165FS support interface to turn on / off channels through PMBus. 1’b1: Output on (if enable high) 1’b0: Output off
6:0	R/W	RESERVED	Reserved.

CLEAR_FAULTS (03h)

This command is used to clear any fault bit in all status registers: STATUS_BYTE (78h), STATUS_WORD (79h), STATUS_VOUT (7Ah), STATUS_TEMPERATURE (7Dh), STATUS_CML (7Eh), STATUS_MFR_SPECIFIC (80h), WATCHDOG_STATUS_1 (BAh), MFR_OTP_MEM_STATUS (DBh) and STATUS_FREQ (E7h)

This command is written only. There is no data byte for this command.

WRITE_PROTECT (10h)

Format: Unsigned binary

The WRITE_PROTECT command is used to control writing to the converter. The intent of this command is to provide protection against accident changes. It’s not intended to provide protection against deliberate or malicious changes to the converter’s configuration or operation. All the supported commands may have their parameters read, regardless of the WRITE_PROTECT settings.

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Bits	Access	Bit Name	Description
7:0	R/W	WRITE_PROTECT	When PMBus interface is available, the register can be written. If host try to write a protected command, MPQ70165FS will report invalid data fault and ALERT# will be pulled down. CLEAR FAULT, STORE USER ALL, RESTORE USER ALL are still available, regardless of the WRITE_PROTECT settings. 8'b10000000: Disable all writes except to the WRITE_PROTECT command 8'b01000000: Disable all writes except to the WRITE_PROTECT, OPERATION and PAGE commands 8'b00100000: Disable all writes except to the WRITE_PROTECT, OPERATION, PAGE and VOUT_COMMAND commands 8'b00000000: Enable writes to all commands. Others: Invalid commands

STORE_USER_ALL (15h)

This command instructs the MPQ70165FS to copy the contents of the operating memory to the matching locations in the OTP, except the internal trim registers. This process will operate when MPQ70165FS receives a STORE_USER_ALL command from the PMBus interface.

This command is written only. There is no data byte for this command.

RESTORE_USER_ALL (16h)

This command instructs the MPQ70165FS to copy the contents from the OTP and over-write the matching locations in the operating memory. Trim registers are not over-written by this process. Any items in OTP that do not have matching locations in the operating memory are ignored.

It's permitted to use the RESTORE_USER_ALL command while the MPQ70165FS is operating. However, the MPQ70165FS may be unresponsive during the operation with unpredictable, undesirable or even catastrophic results.

This command is written only. There is no data byte for this command.

CAPABILITY (19h)

Format: Unsigned binary

This command provides one byte to return key PMBus features that MPQ70165FS can support.

Bits	Access	Bit Name	Description
7	R	PACKET_ERR_CHECKING	1'b1: Packet Error Checking is supported Others: Invalid commands
6:5	R	MAX_BUS_SPEED	2'b10: Maximum supported bus speed is 1 MHz Others: Invalid commands
4	R	SMBALERT#	1'b0: The device does not have a SMBALERT# pin and does not support the SMBus ALERT# Response protocol Others: Invalid commands
3	R	NUMERIC_FORMAT	1'b0: Numeric data is in LINEAR11, ULINEAR16, SLINEAR16, or DIRECT format Others: Invalid commands
2	R	AVSBUS_SUPPORT	1'b0: AVSBus Not supported Others: Invalid commands
1:0	R	RESERVED	Reserved.

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VOUT_MODE (20h)

Format: Unsigned binary

The VOUT_MODE command is used to command and read the output voltage mode. The 3 MSB bits are used to determine the data format (only direct format is supported in MPQ70165FS).

Bits	Access	Bit Name	Description
7:0	R	VOUT_MODE	8'b01000000: Direct mode. Coefficients are m=160, R=0, b=-33.

VOUT_COMMAND (21h)

Format: Unsigned binary

The VOUT_COMMAND command sets the output voltage of all channels.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_CMD	The command is used to set output voltage. Steps as below: 13. Select page x by PAGE (00h). 14. Change V_{OUT} by VOUT_COMMAND. V_{OUT} is influenced by VOUT_SCALE_LOOP, see the below description for detail V_{OUT} range. 13. When VOUT_SL = 1, V_{OUT} can be set from 0.20625V to 1.8V, 6.25mV/step. 14. When VOUT_SL = 2, V_{OUT} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT} = (VOUT_CMD x 6.25mV + 206.25mV) x VOUT_SL

VOUT_MAX (24h)

Format: Unsigned binary

The VOUT_MAX command sets an upper limit on the output voltage that all channels can command regardless of any other commands or combinations. The intent of this command is to provide a safeguard against a user accidental setting the output voltage to a possible destructive level rather than to be the primary output overvoltage protection.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_MAX	When VOUT_SL = 1, V_{OUT_MAX} can be set from 0.20625V to 1.8V, 6.25mV/step. When VOUT_SL = 2, V_{OUT_MAX} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT_MAX} = (VOUT_MAX x 6.25mV + 206.25mV) x VOUT_SL Attempting to write a higher value to VOUT_COMMAND causes VOUT_COMMAND to be set to VOUT_MAX and asserts a VOUT_MAX_MIN warning.

VOUT_SCALE_LOOP (29h)

Format: Unsigned binary

The VOUT_SCALE_LOOP command sets the scale for the V_{OUT}, V_{OUT_MAX}, V_{OUT_MIN}, V_{OUT} start-up slew rate, V_{OUT} shutdown slew rate and V_{OUT} slew rate on all Pages.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	VOUT_SC_L	According to application requirement, set adjust scale. The device should be off via the OPERATINO command before change the scale. 1'b0: VOUT_SL= 1 1'b1: VOUT_SL= 2

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VOUT_MIN (2Bh)

Format: Unsigned binary

The VOUT_MIN command sets a lower limit on the output voltage that all channels can command regardless of any other commands or combinations. The intent of this command is to provide a safeguard against a user accidentally setting the output voltage to a possible destructive level rather than to be the primary output under-voltage protection.

Bits	Access	Bit Name	Description
7:0	R/W	VOUT_MIN	When VOUT_SL = 1, V_{OUT_MIN} can be set from 0.20625V to 1.8V, 6.25mV/step. When VOUT_SL = 2, V_{OUT_MIN} can be set from 0.4125V to 3.6V, 12.5mV/step. It can be calculated with the following equation: V_{OUT_MIN} = (VOUT_MIN x 6.25mV + 206.25mV) x VOUT_SL Attempting to write a lower value to VOUT_COMMAND causes VOUT_COMMAND to be set to VOUT_MIN, and asserts a VOUT_MAX_MIN warning

TON_DELAY (60h)

Format: Unsigned binary

The TON_DELAY command sets the turn-on delay time for all channels.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	TON_DELAY	Every buck's turn on delay time can be set independently. Please change page vis PAGE command, and then set the register to configure turn on delay time. It can be calculated with the following equation: Startup Delay = TON_DELAY x 0.25ms. (0-7.75ms)

TOFF_DELAY (64h)

Format: Unsigned binary

The TOFF_DELAY command sets the turn-off delay time for all channels.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	TOFF_DELAY	Every buck's turn off delay time can be set independently. Please change page vis PAGE command, and then set the register to configure turn off delay time. Only when enable is low or PMBus off command or V_{IN} < UVLO, the device can perform the turn off sequence. It can be calculated with the following equation: Shutdown Delay = TOFF_DELAY x 0.25ms. (0-7.75ms)

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STATUS_BYTE (78h)

Format: Unsigned binary

The STATUS_BYTE command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	BUSY	While MPQ70165FS is performing OTP or restore OTP, host send PMBus write command. MPQ70165FS will report the Busy fault and ALERT# will be pulled down (see Table 1 at page 39). The bit is set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No busy fault 1'b1: A fault was declared because the device was busy and unable to respond
6	R	OFF	The bit can indicate channels' current status. For different channels' status, only change page via PAGE command before reading this bit. When the channel's status in between t_{ON_DELAY} ending and t_{OFF_DELAY} ending, the bit will be set to 0. Besides, this bit will be set to 1. 1'b0: Device is on 1'b1: Device is off
5	R	VOUT_OV_FAULT	When one channel's V_{OUT} exceeds V_{OUT} OV threshold duration ~10us, the channel will happen V_{OUT} OV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 7Ah bit7. 1'b0: No output over voltage fault has occurred 1'b1: An output over voltage fault has occurred
4	R	VOUT_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 79h bit6 and 7Ah bit4. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
3	R	RESERVED	Reserved.
2	R	TEMPERATURE	When MPQ70165FS's junction temperature exceeds TWN threshold or TSD threshold, the bit will be set to 1 at all pages. But the bit cannot accurately determine which thermal fault occurs. Please read 7Dh to ger the detail thermal fault information. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No TSD fault and TWN fault occurred 1'b1: A TSD fault or TWN fault has occurred
1	R	CML	When MPQ70165FS happens PMBus fault or OTP CRC (during 2nd restore OTP) fault, this bit will be set to 1 at all pages. PMBus fault include unsupported command fault, unsupported data fault, PEC fault and communication fault (include start error, stop error and long error). Please read 7Eh to ger the detail fault information. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No PMBus or OTP CRC (during 2nd restore OTP) fault has occurred. 1'b1: A PMBus or OTP CRC (during 2nd restore OTP) fault has occurred.

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0	R	FAULT_OTHER	In a channel, when VOUT COMMAND exceeds VOUT_MAX COMMAND or VOUT COMMAND lower than VOUT_MIN COMMAND, the bit will be set to1 under corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No VOUT MAXMIN fault has occurred. 1'b1: VOUT MAXMIN fault has occurred.

STATUS_WORD (79h)

Format: Unsigned binary

This command returns two bytes of information with a summary of the device fault/warning condition. The higher byte gives more detailed information of the fault conditions. The higher byte shares with register STATUS_BYTE.

Bits	Access	Bit Name	Description
15:8	R	STATUS_BYTE	The same as STATUS BYTE.
7	R	VOUT_FAULT	When one channel happens V_{OUT} related fault, this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No output voltage fault has occurred 1'b1: An output voltage fault has occurred. (including OV fault, UV fault, VOUT MAXMIN fault, PG_OV fault and PG_UV fault)
6	R	VO_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4 and 7Ah bit4. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
5	R	RESERVED	Reserved.
4	R	MANUFAC_FAULT	When MPQ70165FS happens reference fault, ABIST fault or watchdog fail fault, this bit will be set to 1 at all pages. Please read 80h to ger the detail fault information. This fault flag will be cleared once the device performs ABIST if only reference fault happened. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No manufacturer specific fault has occurred 1'b1: A manufacturer specific fault has occurred. (Including Reference fault, ABIST fault and Watchdog fail event)
3	R	PGOOD	The bit can indicate if channels' output exceeds the PGOOD range. For different channels' PGOOD status, only change page via PAGE command before reading this bit. When channel's output is not within PGOOD range, the bit will be set to 1 at corresponding page. Other page doesn't set this bit to 1. When channel's output is within PGOOD range, the bit will be set to 0 automatically at corresponding page. 1'b0: Power Good is high 1'b1: Power Good is low
2:0	R	RESERVED	Reserved.

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STATUS_VOUT (7Ah)

Format: Unsigned binary

The STATUS_VOUT command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	VOUT_OV_FAULT	When one channel's V_{OUT} exceeds V_{OUT} OV threshold duration ~10us, the channel will happen V_{OUT} OV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4. 1'b0: No output over voltage fault has occurred 1'b1: An output over voltage fault has occurred
6:5	R	RESERVED	Reserved.
4	R	VOUT_UV_FAULT	When one channel's V_{OUT} is lower than V_{OUT} UV threshold duration ~10us, the channel will happen V_{OUT} UV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. This bit is same with 78h bit4 and 79h bit6. 1'b0: No output under voltage fault has occurred 1'b1: An output under voltage fault has occurred
3	R	RESERVED	Reserved.
2	R	PG_OV	When one channel's V_{OUT} exceeds PG upper threshold, the channel will happen PGOV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No pgova fault has occurred 1'b1: A pgova fault has occurred
1	R	PG_UV	When one channel's V_{OUT} is lower than PG lower threshold, the channel will happen PGUV fault, and this bit is set to 1 only at corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No pguva fault has occurred 1'b1: A pguva fault has occurred
0	R	VOUT_MAX_MIN	In a channel, when VOUT COMMAND exceeds VOUT_MAX COMMAND or VOUT COMMAND lower than VOUT_MIN COMMAND, the bit will be set to1 under corresponding page. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No V_{OUT_CMD} maxmin fault has occurred 1'b1: An V_{OUT_CMD} maxmin fault has occurred

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STATUS_ TEMPERATURE (7Dh)

Format: Unsigned binary

The STATUS_TEMPERATURE command returns the value of a number of flags indicating the state of the MPQ70165FS junction temperature.

Bits	Access	Bit Name	Description
7	R	TEMP_OT_FAULT	When MPQ70165FS's junction temperature exceeds TSD threshold, the bit will be set to 1 at all pages. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No over temperature fault has occurred 1'b1: An over temperature fault has occurred
6	R	TEMP_OT_WARNING	When MPQ70165FS's junction temperature exceeds TWN threshold, the bit will be set to 1 at all pages. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No over temperature warning has occurred 1'b1: An over temperature warning has occurred
5:0	R	RESERVED	Reserved.

STATUS_CML (7Eh)

Format: Unsigned binary

The STATUS_CML command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	INVALID_CMD	If host sends any write/read command to a register that MPQ70165FS does not define, the device will report invalid command fault and this bit set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No invalid or unsupported commands received 1'b1: An invalid or unsupported command has been received
6	R	INVALID_DATA	If host send write command to an only read register or a protected register by WRITE PROTETCION or a lock register by SECURE_LOCKOUT, the device will report invalid data fault and this bit set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No invalid data received 1'b1: Invalid data has been received
5	R	PEC_ERROR	Under PEC enabled, only when host send a write command with incorrect PEC value, MPQ70165FS will report PEC fault and this bit is set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: Packet Error Check not failed 1'b1: Packet Error Check failed
4	R	CRC_ERROR	In 2nd RESTORE OTP, if OTP CRC check value is incorrect, MPQ70165FS will report start up load OTP CRC fail, and this bit will be set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: CRC did not fail 1'b1: CRC failed (OTP CRC check fail)
3:1	R	RESERVED	Reserved.

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0	R	COM_FAULT	If MPQ70165FS happens start error or stop error or long error, this bit will be set to 1. Long error: When host send a read command with two bytes to an MPQ70165FS register with one byte definition, MPQ70165FS will report long error. Start error: During the process of slave address or command address or data or PEC value, if host send a start signal, MPQ70165FS will report start error. Stop error: During the process of slave address or command address or data or PEC value, if host send a stop signal, MPQ70165FS will report stop error. 1'b0: No communication fault received 1'b1: A communication fault has occurred (START err, STOP err, LONG err)

STATUS_ MFR_SPECIFIC (80h)

Format: Unsigned binary

The STATUS_MFR_SPECIFIC command returns the value of a number of flags indicating the state of the MPQ70165FS.

Bits	Access	Bit Name	Description
7	R	RESERVED	Reserved.
6	R	VMON_OV	After VMON function is active, if VMON voltage exceeds VMON_HIGH_THRESHOLD (D2h), MPQ70165FS will report VMON over-voltage fault. This bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No VMON_OV fault has occurred 1'b1: A VMON_OV fault has occurred
5	R	VMON_UV	After VMON function is active, if VMON voltage drop below of VMON_LOW_THRESHOLD (D1h), MPQ70165FS will report VMON under-voltage fault. This bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No VMON_UV fault has occurred 1'b1: A VMON_UV fault has occurred
4	R	ERRIN_FAULT	When WDI pin is reused to error input pin, if this pin voltage level is high, MPQ70165FS will report ERRIN fault. This bit will be set to 1. 1'b0: No ERRIN fault has occurred 1'b1: An ERRIN fault has occurred
3	R	LBIST_FAULT	During LBIST, if any logic circuits have error, MPQ70165FS will report LBIST fault and this bit will be set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No LBIST fault has occurred 1'b1: A LBIST fault has occurred
2	R	REFERENCE_FAULT	When the difference between the main reference is more than normal range (97% ~ 103% x auxiliary reference), MPQ70165FS will report reference fault and this bit will be set to 1 at all pages. This fault flag will be cleared once the device performs ABIST. This fault flag can be also cleared by writing 03h register without data at anyone defined page when no this fault continue happening. 1'b0: No reference fault has occurred 1'b1: A reference fault has occurred
1	R	ABIST_FAULT	If the output of any of the comparators checked by ABIST is detected as not pulling up or down properly, MPQ70165FS will report an ABIST fault and this bit will be set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No ABIST fault has occurred 1'b1: An ABIST fault has occurred

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0	R	WD_FAULT	If microcontroller doesn't provide good watchdog event to MPQ70165FS cause watchdog fault count being equal to watchdog fault limit, MPQ70165FS will report watchdog fail and this bit will be set to 1 at all pages. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No watchdog fail has occurred 1'b1: A watchdog fail has occurred

SECURE_LOCKOUT (9Dh)

Format: Unsigned binary

The SECURE_LOCKOUT command to lockout write access to most registers.

Bits	Access	Bit Name	Description
7:0	R/W	SECURE_LOCKOUT	Write 01011010 to lockout write access to most registers. Registers then can still be written are listed as N under "Lockout?" This can only be cleared by setting all enables to low.

MULTIPHASE_CONFIG (9Eh)

Format: Unsigned binary

The MULTIPHASE_CONFIG command to set channels work in parallel or independent.

Bits	Access	Bit Name	Description
7:3	R	RESERVED	Reserved.
2	R/W	MULTIPHASE_B5B6	Set parallel or independent work between buck5 and buck6. Please see Multiphase Operation section for detail information. 1'b0: Independent 1'b1: Multiphase, Buck5 runs as the primary.
1	R/W	MULTIPHASE_B3B4	Set parallel or independent work between buck3 and buck4. Please see Multiphase Operation section for detail information. 1'b0: Independent 1'b1: Multiphase, Buck3 runs as the primary.
0	R/W	MULTIPHASE_B1B2	Set parallel or independent work between buck1 and buck2. Please see Multiphase Operation section for detail information. 1'b0: Independent 1'b1: Multiphase, Buck1 runs as the primary.

HICCUP_TIMER (A0h)

Format: Unsigned binary

The HICCUP_TIMER command to set the hiccup time during hiccup state.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	HICCUP_TIMER	When MPQ70165FS happens deep safe event causing state machine to enter deep safe state and maintain hiccup time. Therefore, hiccup time can be regarded as the time when a deep safe state is maintained. 2'b00: 2ms 2'b01: 4ms 2'b10: 6ms 2'b11: 8ms

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STATUS_PIN_STATE (A1h)

Format: Unsigned binary

The STATUS_PIN_STATE command to indicate the pin voltage status.

Bits	Access	Bit Name	Description
7:4	R	RESERVED	Reserved.
3	R	EN_2_STATE	Indicate EN2 pin voltage status is high or low. When EN2 pin voltage exceeds EN rising threshold, this bit will be set to 1. When EN2 pin voltage is lower than EN falling threshold, this bit will be set to 0. 1'b0: Low 1'b1: High
2	R	EN_1_STATE	Indicate EN1 pin voltage status is high or low. When EN1 pin voltage exceeds EN rising threshold, this bit will be set to 1. When EN1 pin voltage is lower than EN falling threshold, this bit will be set to 0. 1'b0: Low 1'b1: High
1	R	PG_MEAS_STATE	Indicate PG pin voltage status is high or low. When PG pin voltage exceeds EN logic on threshold, this bit will be set to 1. When PG pin voltage is lower than EN logic on threshold, this bit will be set to 0. 1'b0: Low 1'b1: High
0	R	ALERT#_MEAS_STATE	Indicate ALERT# pin voltage status is high or low. When ALERT# pin voltage exceeds EN logic on threshold, this bit will be set to 1. When ALERT# pin voltage is lower than EN logic on threshold, this bit will be set to 0. 1'b0: Low 1'b1: High

VOUT_STARTUP_SLEW (A3h)

Format: Unsigned binary

The VOUT_STARTUP_SLEW command to set the rising rate of all channels' output during soft start.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_STARTUP_SLEW	These bits can control the rising rate of channels' output voltage. Please change page vis PAGE command, and then set the register to configure rising rate. To make the output voltage more linear during the soft start period, it is recommended to choose 1.25mV/us or 2.5mV/us. 2'b00: VOUT_SL x 1.25mV/us 2'b01: VOUT_SL x 2.5mV/us 2'b10: VOUT_SL x 5mV/us 2'b11: VOUT_SL x 10mV/us

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VOUT_ SHUTDOWN _SLEW (A5h)

Format: Unsigned binary

The VOUT_SHUTDOWN_SLEW command to set the rising rate of all channels' output during soft stop.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_SHUTDOWN_SLEW	These bits can control the rising rate of channels' output voltage. Please change page vis PAGE command, and then set the register to configure rising rate. To make the output voltage more linear during the soft stop period, it is recommended to choose 1.25mV/us or 2.5mV/us. 2'b00: VOUT_SL x 1.25mV/us 2'b01: VOUT_SL x 2.5mV/us 2'b10: VOUT_SL x 5mV/us 2'b11: VOUT_SL x 10mV/us

VOUT_SLEW (A7h)

Format: Unsigned binary

The VOUT_ SLEW command to set the rising rate of all channels' output during changing output voltage online.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	VOUT_SLEW	These bits can control the rising rate of channels' output voltage during changing output voltage online via PMBus. Please change page vis PAGE command, and then set the register to configure rising rate. 2'b00: VOUT_SL x 2.5mV/us 2'b01: VOUT_SL x 5mV/us 2'b10: VOUT_SL x 10mV/us 2'b11: VOUT_SL x 20mV/us

OUTPUT_DISCHARGE (A9h)

Format: Unsigned binary

The OUTPUT_DISCHARGE command to set the discharge path of all channels enabled or disabled.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1	R/W	SS_EN	This bit can control soft stop function to enabled or disabled. Please change page vis PAGE command, and then set the register to configure rising rate. For detail information, please see Output Discharge and Internal Soft Stop sections. 1'b0: Soft Stop Disable 1'b1: Soft Stop Enable
0	R/W	DISCHARGE_EN	This bit can control 100 Ω discharge resistance to be or not to be connected between VOUT and AGND. Please change page vis PAGE command, and then set the register to configure rising rate. For detail information, please see Output Discharge section. 1'b0: 100 Ω Output Discharge Disable 1'b1: 100 Ω Output Discharge Enable

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OUTPUT_SHUTDOWN_MODE (ABh)

Format: Unsigned binary

The OUTPUT_SHUTDOWN_MODE command to set the shutdown mode of all channels.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	OUTPUT_SHUTDOWN_MODE	MPQ70165FS only support the Time-Based Shutdown Mode. When EN off or PMBus off, the channel will shut down after the corresponding turn off delay time timing ending. 1'b0: Time Based Shutdown Mode

FREQUENCY_DITHER (ADh)

Format: Unsigned binary

The FREQUENCY_DITHER command to enable or disable frequency spread spectrum in MPQ70165FS.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	FREQUENCY_DITHER	Frequency spread spectrum Control. Please see Frequency Spread Spectrum (FSS) section. 1'b0: Disable 1'b1: Enable

COMPENSATION_CONFIG_1 (B0h)

Format: Unsigned binary

The COMPENSATION_CONFIG_1 command configures the compensation network parameters of all channels.

Bits	Access	Bit Name	Description
7:4	R/W	RCOMP	Sets R_{COMP}. The default selection in 1000. 4'b0000: 550K 4'b0001: 600K 4'b0010: 450K 4'b0011: 500K 4'b0100: 750K 4'b0101: 800K 4'b0110: 650K 4'b0111: 700K 4'b1000: 150K 4'b1001: 200K 4'b1010: 50K 4'b1011: 100K 4'b1100: 350K 4'b1101: 400K 4'b1110: 250K 4'b1111: 300K
3	R	RESERVED	Reserved.
2:0	R/W	CCOMP_HP	Sets C_{COMP_HP}. The default selection is 011. 100fF/LSB. 3'b000: 0 x LSB … 3'b111: +7 x LSB

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COMPENSATION_CONFIG_2 (B1h)

Format: Unsigned binary

The COMPENSATION_CONFIG_2 command configures the compensation network parameters of all channels.

Bits	Access	Bit Name	Description
7	R	RESERVED	Reserved.
6:4	R/W	CCOMP_LF	Sets C_{COMP_LF}. The default selection is 011. 3'b000: 44.5pF 3'b001: 49.5pF 3'b010: 54.5pF 3'b011: 59.5pF 3'b100: 24.5pF 3'b101: 29.5pF 3'b110: 34.5pF 3'b111: 39.5pF
3:2	R	RESERVED	Reserved.
1:0	R/W	SLOPE_COMP	Sets SLOPE_COMP. The default selection is 10. 2'b00: I_{SLOPE} 2'b01: I_{SLOPE} x 2 2'b10: I_{SLOPE} x 3 2'b11: I_{SLOPE} x 4

WATCHDOG_CONFIG (B2h)

Format: Unsigned binary

The WATCHDOG_CONFIG command configures watchdog parameters.

Bits	Access	Bit Name	Description
7:6	R	RESERVED	Reserved.
5	R/W	WD_MODE	Set the watchdog mode to Window or Question – Answer mode. If the microcontroller changes the watchdog mode while the watchdog is working, the close/open window time must be reconfigured after the watchdog mode is changed. Make sure the window starts timing from close window in the new mode. 1'b0: Window Watchdog, WDI pin as watchdog input pin. 1'b1: QA Watchdog, WDI pin as external error input pin.
4	R/W	WD_EN	This bit can control watchdog to be enabled or disabled. Configuring other watchdog parameters before enabling watchdog. 1'b0: Watchdog disabled 1: Watchdog enabled
3	R/W	WD_FAULT_CFG	When MPQ70165FS detect a watchdog failure, this bit can control whether MPQ70165FS will perform RESTORE_USER_ALL command automatically or not. 1'b0: Watchdog failure does not perform RESTORE_USER_ALL command. 1'b1: Watchdog failure cause MPQ70165FS to perform RESTORE_USER_ALL command automatically.
2:0	R/W	WD_FAULT_LIMIT	Set an upper limit to indicate watchdog fail. When watchdog fault count is equal to the upper limit, MPQ70165FS will report watchdog fail and 80h bit0 will be set to 1. Watchdog Fault Limit= WD_FAULT_LIMIT [2:0] + 1

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WATCHDOG_WINDOW_CONFIG (B4h)

Format: Unsigned binary

The WATCHDOG_WINDOW_CONFIG command configures watchdog Close / Open window time.

Bits	Access	Bit Name	Description
7:6	R	RESERVED	Reserved.
5:3	R/W	WD_CLOSE_WINDOW	Set Close Window time during watchdog operation. 3'b000: 1ms 3'b001: 2ms 3'b010: 3ms 3'b011: 4ms 3'b100: 6ms 3'b101: 8ms 3'b110: 10ms 3'b111: 12ms
2:0	R/W	WD_OPEN_WINDOW	Set Open Window time during watchdog operation. 3'b000: 1ms 3'b001: 2ms 3'b010: 3ms 3'b011: 4ms 3'b100: 6ms 3'b101: 8ms 3'b110: 10ms 3'b111: 12ms

WATCHDOG_TOKEN_FDBCK (B6h)

Format: Unsigned binary

The WATCHDOG_TOKEN_FDBCK command configures watchdog parameters.

Bits	Access	Bit Name	Description
7:5	R/W	WD_TOKEN_FDBCK	These bits are only available for Q-A watchdog mode. They can control the sequence of generated questions and respective answers.
4:0	R/W	WD_TOKEN_SEED	These bits are only available for Q-A watchdog mode. They are used to generate a set of new questions.

WATCHDOG_TOKEN_VALUE (B8h)

Format: Unsigned binary

The WATCHDOG_TOKEN_VALUE command indicates watchdog question during one Q-A watchdog sequence.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R	WD_TOKEN_VALUE	These bits are only available for Q-A watchdog mode. They return to watchdog question. It is recommended to read the question when a new watchdog sequence just begins.

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WATCHDOG_STATUS_1 (BAh)

Format: Unsigned binary

The WATCHDOG_STATUS_1 command indicates some watchdog faults.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4	R	WD_ANS_ERR	This is only available in Q-A watchdog mode. If any of the answers received by MPQ70165FS is incorrect, this bit is set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: Answer Correct 1'b1: Answer Incorrect
3	R	WD_CFG_CHG	This is available in Window and Q-A watchdog mode. If the microcontroller writes to the B4h register of the MPQ70165FS, this bit will be set to 1 regardless of the value written. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: B4h register is not written 1'b1: B4h register is written
2	R	WD_SEQU_ERR	This is only available in Q-A watchdog mode. If any of the answers received by MPQ70165FS is incorrect or correct Answer-0 is sent in close window or happens time-out, this bit is set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No error 1'b1: Token early or wrong answer or time out
1	R	WD_TO_ERR	This is available in Window and Q-A watchdog mode. If MPQ70165FS don't receive valid trigger signal during window mode or MPQ70165FS receives less than four answers during Q-A mode, this bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No watchdog time-out event occurred 1'b1: Watchdog time-out event occurred
0	R	WD_TOKEN_ERR	This is available in Window and Q-A watchdog mode. In Q-A mode, if the Answer-0 is sent in close window, this bit is set to 1 regardless of incorrect or correct answer. In Window mode, if WDI valid pulse is identified in close window, this bit is set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No early watchdog error 1'b1: An early watchdog error occurs

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WATCHDOG_STATUS_2 (BCh)

Format: Unsigned binary

The WATCHDOG_STATUS_2 command indicates watchdog answer order and watchdog fault count.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:3	R	WD_ANS_CNT	This is only available in Q-A watchdog mode. These two bits indicate the current answer order that needs to be written. It is recommended to read this bit before the microcontroller writes the answer to ensure that the answer being written is required by MPQ70165FS. Default value is 2'b11. Each time MPQ70165FS receives an answer, the two bits are updated. The order of update is always 2'b11 -> 2'b10 -> 2'b01 -> 2'b00. 2'b00: 1st watchdog answer needs to be written 2'b01: 2nd watchdog answer needs to be written 2'b10: 3rd watchdog answer needs to be written 2'b11: 4th watchdog answer needs to be written
2:0	R	WD_ERR_CNT	This is available in Window and Q-A watchdog mode. These bits record the number of watchdog fault. But when MPQ70165FS perform a good watchdog event, the counter will be decreased 1. Therefore, the value of counter does not represent the number of bad watchdog events detected by MPQ70165FS in the normal state. The maximum value of this counter is 7. Therefore, with the upper limit set to 8, when the counter is 7, MPQ70165FS then detects a bad event or time-out, and the counter does not increase. Watchdog Fault Counter = WD_ERR_CNT [2:0]

WATCHDOG_ANSWER (BEh)

Format: Unsigned binary

The WATCHDOG_ANSWER command receives watchdog answers.

Bits	Access	Bit Name	Description
7:0	R/W	WD_ANS	This is only available in Q-A watchdog mode. Microcontroller send Watchdog Answer to MPQ70165FS. Please refer to Figure 27 and 28 for the Answer calculation formula.

ALERT#_PG_FORCE_ASSERT (C0h)

Format: Unsigned binary

The ALERT#_PG_FORCE_ASSERT command forces PG and ALERT# to assert.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1	R/W	ALERT#_FORCE_ASSERT	Write 1'b1 to this bit, ALERT# will be low. Write 1'b0 to this bit, ALERT# will be pulled high if no other fault happens. 1'b0: Do not force ALERT# to be asserted 1'b1: Force ALERT# to be asserted
0	R/W	PG_FORCE_ASSERT	Write 1'b1 to this bit, PG will be low. Write 1'b0 to this bit, PG will be pulled high if no other fault happens. 1'b0: Do not force PG to be asserted 1'b1: Force PG to be asserted

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SAFETY_CONFIG (C2h)

Format: Unsigned binary

The SAFETY_CONFIG command enables certain functions.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4	R/W	OVP	The MPQ70165FS always provides over-voltage protection for the output voltage. 1'b1: Over Voltage Protection Enable (don't support disable OVP)
3	R/W	UVP	The MPQ70165FS always provides over-voltage protection for the output voltage. 1'b1: Under voltage Protection Enable (don't support disable UVP)
2	R/W	BIST_EN	The bit can control to enable or disable BIST function. BIST here includes PRE-DIANOSTIC and Memory CRC check. 1'b0: BIST Disabled 1'b1: BIST Enabled (Default)
1	R/W	PG_REACTION	Set PG reaction to 1, MPQ70165FS will support these faults causing state machine to enter deep safe state except pulling-down PG signal. If PG reaction is set to 0, these faults only pull-down PG signal. These faults include PG OV, PG UV, VMON OV and VMON UV. 1'b0: PG low fault (only for PGUV/OV, VMON OV/UV fault) does not move to safe state (Default) 1'b1: PG low fault (only for PGUV/OV, VMON OV/UV fault) causes safe state (all outputs off)
0	R/W	PEC_REQ	This bit can enable or disable PMBus interface communication CRC check, here we call it the PEC function. If PEC is disable, the host don't need send PEC value during write command and the MPQ70165FS don't turn PEC value during read command. If PEC is optional, the host can choose to send or not send PEC value during write command and the MPQ70165FS will definitely send PEC value during read command. 1'b0: PEC Disable (Default) 1'b1: PEC Optional

ALTERT#_MAPPING (C4h)

Format: Unsigned binary

The ALERT#_MAPPING command can configure ALERT# reaction when some faults happen.

Bits	Access	Bit Name	Description
7:6	R	RESERVED	Reserved.
5	R/W	WD_ALERT#_MAPPING	This bit can control whether watchdog failure resets ALERT# signal. When the bit is set to 1, ALERT# isn't also reset and only the register report fault information. 1'b0: Watchdog Failure causes ALERT# reset & register assertion (default) 1'b1: Watchdog Failure cause register assertion only
4	R/W	CRC_ALERT#_MAPPING	This bit can control whether OTP CRC check (during 2nd restore OTP) fail pulls down ALERT# signal. When the bit is set to 1, although MPQ70165FS happens OTP CRC check fail during 2nd restore OTP, ALERT# isn't also pulled down and only the register report fault information. 1'b0: OTP CRC check (during 2nd restore OTP) Failure causes deep safe state and ALERT# low & register assertion (default) 1'b1: OTP CRC check (during 2nd restore OTP) Failure cause deep safe state and register assertion

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3	R/W	PMBus _CML_ALERT#_MAPPING	This bit can control whether ALERT# is pulled down when MPQ70165FS detect unsupported command fault or unsupported data fault or communication fault (include start error, stop error, long error). 1'b0: PMBus Communication Failure causes ALERT# low & register assertion (default) 1'b1: PMBus Communication Failure cause register assertion only
2	R/W	PMBus _CRC_ALERT#_MAPPING	This bit can control whether ALERT# is pulled down when MPQ70165FS detect PEC fault. 1'b0: PMBus CRC Failure causes ALERT# low & register assertion(default) 1'b1: PMBus CRC Failure cause register assertion only
1	R/W	REF_FAULT_ALERT #_MAPPING	This bit can control whether ALERT# is pulled down when MPQ70165FS detect reference fault. 1'b0: Reference check Failure causes ALERT# low & register assertion (default) 1'b1: Reference check Failure only causes register assertion
0	R/W	BIST_FAULT_ALERT #_MAPPING	This bit can control whether ALERT# is pulled down when MPQ70165FS detect BIST failure. BITS failure includes ABIST fault, Memory CRC (in Normal state) fault, OTP CRC (in 2nd Restore OTP, Pre-Diagnostic and Normal state) fault and Clock (in Pre-Diagnostic state) fault. 1'b0: BIST Failure causes ALERT# low & register assertion (default) 1'b1: BIST Failure only causes register assertion

PG_MAPPING (C6h)

Format: Unsigned binary

The PG_MAPPING command can configure PG reaction and state machine reaction when some faults happen.

Bits	Access	Bit Name	Description
7:6	R	RESERVED	Reserved.
5	R/W	WD_PG_MAPPING	This bit can control whether watchdog fail pulls down PG signal. When the bit is set to 1, although MPQ70165FS happens watchdog fail because watchdog fault count is equal to watchdog fault limit, PG isn't also pulled down and only the register report fault information. 1'b0: Watchdog Fail causes PG low & register assertion (default) 1'b1: Watchdog Fail cause register assertion only
4	R/W	CRC_PG_MAPPING	This bit can control whether MPQ70165FS masks the OTP CRC check failure during 2nd Restore OTP. If this bit is set to1, state machine doesn't stuck in DEEP SAFE state with hiccup time even though MPQ70165FS detect OTP CRC (during 2nd Restore OTP) check failure. 1'b0: OTP CRC Failure causes deep safe state & register assertion (default) 1'b1: OTP CRC Failure cause register assertion only
3	R/W	PMBus_CML_PG_MAPPING	This bit can control whether PG is pulled down when MPQ70165FS detect unsupported command fault or unsupported data fault or communication fault (include start error, stop error, long error). 1'b0: PMBus Communication Failure causes PG low & register assertion (default) 1'b1: PMBus Communication Failure cause register assertion only
2	R/W	PMBus_CRC_PG_MAPPING	This bit can control whether PG is pulled down when MPQ70165FS detect PEC failure. 1'b0: PMBus CRC Failure causes PG low & register assertion(default) 1'b1: PMBus CRC Failure cause register assertion only

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1	R/W	REF_FAULT_PG_MAPPING	This bit can control whether PG is pulled down and state machine enters deep safe state when MPQ70165FS detect reference failure. 1'b0: Reference Failure causes deep safe state and PG low & register assertion (default) 1'b1: Reference Failure cause register assertion only
0	R/W	BIST_FAULT_PG_MAPPING	This bit can control whether PG is pulled down and state machine enters deep safe state when MPQ70165FS detect BIST failure. BITS failure includes ABIST fault, Memory CRC (in Normal state) fault, OTP CRC (in 2nd Restore OTP, Pre-Diagnostic and Normal state) fault and Clock (in Pre-Diagnostic state) fault. But for the process of pre-diagnostic, once ABIST fault or Clock check fault or OTP CRC check fault is detected, MPQ70165FS will report pre-diagnostic failure at pre-diagnostic ending. The pre-diagnostic failure will cause state machine entering deep safe state when this bit is set to 1. 1'b0: BIST Failure causes deep safe state and PG low & register assertion (default) 1'b1: BIST Failure cause register assertion only

PG_ALERT#_DELAY (C8h)

Format: Unsigned binary

The PG_ALERT#_DELAY command can configure PG and ALERT# delay time.

Bits	Access	Bit Name	Description
7:4	R	RESERVED	Reserved.
3:2	R/W	ALERT#_DELAY	When all faults disappear, ALERT# can be pulled high again through write clear fault command. After writing clear fault command, timing ALERT# delay time, ALERT# will be pulled high. 2'b00: Immediate 2'b01: 2ms 2'b10: 5ms 2'b11: 10ms
1:0	R/W	PG_DELAY	When all faults disappear and state enter normal state, PG can be pulled high again automatically after waiting PG delay time. 2'b00: Immediate 2'b01: 2ms 2'b10: 5ms 2'b11: 10ms

PG_CONFIG (CAh)

Format: Unsigned binary

The PG_CONFIG command can configure PG OV / UV threshold of all channels.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	PG_THRESHOLD	This bit can set different PG OV rising threshold and PG UV falling threshold for different channel. Please change page vis PAGE command, and then set the register to configure PG threshold. 1'b1: +4.5% / - 3.5% 1'b0: +6.5% / - 5.5%

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LIGHT_LOAD (CCh)

Format: Unsigned binary

The LIGHT_LOAD command can configure the work mode to FCCM at light load.

Bits	Access	Bit Name	Description
7:1	R	RESERVED	Reserved.
0	R/W	MODE	The bit can set the work mode is FCCM at light load. 1'b0: Invalid command. 1'b1: FCCM

ILIM_SCALE (CEh)

Format: Unsigned binary

The ILIM_SCALE command can configure the current limit scale of all channels.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1:0	R/W	ILIM_SCALE	These bits can set the scale of peak / valley / reverse current limit, default value is 100% (2'b00). Please change page vis PAGE command, and then set the register to configure current limit scale. 2'b00: 100% 2'b01: 75% 2'b10: 50%

ENABLE_GROUP (CFh)

Format: Unsigned binary

The ENABLE_GROUP command can configure these functions are controlled to EN1 or EN2.

Bits	Access	Bit Name	Description
7	R/W	MON_EN_GUP	Setting EN1 or EN2 controls whether the VMON function is enabled or disabled. 1'b0: Monitor in Enable Group 1 1'b1: Monitor in Enable Group 2
6	R/W	SEQ_EN_GUP	Setting EN1 or EN2 controls whether the SEQ function is enabled or disabled. 1'b0: Sequencer in Enable Group 1 1'b1: Sequencer in Enable Group 2
5	R/W	BK6_EN_GUP	Setting EN1 or EN2 controls whether the buck6 is enabled or disabled. 1'b0: Buck 6 in Enable Group 1 1'b1: Buck 6 in Enable Group 2
4	R/W	BK5_EN_GUP	Setting EN1 or EN2 controls whether the buck5 is enabled or disabled. 1'b0: Buck 5 in Enable Group 1 1'b1: Buck 5 in Enable Group 2
3	R/W	BK4_EN_GUP	Setting EN1 or EN2 controls whether the buck4 is enabled or disabled. 1'b0: Buck 4 in Enable Group 1 1'b1: Buck 4 in Enable Group 2
2	R/W	BK3_EN_GUP	Setting EN1 or EN2 controls whether the buck3 is enabled or disabled. 1'b0: Buck 3 in Enable Group 1 1'b1: Buck 3 in Enable Group 2

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1	R/W	BK2_EN_GUP	Setting EN1 or EN2 controls whether the buck2 is enabled or disabled. 1'b0: Buck 2 in Enable Group 1 1'b1: Buck 2 in Enable Group 2
0	R/W	BK1_EN_GUP	Setting EN1 or EN2 controls whether the buck1 is enabled or disabled. 1'b0: Buck 1 in Enable Group 1 1'b1: Buck 1 in Enable Group 2

SEQ_DELAY (D0h)

Format: Unsigned binary

The SEQ_DELAY command can configure SEQ output delay time.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	SEQ_DELAY	After PRE-DIAG state, when related enable pin voltage exceeds EN rising threshold, MPQ70165FS begin to time the SEQ delay time. Once the delay timing ending, SEQ will be pulled high. When related enable pin voltage ramp down to the below of EN falling threshold, MPQ70165FS begin to time the SEQ delay time. Once the delay timing ending, SEQ will be pulled low. Sequence Output Startup Delay = SEQ_DELAY [4:0] x 0.25ms. (0 – 7.75ms)

VMON_LOW_THRESHOLD (D1h)

Format: Unsigned binary

The VMON_LOW_THRESHOLD command can set the lower limit of voltage monitor range.

Bits	Access	Bit Name	Description
7:0	R/W	VMON_LOW_THR	MON_RANGE[0] = 0: V_{MON_LOW} setting, from 0.4125V to 3.6V V_{MON_LOW} = VMON_LOW_THR [7:0] x 12.5mV + 412.5mV MON_RANGE[0] = 1: V_{MON_LOW} setting, from 0.2V to 3.3875V V_{MON_LOW} = VMON_LOW_THR [7:0] x 12.5mV + 200mV

VMON_ HIGH _THRESHOLD (D2h)

Format: Unsigned binary

The VMON_HIGH_THRESHOLD command can set the upper limit of voltage monitor range.

Bits	Access	Bit Name	Description
7:0	R/W	VMON_HIGH_THR	MON_RANGE[0] = 0: V_{MON_HIGH} setting, from 0.4125V to 3.6V V_{MON_HIGH} = VMON_HIGH_THR [7:0] x 12.5mV + 412.5mV MON_RANGE[0] = 1: V_{MON_HIGH} setting, from 0.2V to 3.3875V V_{MON_HIGH} = VMON_HIGH_THR [7:0] x 12.5mV + 200mV

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VMON_DELAY (D3h)

Format: Unsigned binary

The VMON_DELAY command can set the delay time of VMON available.

Bits	Access	Bit Name	Description
7:5	R	RESERVED	Reserved.
4:0	R/W	VMON_DELAY	After PRE-DIAG state, when related enable pin voltage exceeds EN rising threshold, MPQ70165FS begin to time the VMON delay time. Once the delay timing ending, VMON function will be active. MPQ70165FS only detect VMON voltage during VMON active state. MON Block on/off Delay = VMON_DELAY[4:0] x 0.25ms. (0 – 7.75ms)

MON_RANGE (D4h)

Format: Unsigned binary

The MON_RANGE command can set the voltage monitor range.

Bits	Access	Bit Name	Description
7:2	R	RESERVED	Reserved.
1	R/W	MON_ENABLE	This bit is used to enable or disable VMON function when the corresponding enable pin is high level. 1'b0: Disable 1'b1: Enable
0	R/W	MON_RANGE	This bit can set the VMON range. Please see the below formula. 1'b0: $V_{MON_REF_HIGH}$ = (VMON_HIGH_THR [7:0] + 33) x 12.5mV $V_{MON_REF_LOW}$ = (VMON_LOW_THR [7:0] + 33) x 12.5mV 1'b1: $V_{MON_REF_HIGH}$ = (VMON_HIGH_THR [7:0] + 16) x12.5mV $V_{MON_REF_LOW}$ = (VMON_LOW_THR [7:0] + 16) x 12.5mV

ADDRESS (D5h)

Format: Unsigned binary

The ADDRESS command can set the PMBus address.

Bits	Access	Bit Name	Description
7	R	RESERVED	Reserved.
6:0	R/W	ADDRESS	MPQ70165FS support to set different PMBus address. (0x00 ~ 0x7F)

CONFIG_CODE (D6h)

Format: Unsigned binary

The CONFIG_CODE command can indicate whether user page 1 / 2 are OTP programmed.

Bits	Access	Bit Name	Description
7:0	R/W	CONFIG_CODE	For MPS provided MPQ70165FS, if the register value is 0x00, MPQ70165FS has two OTP chance. If the register value is 0x01, MPQ70165FS has one OTP chance. If the register value is 0x03, MPQ70165FS has no OTP chance.

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OTP_CRC (D7h)

Format: Unsigned binary

The OTP_CRC command can return OTP CRC value.

Bits	Access	Bit Name	Description
15:0	R	OTP_CRC	The register stores the CRC value of user OTP data. Only when user page 1 is programmed or user page 1 and page 2 are programmed, the register will return to a valid value. Otherwise, the register value is always 0x0000.

MEMORY _CRC (D9h)

Format: Unsigned binary

The MEMORY_CRC command can return Memory CRC value.

Bits	Access	Bit Name	Description
15:0	R	MEMORY_CRC	After Memory CRC check is activated, the register stores the CRC value of the current registers' value. Please refer to register map, which registers participate in memory CRC check.

MFR_OTP_MEM_CHK (DAh)

Format: Unsigned binary

The MFR_OTP_MEM_CHK command can configure the work mode of OTP CRC check and Memory CRC check.

Bits	Access	Bit Name	Description
7:3	R	RESERVED	Reserved.
2	R/W	MFR_OTP_MEM_CHK	The bit can set whether OTP CRC check and Memory CRC check are continuous check at Normal state. OTP CRC check: If the bit is set to 1, the OTP CRC check is always working when DAh bit1 = 1'b1. If the bit is set to 0, the OTP CRC check will be performed when DAh bit1 is changed from 1'b0 to 1'b1. Memory CRC check: If the bit is set to 1, the Memory CRC check is always working when DAh bit0 = 1'b1. If the bit is set to 0, the Memory CRC check will perform when DAh bit0 is changed from 1'b0 to 1'b1. 1'b1: continuous check after OTP CRC check and Memory CRC check config enable. 1'b0: check once after OTP CRC check and Memory CRC check changed from disable to enable.
1	R/W	MFR_OTP_CRC_CHK_EN	The bit can control whether OTP CRC check is enabled or disabled. 1'b0: Disable 1: Enable
0	R/W	MFR_MEM_CRC_CHK_EN	The bit can control whether Memory CRC check is enabled or disabled. 1'b0: Disable 1'b1: Enable

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MFR_OTP_MEM_STATUS (DBh)

Format: Unsigned binary

The MFR_OTP_MEM_STATUS command can indicate some related faults of OTP CRC and Memory CRC.

Bits	Access	Bit Name	Description
7:6	R	RESERVED	Reserved.
5	R	STRUP_OTP_CRC_ERR	When MPQ70165FS detect OTP CRC error during 2nd Restore OTP, the bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No start up load OTP CRC error 1'b1: Start up load OTP CRC error
4	R	STRUP_OTP_IND_ERR	There are 3 bits inside the MPQ70165FS to indicate the usage of OTP. 3 'b000 means no page is programmed. 3' b001 means internal trim page is programmed. 3 'b011 means trim page and 1st user page are programmed. 3 'b111 for trim page, 1st user page and 2nd user page are both burned. On 2nd Restore OTP, if MPQ70165FS detects anything other than the four case above, the bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No start up load OTP indicator error 1'b1: Start up load OTP indicator error
3	R	STORE_OTP_ERR	When MPQ70165FS has one or two OTP chance and then host send Store User All completed, if the 3 bits OTP indicator don't be changed, the bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No store OTP fail 1'b1: Store OTP fail
2	R	BIST_CHK_MEM_CRC_ERR	When MPQ70165FS detect Memory CRC error during Normal state, the bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No BIST check memory CRC error 1'b1: BIST check memory CRC error
1	R	BIST_CHK_OTP_CRC_ERR	When MPQ70165FS detect OTP CRC error during Pre-Diagnostic and Normal state, the bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No BIST check OTP CRC error 1'b1: BIST check OTP CRC error
0	R	BIST_CHK_OTP_IND_ERR	There are 3 bits inside the MPQ70165FS to indicate the usage of OTP. 3 'b000 means no page is programmed. 3' b001 means internal trim page is programmed. 3 'b011 means trim page and 1st user page are programmed. 3 'b111 for trim page, 1st user page and 2nd user page are both burned. On OTP CRC check (during Pre-diagnostic and Normal state), if MPQ70165FS detects anything other than the four case above, the bit will be set to 1. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No BIST check OTP indicator error 1'b1: BIST check OTP indicator error

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STATUS_FREQ (E7h)

Format: Unsigned binary

The STATUS_FREQ command can indicate clock check fault.

Bits	Access	Bit Name	Description
7:3	R	RESERVED	Reserved.
2	R	REG_W_CHK_ERR	When a written data of register isn't same as the intended written value, MPQ70165FS will report register write check error. ALERT# is pulled down. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No register write check error occurred 1'b1: A register write check error occurred
1	R	RESERVED	Reserved.
0	R	CLK_CHK_ERR	When MPQ70165FS internal clock error counter is equal to 8, the device will report clock check fault and the bit is set to 1. PG and ALERT# are pulled down, all channels are shutdown. Clear the fault flag is only through writing 03h register without data at anyone available page. 1'b0: No clock check error occurred 1'b1: A clock check error occurred

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APPLICATION INFORMATION

Figure 33 shows the MPQ70165FS's typical application circuit.

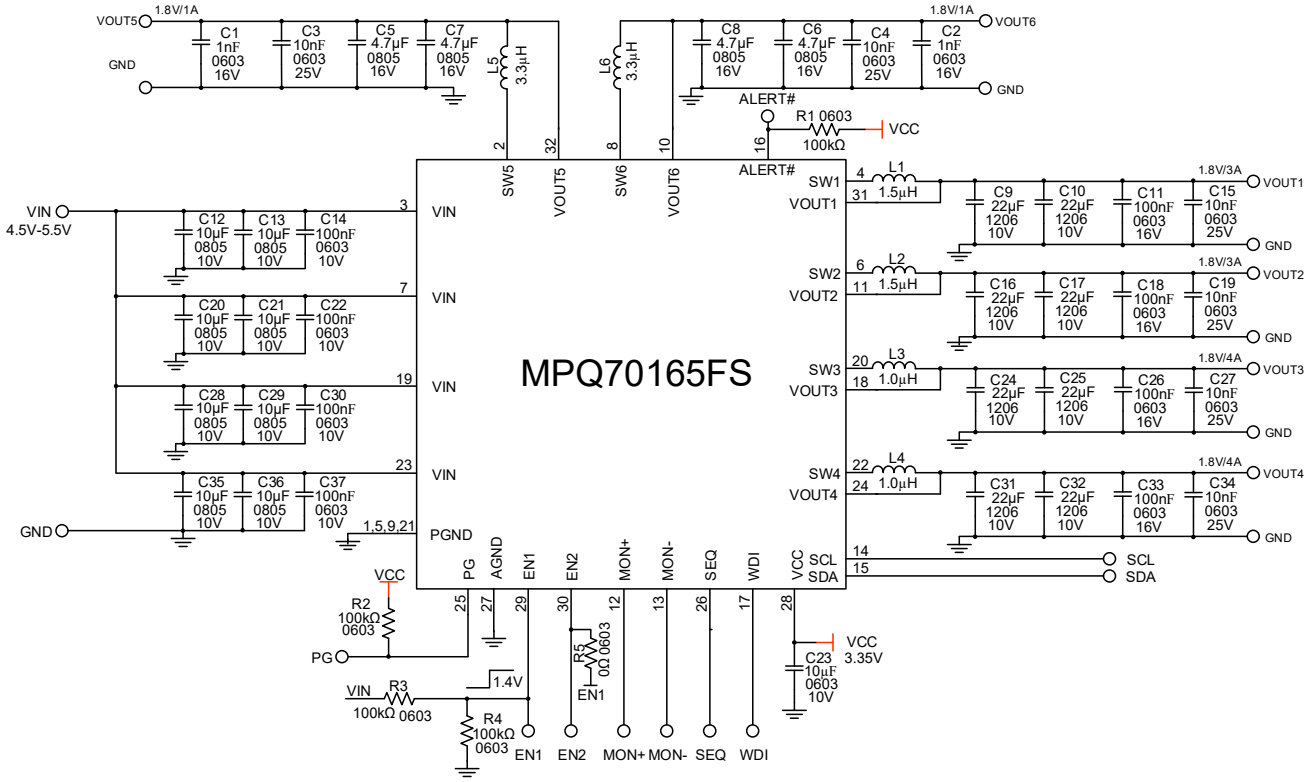


Figure 33: Typical Application Circuit (V_{OUT1-6} = 1.8V, f_{SW} = 2MHz)

Table 4: Design Guide Index

Pin #	Pin Name	Component	Design Guide Index
1,5,9,21	PGND	-	GND Connection (PGND, Pins 1,5,9 and 21; AGND, Pin27)
2	SW5	L5, C1, C3, C5, C7	Selecting the Output Capacitors (SW1, Pin 4; SW2, Pin 6; SW3, Pin20; SW4, Pin22; SW5, Pin2; SW6, Pin8); Selecting the Inductors (SW1, Pin 4; SW2, Pin 6; SW3, Pin20; SW4, Pin 22; SW5, Pin2; SW6, Pin8)
3,7,19,23	VIN	C12, C13, C14, C20, C21, C22, C28, C29, C30, C35, C36, C37	Selecting the Input Capacitors (VIN, Pins 3,7,19 and 23)
4	SW1	L1, C9, C10, C11, C15	Selecting the Output Capacitors (SW1, Pin 4; SW2, Pin 6; SW3, Pin 20; SW4, Pin 22; SW5, Pin2; SW6, Pin8); Selecting the Inductors (SW1, Pin 4; SW2, Pin 6; SW3, Pin20; SW4, Pin 22; SW5, Pin2; SW6, Pin8)

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Table 4: Design Guide Index *(continued)*

Pin #	Pin Name	Component	Design Guide Index
6	SW2	L2, C16, C17, C18, C19	Selecting the Output Capacitors (SW1, Pin 4; SW2, Pin 6; SW3, Pin 20; SW4, Pin 22; SW5, Pin2; SW6, Pin8); Selecting the Inductors (SW1, Pin 4; SW2, Pin 6; SW3, Pin20; SW4, Pin 22; SW5, Pin2; SW6, Pin8)
8	SW6	L6, C2, C4, C6, C8	Selecting the Output Capacitors (SW1, Pin 4; SW2, Pin 6; SW3, Pin 20; SW4, Pin 22; SW5, Pin2; SW6, Pin8); Selecting the Inductors (SW1, Pin 4; SW2, Pin 6; SW3, Pin20; SW4, Pin 22; SW5, Pin2; SW6, Pin8)
10	VOUT6	-	Setting the Output Voltage (VOUT1, Pin31; VOUT2, Pin11; VOUT3, Pin18; VOUT4, Pin24; VOUT5, Pin32; VOUT6, Pin10)
11	VOUT2	-	Setting the Output Voltage (VOUT1, Pin31; VOUT2, Pin11; VOUT3, Pin18; VOUT4, Pin24; VOUT5, Pin32; VOUT6, Pin10)
12	MON+	-	Monitor Different Input (MON+, Pin12; MON-, Pin13)
13	MON-	-	Monitor Different Input (MON+, Pin12; MON-, Pin13)
14	SCL	-	PMBus Interface (SCL, Pin14; SDA, Pin15)
15	SDA	-	PMBus Interface (SCL, Pin14; SDA, Pin15)
16	ALERT#	R1	Power Good and ALERT# Indicators (ALERT#, Pin16; PG, Pin25)
17	WDI	-	Window Watchdog Input(WDI, Pin17)
18	VOUT3	-	Setting the Output Voltage (VOUT1, Pin31; VOUT2, Pin11; VOUT3, Pin18; VOUT4, Pin24; VOUT5, Pin32; VOUT6, Pin10)
20	SW3	L3, C24, C25, C26, C27	Selecting the Output Capacitors (SW1, Pin 4; SW2, Pin 6; SW3, Pin 20; SW4, Pin 22; SW5, Pin2; SW6, Pin8); Selecting the Inductors (SW1, Pin 4; SW2, Pin 6; SW3, Pin20; SW4, Pin 22; SW5, Pin2; SW6, Pin8)
22	SW4	L4, C31, C32, C33, C34	Selecting the Output Capacitors (SW1, Pin 4; SW2, Pin 6; SW3, Pin 20; SW4, Pin 22; SW5, Pin2; SW6, Pin8); Selecting the Inductors (SW1, Pin 4; SW2, Pin 6; SW3, Pin20; SW4, Pin 22; SW5, Pin2; SW6, Pin8)
24	VOUT4	-	Setting the Output Voltage (VOUT1, Pin31; VOUT2, Pin11; VOUT3, Pin18; VOUT4, Pin24; VOUT5, Pin32; VOUT6, Pin10)
25	PG	R2	Power Good and ALERT# Indicators (ALERT#, Pin16; PG, Pin25)
26	SEQ	-	Sequence Output (SEQ, Pin26)
27	AGND	-	GND Connection (PGND, Pins 1,5,9 and 21; AGND, Pin27)

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Table 4: Design Guide Index *(continued)*

Pin #	Pin Name	Component	Design Guide Index
28	VCC	C23	Internal VCC (VCC,Pin28)
29	EN1	R3, R4, R5	Enable Group (EN1, Pin29; EN2, Pin30)
30	EN2	R3	Enable Group (EN1, Pin29; EN2, Pin30)
31	VOUT1	-	Setting the Output Voltage (VOUT1, Pin31; VOUT2, Pin11; VOUT3, Pin18; VOUT4, Pin24; VOUT5, Pin32; VOUT6, Pin10)
32	VOUT5	-	Setting the Output Voltage (VOUT1, Pin31; VOUT2, Pin11; VOUT3, Pin18; VOUT4, Pin24; VOUT5, Pin32; VOUT6, Pin10)

Setting the Output Voltage (VOUT1, Pin31; VOUT2, Pin11; VOUT3, Pin18; VOUT4, Pin24; VOUT5, Pin32; VOUT6, Pin10) (16)

OTP registers VOUT_COMMAND and VOUT_SCALE_LOOP set the output voltage. While V_{OUT_SL} = 1, buck3 / 4 output voltage range is from 0.20625V to 1.8V, and buck1 / 2 / 5 / 6 output voltage range is from 0.20625V to 40%(16) x V_{IN}. While V_{OUT_SL} = 2, buck3 / 4 output voltage range is from 0.4125V to 3.6V, and buck1 / 2 / 5 / 6 output voltage range is from 0.4125V to 40%(16) x V_{IN}. Please refer to the Register Map section for detail setting method.

Write the VOUT_CMD [7:0] in register VOUT_COMMAND, and the VOUT_SC_L [0] in register VOUT_SCALE_LOOP. Output voltage can be calculated with Equation (1):

$$V_{OUT}(mV) = (VOUT_CMD*6.25+206.25)*VOUT_SL \quad (1)$$

Selecting the Input Capacitor (VIN, Pins 3,7,19 and 23)

The input current to the step-down converter is discontinuous and therefore requires a capacitor to supply AC current to the converter while maintaining the DC input voltage. For the best performance, use low ESR capacitors. Ceramic capacitors with X5R or X7R dielectrics are highly recommended because of their low ESR and small temperature coefficients.

It is strongly recommended to use another lower-value capacitor (e.g.: 0.1µF) with a small package size (0603) to absorb high-frequency switching noise. Place the smaller capacitor as close to VIN and GND as possible.

Since C_{IN} absorbs the input switching current, it requires an adequate ripple current rating. The RMS current in the input capacitor can be estimated with Equation (2):

$$I_{CIN} = I_{LOAD} \times \sqrt{\frac{V_{OUT}}{V_{IN}} \times (1 - \frac{V_{OUT}}{V_{IN}})} \quad (2)$$

The worst-case condition occurs at VIN = $2V_{OUT}$, shown in Equation (3):

$$I_{CIN} = \frac{I_{LOAD}}{2} \quad (3)$$

For simplification, choose an input capacitor with an RMS current rating greater than half of the maximum load current.

The input capacitor can be electrolytic, tantalum, or ceramic. When using electrolytic or tantalum capacitors, add a small, high-quality ceramic capacitor (e.g.: 0.1µF) as close to the IC as possible. When using ceramic capacitors, ensure that they have enough capacitance to provide a sufficient charge to prevent excessive voltage ripple at the input. The input voltage ripple caused by the capacitance can be estimated with Equation (4):

$$\Delta V_{IN} = \frac{I_{LOAD}}{f_{SW} \times C_{IN}} \times \frac{V_{OUT}}{V_{IN}} \times (1 - \frac{V_{OUT}}{V_{IN}}) \quad (4)$$

Selecting the Output Capacitor (SW1, Pin 4; SW2, Pin 6; SW3, Pin20; SW4, Pin 22; SW5, Pin2; SW6, Pin8)

The output capacitor maintains the DC output voltage. Use ceramic, tantalum, or low-ESR electrolytic capacitors. For best results, use low ESR capacitors to keep the output voltage ripple low. The output voltage ripple can be estimated with Equation (5):

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$$\Delta V_{OUT} = \frac{V_{OUT}}{f_{SW} \times L} \cdot (1 - \frac{V_{OUT}}{V_{IN}}) \cdot (R_{ESR} + \frac{1}{8 f_{SW} \times C_{OUT}}) \quad (5)$$

Where L is the inductor value, and R_{ESR} is the equivalent series resistance (ESR) value of the output capacitor.

For ceramic capacitors, the capacitance dominates the impedance at the switching frequency and causes the majority of the output voltage ripple.

For simplification, the output voltage ripple can be estimated with Equation (6):

$$\Delta V_{OUT} = \frac{V_{OUT}}{8 \times f_{SW}^2 \times L \times C_{OUT}} \times (1 - \frac{V_{OUT}}{V_{IN}}) \quad (6)$$

For tantalum or electrolytic capacitors, the ESR dominates the impedance at the switching frequency. For simplification, the output ripple can be approximated with Equation (7):

$$\Delta V_{OUT} = \frac{V_{OUT}}{f_{SW} \times L} \times (1 - \frac{V_{OUT}}{V_{IN}}) \times R_{ESR} \quad (7)$$

The characteristics of the output capacitor also affect the stability of the regulation system. The MPQ70165FS can be optimized for a wide range of capacitance and ESR values.

Selecting the Inductor (SW1, Pin 4; SW2, Pin 6; SW3, Pin20; SW4, Pin 22; SW5, Pin2; SW6, Pin8)

A 0.22μH to 3.3μH inductor with a DC current rating at least 25% higher than the maximum load current is recommended for most applications. For higher efficiency, choose an inductor with a lower DC resistance. A larger-value inductor results in less ripple current and a lower output ripple voltage, but also has a larger physical size, higher series resistance, and lower saturation current. A good rule for determining the inductor value is to allow the inductor ripple current to be approximately 30% of the maximum load current. The inductance value can then be calculated with Equation (8):

$$L = \frac{V_{OUT}}{f_{SW} \times \Delta I_L} \times (1 - \frac{V_{OUT}}{V_{IN}}) \quad (8)$$

Where ΔI_L is the peak-to-peak inductor ripple current.

Choose the inductor ripple current to be approximately 30% of the maximum load current. The maximum inductor peak current can be calculated with Equation (9):

$$I_{LP} = I_{LOAD} + \frac{V_{OUT}}{2 f_{SW} \times L} \times (1 - \frac{V_{OUT}}{V_{IN}}) \quad (9)$$

Internal VCC (VCC, Pin28)

The VCC capacitor (C23) is recommended to be 10μF.

Most of the internal circuitry is powered by the internal 3.35V VCC regulator. This regulator uses VIN as its input and operates across the full V_{IN} range. When V_{IN} exceeds 3.35V, VCC is in full regulation. When V_{IN} drops below 3.35V, the VCC output degrades.

Monitor Different Input (MON+, Pin12; MON-, Pin13)

Monitor over-voltage and under-voltage for external voltages with programmable threshold. OTP registers **VMON_LOW_THRESHOLD**, **VMON_HIGH_THRESHOLD** and **MON_RANGE** can set the monitor range. ±2.5% monitoring accuracy is guaranteed when V_{MON} threshold is not lower than 1.3V. If V_{MON} threshold is below of 1.3V, the ±32.5mV max error is allowed.

When EN1 and EN2 are both low, if MON+ or MON- voltage exceeds the EN on logic threshold and V_{IN} is ready, V_{CC} will be ready and then enable digital block. And state machine will stop in start-up state if no fault happens. But bucks don't start-up and also not timing turn on delay time. SEQ and VMON function don't be activated and also not timing turn on delay time.

PMBus Interface (SCL, Pin14; SDA, Pin15)

The MPQ70165FS works as a slave-only device which supports both the standard mode (100kbit/s) and fast mode (400kbit/s) bidirectional data transfer, adding flexibility to the power supply solution. Refer to the PMBus Interface section for details.

The SCL and SDA lines are externally pulled to a bus voltage with resistor (e.g. 1.5kΩ).

Power Good and ALERT# Indicators (ALERT#, Pin16; PG, Pin25)

The PG and ALERT pins' structure is same with integrate an internal, open-drain N-channel

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Mosfet. Through a pulled-up resistance (e.g. 100kΩ) is connected to bus supply (e.g. V_{CC}). If the bus supply is other power supply rail, its voltage doesn't exceed current V_{CC}. PG and ALERT have different behaviors for different faults (see Table 1 on page 39).

Window Watchdog Input (WDI, Pin17)

The WDI pin is an input pin. In the window watchdog mode, the MCU will periodically send a pulse to this pin. MPQ70165FS judges whether the MCU software program is normal by detecting the width of the pulse and the sending time. Refer to the Watchdog Operation section for details.

When EN1 and EN2 are both low, if WDI voltage exceeds the EN on logic threshold and V_{IN} is ready, V_{CC} will be ready and then enable digital block. And state machine will stop in start-up state if no fault happens. But bucks don't start-up and also not timing turn on delay time. SEQ and VMON function don't be activated and also not timing turn on delay time.

Sequence Digital Output (SEQ, Pin36)

SEQ is an output pin. After the PRE-DIAG is completed, follow the configured EN signal. If the EN signal is high, the SEQ pin will also be pulled high after the T_{SEQ_DELAY} time has elapsed. The delay time can be set by the SEQ_DELAY register.

When EN1 and EN2 are both low, if SEQ voltage exceeds the EN on logic threshold and V_{IN} is ready, V_{CC} will be ready and then enable digital block. And state machine will stop in start-up state if no fault happens. But bucks don't start-up and also not timing turn on delay time. SEQ and VMON function don't be activated and also not timing turn on delay time.

Enable Group (EN1, Pin29; EN2, Pin30)

MPQ70165FS provides two EN pins. The Buck_x, SEQ and Monitor functions that can be controlled by a certain EN pin. The OTP register **ENABLE_GROUP** can set different configuration for actual demand.

GND Connection (PGND, Pins 1, 5, 9 and 21; AGND, Pin27)

See the PCB Layout Guidelines for more details.

PCB Layout Guidelines [17]

Efficient PCB layout, especially for input capacitor placement, is critical for stable operation. A four-layer layout is strongly recommended to achieve better thermal performance. For best results, refer to Figure 28 and follow the guidelines below.

1. Place input capacitors as close to VIN and GND as possible.
2. Use a large ground plane to connect to PGND directly.
3. Add vias near PGND if the bottom layer is a ground plane.
4. Ensure that the high-current paths at GND and VIN have short, direct, and wide traces.
5. Place the ceramic input capacitor, especially the small package size (0603) input bypass capacitor, as close to VIN and PGND as possible to minimize high frequency noise.
6. Keep the connection of the input capacitor and VIN as short and wide as possible.
7. Place the VCC capacitor as close to VCC and AGND as possible.
8. Route SW away from sensitive analog areas, such as FB.
9. Ensure that the trace between FB and output is as short as possible.
10. Use multiple vias to connect the power planes to the internal layers.
11. Connect all VIN through vias before input capacitors.

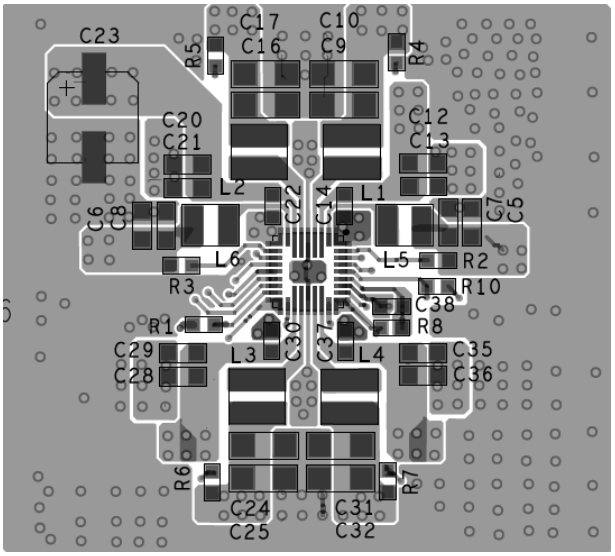
Notes:

18) Buck 1 / 5 share the same VIN. When their HS-FET on time overlaps, it may cause SW unstable. The phase shifts 180° between Buck1 and Buck5, so their duty cycle shall be smaller than 40%. If Buck1 or Buck5 is disabled, another buck can ignore the limitation. The same goes for Buck2 / 6.

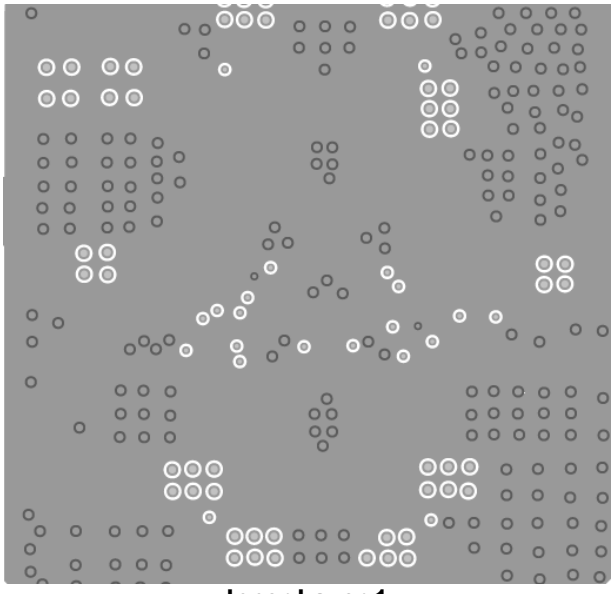
19) The recommended PCB layout is based on Figure 33.

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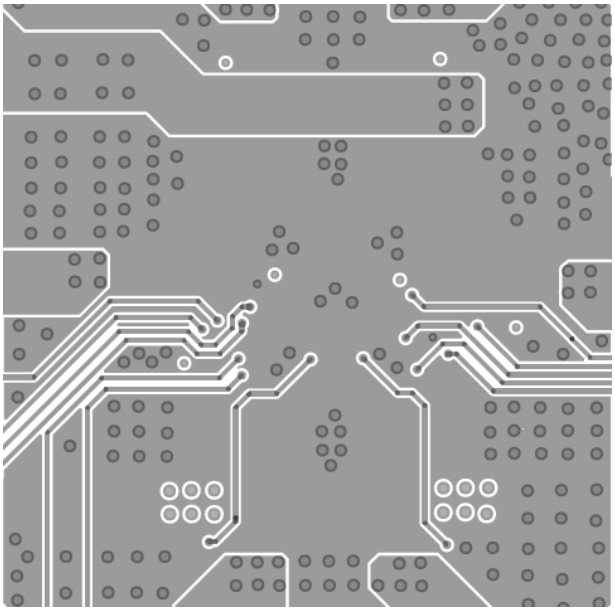
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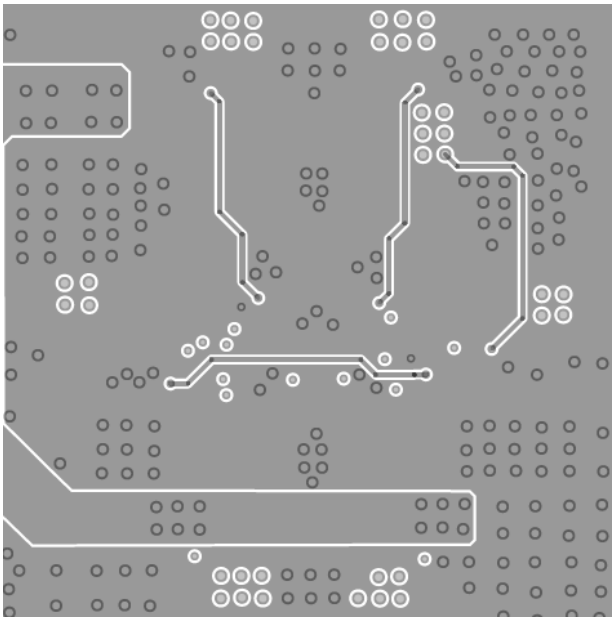
Top Layer



Inner Layer 1



Inner Layer 2



Bottom Layer

Figure 34: Recommended PCB Layout

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TYPICAL APPLICATIONS

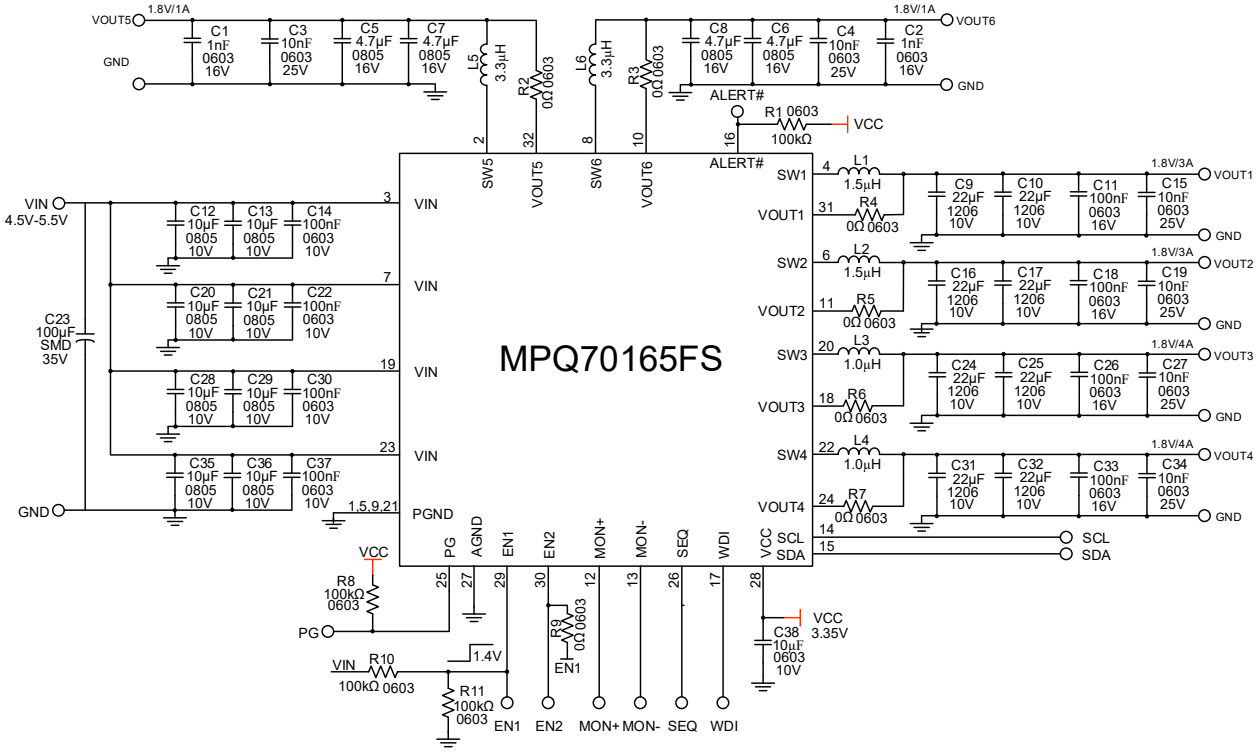


Figure 35: V_{OUT1-6} = 1.8V, f_{SW} = 2MHz of Typical Application Circuit

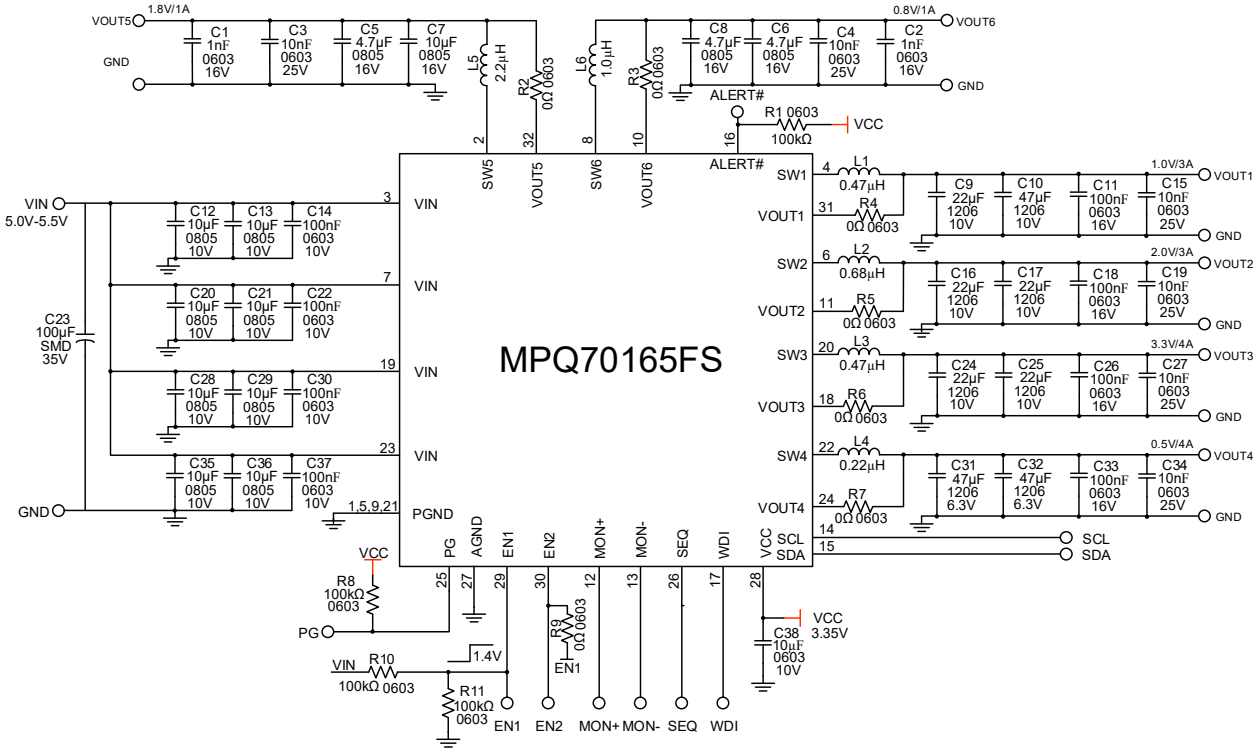


Figure 36: V_{OUT1} = 1.0V, V_{OUT2} = 2.0V, V_{OUT3} = 3.3V, V_{OUT4} = 0.5V, V_{OUT5} = 1.8V, V_{OUT6} = 0.8V, $f_{SW1,2,3,5,6}$ = 2MHz, f_{SW4} = 1.3MHz, of Typical Application Circuit

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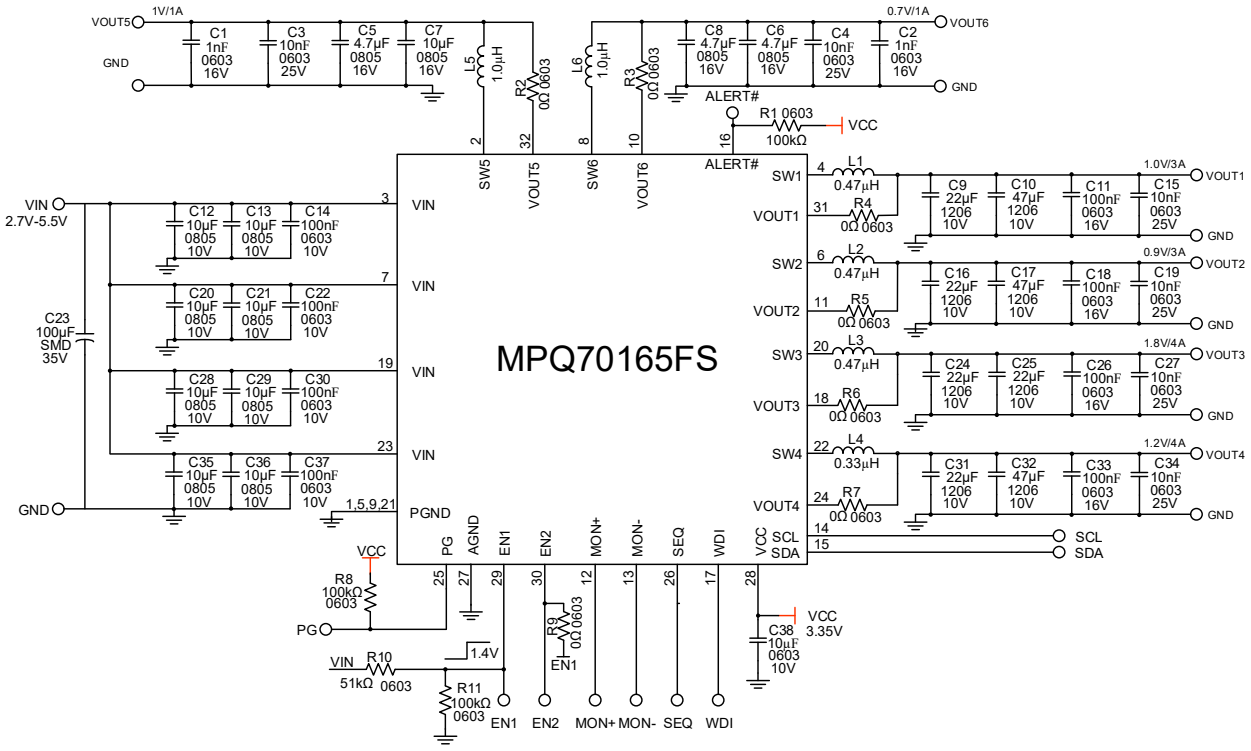


Figure 37: V_{OUT1} = 1.0V, V_{OUT2} = 0.9V, V_{OUT3} = 1.8V, V_{OUT4} = 1.2V, V_{OUT5} = 1.0V, V_{OUT6} = 0.7V, $f_{SW1,2,3,4,5,6}$ = 2MHz, of Typical Application Circuit

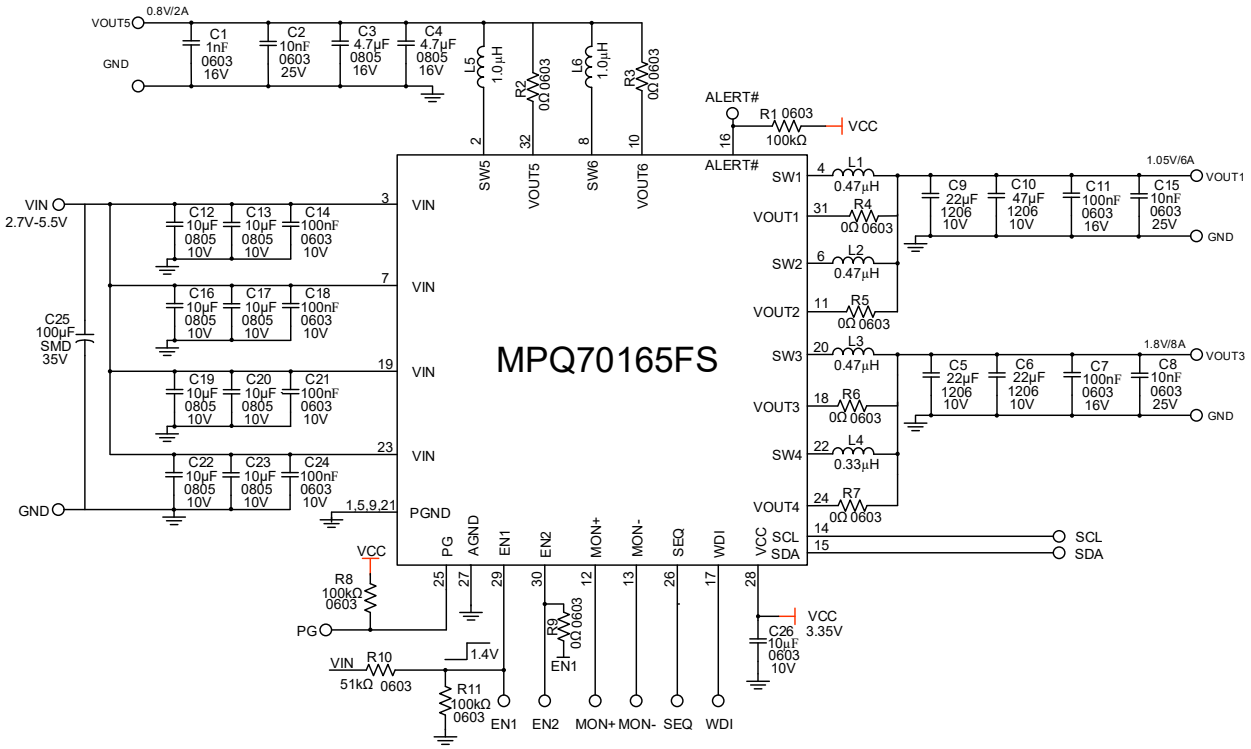


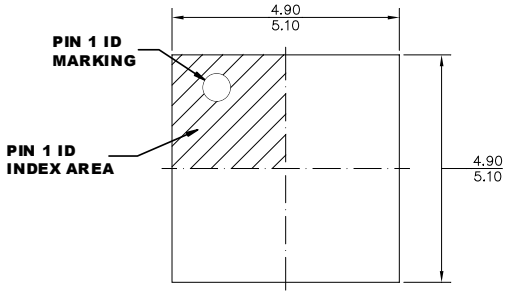
Figure 38: $V_{OUT1/2}$ = 1.05V, $V_{OUT3/4}$ = 1.8 V, $V_{OUT5/6}$ = 0.8V, $f_{SW1,2,3,4,5,6}$ = 2MHz, Multiphase Operation of Typical Application Circuit

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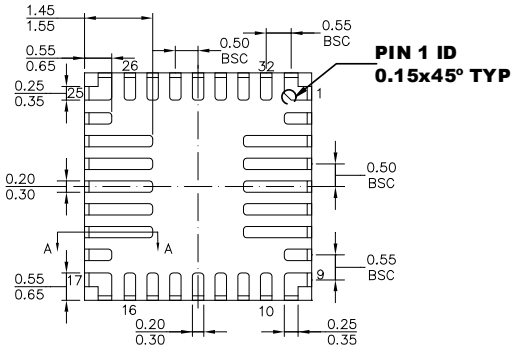
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PACKAGE INFORMATION

TQFN-32 (5mmx5mm)
Wettable Flank



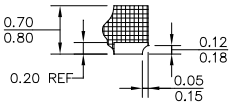
TOP VIEW



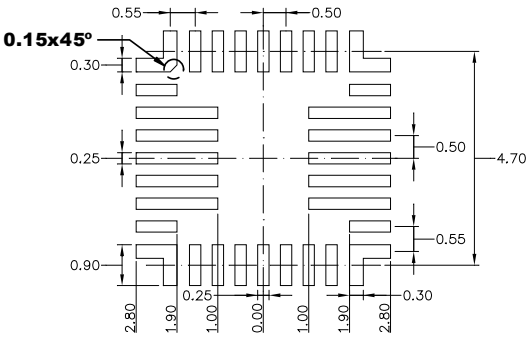
BOTTOM VIEW



SIDE VIEW



SECTION A-A



RECOMMENDED LAND PATTERN

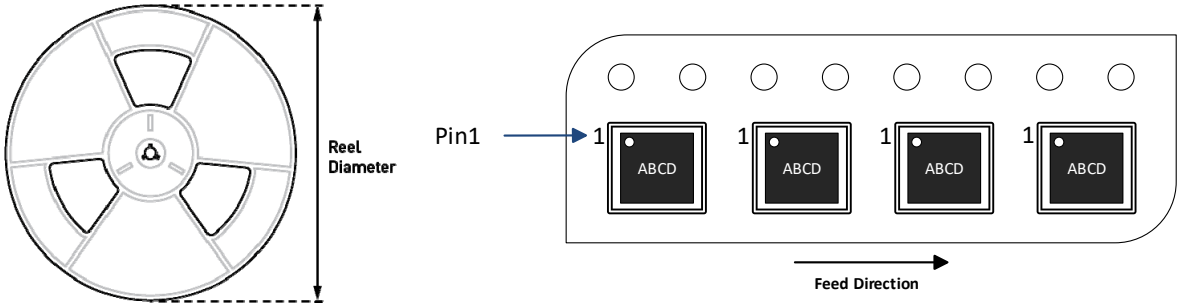
NOTE:

1) THE LEAD SIDE IS WETTABLE.
2) ALL DIMENSIONS ARE IN MILLIMETERS.
3) LEAD COPLANARITY SHALL BE 0.08 MILLIMETERS MAX.
4) JEDEC REFERENCE IS MO-220.
5) DRAWING IS NOT TO SCALE.

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CARRIER INFORMATION



Part Number	Package Description	Quantity/ Reel	Quantity/ Tube	Quantity/ Tray	Reel Diameter	Carrier Tape Width	Carrier Tape Pitch
MPQ70165FSGUTE -xxxxAEC1-Z	TQFN-32 (5mmx5mm)	5000	N/A	N/A	13 in.	12 mm	8 mm

* N/A indicates “not available” in tubes. For 500 pieces tape & reel prototype quantities, see factory. (Order code for 500 pieces partial reel is “-P”, tape & reel dimensions same as full reel.)

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